

ENERGY IN IRELAND

2025 Report





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Sustainable Energy Authority of Ireland

SEAI is the expert authority on decarbonising Ireland's energy system, using robust, objective data and an evidence-based approach to help inform policy development and energy user behaviours. SEAI empowers households, businesses, motorists and the public sector to move away from fossil fuels with practical advice and financial supports. We encourage positive changes in how Ireland uses energy to deliver a more prosperous, energy efficient and cleaner environment.

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Foreword

Understanding how and where we use energy is essential to supporting our ambitions for Ireland. It's at the heart of how we achieve a healthier, secure and more competitive economy and society – one that's powered by clean, affordable energy.

SEAI publishes Energy in Ireland each year to provide the definitive data on Ireland's energy supply, demand and related emissions. This publication provides a look back on 2024, with provisional estimates for 2025. This year's data are particularly significant as we are now drawing to the end of Ireland's first carbon budget, our national commitment to a pathway of reduced emissions. With over 20 years of data now reported by SEAI, this information helps us understand if we are on track in meeting those commitments and allows us to look at trends in performance and achievements, as we move through the mid-point of this crucial decade.



Since 2018, the baseline year for the national carbon budgets, total energy related emissions have fallen by 16%, demonstrating that we are making some progress while also growing our population and economy. It shows that significant reductions are possible when backed by strong policy, investment and public support and participation working together. But it also tells us that we need to pick up the pace and do a lot more. Importantly we haven't broken the link between economic development and fossil fuels in a structural, meaningful way yet and that is a concern.

Ireland's energy emissions are falling at a rate that is short of what is needed. But the insights from this report show us what is possible once we put our shoulder to the wheel. In 2024, we had 41% of our electricity provided by wind, solar and other renewables. We're making progress, but we must accelerate delivery of renewables and major infrastructure such as grid enhancements, public transport and district heating – piping heat from waste and renewable source into people's homes and businesses. We also need to ensure that we are strategic with energy consumption and growth in our national decision-making. This is critical if we are to make headway with efforts to reduce our energy demand overall and increase the share of that demand met through renewable technology.

Insights from the report highlight that we have seen growth across a range of renewable technologies. Ireland's heat pumps delivered more renewable energy than all its solar farms and rooftop solar PV combined, and that's while also recognising the rapid increase in solar PV generation. This is a significant step forward for heat pump technology which has such an important role to play in meeting our heat energy needs.

We saw growth in biofuel blending at garage forecourts meaning less emissions from petrol and diesel cars, but we still need a significant shift away from private car use to more walking and cycling and public transport. This shift will be easier for some than others. It is critically important that we drive ahead with projects such as local bus, train and commuter links and the long-awaited Metrolink.

Our journey requires effort from everyone, but it will be worth it. From Government delivering policy, supports and investment, to industry, commercial and public sectors being strategic with their energy demand growth and ensuring it is efficient and met with renewables, to each and every one of us supporting renewable and infrastructural projects for the greater good and making energy efficient

decisions. We all have a part to play. A just transition and helping those least able to make the necessary changes, must continue to be central to everything we do.

If we succeed, people will be living in warmer, more comfortable homes. They will commute with ease, walking and cycling in cleaner air or through electrified transport to more energy efficient workplaces. Our businesses will be more competitive with all of society holding more control and certainty over energy costs.

At SEAI, we support householders, businesses and communities who want to reduce their reliance on fossil fuel every year. We will continue to do all we can to support you to play your part in moving to clean, more secure energy. Our future together will be bright, with so many benefits for our lives, as we all work together on this movement. There are so many reasons now to make a change. We haven't a moment to lose.

William Walsh, CEO

Contents

Acknowledgements.....	2
Foreword	3
Contents	5
Technical Highlights	9
1 Ireland's Key Energy Trends in 2024 and 2025.....	10
1.1 Energy-Related Emissions	11
1.2 Supply and Demand of Energy	15
1.2.1 Energy Supply Trends	15
1.2.2 Energy Demand Trends	18
1.3 Supply and Demand of Electricity	21
1.3.1 Electricity Supply Trends.....	21
1.3.2 Electricity Demand Trends	28
1.4 Transport Energy Demand	30
1.5 Heat Demand	33
1.5.1 Overall Heat Demand	33
1.5.2 Residential Heat Demand.....	35
1.6 Ireland's EU Energy Targets.....	38
1.6.1 Renewable Energy Directive Targets.....	38
1.6.2 Energy Efficiency Directive Targets	39
2 Overview of energy data	41
2.1 The National Energy Balance	41
2.2 Additional data.....	43
2.3 Data on energy-related emissions	44
3 Primary energy	45
3.1 Primary energy production	45
3.2 Primary energy imports and exports.....	48
3.3 Primary energy supply and requirement.....	49
3.4 Security of supply and import dependency	52
4 Energy transformation	54
4.1 Overview of transformation.....	54

4.2	Electricity generation and supply	55
4.2.1	Energy input to electricity generation.....	55
4.2.2	Gross electricity production from generation	58
4.2.3	Electricity imports and exports.....	65
4.2.4	Gross electricity supply.....	65
4.2.5	Energy storage	69
4.2.6	Combined heat and power generation	71
4.2.7	Solar, wind and hydro capacity.....	74
4.3	Oil refining	79
4.4	Other transformation processes	80
4.4.1	Biomethane in natural gas network	81
4.4.2	Peat briquetting	81
5	Final energy	83
5.1	Final energy consumption by energy type	83
5.2	Final energy consumption by sector	87
5.2.1	Final energy consumption in industry	89
5.2.2	Final energy consumption in transport	92
5.2.3	Final energy consumption in residential sector	100
5.2.4	Final energy consumption in services.....	103
5.2.5	Final energy consumption in agriculture and fisheries	108
6	Energy modes	110
6.1	Final energy consumption by mode	110
6.2	Heat mode	112
6.2.1	Final energy in heat mode by sector.....	112
6.2.2	Final energy in heat mode by fuel.....	113
6.3	Transport mode	116
6.3.1	Final energy in transport mode by subsector	116
6.3.2	Final energy use in transport mode by energy type.....	117
6.4	Electricity mode.....	119
6.4.1	Final energy use in electricity mode by sector	119
6.5	Primary energy requirement by mode	120
7	Energy-related greenhouse gas (GHG) emissions	122
7.1	GHG intensity of electricity total consumption.....	122

7.2	Energy-related GHG emissions by sector	123
7.2.1	Industry emissions	126
7.2.2	Transport emissions.....	127
7.2.3	Residential emissions.....	129
7.2.4	Commercial and public services emissions	130
7.3	Energy-related GHG emissions by mode	131
7.4	Avoided emissions from renewable energy	131
8	Energy targets and policy	135
8.1	Greenhouse gas emissions	135
8.1.1	Greenhouse gas emissions reductions targets	136
8.1.2	Carbon budgets and sectoral emission ceilings.....	140
8.2	Renewable energy targets.....	144
8.2.1	Renewable Energy Directive and targets.....	144
8.2.2	Biomass sustainability	145
8.2.3	RES Results	146
8.2.4	Electricity from renewable energy sources (RES-E).....	151
8.2.5	Heat from renewable energy sources (RES-H)	153
8.2.6	Transport energy from renewable sources (RES-T).....	155
8.3	Energy prices to industry.....	160
9	Drivers of energy demand	165
9.1	Energy, economy and emissions	165
9.2	Energy and the weather.....	167
9.3	Economic energy intensities	168
10	Further sectoral analysis	170
10.1	Primary energy requirement by sector	170
10.2	Industry energy intensity.....	171
10.3	Transport	172
10.3.1	Private car activity.....	172
10.3.2	CO ₂ intensity of new private cars	173
10.3.3	Penetration of zero emissions vehicles.....	174
10.3.4	Heavy goods vehicle activity	175
10.4	Residential	178
10.4.1	Weather correction	178

10.4.2	Energy consumption per dwelling.....	179
10.5	Commercial and public services.....	181
10.5.1	Energy intensity of the commercial and public services sector	181
10.5.2	Public sector developments.....	182
11	Provisional energy data from monthly surveys	184
11.1	Monthly utility scale electricity data to 2025.....	184
11.1.1	Seasonality in utility scale monthly electricity supply.....	185
11.1.2	Monthly utility scale electricity generated	190
11.1.3	Monthly electricity imported and exported in 2025	193
11.2	Gas.....	194
11.2.1	Seasonality in monthly gas supply	194
11.2.2	Monthly gas supply in 2025	196
11.3	Oil and liquid biofuels	198
11.3.1	Seasonality and COVID impacts in monthly oil and road liquid biofuel deliveries.....	199
11.3.2	Monthly oil and road biofuel supply in 2025	204
	Appendix 1: List of figures	206
	Appendix 2: List of tables	211
	Appendix 3: Glossary of abbreviations	213
	Appendix 4: Glossary of terms	215
	Appendix 5: Energy units and conversion factors	217
	Energy conversion factors.....	217
	Energy units.....	217
	Decimal prefixes.....	217
	Calorific values.....	217
	Appendix 6: Data sources	219
	Appendix 7: Bibliography	220
	Appendix 8: Energy statistics revisions	223
	Appendix 9: Electricity and gas price tables	225
	Appendix 10: Energy in Ireland data and charts.....	231
	Appendix 11: Version control	235

Technical Highlights

Energy-related Emissions

- Ireland's national energy-related emissions in 2024 were at their lowest level in over 30 years.
- Energy-related emissions in 2024 were 30.9 MtCO₂eq, down 1.5% on 2023-levels.
- Electricity accounted for 22.5% of energy emissions, transport for 37.7%, and heat for 39.8%.
- 2024 electricity emissions were 6.9 MtCO₂eq, the lowest on record, down 8.3% on 2023-levels.
- 2024 transport emissions were 11.7 MtCO₂eq, down 1.2% on 2023-levels.
- 2024 heat emissions were 12.3 MtCO₂eq, up 2.4% on 2023-levels.
- Energy-related emissions in 2024 were down 16% on 2018-levels, the carbon budget baseline.

Renewable Energy

- 14.6% of Ireland's primary energy was renewable in 2024 – the highest value to date.
- Ireland used 1.31 TWh more renewable energy in 2024 than in 2023.
- Just under three-quarters of renewable energy in 2024 came from wind, biodiesel, and biomass.
- Solar PV generation was up 69% and ambient heat capture from heat-pumps was up 19% in 2024.
- In 2024, Ireland's RES-Overall result was 16.1%, up from 15.2% in 2023 – the highest value to date.
- Ireland's RES-Overall result is now above its 2020 16% baseline target, with a 2030 target of 43%.

Transport Energy

- Overall, 93.0% of Ireland's transport energy came from fossil fuels in 2024.
- Road and air accounted for 75.7% and 21.5% of transport energy demand, respectively.
- Energy demand from private car use increased by 3.2% in 2024.
- Energy demand from HGV road freight increased by 7.9% in 2024.
- Energy demand from both private cars and HGVs in 2024 was higher than pre-COVID 2019-levels.
- 9.0% of energy in road diesel, and 6.0% in road petrol, came from renewable biofuels in 2024.

Electricity Supply

- Wind (41.8%), gas (32.1), and net imports (13.9%) led Ireland's gross electricity supply in 2024.
- Fossil generation accounted for 45.3% of electricity supply in 2024 - the lowest value to date.
- Ireland's 2024 RES-Electricity result was 41.3%, up from 40.4% in 2023 – the highest value to date.
- Ireland's RES-Electricity target for 2030 is 80%.

Electricity Demand

- Electricity demand in Ireland was 32.9 TWh in 2024, up 4.1% on 2023-levels
- Commercial services, which includes the ICT sub-sector, accounted for 41.2% of electricity demand.
- The residential sector accounted for 25.5% of electricity demand in 2024.
- Data centres accounted for 21.2% of all electricity demand in 2024.
- Data centres account for 88.2% of the increase observed in Ireland's electricity demand since 2015.

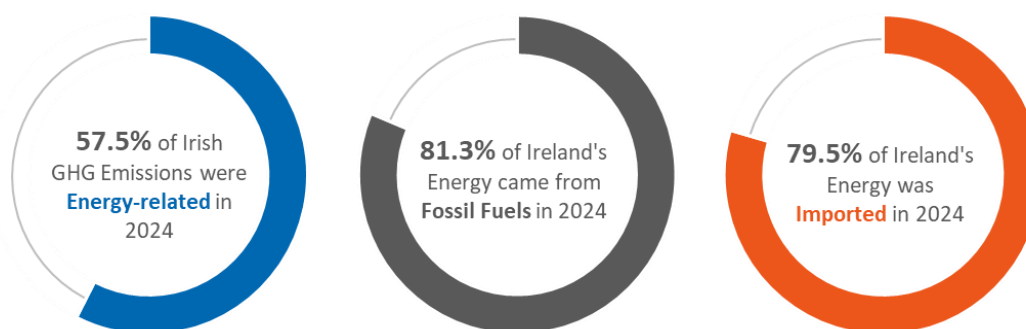
Heat Demand

- Heat demand accounted for 35.1% of Ireland's energy demand in 2024.
- Ireland's heat demand in 2024 was 50.6TWh, up 4.7% on 2023-levels.
- The residential (44.1%) and industry (36.5%) sectors accounted for over 80% of heat demand.
- In 2024, 89.8% of heat demand was satisfied using fossil fuels.
- Ireland's residential heat demand in 2024 was 22.3 TWh, up 6.0% on the previous year.

1 Ireland's Key Energy Trends in 2024 and 2025

SEAI's *Energy in Ireland 2025* report delivers a comprehensive statistical analysis of Ireland's energy system, covering supply, transformation, and end-use demand. It provides clear quantitative breakdowns of the fuels and economic sectors driving energy-related emissions, the composition of Ireland's energy mix, and the supply-streams satisfying our national energy requirements.

Figure 1.1: Key high-level insights on Ireland's greenhouse gas (GHG) emissions, the national energy mix, and Ireland's energy import dependency



The publication of this year's *Energy in Ireland* report comes at a key moment in Ireland's energy transition:

1. **National Carbon Budgets** - December 2025 marks the end of Ireland's first national carbon budget (2021–2025), with clear sectoral emissions ceilings (SECs) for energy-related emissions in electricity, transport, built environment, etc.
2. **Renewable Energy Directive Targets** - the end of 2025 is the precise midpoint of the 2020–2030 decade in which Ireland has committed to raising its renewable energy share to 43.0%, in line with the requirements of the third EU Renewable Energy Directive (RED-III).
3. **Energy Efficiency Directive Targets** – from the end of 2025, Ireland has five years to make sincere cooperative efforts to limit its total final energy consumption to 10.451 Mtoe by 2030, under Article 4 of the EU Energy Efficiency Directive (EED) – a reduction of nearly 15% compared to 2024 levels.

SEAI's *Energy in Ireland 2025* report delivers both high-level summaries and detailed data. **Section 1** provides a summary of key energy trends and observations from 2024 and 2025, blending definitive annual data from the 2024 national energy balance with provisional monthly data from January to September 2025. **Sections 2 to 11** and accompanying **Appendices** provide more detailed quantitative descriptions of Ireland's primary energy, energy transformations, final energy demand, energy-related emissions, as well as technical details on active energy targets and policy, glossaries, conversion factors, etc.

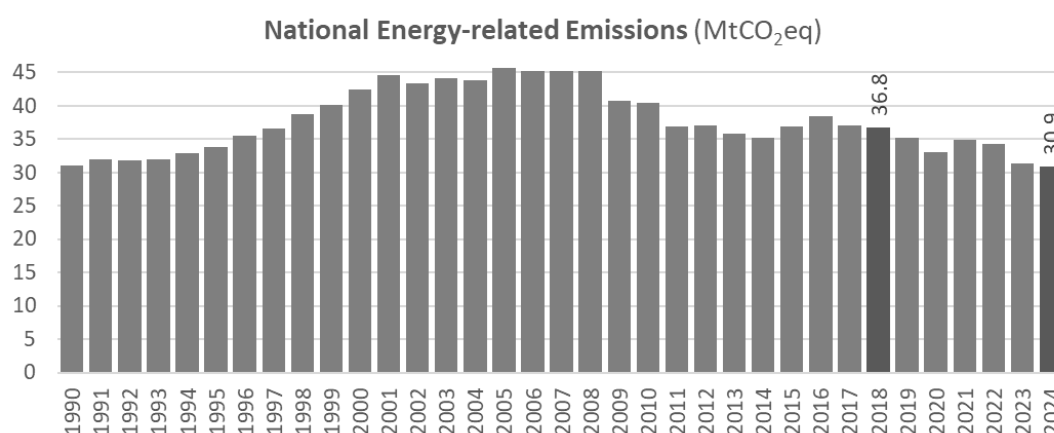
As introduced last year, the *Energy in Ireland 2025* report now quantifies energy-related emissions data in terms of mega tonnes of carbon dioxide equivalent (MtCO₂eq). This ensures strong alignment between SEAI's reporting of energy-related emissions and the SECs in the national carbon budgets, and to the EPA's Greenhouse Gas (GHG) Emissions Inventory, compiled in accordance with IPCC guidelines. Emissions quantified in units of MtCO₂eq provide a fuller assessment of global warming potential, because they capture the impact of methane and nitrous oxide, as well as those from carbon dioxide.

1.1 Energy-Related Emissions

Ireland's national energy-related emissions in 2024 were 30.9 MtCO₂eq, down 1.5% on the previous year. Ireland's national energy-related emissions, as reported in the EPA's Greenhouse Gas (GHG) Inventory and as used in the carbon budgets, do not include emissions associated with international aviation or international maritime transport¹.

Ireland's national energy-related emissions in 2024 were at a record low - their lowest level in over 30 years. Energy-related emissions in 2024 were down 16.0% on 2018 levels, the base year for the calculation of the national carbon budgets and Ireland's overall 2030 emission reduction target.

Figure 1.2: Time-series of Ireland's national energy-related emissions since 1990



To gain deeper insights, the total energy-related emissions can be broken out into the high-level climate action plan (CAP) sectors of transport emissions and electricity emissions, as those are defined in the national carbon budgets, and all other remaining 'heat' emissions². Noting that while the transport and electricity sectors have well defined sectoral emission ceilings (SECs) within the carbon budgets, there is no explicit target for energy-related emissions or the heat emissions³.

Using this breakout, we see that the largest decrease in annual energy-related emissions between 2018 and 2024 came from electricity. Electricity emissions in 2024 were 6.9 MtCO₂eq, down 3.3 MtCO₂eq or 32.1% on 2018 levels. In contrast, the smallest change in annual energy-related emissions between 2018 and 2024 came from transport. Transport emissions in 2024 were 11.7 MtCO₂eq, down 0.7 MtCO₂eq or 5.3% on 2018 levels.

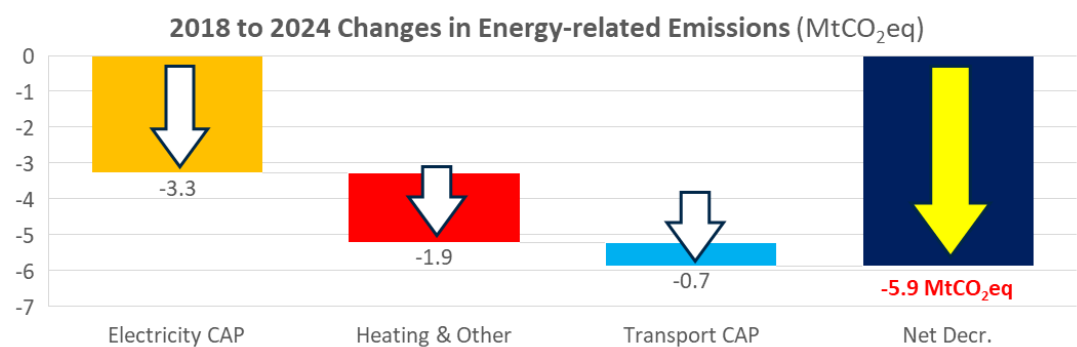
While overall energy-related emissions have fallen steadily in recent years, SEAI's National Energy Projections report 2025 indicates that the pace of these reductions is not yet sufficient to deliver on Ireland's legally binding carbon budgets, even under the most optimistic scenarios considered under SEAI's modelling.

¹ These international emissions are calculated and reported separately in accordance with guidance from United Nations Framework Convention on Climate Change (UNFCCC) and the Intergovernmental Panel on Climate Change (IPCC).

² All other remaining 'heat' emissions are defined as total energy-related emissions, less electricity emissions, less transport emissions.

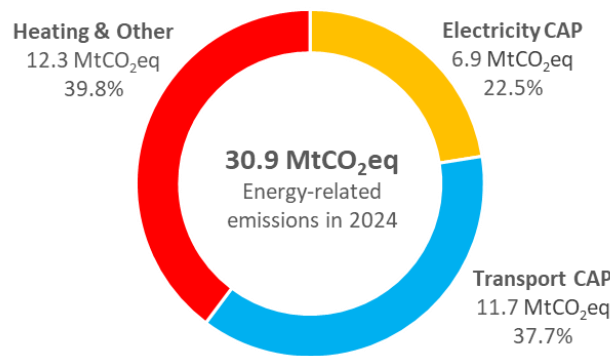
³ As per the EPA's GHG inventory, electricity sector emissions under the carbon budgets are given by the sum of emissions from (1) public electricity and heat production (IPCC code 1.A.1.a), (2) solid fuel and other energy industries (IPCC code 1.A.1.c), and (3) fugitive emissions (IPCC code 1.B.1 & 1.B.2). As a result, the electricity emission in the carbon budgets include no emissions from auto-producers (including CHP plants), but do include the emissions associated with peat-briquetting, and fugitive emissions, such as the emissions associated with the production, processing, transmission, and storage of natural gas, etc.

Figure 1.3: Waterfall plot showing 2018 to 2024 changes in national energy-related emissions



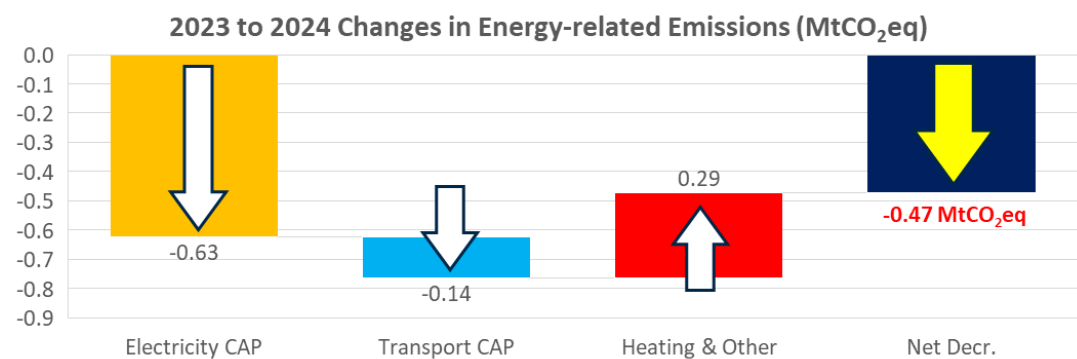
The high-level breakout of Ireland’s national energy-related emissions in 2024 are 6.9 MtCO₂eq (22.5%) from electricity, 11.7 MtCO₂eq (37.7%) from transport, and 12.3 MtCO₂eq (39.8%) from heat.

Figure 1.4: Breakout of 2024 national energy-related emissions across high-level sectors



In absolute terms, national energy-related emissions in 2024 were down 0.47 MtCO₂eq or 1.5% on the previous year. However, not all energy-related emissions decreased in 2024. While electricity emissions decreased by 0.63 MtCO₂eq (-8.3%), and transport emissions decreased by 0.14 MtCO₂eq (-1.2%) in 2024, heat emissions increased by 0.29 MtCO₂eq (+2.4%). For more detail on changes in heat demand, see sections 1.5 and 6.2. For more details on energy-related emissions, see section 7.

Figure 1.5: Waterfall plot showing 2023 to 2024 changes in national energy-related emissions

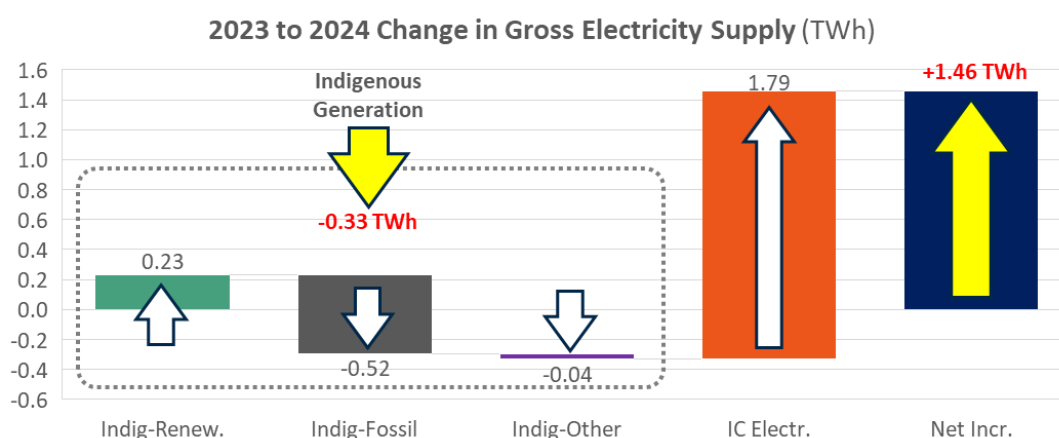


The largest contributor to decreased energy-related emissions in 2024 was the decrease observed in electricity emissions. Despite an increase in electricity demand in 2024, electricity emissions decreased to the lowest level on record. Electricity emissions in 2024 reached this record low due to several factors:

1. An overall decrease in indigenous generation of electricity in Ireland.
2. A decrease in indigenous generation of electricity from fossil fuels.
3. An increased use of net-imported electricity across international interconnectors with Northern Ireland and Great Britain, the emissions from which are not countable towards Ireland's national emissions.

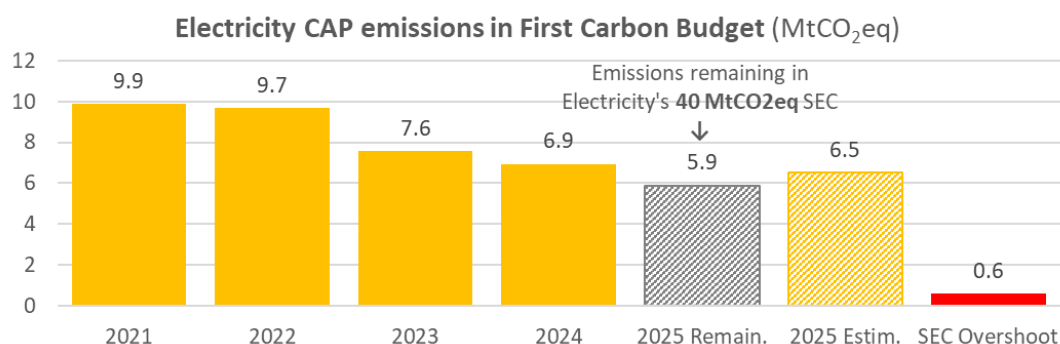
Ireland's gross electricity supply, which is the electricity supply needed to deliver on end-user electricity demand, increased by 1.46 TWh or 4.2% in 2024. However, Ireland's indigenous generation of electricity decreased by 0.33 TWh or 1.0% in 2024. Ireland made increased use of net imports across international interconnectors (IC Electr.) in 2024 to increase supply, with net imports of electricity up 1.79 TWh or 54.5% compared to the previous year. The generation of electricity from renewables in Ireland increased by 0.23 TWh in 2024, while the generation of electricity in Ireland from fossil fuels decreased by 0.52 TWh. It is this decrease in generation from fossil fuels that resulted in the decrease in electricity emissions.

Figure 1.6: Waterfall plot showing 2023 to 2024 changes in Ireland's gross electricity supply



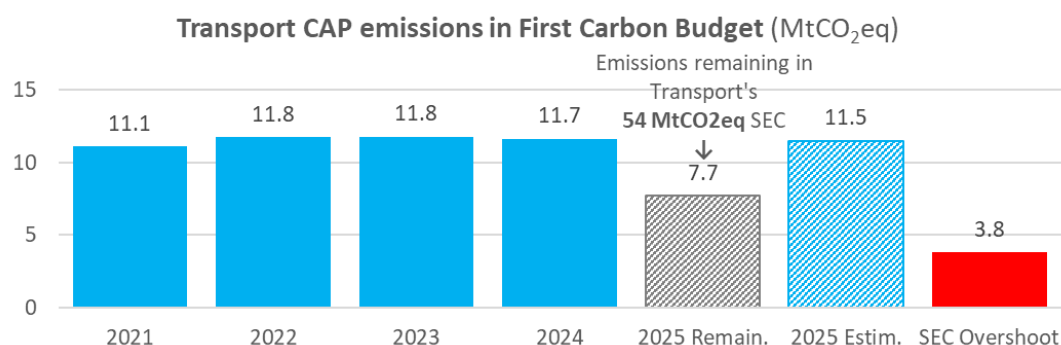
Ireland's electricity emissions under the national carbon budgets are officially determined by the Environment Protection Agency (EPA) through the Greenhouse Gas (GHG) Inventory. The EPA have reported electricity GHG emissions of 9.9 MtCO₂eq, 9.7 MtCO₂eq, 7.6 MtCO₂eq and 6.9 MtCO₂eq for the first 4-years of the first carbon budget (2021-2025), respectively. This leaves 5.9 MtCO₂eq of 'allowable' emissions for electricity in the first carbon budget.

Based on its monthly surveys of energy suppliers, SEAI can make a provisional extrapolation of Ireland's electricity emissions to the end of 2025. This best estimate indicates 6.5 MtCO₂eq of electricity emissions in 2025, which would be down 6% on 2024 levels.

Figure 1.7: Reported and estimated electricity sector emissions in the first carbon budget

If SEAI's estimate for 2025 proves accurate, then the electricity sector will overshoot its sectoral emission ceiling (SEC) of 40 MtCO₂eq by 0.6 MtCO₂eq of emissions. For reference, the electricity sector's SEC in the second carbon budget (2026-2030) is 20 MtCO₂eq – precisely half of that in the first budget – which is equivalent to a nominal annual emission of just 4.0 MtCO₂eq.

The EPA have reported transport GHG emissions of 11.1 MtCO₂eq, 11.8 MtCO₂eq, 11.8 MtCO₂eq and 11.7 MtCO₂eq for the first 4-years of the first carbon budget (2021-2025), respectively. This leaves 7.7 MtCO₂eq of 'allowable' emissions for transport in the first carbon budget. Based on its monthly surveys of energy suppliers, SEAI can make a provisional extrapolation of Ireland's transport emissions to the end of 2025. This best estimate indicates 11.5 MtCO₂eq of transport emissions in 2025, which would be down 1% on 2024 levels.

Figure 1.8: Reported and estimated transport sector emissions in the first carbon budget

If SEAI's estimate for 2025 proves accurate, then the transport sector will overshoot its SEC of 54 MtCO₂eq by 3.8 MtCO₂eq of emissions. For reference, the transport sector's SEC in the second carbon budget is 37 MtCO₂eq, which is equivalent to nominal annual emissions of just 7.4 MtCO₂eq.

While SEAI can make extrapolations of electricity and transport emissions to the end of 2024, it cannot confidently generate an estimate for heat emissions in 2025. This uncertainty comes from the unknown impact of the Q4-2025 heating season on demand, and the relatively poor rate of responses that SEAI receives from its monthly surveys to solid fuel suppliers. Once its annual survey returns from suppliers are received, and public administrative data from the EPA, the Central Statistics Office (CSO), and Revenue Commissioners, becomes available, SEAI will determine Ireland's 2025 heat emissions and total energy-related emissions.

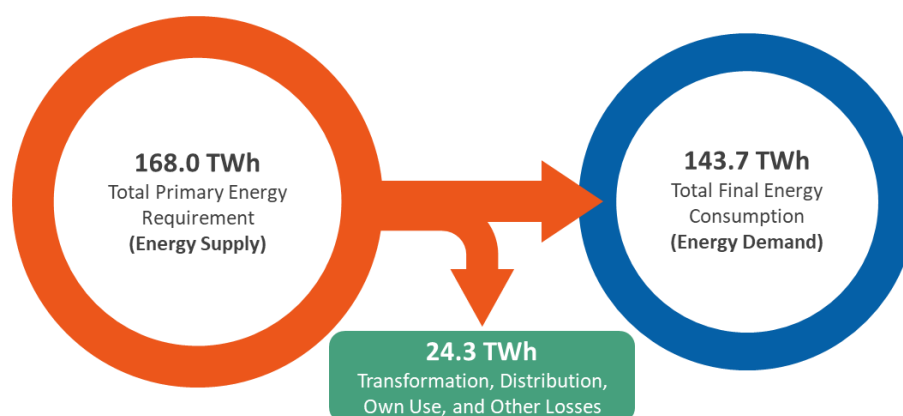
For more details on the national carbon budgets and SECs, see section 8.1.2.

1.2 Supply and Demand of Energy

Ireland's total primary energy requirement in 2024 was 168.0 TWh, up 2.6% on the previous year. Total primary energy requirement can be thought of as **Ireland's energy supply**. It is the amount of 'raw' energy that Ireland needs to satisfy end-user demand across the economy. It includes all the energy that goes directly to end-users, as well as crude oil that is first processed in Ireland into oil-products, like diesel and petrol used on our roads, and the gas and coal fed into power plants to generate electricity.

Ireland's total final energy consumption in 2024 was 143.7 TWh, up 2.4% on the previous year. Total final energy consumption can be thought of as **Ireland's energy demand**. It is the amount of energy directly consumed by end-users in the Irish economy. It excludes all transformation, distribution, and other losses associated with the processing and delivery of primary energy to final end-users.

Figure 1.9: Illustration of the relationship between Ireland's total primary energy requirement (i.e. energy supply) and Ireland's total final energy consumption (i.e. energy demand)

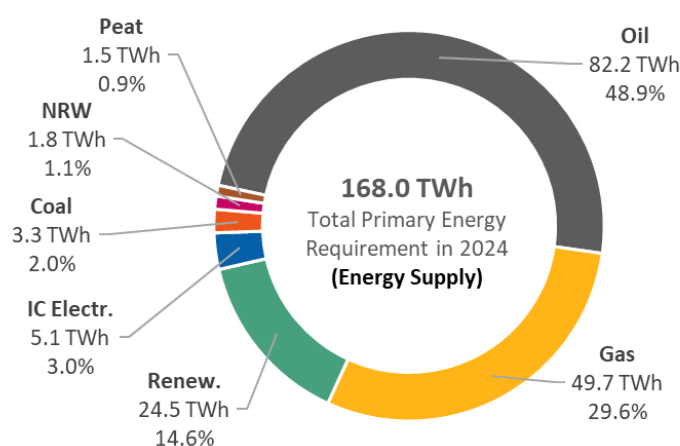


Understanding the distinction between energy supply and energy demand, and how they influence each other, helps interpret the dynamics of Ireland's energy system and provides a more complete picture of the national energy landscape.

1.2.1 Energy Supply Trends

Ireland's 168.0 TWh energy supply in 2024 remains dominated by fossil fuels. Fossil fuels accounted for 81.3% of the national energy supply in 2024. However, the contribution of fossil fuels to the supply mix is falling slowly - down from 97.2% in 2004 and 90.5% in 2014. In 2024, the three largest energy products contribution to the national energy supply were oil (48.9%), gas (29.6%), and renewable energy (14.6%).

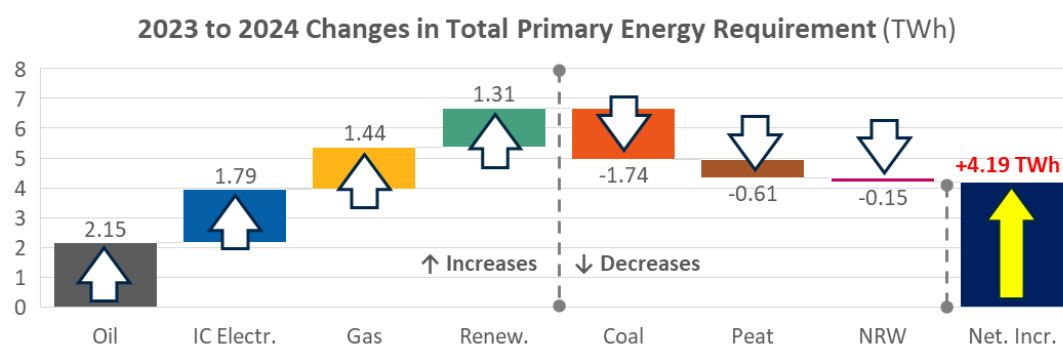
Figure 1.10: Breakout of 2024 energy supply across energy products. The term 'NRW' stands for non-renewable wastes



In absolute terms, Ireland's energy supply in 2024 was 4.19 TWh higher than the previous year. There were increases in oil, net imports of electricity across international interconnectors (IC Electr.), gas, and renewable energy. These increases were coupled with decreases in coal, peat, and non-renewable wastes (NRW).

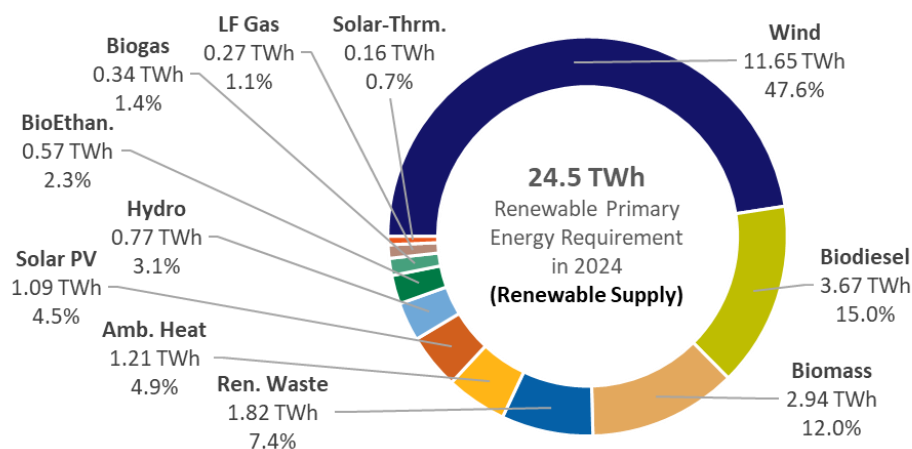
In 2024, both the use of oil and gas in Ireland's energy supply increased faster than its renewable energy supply. The 1.31 TWh increase in renewable energy in 2024 accounted for only 31.2% of the net increase in energy supply, with the remaining two-thirds of the net increase satisfied mainly through increased use of net imports of electricity across international interconnectors and increased use of fossil fuels.

Figure 1.11: Waterfall plot showing 2023 to 2024 changes in Ireland's energy supply. The term 'NRW' stands for non-renewable wastes



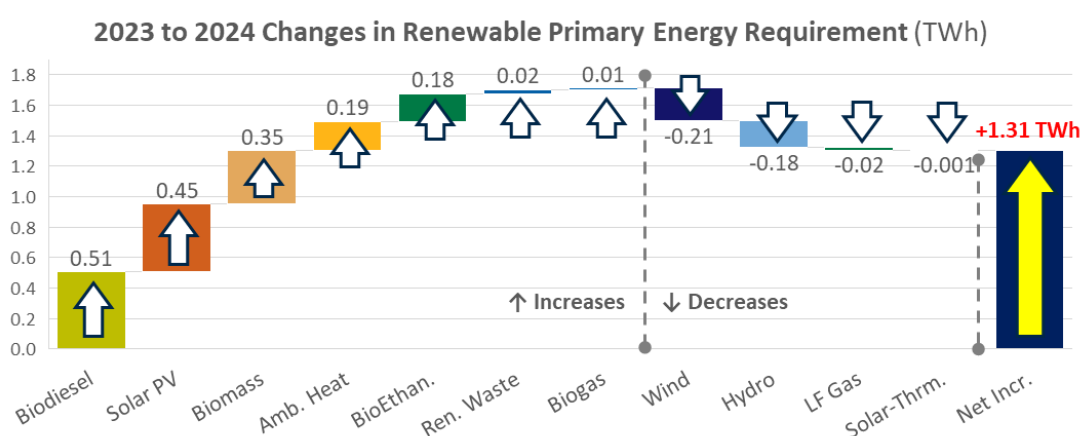
Ireland's renewable energy supply in 2024 was 24.5 TWh, up 5.6% on the previous year. Ireland's renewable energy supply in 2024 was the highest on record – both in absolute and relative terms. For more detailed breakdown of energy supply, see section 3.

Ireland's renewable energy supply in 2024 came from a broad portfolio of technologies and sustainable energy products. The three largest sources of renewable energy supply were wind generation of electricity (47.6%), the biodiesel blended into the 'road diesel' sold in garage forecourts (15.0%), and biomass (12.0%). Together, these three sources account for just under three-quarters of all renewable energy used in Ireland in 2024.

Figure 1.12: Breakout of 2024 renewable energy supply across energy sub-products

The national fleet of heat pumps captured renewable ambient heat accounting for 4.9% of Ireland's renewable energy supply in 2024. In 2024, Ireland's heat pumps delivered more renewable energy than all its solar-farms and rooftop solar PV combined. Heat pumps remain one of the fastest growing sources of renewable energy in Ireland, with ambient heat capture in 2024, up 18.7% on the previous year.

In absolute terms, Ireland's renewable energy supply in 2024 was 1.31 TWh higher than the previous year. There were increases in biodiesel, solar PV, biomass, ambient heat capture from heat-pumps, bioethanol, renewable waste and biogas. These increases were coupled with decreases in hydro generation, wind generation, landfill (LF) gas and solar-thermal heating.

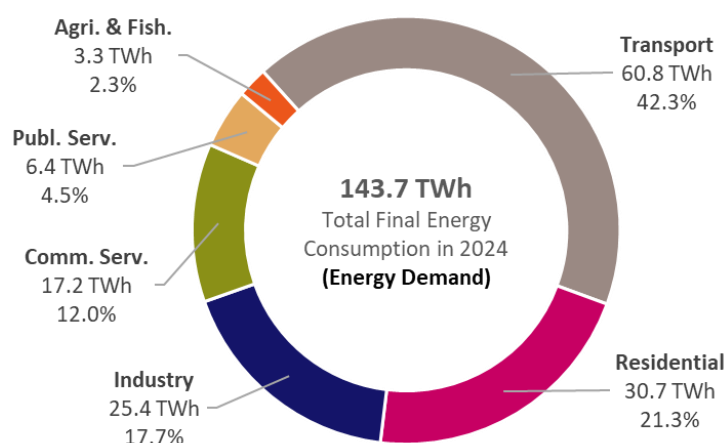
Figure 1.13: Waterfall plot showing 2023 to 2024 changes in Ireland's renewable energy supply

The largest sub-product increase in renewable energy in 2024 came from biodiesel. Ireland's biodiesel supply increased by 0.51 TWh or 16.0% on the previous year, driven by increased biofuel blending into road diesel, which increased from 7.9% in 2023 to 9.0% in 2024. The largest sub-product decrease in renewable energy in 2024 came from wind generation. Ireland's electricity generation from wind decreased by 0.21 TWh or 1.8% on the previous year, despite installed wind capacity increasing. For more detailed breakdowns on wind, solar, and other renewable energy supply streams, see sections 1.3 and 3.

1.2.2 Energy Demand Trends

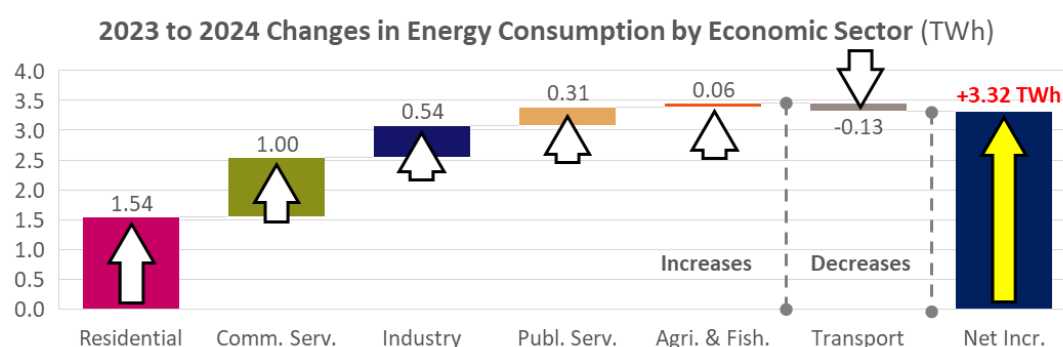
Ireland's total end-user energy demand of 143.7 TWh in 2024 can be broken out into sectoral demand. In 2024, the three economic sectors with greatest energy demand were transport (42.3%), residential (21.3%), and industry (17.7%). Combined, the transport and residential sectors account for almost two-thirds of national energy demand.

Figure 1.14: Breakout of 2024 energy demand across economic sectors



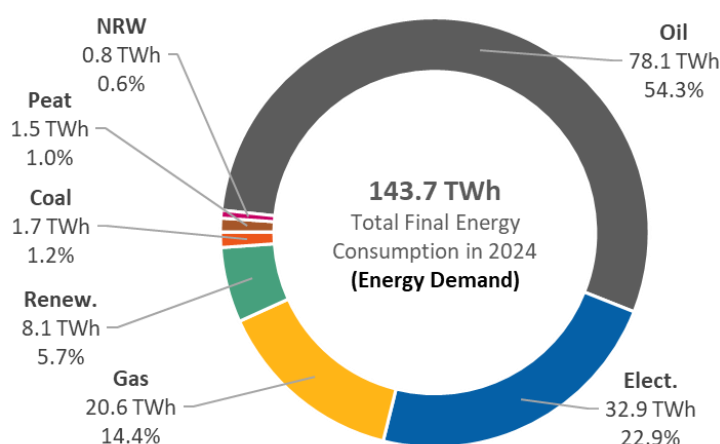
In absolute terms, Ireland's energy demand in 2024 was 3.32 TWh higher than the previous year. There were increases in energy demand from all economic sectors, except transport, which was down 0.13 TWh or 0.2% on the previous year. In 2024, the largest sectoral increase in energy demand came from the residential sector, which increased by 1.54 TWh or 5.3% on the previous year. The second largest came from the commercial services sector, which increased by 1.00 TWh or 6.2% on the previous year.

Figure 1.15: Waterfall plot showing 2023 to 2024 changes in Ireland's energy demand by sector



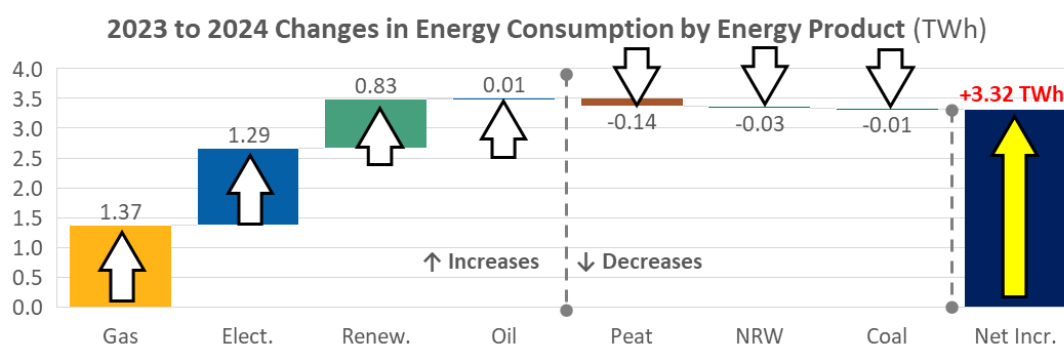
In terms of energy product breakout, just over half (54.3%) of energy demand in 2024 was satisfied through end-user consumption of oil products, mainly petrol, diesel, and heating oils. Electricity accounted for just under a quarter (22.9%) of end-user energy demand in 2024, followed by gas demand (14.4%). For more detailed breakdowns on energy demand by sector, see section 5.2.

Figure 1.16: Breakout of 2024 energy demand across energy products. The term 'NRW' stands for non-renewable wastes



As noted previously, Ireland's energy demand in 2024 was 3.32 TWh higher than the previous year. At the energy product level, there was increased end-user demand for gas, electricity, direct-use renewable energy⁴, and oil. These increases were coupled with decrease demand for peat, non-renewable waste, and coal. For more detailed breakdowns on energy demand by product, see section 5.1.

Figure 1.17: Waterfall plot showing 2023 to 2024 changes in Ireland's energy demand by product. The term 'NRW' stands for non-renewable wastes

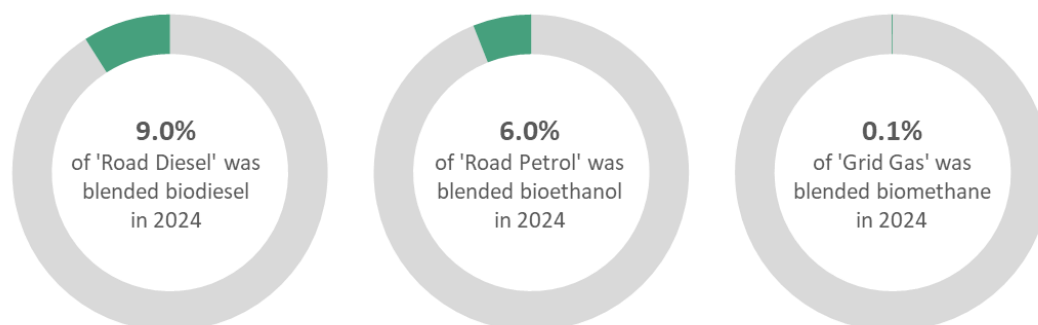


Ireland's end-user demand for renewable energy is satisfied using both 'standalone' renewable energy products, like biomass (e.g. wooden logs and wood pellets), and 'blended' renewable energy products. An example of a blended energy product is 'road diesel', which is the diesel fuel mix available for purchase at garage forecourts across the country. Road diesel is a blend of 100% fossil-based petrochemical diesel and some fraction of sustainable biofuel components, including fatty acid methyl esters (FAME), hydro-treated vegetable oil (HVO), and co-processed HVO. The biofuel blending of these sustainable compounds into road diesel helps reduce transport emissions by substituting-out some fraction of the fossil-based diesel.

⁴ Direct-use renewable energy refers to renewable energy sub-products that are directly consumed by an end-user, such as biomass burned for heating in a home, or biodiesel used as fuel in a vehicle. It specifically excludes the renewable share of electricity consumed by end users, as that is classified under renewable electricity supply rather than direct end-use of renewable energy.

Similarly, Ireland's road petrol also consists of a blend of fossil-based petrol and bioethanol, and our 'grid gas' has recently contained a blend of fossil gas and biomethane, all helping reduce emissions from those fuels. The renewable energy content of road diesel, road petrol, and grid gas in 2024 were 9.0%, 6.0%, and 0.1% respectively.

Figure 1.18: Renewable energy content of road diesel, road petrol, and grid gas in 2024



Biofuel blending into road diesel and road petrol has taken place for over a decade in Ireland, but blending biomethane into grid gas is relatively new. Here, the REPowerEU-funded capital grants scheme, launched by the *Department of Agriculture, Food, and the Marine* in September 2024, aims to support the development of new biomethane plants and the conversion of biogas CHP plants for gas upgrading and grid injection. The scheme is expected to enable 5–7 plants, increasing biomethane capacity by over 0.4 TWh annually in the coming years. It is expected that the biomethane produced will be purchased primarily by suppliers to meet their obligations under the upcoming Renewable Heat Obligation⁵.

The reduction in transport and heating emissions delivered by blending renewable energy into fossil fuels is welcome. However, it is important to acknowledge that 91% of Ireland's road diesel, 94% of its road petrol, and 99.9% of its grid gas still came from fossil fuels in 2024 and acted as considerable sources of Ireland's emissions. For more details on blended road diesel and road petrol, see sections 1.4 and 5.2.2. For more details on biomethane in grid gas, see section 4.4.1.

⁵ www.gov.ie/en/department-of-climate-energy-and-the-environment/publications/renewable-heat-obligation/

1.3 Supply and Demand of Electricity

Electricity accounted for just under a quarter (22.9%) of Ireland's energy demand in 2024, with just over 41% of that electricity coming from verified sustainable renewable sources. Renewable electricity is by far the largest vector of renewable energy into Ireland's homes, offices, and businesses.

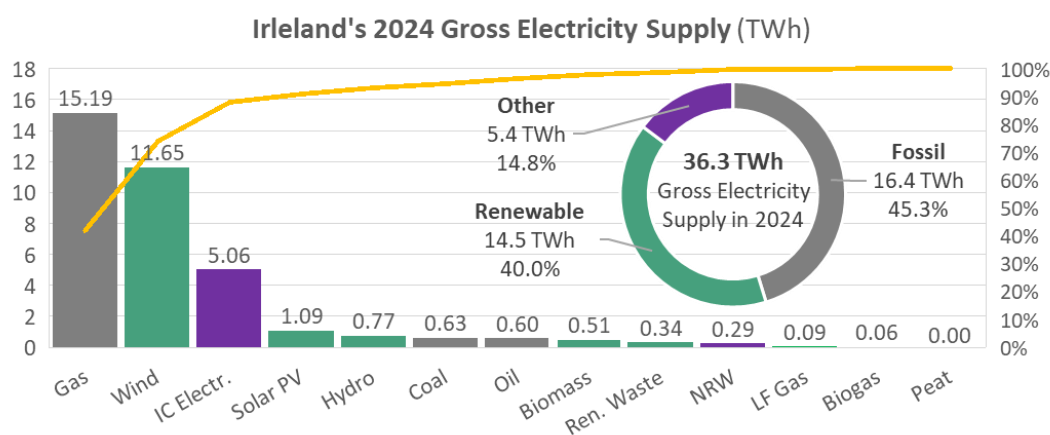
1.3.1 Electricity Supply Trends

Ireland's gross electricity supply in 2024 was 36.3 TWh, up 4.2% on 2023 levels. Gross electricity supply includes all electricity generation by utility-scale power plants (public thermal and CHP) as well as all micro-to-small scale electricity generation, including rooftop solar PV. Gross supply includes the net import of electricity across international interconnectors but excludes the transmission and distribution losses associated with moving electricity across the national grid, and the cycle-losses incurred from energy storage, *i.e.* pumped or battery-storage.

In 2024, 41.8% of Ireland's gross electricity supply came from natural gas, followed by 32.1% from wind generation, and 13.9% from the net-import of electricity from Northern Ireland and Great Britain. Combined these three sources accounted for 87.9% of Ireland's electricity supply.

In total, fossil fuel generation accounted for 45.3% of the electricity supply in 2024 – the lowest level on record - with renewable generation accounting for 40.0% of electricity supply. Other sources, consisting of net-imports of electricity across international interconnectors and generation from non-renewable wastes, accounted for the remaining 14.8%. On a percentage basis, Ireland's electricity supply has never had less fossil fuel generation or used more net-imported electricity.

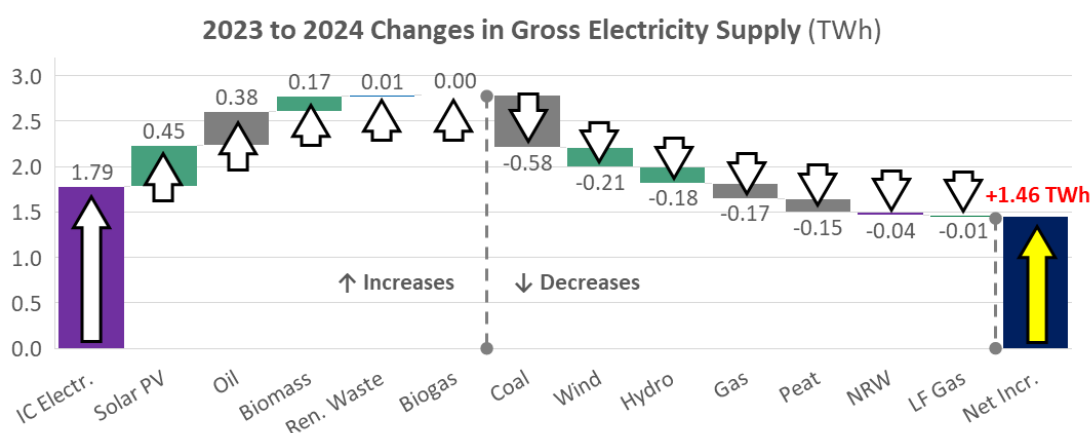
Figure 1.19: Breakout of 2024 gross electricity supply by energy sub-products and type. The 'other' energy type includes non-fossil, non-renewable energy, like non-renewable wastes (NRW) and net imports of electricity across international interconnectors



In absolute terms, Ireland's gross electricity supply in 2024 was 1.46 TWh higher than the previous year. There was a significant increase in net imports across international interconnectors (IC Electr.), and smaller increases in PV, oil, biomass, renewable wastes, and biogas. These increases were coupled with decreases in generation from coal, wind, hydro, gas, peat, non-renewable wastes, and landfill (LF) gas.

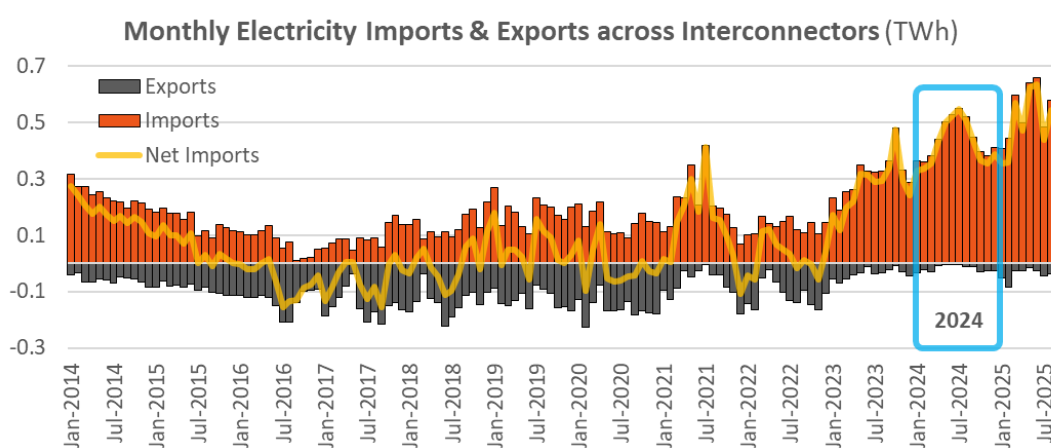
Ireland's renewable electricity supply in 2024 increased by 0.23 TWh or 1.6% in 2024. This net increase in renewable electricity supply from increases in solar PV (+0.45 TWh), biomass (+0.17 TWh), renewable wastes, and biogas, coupled with decreases from wind (-0.21 TWh), hydro (-0.18 TWh), and landfill gas.

Figure 1.20: Waterfall plot showing 2023 to 2024 changes in gross electricity supply by sub-product. The term 'NRW' stands for non-renewable wastes



As noted above, a significant fraction of Ireland's gross electricity supply in 2024 came from net-imported electricity across interconnectors. Based on monthly data from January to September 2025, it is apparent that the fraction of electricity supply from net imports has further increased in 2025. The best estimate from SEAI, based on an extrapolation of January-September data to the end of 2025, is that net-imported electricity may account for over 16% of Ireland's utility scale electricity supply in 2025.

Figure 1.21: Monthly imports, exports, and net imports of electricity across international interconnectors with Northern Ireland and Great Britain



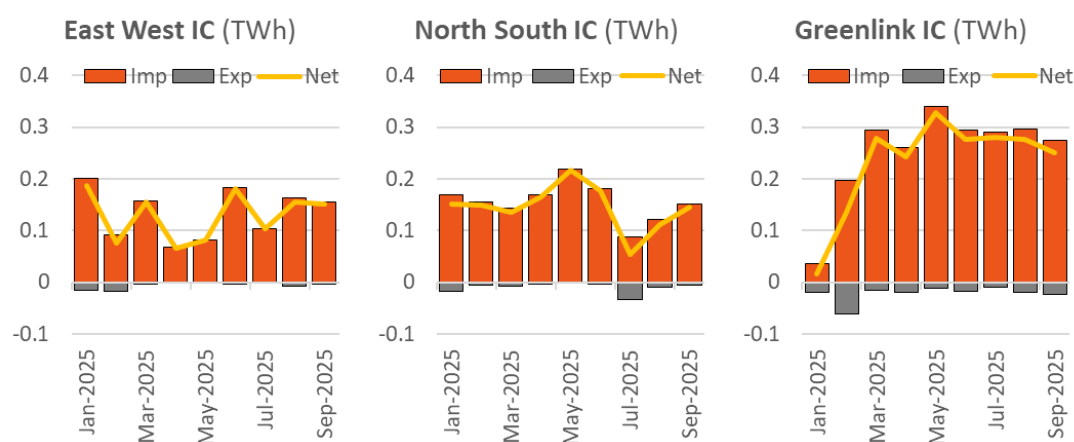
The Greenlink interconnector, connecting Wexford with Wales, entered its 'commercial operations' phase at the beginning of 2025, with capacity available through the implicit SEM-GB 'Intra Day 1 Auction' from 29th January 2025. Since then, Ireland has three international interconnectors to Northern Ireland and Great Britain:

- East-West interconnector
- North-South interconnector⁶
- Greenlink interconnector

⁶ The North-South interconnector is the aggregate of four circuits that connect the two systems.

These interconnectors allow for international trade and balancing of electricity flows to-and-from Ireland and Northern Ireland (within the integrated Single Electricity Market, or I-SEM) and mainland Great Britain (outside the I-SEM)⁷. In the period of January to September 2025, 45.9% of Ireland's net imports of electricity flowed across the Greenlink interconnector, with the remaining 28.7% and 25.4% of net imports flowing across the North-South and East-West interconnectors, respectively.

Figure 1.22: Monthly imports, exports, and net imports of electricity in 2025 across the North-South, East-West, and Greenlink interconnectors

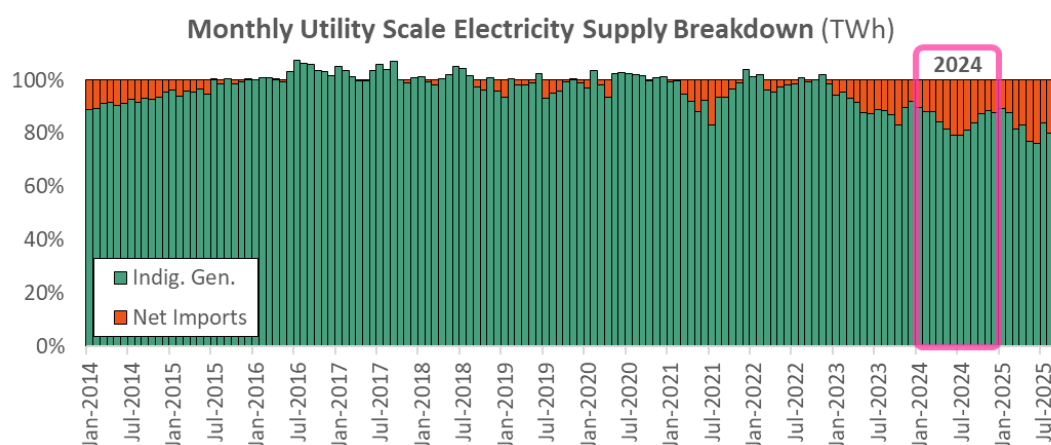


Trends in the relative contribution of net-imported electricity towards Ireland's electricity supply can be more fully explored using utility-scale gross electricity supply data, which is available monthly. Utility-scale gross supply covers electricity from large-scale grid-connected generators and accounts for approximately 95% of all electricity supply, so it is an excellent proxy for total electricity supply.

This monthly data shows that Ireland has consistently made use of relatively high levels of net imports since Q1-2023. In 23 of the 24 calendar months from October 2023 to September 2025, Ireland has net imported 10% or more of its gross electricity supply. In June 2025, net imports accounted for almost one quarter (23.8%) of Ireland's utility-scale gross electricity supply, and in May 2025 net imports accounted for more electricity supply than wind generation.

⁷ The Moyle interconnector connects the I-SEM in Northern Ireland with the mainland Great Britain market. As the endpoints of this interconnector are not within the national boundary of the Republic of Ireland, electricity flows across this interconnector are not explicitly captured in Irish national statistics.

Figure 1.23: Monthly utility scale electricity supply in Ireland broken out by indigenous generation and net imports across international interconnectors

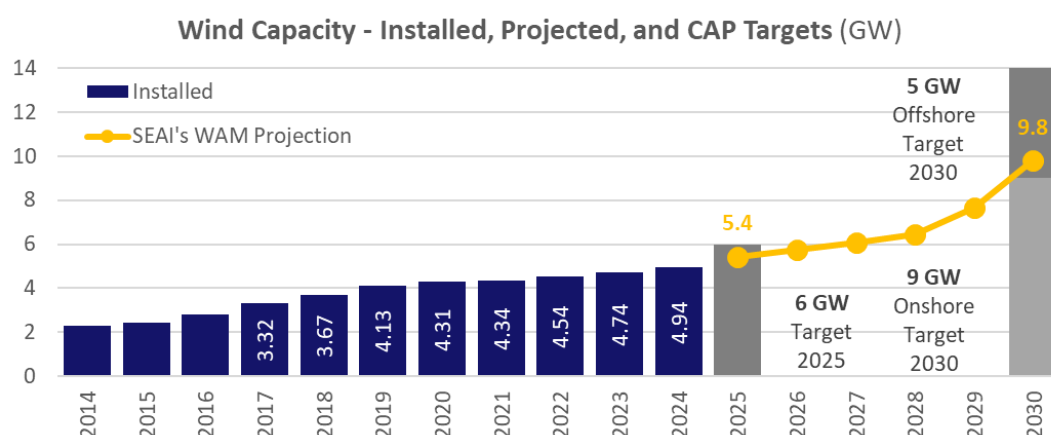


In the above figure, where the data shows values greater than 100%, this indicates that Ireland was a net-exporter of electricity to Northern Ireland and Great Britain in that month, *i.e.* on a net basis, it produced all its own electricity supply and more.

1.3.1.1 Wind Supply

Ireland's wind generation comes from the thousands of individual wind turbines installed across the country. Ireland's installed wind capacity in 2024 was 4.94 GW, up 4.3% on the previous year. Ireland's 2025 CAP target for installed wind capacity is 6 GW, and its 2030 CAP targets for onshore and offshore capacity are 9 GW and 5 GW, respectively. SEAI's projections under the 'with additional measures' (WAM) scenario indicate a total installed capacity of 9.8 GW by the end of 2030. This WAM scenario is the most optimistic reporting scenario modelled, achieving 68% RES-E by 2030⁸.

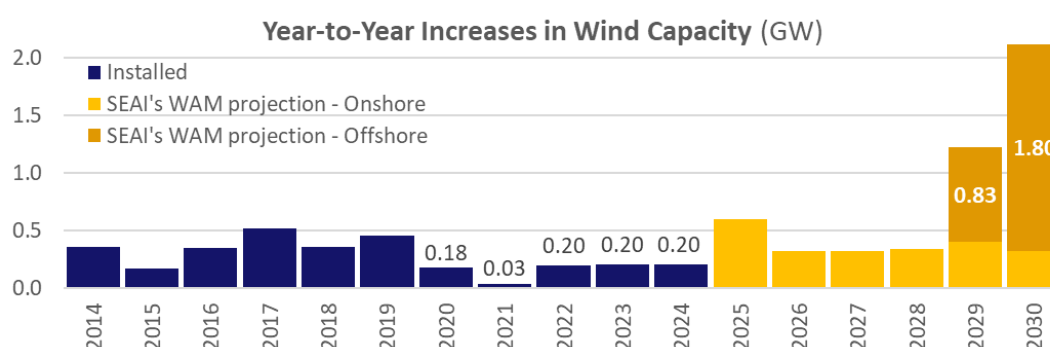
Figure 1.24: Installed wind capacity with climate action plan (CAP) targets for 2025 and 2030, and 'with additional measures' (WAM) capacity projections



⁸ www.seai.ie/sites/default/files/publications/National-Energy-Projections-Report-2025.pdf

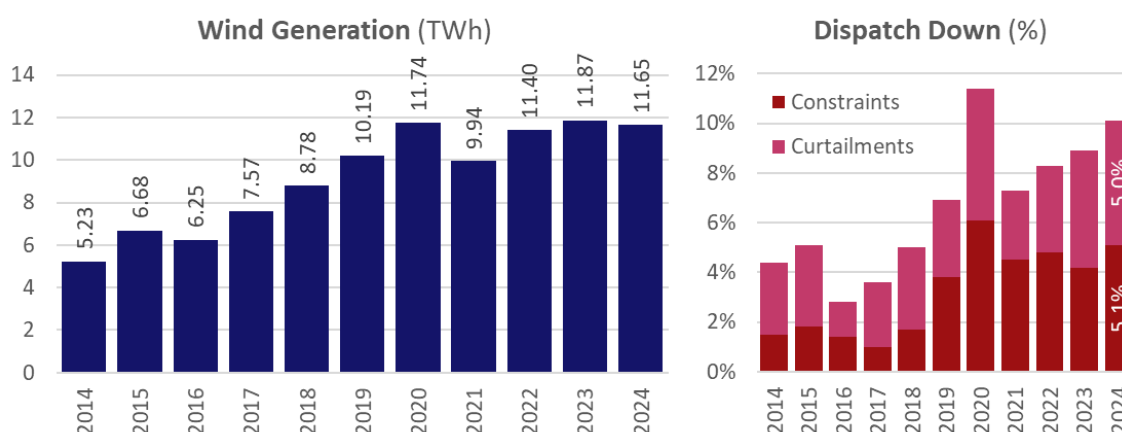
Over the last 10-years, Ireland has added wind capacity at an average rate of 0.27 GW per annum, although this has dropped to a rate of 0.16 GW over the last 5-years. To align to the pace of the WAM scenario, the roll-out rate of onshore wind capacity needs to return to that previously achieved between 2016 and 2019. In addition, offshore wind projects delivering annual increases of between 0.8 GW and 1.4 GW in installed capacity need to be in place before the end of 2030. For more details on wind capacity, see section 4.2.7.

Figure 1.25: Year-to-year increases in installed wind capacity to 2024, and year-to-year increases in WAM capacity projections of capacity from 2025 to 2030



Ireland generated 11.65 TWh of renewable energy in 2024 from wind, down 1.8% on the previous record of 11.87 TWh set in 2023. Annual wind generation values over the last 5 years (2020 to 2024) have been relatively consistent. For four of the last five years, Ireland's wind generation has varied between 11.40 TWh and 11.87 TWh, which is only a $\pm 2\%$ variation. In contrast, wind generation in the preceding 5 years (2016 to 2020) showed an average annual increase of over 20%. For more details on wind generation, see section 4.2.

Figure 1.26: Annual wind generation and dispatch down values figures



Multiple factors can impact the level of wind generation each year – the installed capacity, the average quality of the wind, and the ‘dispatch down’ applied to wind generation due to constraint and curtailment on the local and national grid. While a nationally representative wind quality figure is difficult to determine, EirGrid provides precise monthly constraint and curtailment statistics⁹ for wind generation, where:

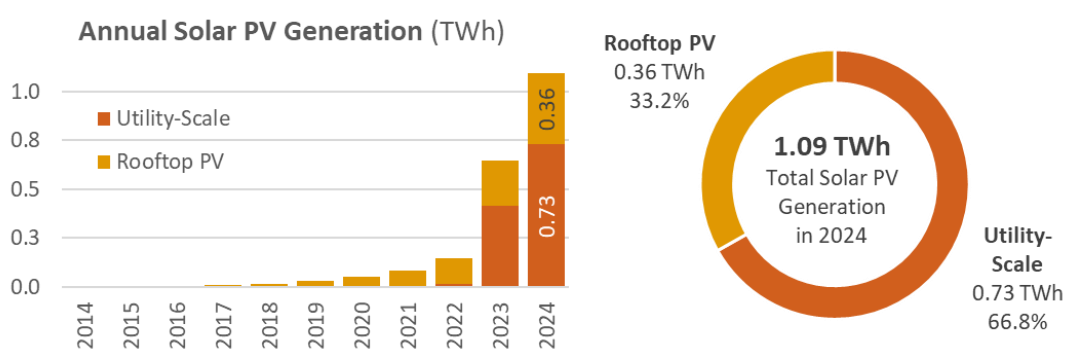
- **Constraint** is a reduction in allowed wind output due to local or network-specific grid limitations (e.g. when part of the system cannot safely carry the wind power available, due to line, transformer, or node limits)
- **Curtailment** is a reduction in allowed wind output due to system wide limitations (e.g. when too much non-synchronous generation would be online relative to demand or stability limits, *i.e.* the safe system non-synchronous penetration level)

This data from EirGrid shows an approximately 50:50 split across constraint and curtailment dispatch down in the last 5 years, and a generally increasing trend in dispatch down. The average annual level of dispatch down for the last 5 years (2020 to 2024) was 9.2%, compared to 4.7% for the preceding 5 years (2015 to 2019). Future levels of dispatch down could be reduced through upgrades and expansions of the electricity grid, increased flexibility in electricity demand-side management, and increased capacity for energy storage and deployment.

1.3.1.2 Solar PV Supply

Ireland generated 1.09 TWh of renewable energy from solar PV in 2024, up 69.1% on the previous record set in 2023. The rapid increase in solar PV generation since 2022 has come from a step-change increase in the connection of utility-scale ‘solar farms’ to the national grid, coupled with near exponential increases in embedded ‘rooftop’ solar PV on homes, businesses, and farms. In 2024, almost exactly two-thirds (66.8%) of Ireland’s solar PV generation came from utility-scale solar-farms.

Figure 1.27: Time-series of Ireland’s solar PV generation and a breakout of 2024 solar PV generation

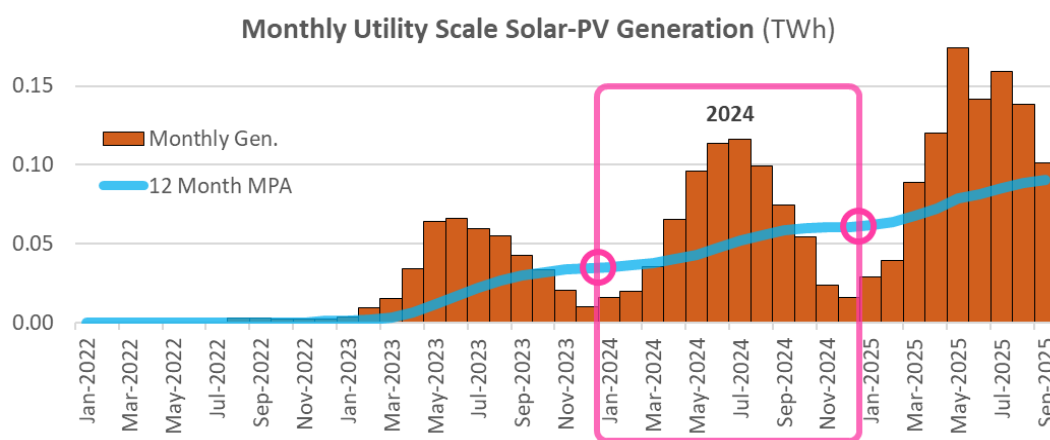


The strong seasonal variation of solar PV is clearly apparent in the monthly generation data from utility-scale solar farms. The seasonal profile of solar PV is complementary to that of wind generation. Solar PV peaks during the summer months, when wind is at its lowest, and wind peaks during the winter months, when solar is at its lowest. A balanced blend of solar PV and wind capacity therefore acts to smooths the seasonal-level variation associated with renewable electricity generation, reducing the need for dispatchable fossil fuel generation.

⁹ Annual Renewable Energy Constraint and Curtailment Report 2024 from EirGrid -

<https://cms.eirgrid.ie/sites/default/files/publications/Annual-Renewable-Constraint-and-Curtailment-Report-2024-V1.0.pdf>

Figure 1.28: Monthly solar PV generation from utility-scale grid-connected solar farms. The blue line is a 12-month moving point average (MPA) to illustrate the average trend



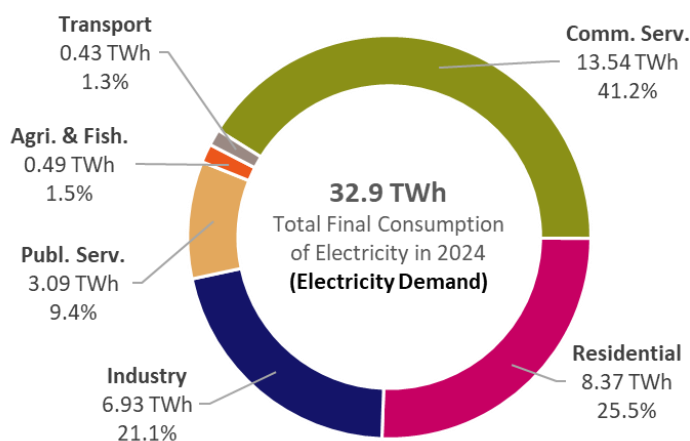
Based on monthly generation data from utility-scale solar-farms, solar PV generation in 2025 will be higher again than in 2024. Extrapolating January-September 2025 data to the end of 2025, SEAI's best estimate is that utility-scale solar PV generation will be 1.1 TWh in 2025, up over 50% on 2024 levels. Total solar PV generation in 2025 could then reach approximately 1.6 TWh.

For more details on solar PV capacity and generation, see sections 4.2.

1.3.2 Electricity Demand Trends

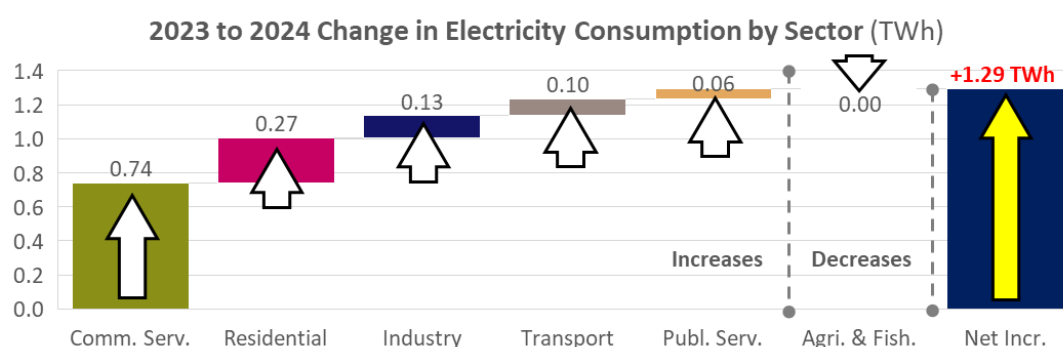
Ireland's electricity demand in 2024 was 32.9 TWh, up 4.1% on the previous year. The three leading demand sectors for electricity in 2024 were commercial services (41.2%), residential (25.5%), and industry (21.1%). Only 1.3% of electricity demand in 2024 came from the transport sector – from the national fleet of electric vehicles, the LUAS, and DART commuter services.

Figure 1.29: Breakout of 2024 electricity demand across economic sectors



In absolute terms, Ireland's electricity demand in 2024 was 1.29 TWh higher than the previous year. There were increases in electricity demand from all economic sectors, except agriculture and fisheries, which was down 0.001 TWh or 0.2% on the previous year. The largest sectoral increase in electricity demand in 2024 came from the commercial services sector, which increased by 0.74 TWh or 5.8% on the previous year. The commercial services sector includes the ICT sub-sector, which includes data centre demand.

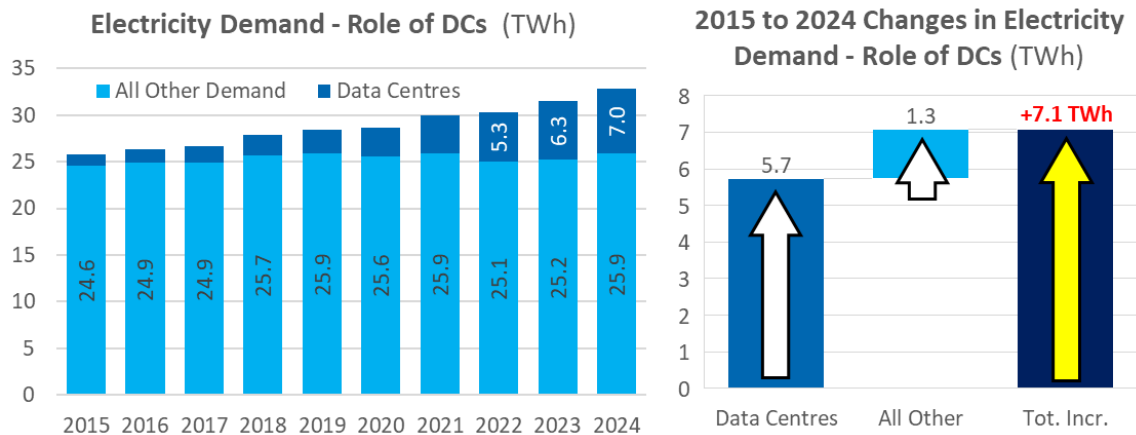
Figure 1.30: Waterfall plot showing 2023 to 2024 changes in electricity demand by sector



Ireland's electricity demand has grown every year for the last 10 years. A key driver of this increased electricity demand has been Ireland's data centres. Data published by the CSO¹⁰ shows that Ireland's data centre electricity demand in 2024 was 7.0 TWh, up 10.0% on the previous year. In 2024, data centres accounted for just over one-fifth (21.2%) of all electricity demand in Ireland. For reference, when it announced a new scheme for rating the sustainability of data centres in 2024, the European Commission noted that data centres accounted for approximately 3% of the EU's electricity demand.

¹⁰ <https://www.cso.ie/en/releasesandpublications/ep/p-dcmec/datacentresmeteredelectricityconsumption2024/keyfindings/>

Figure 1.31: Breakout of Ireland’s electricity demand to illustrate the role of data centre (DC) electricity demand



Outside of data centres, net electricity demand from all other sectors of the Irish economy has been relatively constant, varying only from 24.6 TWh to 25.9 TWh in the last 9 years – a variance of only $\pm 2.7\%$. Comparing the change in annual electricity demand from 2015¹¹ to 2024, it is apparent that data centres are responsible for 88.2% of increased electricity demand in the Irish economy since 2015.

For more details on electricity demand, see sections 5.1 and 6.4.

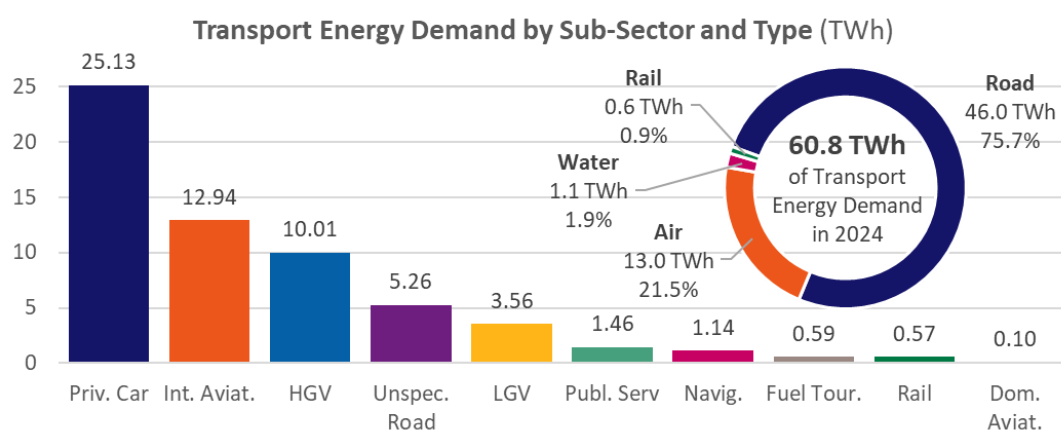
¹¹ 2015 is the first year for which the CSO publish metered electricity consumption of data centres.

1.4 Transport Energy Demand

Transport has the highest energy demand of any sector of the Irish economy. In 2024, the transport sector accounted for 42.3% of Ireland's energy demand, 37.7% of energy-related emissions, and 21.7% of Ireland's total GHG emissions¹².

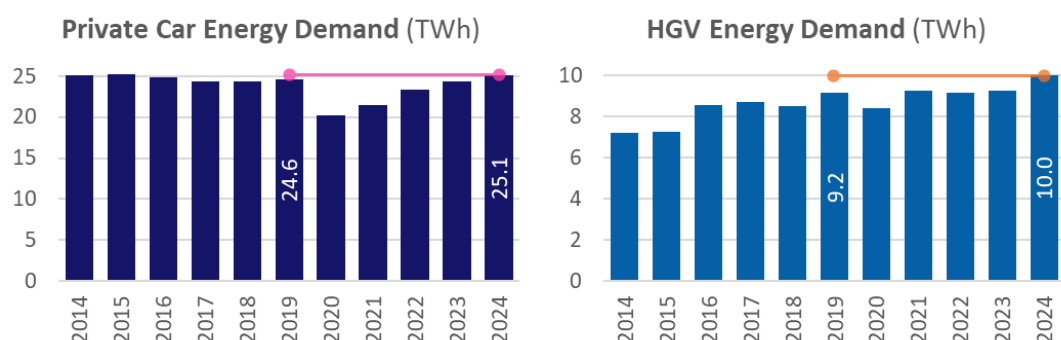
Private car use accounted for 41.4% of transport energy demand in 2024. International aviation accounted for a further 21.3% of transport energy demand, followed by heavy good vehicle (HGV) road freight demand at 16.5%. In total, 'road transport' accounted for just over three-quarters (75.7%) of Ireland's transport energy demand.

Figure 1.32: Breakout of 2024 transport demand by sub-sector¹³ and type



Energy demand from all three of the largest sub-sectors of transport – private car use, international aviation, and HGV road freight – are increasing. Energy demand from private car use was 25.1 TWh in 2024, up by 3.2% on the previous year. Energy demand from the national fleet of HGVs was 10.0 TWh in 2024, up 7.9% on the previous year. Energy demand from both private car and HGVs was higher in 2024 than in 2019, the last calendar year before COVID impacts. This indicates that their energy demand has now fully rebounded from COVID impacts.

Figure 1.33: Time-series of Ireland's energy demand for private car use and road freight

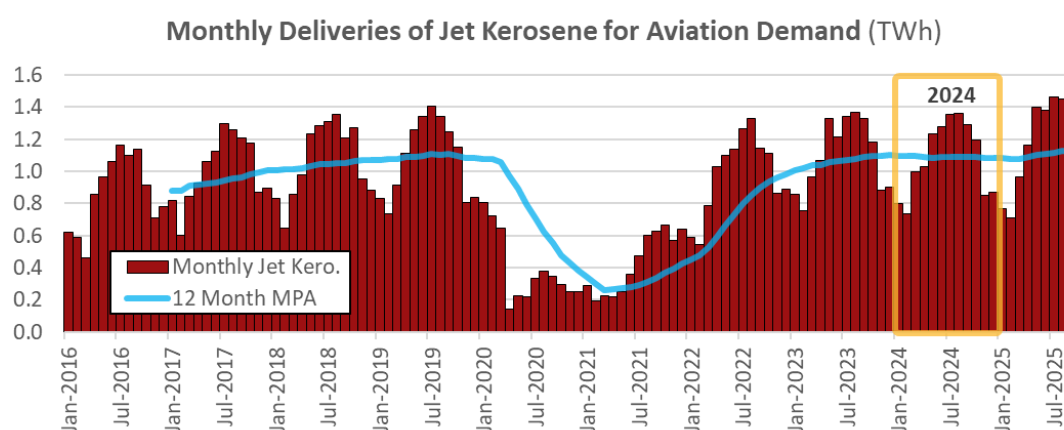


¹² EPA's Ireland's Provisional Greenhouse Gas Emissions (1990-2024) - <https://www.epa.ie/publications/monitoring--assessment/climate-change/air-emissions/EPA-Provisional-1990-2024-GHG-Report-1716.pdf>

¹³ The 'unspecified road' category is road transport fuel consumption not attributable to the other listed road transport sub-sectors. For more detail, see section 5.2.2.

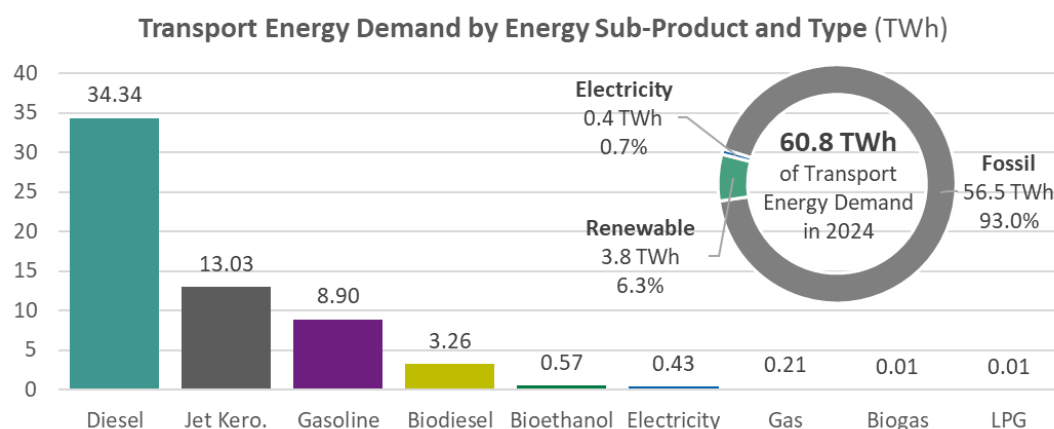
Energy demand for aviation was 13.0 TWh in 2024, relatively unchanged compared to 13.3 TWh in 2023. However, based on monthly data from January to September 2025, it is apparent that deliveries of jet kerosene for aviation in 2025 will be higher than in 2024. Based on an extrapolation of January-September 2025 data to the end of 2025, SEAI's best estimate is that deliveries of jet kerosene for aviation in 2025 may be 5% higher than in 2024. If this estimate is confirmed by annual survey data, then Ireland's aviation energy demand in 2025 will be higher than in 2019, indicating that aviation energy demand has now fully rebounded from COVID impacts.

Figure 1.34: Monthly jet kerosene deliveries for aviation energy demand. The blue line is a 12-month moving point average (MPA) to illustrate the average trend



In terms of energy product breakout, over half (56.5%) of transport energy demand in 2024 was satisfied using petrochemical diesel, *i.e.* the 100% fossil fuel component in blended road diesel. A further 21.5% of transport energy demand was satisfied using jet kerosene, which is currently 100% fossil fuel based. Overall, 93.0% of Ireland's transport energy demand in 2024 was satisfied using fossil fuels in 2024, with renewable energy and electricity only satisfying 6.3% and 0.7% of that demand, respectively.

Figure 1.35: Breakout of 2024 transport demand by energy sub-product and type

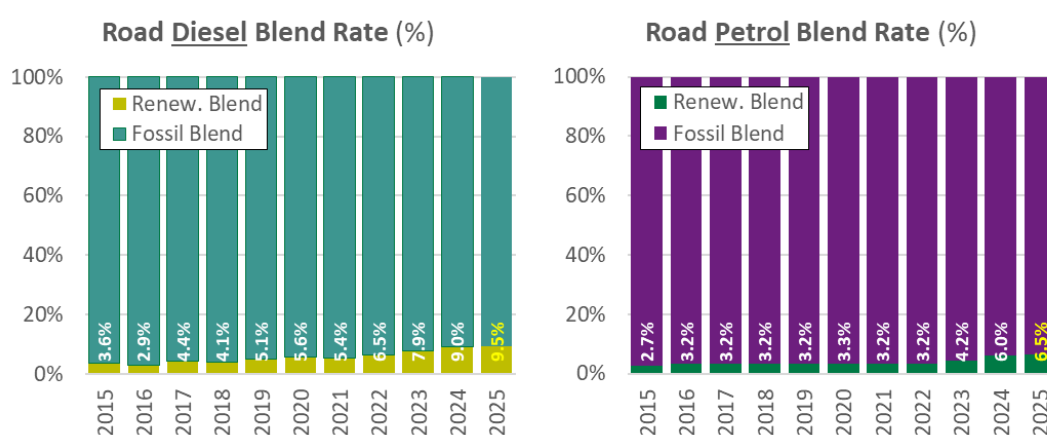


Renewable energy in transport is mainly delivered through the blending of fatty acid methyl esters (FAME), hydro-treated vegetable oil (HVO), and co-processed HVO into petrochemical diesel, and bioethanol in petrochemical gasoline. The resulting blended products of 'road diesel' and 'road petrol' are the fuels available for purchase at garage forecourts across the country.

In 2024, 9.0% of road diesel's energy content came from biofuels, up from 7.9% in 2023. Based on monthly survey returns from January to September 2025, SEAI's best estimate is that 9.5% of road diesel energy content will come from biofuel in 2025, setting a further record high.

After remaining relatively constant around 3.2% from 2016 to 2022, 6.0% of Ireland's road petrol energy content came from biofuels in 2024, up from 4.2% in 2023. Based on monthly surveys returns from January to September 2025, SEAI's best estimate is that 6.5% of road petrol energy content will come from biofuel in 2025, setting a further record high.

Figure 1.36: Ireland's annual biofuel energy blending rates into road diesel and road petrol. The 2025 value, highlighted in yellow text, is a best estimate from SEAI, based on monthly data from January to September 2025



While biofuel blending can help reduce transport emissions, it is only a partial solution. Internationally, there is limited availability of sustainable biofuel feedstock, and unchecked demand for biofuel production crops can act to displace crops needed for food production. Additionally, there are technical limits on the blend-rate of certain biofuels in standard road vehicles. For example, bioethanol is limited to 10% by volume in EN 228 gasoline, and conventional FAME-based biodiesel is limited to 7% by volume in EN 590 diesel.

For more details on transport demand, see sections 5.2.2 and 6.3.

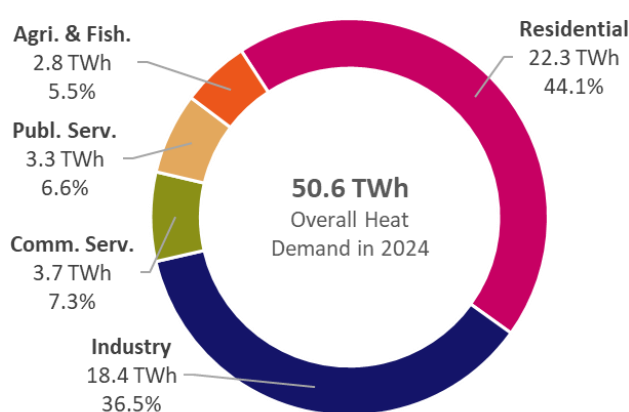
1.5 Heat Demand

For the purposes of national energy statistics, and to align with European energy reporting, heat demand is defined as total energy demand, less electricity demand, less transport demand. Heat demand accounted for over 35.1% of Ireland's energy demand in 2024.

1.5.1 Overall Heat Demand

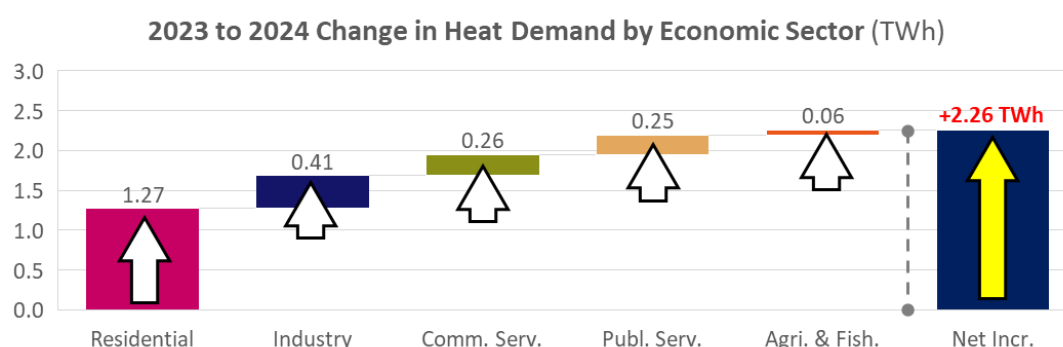
Ireland's heat demand in 2024 was 50.6 TWh, up 4.7% on the previous year. In 2024, the three economic sectors with greatest heat demand were residential (44.1%), industry (36.5%), and commercial services (7.3%). Combined, the residential and industry sectors account for 80.6% of national heat demand.

Figure 1.37: Breakout of 2024 overall heat demand across economic sectors

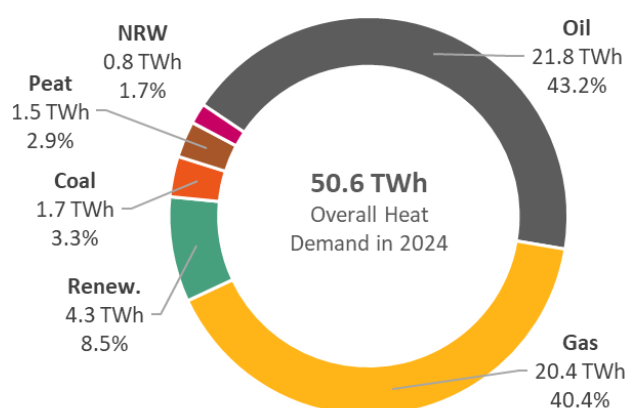


In absolute terms, Ireland's heat demand in 2024 was 2.26 TWh higher than the previous year. There were increases in heat demand across all sectors of the economy, led by an increase of 1.27 TWh from the residential sector.

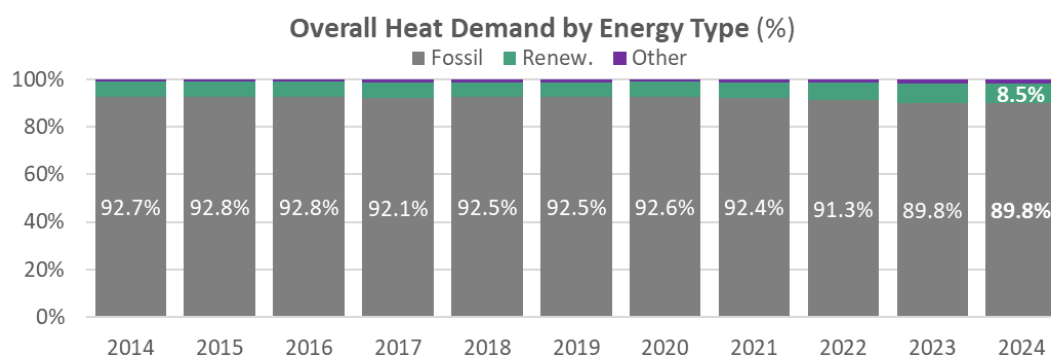
Figure 1.38: Waterfall plot showing 2023 to 2024 changes in heat demand by sector



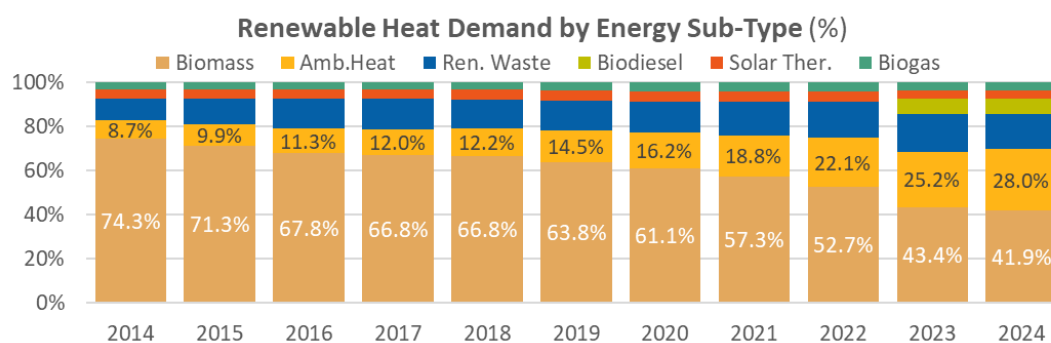
In terms of energy product breakout, 43.2% of heat demand in 2024 was satisfied through end-user consumption of oil products like kerosene and gasoil. Gas accounted for another 40.4% of heat demand in 2024. In 2024, only 8.5% of Ireland's heat demand was satisfied using renewable energy.

Figure 1.39: Breakout of 2024 overall heat demand across energy products

Ireland's heat demand remains dominated by fossil fuels. In 2024, 89.8% of heat demand was satisfied using fossil fuels. However, the proportion of fossil fuels satisfying heat demand is falling slowly - down from 92.7% in 2014.

Figure 1.40: Time-series breakout of Ireland's overall heat demand by energy type. The 'other' energy type includes non-fossil, non-renewable energy, like non-renewable wastes

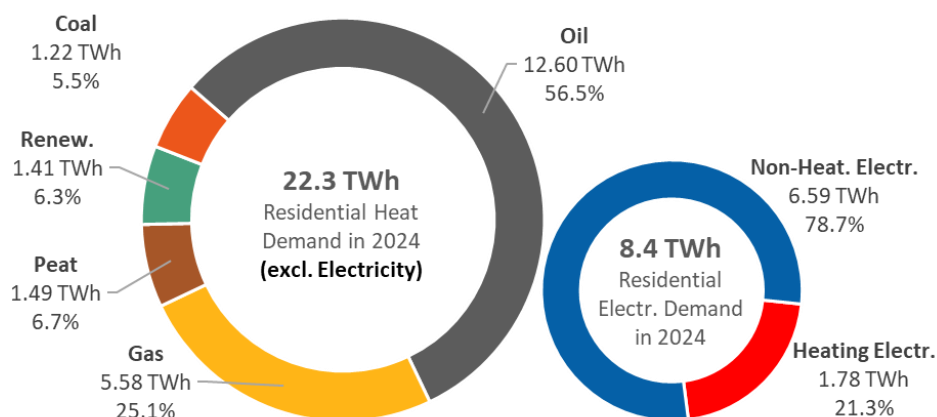
Historically, most renewable energy satisfying Ireland's heat demand came from biomass. In 2014, biomass accounted for almost three-quarters (74.3%) of the renewable energy satisfying heat demand. In 2024, biomass accounted for only 41.9% of renewable energy satisfying heat demand, with 28.0% coming from the ambient heat captured by heat pumps, and 7.0% coming from biodiesel.

Figure 1.41: Time-series breakout of Ireland's renewable heat demand by energy sub-type

1.5.2 Residential Heat Demand

Ireland's residential heat demand in 2024 was 22.3 TWh, up 6.0% on the previous year. In terms of energy product breakout, well over half (56.5%) of residential heat demand in 2024 was satisfied through oil products, mainly kerosene. Gas accounted for almost exactly one-quarter (25.1%) of residential heat demand in 2024, followed by peat (6.7%).

Figure 1.42: Breakout of 2024 residential heat demand across energy products, and a breakout of residential electricity demand by consumption activity



Since heat demand is defined as total energy demand, less electricity demand, less transport demand, the 2024 residential heat demand value of 22.3 TWh does not include the 2024 residential electricity demand value of 8.4 TWh. Based on SEAI's modelling of energy end use in households using the dwelling energy assessment procedure (DEAP) and CSO weightings of housing archetypes, SEAI estimates that 21% of residential electricity demand was used for heating in 2024.

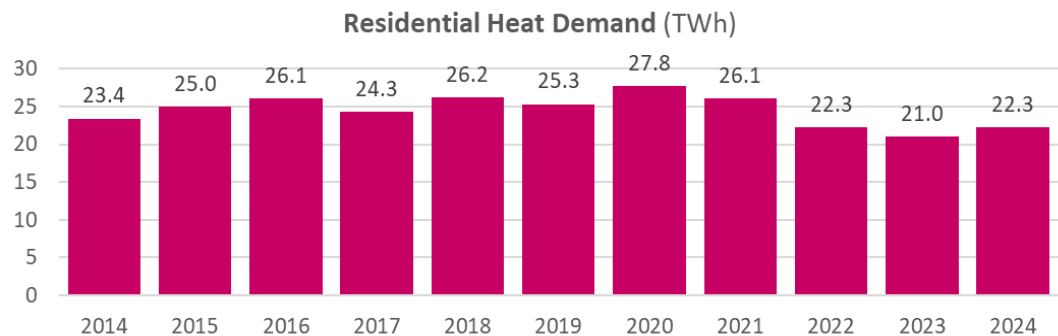
Renewable energy satisfied only 6.3% of residential heat demand in 2024. Ambient heat captured by heat pumps accounted for almost two-thirds (64.7%) of the renewable residential heat demand in 2024, followed by 23.8% from biomass (e.g. wood logs, wood pellets, etc.), and 11.5% from solar thermal heating. At the end of 2024, there were approximately 153,000 homes with heat pumps, up from 119,000 at the end of 2024¹⁴.

As noted previously, Ireland's residential heat demand in 2024 was up 6.0% on the previous year, having previously decreased every year for the last 3 years. Residential heat demand is driven by many different interconnected factors:

1. The affordability of energy prices, particularly oil and gas
2. Weather effects, particular the length and average temperature of the winter season
3. Societal and behavioural shifts, such as the return-to-office vs. work-from-home balance
4. The impact of new builds and energy upgrades on the heating efficiency of the housing stock
5. Demographic shifts in occupation of housing stock

¹⁴ Figures extracted from SEAI's BER database.

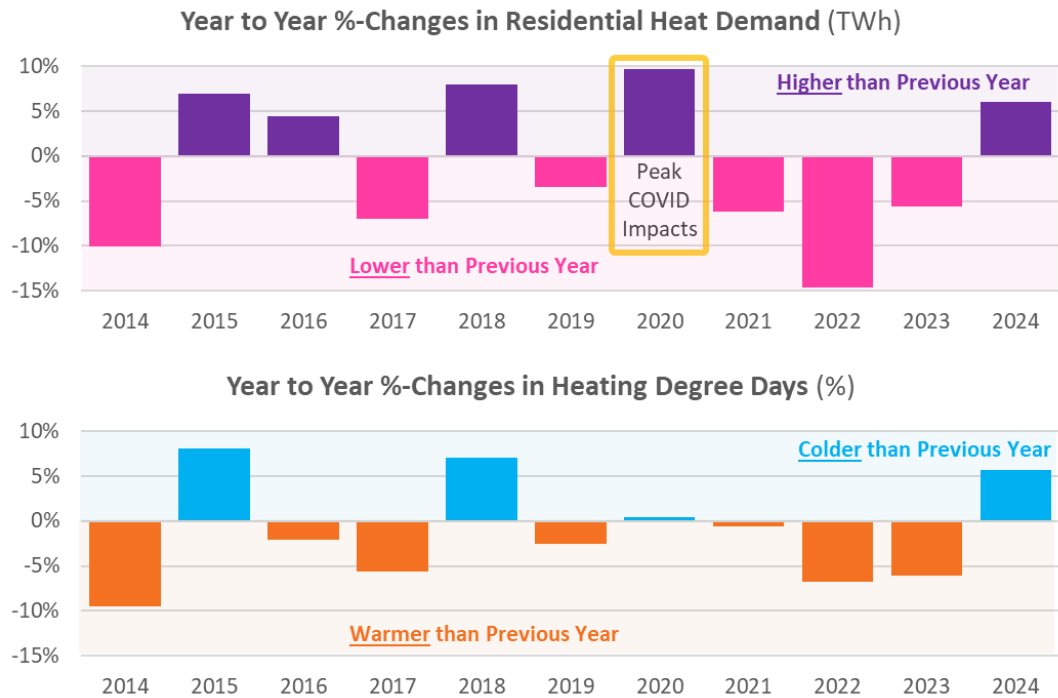
Figure 1.43: Time-series of Ireland's residential heat demand



Weather effects play a particularly strong role in driving year-to-year changes in residential heat demand, because they influence how often and how long home-heating systems are run. A simple high-level metric for quantifying how often and how intensely homes around Ireland need to be heated during the year is the sum of heating degree days (HDDs) below 15.5°C in that year. Met Éireann publishes monthly HDD data from multiple weather stations around Ireland.

There is a strong correlation between year-to-year changes in residential heat demand and year-to-year changes in the number of HDDs. For example, the percentage changes in the number of HDDs for 2023 and 2024 were -6.1% and +5.7%, respectively, while the corresponding changes in residential heat demand were -5.6% and +6.0%, respectively. In general, the number of HDDs in a year is a strong predictor for residential heat demand in that year.

Figure 1.44: Comparison of year-to-year changes in residential heat demand and year-to-year changes in the number of heating degree days (HDDs)



However, the simple correlation between the number of HDDs and residential heat demand is not perfect, especially where other factors strongly impact on heat demand. For example, 2020 was only marginally colder having only 0.4% more HDDs compared to 2019, residential heat demand in 2020 increased by 9.7%. This was likely due to the strong impact of work-from-home and school-from-home mandates with the onset of COVID. Conversely, residential energy demand decreased by 14.6% in 2022, despite the number of HDDs decreasing by only 6.8%. This was likely due to the impact of increased energy prices in 2022, when the cost of kerosene and gas approximately doubled.

Based on the 2025 HDD data available from Met Éireann, and the January to September 2025 monthly data on residential gas demand, and monthly data on kerosene deliveries¹⁵ available to SEAI, it seems likely that residential heat demand in 2025 will be lower than in 2024. However, given the uncertainty of the heating season in Q4-2025, it is not currently possible to confidently generate a best estimate value.

For more details on heat demand, see section 6.2.

¹⁵ Approximately 95% of kerosene deliveries go to the residential market

1.6 Ireland's EU Energy Targets

Ireland has 2030 EU energy targets on both the share of renewable energy in its energy portfolio, and the total energy demand of that portfolio. These dual targets act together to improve the **quality** of energy in Ireland, by increasing the share of renewable energy being used, and to reduce the **quantity** of energy needed in Ireland, by increasing efficient use of energy through better technology and behaviours.

1.6.1 Renewable Energy Directive Targets

Ireland's renewable energy share, and that of all EU member states, is calculated and monitored by the European Commission under different iterations of the Renewable Energy Directive (RED). The renewable energy share of EU member states is monitored through four key metrics:

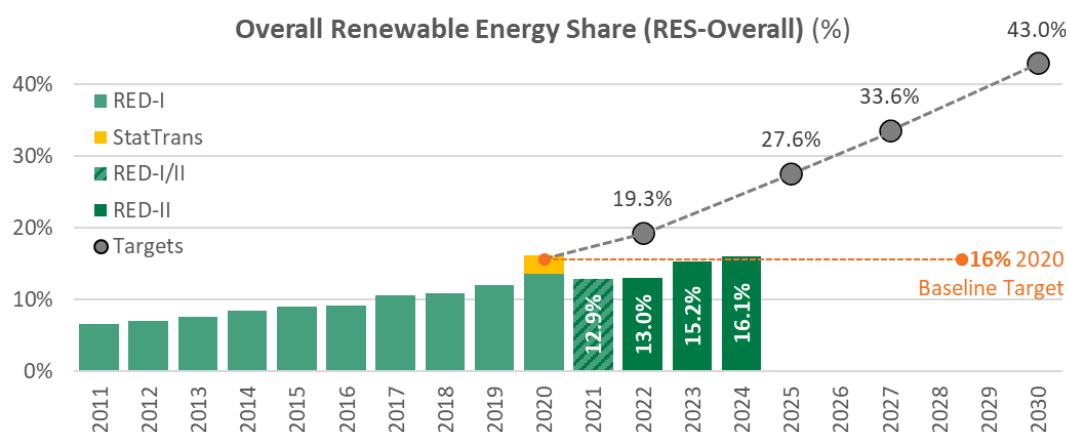
- RES-Overall Overall renewable energy share.
- RES-E Renewable energy share in electricity.
- RES-T Renewable energy share in transport.
- RES-H Renewable energy share in heating (and cooling).

The methodology for calculating the four RES results and setting their targets, which can include applying multipliers, limits, normalisations, and sustainability-checks on specific energy sub-products, is specified through the relevant iteration of the Directive for a given year. Ireland's RES-overall targets out to 2030 are specified in the National Energy Climate Plan (NECP), prepared under the EU Regulation on the Governance of the Energy Union, RED II and RED III. In addition to a 2020 'baseline' RES-Overall target of 16% in all years to 2030, Ireland has a series of reference points for specific milestone years:

- 2022 19.3% (RED-II).
- 2025 27.6% (RED-III).
- 2027 33.6% (RED-III).
- 2030 43.0% (RED-III).

In 2024, Ireland's RES-Overall result was 16.1%, up from 15.2% in 2023. Ireland's RES-overall result for 2024 was the highest on record, without the use of statistical transfers, and just over our 2020 'baseline' target of 16%. However, Ireland's RES-overall result for 2024 is significantly below the indicative trajectory set by the milestone years reference points set out in the NECP.

Figure 1.45: Ireland's overall renewable energy share (RES-Overall) results and targets

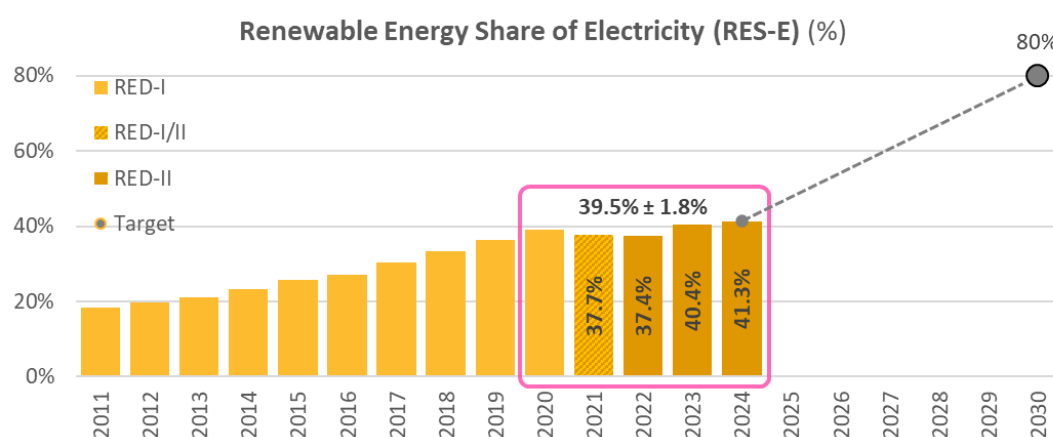


Although Ireland has no binding national target for RES-E for 2030, high levels of renewable energy share in electricity (RES-E) play a key part in Ireland's strategy to achieve its RES-overall target for 2030. Ireland's Climate Action Plan 2025 (CAP 2025) maintains a long-standing target to increase the RES-E to 80% by 2030. In 2024, Ireland's RES-E result was 41.3%, up from 40.4% in 2023. Ireland's RES-E result in 2024 was the highest on record but needs to almost double by 2030 to reach the 80% target.

Over the last 5 years (2020 to 2024), Ireland's RES-E results have been relatively consistent, varying by just ± 1.8 percentage points around a central value of 39.5%. In contrast, over the preceding 5 years (2016 to 2020), Ireland's RES-E result increased by 12.0 percentage points. The apparent plateauing of the RES-E results over the last 5-years is due to several factors:

1. A plateauing of wind generation, which is the largest contributor to renewable electricity, over the last 5 years.
2. A methodology-change in the RES-E calculation applied in 2021 under RED-II, introducing formal sustainability checks on specific renewable energy products, *i.e.* biomass and biogas.
3. Demand for electricity increasing faster than the roll-out rate of renewable generation capacity, *i.e.* increases in the denominator of the RES-E calculation (total electricity supply) offsetting increases in the numerator of the RES-E calculation (renewable electricity supply).

Figure 1.46: Ireland's renewable energy share of Electricity (RES-E) results and target



For more details on Ireland's RES results and targets, including RES-Transport and RES-Heating, see section 8.2.

1.6.2 Energy Efficiency Directive Targets

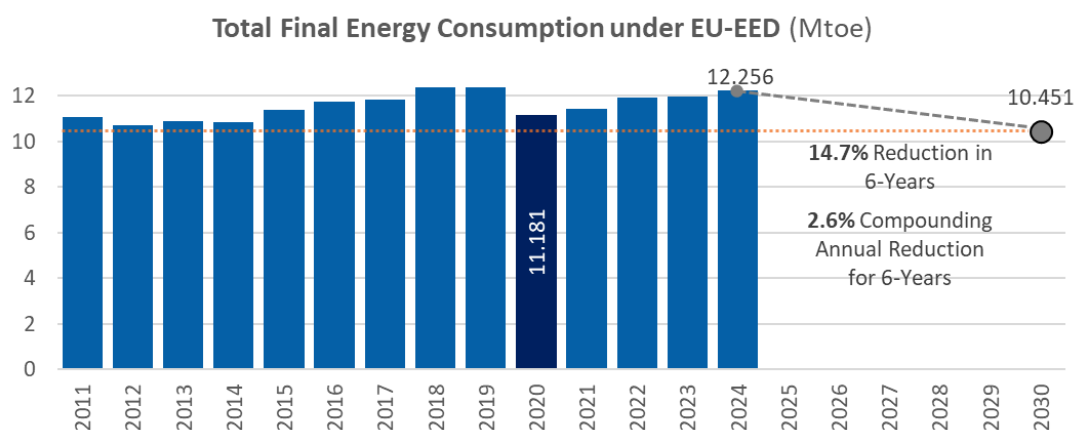
Under Article 4 of the Energy Efficiency Directive (EED), the EU has set itself a binding target of reducing total final energy consumption by 763 Mtoe by 2030. In February 2024, the Irish government approved Ireland's Article 4 target under the EED and submitted this as part of Ireland's NECP. Under its approved Article 4 EED target, Ireland is required to make sincere cooperative efforts to limit its total final energy consumption¹⁶ (TFEC-EED) to 10.451 Mtoe by 2030.

Ireland's TFEC-EED result in 2024 was 12.256 Mtoe, up 2.2% on the previous year. To achieve the 2030 TFEC-EED target, Ireland will need to reduce its energy demand by 14.7% in the next 6 years, which is equivalent to a 2.6% compounding annual reduction for the next 6 years.

¹⁶ The definition of the total final energy consumption used for Article 4 of the EED excludes the use of ambient heat captured by heat-pumps and so differs slightly from the total final energy consumption cited in other statistics.

The 2030 TFEC-EED target requires Ireland to reduce its energy demand to its lowest level in 25 years, below that observed in 2020 during the height of COVID impacts on the economy. Ireland's progress against its EED target is subject to monitoring by the European Commission through Ireland's NECP.

Figure 1.47: Ireland's total final energy consumption under the EU's energy efficiency directive (EED) and Ireland's 2030 Article 4 target



In addition to the Article 4 EED target, which seeks to reduce absolute energy demand, Article 8 of the EED requires Member States to achieve 'new energy savings' targets each year that are equivalent to:

- 2021-2023: 0.8% of TFEC
- 2024-2025: 1.3% of TFEC
- 2026-2027: 1.5% of TFEC
- 2028-2030: 1.9% of TFEC

The Article 8 EED targets are binding on Member States, and the measures and savings must achieve certain criteria to be deemed countable towards those targets:

- **Enduring** – The energy saving must last over time.
- **Material** – The measure employed reported must be the reason for the energy savings.
- **Additional** – The measure must be over-and-above minimum standards required by EU law.
- **Verifiable** – The measure and savings must be monitored and recorded.

2 Overview of energy data

2.1 The National Energy Balance

SEAI's Energy Statistics team, within the Research and Policy Insights Directorate, is responsible for collecting, analysing and reporting Ireland's national energy statistics. It fulfils many of Ireland's obligations for reporting energy statistics to both Eurostat (EU Commission) and the International Energy Agency (IEA).

Regulation (EC) No 1099/2008 on energy statistics, referred to as the Energy Statistics Regulation or ESR, is the key EU legislation that sets out the requirements for compiling and reporting annual and monthly national energy statistics.

SEAI Energy Statistics produces the National Energy Balance, the definitive source of annual data on the supply, transformation, and demand of energy in Ireland. The balance is based on the direct surveying of hundreds of energy suppliers or operators, as well as public administrative data from the Central Statistics Office (CSO), Environmental Protection Agency (EPA), Department of Climate, Energy and the Environment (DCEE), Revenue Commissioners, and others.

The NEB fulfils three primary roles:

- Provides key data to assist in policy making, investment decisions, and research.
- Is a key input to the EPA's Greenhouse Gas (GHG) Inventory, and so directly informs national emissions under the national carbon budgets, sectoral emission ceilings, EU Effort Sharing Regulation and Ireland's reporting to the UNFCCC.
- Provides the basis of Ireland's reporting under the Energy Statistics Regulation, including measurement of Ireland progress towards several targets under the Energy Efficiency Directive and Renewable Energy Directive.

The information in this report is largely based on Ireland's 2024 National Energy Balance [1] and the preceding annual balances back to 1990. These balances quantify the flow and transformation of energy sources from primary supply to final consumption, and profile that consumption across different sectors of the economy.

A simplified schematic for the energy balance flow is illustrated in Figure 2.1. The National Energy Balances are the definitive sources for constructing other normalised national energy indicators, such as energy intensity, energy efficiency, and energy-related greenhouse gas emissions. A Sankey diagram showing the flow of energy in Ireland 2024 is provided in Figure 2.2.

Figure 2.1: Main energy flows in Ireland

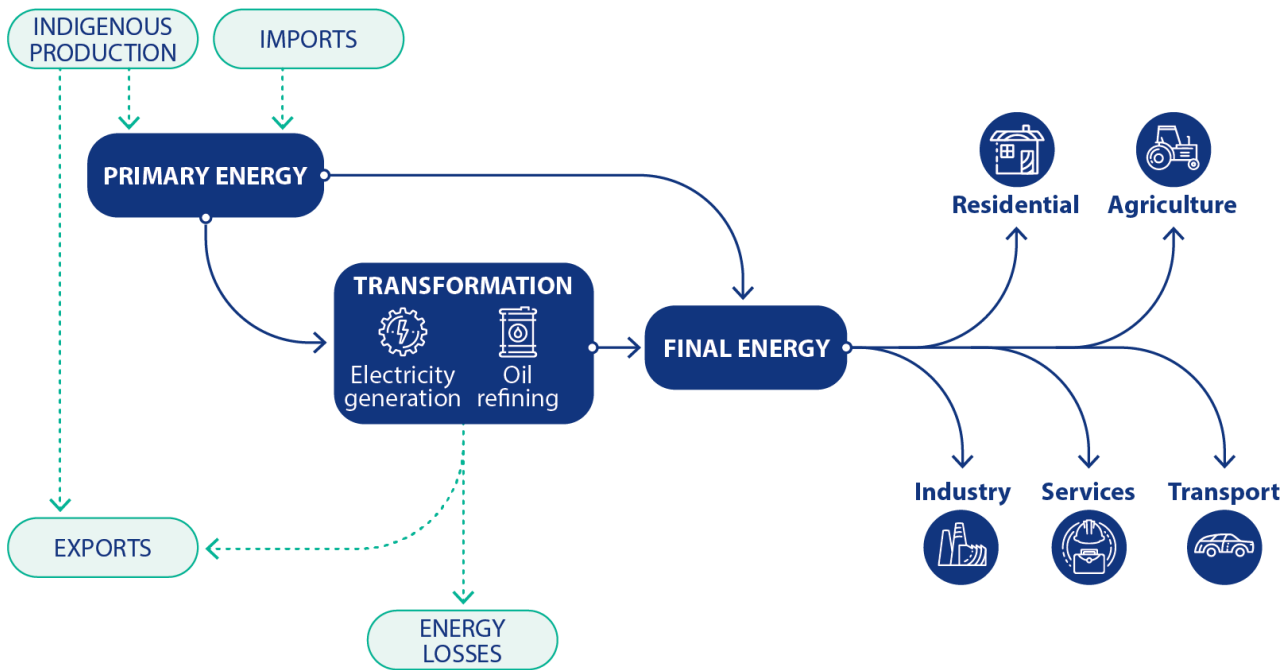
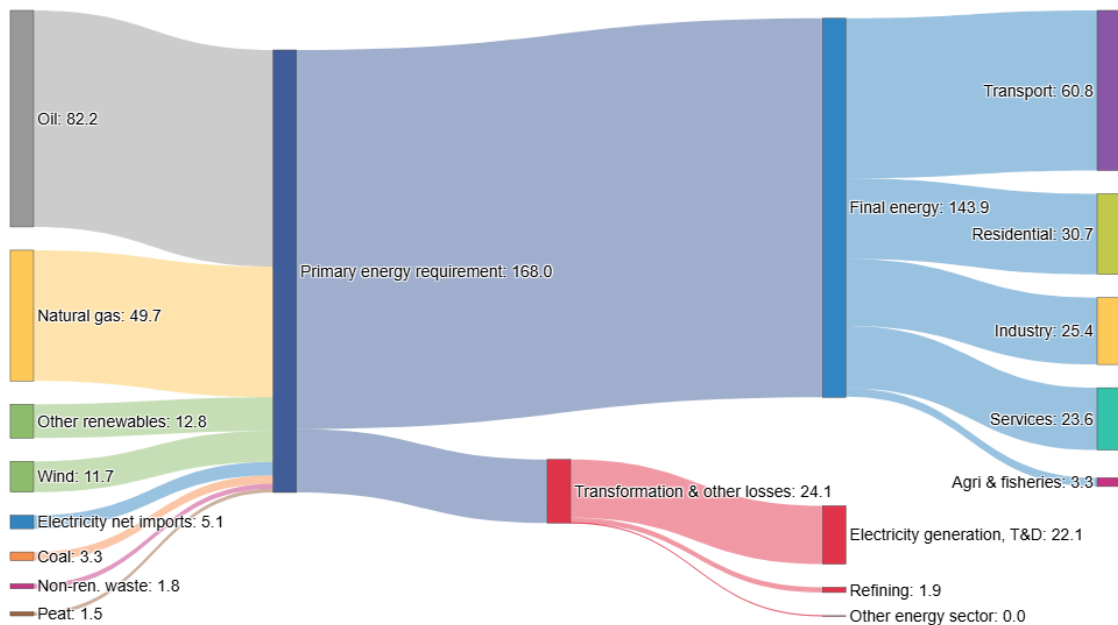


Figure 2.2: Energy Flow in Ireland in 2024 (TWh)



The energy balance is organised into three main parts, which are discussed in the following subsections:

- Primary energy – section 3 provides data on the production (*i.e.* extraction of energy from natural sources), imports, export and stock changes of energy in Ireland.
- Energy transformation – section 4 provides data on the transformation of energy from one to another (*e.g.* electrical power generation, oil refining)
- Final energy consumption – section 5 provides data on the type and quantities of energy consumed in each sector of the economy.

SEAI constructs the expanded National Energy Balance for each fuel or energy product, such as gasoline, lignite, biomass and electricity. There are currently 36 energy products listed in the expanded balance, although some are no longer used in Ireland. The number of fuels and energy products included in the balance has increased over the years as Ireland's energy mix has diversified. For analysis and presentation purposes, energy products can be combined into seven aggregated products or energy types:

- Coal
- Peat
- Oil
- Natural gas
- Renewables
- Non-renewable waste
- Electricity

A complete list of the energy products / sub-types that appear in the expanded National Energy Balance and this report is provided in Appendix 10.

In general, the data in this report is presented for aggregated energy types; however, where appropriate a more complete breakdown is provided for each sub-type. Short versions of the energy balance in units of terawatt-hours (TWh) and kilotonnes of oil equivalent (ktoe) are presented in Appendix 10. A more detailed, expanded balance showing data on each sub-energy type is available on the SEAI website at <https://www.seai.ie/data-and-insights/seai-statistics/key-publications/national-energy-balance/>.

The data in the national energy balances are informed by survey responses received from approximately 300 organisations, including energy producers, import/export companies and energy supply companies. Besides populating the energy balance, SEAI uses these surveys to fulfil Ireland's energy statistics reporting obligations to Eurostat (EU Energy Statistics Regulation (EC 1099/2008)) [2] and the International Energy Agency (IEA).

To ensure that the best possible information is always available to users, SEAI employs a policy of continuous improvement regarding historic energy balances, updating these with data revisions and new methodologies as they become available. An overview of the revisions and amendments to energy statistics made during 2025 is provided in Appendix 8.

2.2 Additional data

To help policymakers, analysts, researchers, and the public better understand Ireland's energy data, SEAI has expanded its use of data visualisations and interactive dashboards. SEAI's interactive graphics and dashboards allow users to explore monthly updates on energy delivery; quarterly updates on energy and fuel costs; and geographical analysis. Please visit SEAI's website at <https://www.seai.ie/data-and-insights/seai-statistics/monthly-energy-data/electricity-monthly/> to access the data.

SEAI currently publishes the energy balance for each year from 1990 to 2024. However, for the purposes of lending clarity to the data of most interest, the charts in this report generally only show

data from 2004 onwards, while the tables show comparisons back to 2014. A spreadsheet containing complete timeseries from 1990 to 2024 for almost all graphs and charts in this report will be made available for download via the link in Appendix 10 shortly after launch.

Feedback and comments on this report are welcome. Contact details are available on the back cover of this report.

2.3 Data on energy-related emissions

For Energy in Ireland 2024, SEAI moved its energy-related emission reporting from mega tonnes of carbon dioxide (MtCO₂) to mega tonnes of carbon dioxide equivalent (MtCO₂eq). The greenhouse gases (GHGs) covered within these CO₂eq estimates include carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O), gases which are known to contribute to the warming of the earth's atmosphere to varying extents over a given timeframe. GHG quantities from fuel combustion have been calculated using sector and fuel specific data from the EPA's National Inventory Submissions [3]. To calculate a total CO₂eq value, the emission metric used to quantify the differences in cumulative warming effects attributed to each of these gases is the 100-year global warming potential (GWP₁₀₀) and has been adapted from the IPCC Fifth Assessment Report. [4]

This change in methodology better aligns SEAI's reporting of energy-related emissions to those in national carbon budgets, and the EPA's GHG National Inventory Submissions, compiled in accordance with IPCC guidelines.

3 Primary energy

Primary energy is the total amount of energy that is required to satisfy the down-stream final energy use of end-users. It describes the initial energy sources that are transformed, transmitted, and distributed, before that energy is made available to end-users for final consumption. Primary energy therefore describes the energy inputs of processes like electricity generation and crude oil refining, rather than the consumption of that electricity or refined oil products.

The primary energy supply depends on the final energy use, as well as the efficiencies of the various transformation processes needed to ensure that end-users receive final energy in the form that is needed. Just like final energy use, primary energy supply can be usefully analysed by fuel, sector and mode.

3.1 Primary energy production

Primary energy production refers to the production of energy products from natural sources within Ireland's national boundaries. Examples of production include extraction of peat, crude oil or natural gas from natural deposits; recovery of waste for energy purposes; electricity generation by wind, solar or hydro; thermal energy collection by solar thermal systems; and production of biomass or biogas.

Figure 3.1 shows the latest breakdown of Ireland's annual primary energy production by energy type. Figure 3.2 shows Ireland's annual primary energy production by energy type and the ratio of primary energy production to supply.

Figure 3.1: Share of primary energy production by energy type

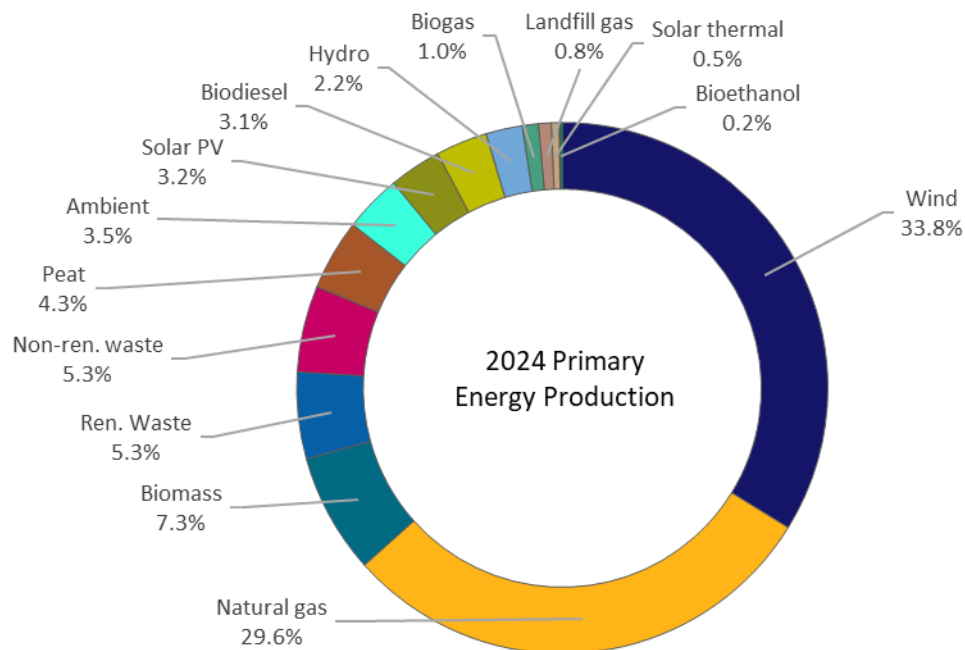


Figure 3.2: Indigenous primary energy production by energy type with production / supply ratio

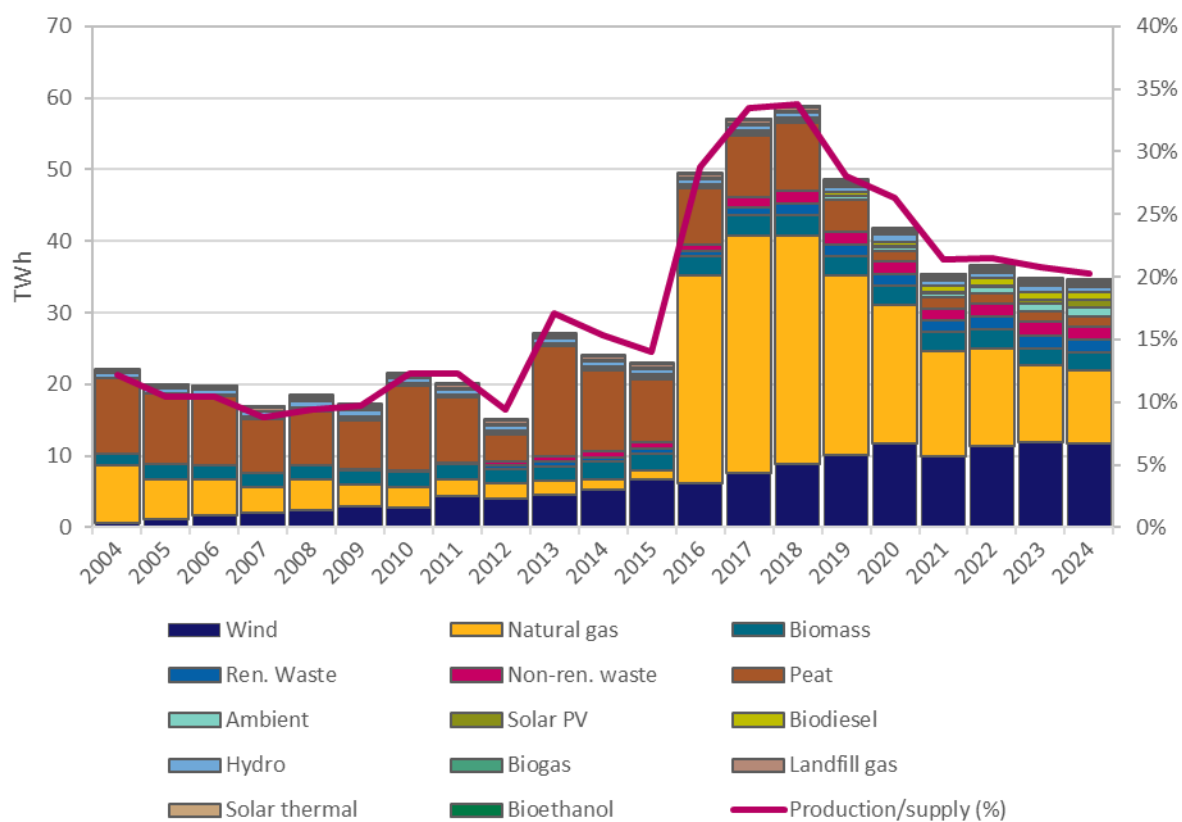


Table 3.1: Primary energy production by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Wind	5.23 (21.8%)	6.68 (29.1%)	6.25 (12.6%)	7.57 (13.3%)	8.78 (14.9%)	10.19 (21.0%)	11.74 (28.2%)	9.94 (28.2%)	11.40 (31.3%)	11.87 (34.2%)	11.65 (33.8%)
Natural gas	1.53 (6.4%)	1.31 (5.7%)	28.99 (58.6%)	33.19 (58.3%)	32.00 (54.5%)	24.97 (51.5%)	19.38 (46.6%)	14.63 (41.4%)	13.55 (37.2%)	10.87 (31.4%)	10.19 (29.6%)
Biomass	2.45 (10.2%)	2.36 (10.3%)	2.67 (5.4%)	2.88 (5.1%)	2.90 (4.9%)	2.76 (5.7%)	2.64 (6.3%)	2.70 (7.6%)	2.81 (7.7%)	2.18 (6.3%)	2.52 (7.3%)
Ren. Waste	0.60 (2.5%)	0.66 (2.9%)	0.74 (1.5%)	1.09 (1.9%)	1.63 (2.8%)	1.59 (3.3%)	1.69 (4.1%)	1.67 (4.7%)	1.76 (4.8%)	1.79 (5.2%)	1.82 (5.3%)
Non-ren. waste	0.77 (3.2%)	0.80 (3.5%)	0.78 (1.6%)	1.32 (2.3%)	1.69 (2.9%)	1.69 (3.5%)	1.71 (4.1%)	1.66 (4.7%)	1.73 (4.7%)	1.96 (5.6%)	1.81 (5.3%)
Peat	11.42 (47.7%)	8.95 (39.0%)	7.90 (16.0%)	8.65 (15.2%)	9.49 (16.1%)	4.54 (9.4%)	1.49 (3.6%)	1.49 (4.2%)	1.49 (4.1%)	1.49 (4.3%)	1.49 (4.3%)
Ambient	0.27 (1.1%)	0.31 (1.4%)	0.38 (0.8%)	0.41 (0.7%)	0.43 (0.7%)	0.50 (1.0%)	0.58 (1.4%)	0.67 (1.9%)	0.83 (2.3%)	1.02 (2.9%)	1.21 (3.5%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.03 (0.1%)	0.06 (0.1%)	0.08 (0.2%)	0.15 (0.4%)	0.65 (1.9%)	1.09 (3.2%)
Biodiesel	0.26 (1.1%)	0.28 (1.2%)	0.28 (0.6%)	0.33 (0.6%)	0.36 (0.6%)	0.48 (1.0%)	0.63 (1.5%)	0.91 (2.6%)	1.17 (3.2%)	1.06 (3.1%)	1.08 (3.1%)
Hydro	0.71 (3.0%)	0.81 (3.5%)	0.68 (1.4%)	0.69 (1.2%)	0.69 (1.2%)	0.89 (1.8%)	0.93 (2.2%)	0.75 (2.1%)	0.70 (1.9%)	0.94 (2.7%)	0.77 (2.2%)
Biogas	0.15 (0.6%)	0.16 (0.7%)	0.19 (0.4%)	0.19 (0.3%)	0.20 (0.3%)	0.22 (0.5%)	0.24 (0.6%)	0.26 (0.7%)	0.33 (0.9%)	0.33 (0.9%)	0.34 (1.0%)
Landfill gas	0.45 (1.9%)	0.49 (2.1%)	0.46 (0.9%)	0.45 (0.8%)	0.39 (0.7%)	0.37 (0.8%)	0.34 (0.8%)	0.34 (1.0%)	0.30 (0.8%)	0.28 (0.8%)	0.27 (0.8%)
Solar thermal	0.12 (0.5%)	0.13 (0.6%)	0.14 (0.3%)	0.15 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.4%)	0.16 (0.5%)	0.16 (0.4%)	0.16 (0.5%)	0.16 (0.5%)
Bioethanol	0 (0%)	0 (0%)	0 (0%)	0.00 (0.0%)	0.03 (0.1%)	0.05 (0.1%)	0.04 (0.1%)	0.04 (0.1%)	0.06 (0.2%)	0.08 (0.2%)	0.06 (0.2%)
Total production	23.97 (100%)	22.94 (100%)	49.46 (100%)	56.94 (100%)	58.77 (100%)	48.44 (100%)	41.62 (100%)	35.31 (100%)	36.43 (100%)	34.68 (100%)	34.46 (100%)

3.2 Primary energy imports and exports

Figure 3.3 shows Ireland's net energy imports by fuel type, along with the ratio of net imports to primary energy supply. Import dependency and energy security is discussed further in section 3.4.

Figure 3.3: Imports (positive) and exports (negative) by energy type (with total net imports)

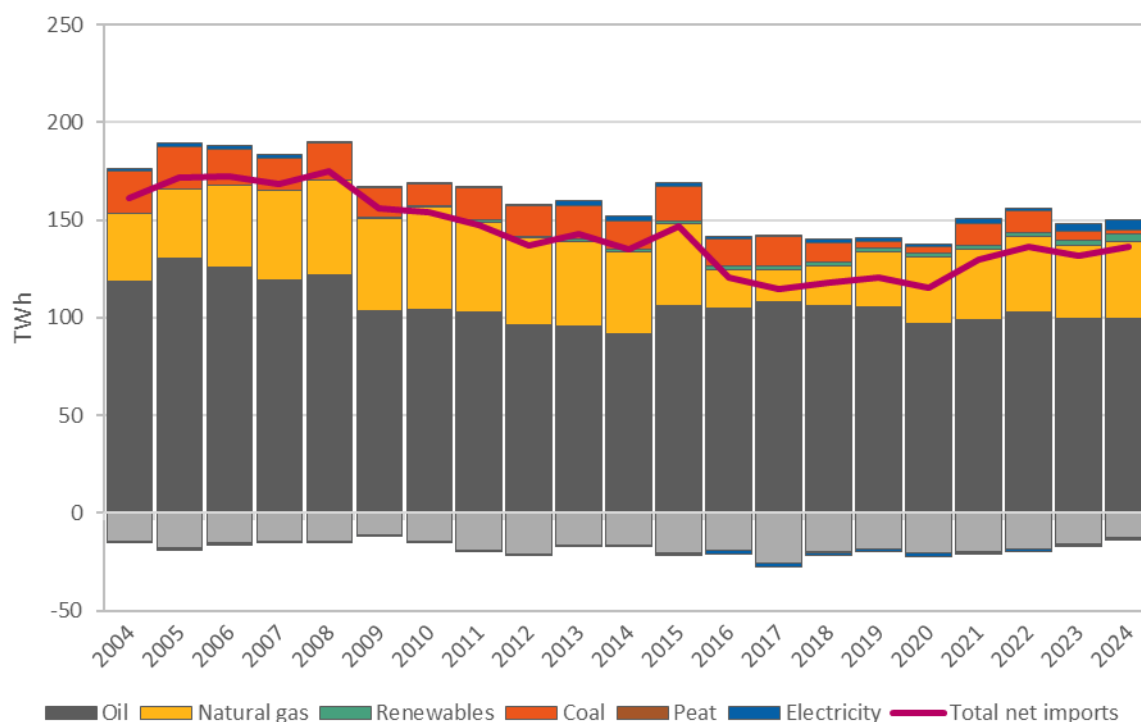


Table 3.2: Imports by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	91.99 (60.4%)	105.94 (62.7%)	104.76 (74.2%)	108.37 (76.1%)	106.26 (75.8%)	105.23 (74.7%)	96.94 (70.3%)	98.62 (65.5%)	102.95 (66.0%)	99.36 (67.1%)	99.47 (66.4%)
Natural gas	41.75 (27.4%)	42.20 (25.0%)	19.82 (14.0%)	16.38 (11.5%)	20.10 (14.3%)	28.20 (20.0%)	33.84 (24.5%)	36.35 (24.1%)	38.45 (24.6%)	37.39 (25.2%)	39.51 (26.4%)
Renewables	1.54 (1.0%)	1.45 (0.9%)	1.62 (1.1%)	1.85 (1.3%)	1.78 (1.3%)	2.15 (1.5%)	2.27 (1.6%)	1.81 (1.2%)	2.18 (1.4%)	3.08 (2.1%)	3.67 (2.5%)
Coal	14.20 (9.3%)	17.53 (10.4%)	14.16 (10.0%)	14.61 (10.3%)	10.35 (7.4%)	3.03 (2.2%)	3.06 (2.2%)	11.35 (7.5%)	10.84 (6.9%)	4.59 (3.1%)	1.90 (1.3%)
Peat	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Non-ren. waste	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Electricity	2.85 (1.9%)	1.75 (1.0%)	0.87 (0.6%)	1.12 (0.8%)	1.62 (1.2%)	2.18 (1.5%)	1.76 (1.3%)	2.45 (1.6%)	1.58 (1.0%)	3.72 (2.5%)	5.27 (3.5%)
Total imports	152.35 (100%)	168.87 (100%)	141.23 (100%)	142.33 (100%)	140.12 (100%)	140.79 (100%)	137.88 (100%)	150.57 (100%)	156.01 (100%)	148.14 (100%)	149.82 (100%)

Table 3.3: Exports by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	16.47 (95.1%)	20.67 (94.1%)	19.10 (91.6%)	25.45 (92.8%)	19.97 (90.8%)	18.23 (91.1%)	20.31 (90.0%)	19.87 (93.9%)	18.19 (91.3%)	15.86 (95.1%)	12.79 (96.2%)
Natural gas	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Renewables	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.11 (0.5%)	0.08 (0.4%)	0.19 (0.9%)	0.20 (1.0%)	0.17 (0.9%)	0.19 (1.1%)	0.14 (1.0%)
Coal	0.12 (0.7%)	0.13 (0.6%)	0.11 (0.5%)	0.11 (0.4%)	0.18 (0.8%)	0.10 (0.5%)	0.08 (0.4%)	0.20 (0.9%)	0.21 (1.1%)	0.18 (1.1%)	0.16 (1.2%)
Peat	0.02 (0.1%)	0.08 (0.4%)	0.05 (0.3%)	0.06 (0.2%)	0.08 (0.4%)	0.07 (0.3%)	0.07 (0.3%)	0.03 (0.1%)	0.01 (0.1%)	0.00 (0.0%)	0 (0%)
Non-ren. waste	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Electricity	0.70 (4.1%)	1.08 (4.9%)	1.58 (7.6%)	1.79 (6.5%)	1.65 (7.5%)	1.54 (7.7%)	1.91 (8.5%)	0.86 (4.1%)	1.33 (6.7%)	0.45 (2.7%)	0.21 (1.6%)
Total exports	17.31 (100%)	21.97 (100%)	20.85 (100%)	27.42 (100%)	21.99 (100%)	20.01 (100%)	22.56 (100%)	21.16 (100%)	19.92 (100%)	16.68 (100%)	13.30 (100%)

Table 3.4: Net imports by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	75.52 (55.9%)	85.26 (58.0%)	85.65 (71.2%)	82.92 (72.2%)	86.29 (73.1%)	87.00 (72.0%)	76.63 (66.5%)	78.75 (60.9%)	84.75 (62.3%)	83.50 (63.5%)	86.68 (63.5%)
Natural gas	41.75 (30.9%)	42.20 (28.7%)	19.82 (16.5%)	16.38 (14.3%)	20.10 (17.0%)	28.20 (23.3%)	33.84 (29.3%)	36.35 (28.1%)	38.45 (28.3%)	37.39 (28.4%)	39.51 (28.9%)
Renewables	1.54 (1.1%)	1.44 (1.0%)	1.62 (1.3%)	1.85 (1.6%)	1.67 (1.4%)	2.07 (1.7%)	2.08 (1.8%)	1.60 (1.2%)	2.01 (1.5%)	2.88 (2.2%)	3.54 (2.6%)
Coal	14.09 (10.4%)	17.40 (11.8%)	14.05 (11.7%)	14.50 (12.6%)	10.17 (8.6%)	2.94 (2.4%)	2.98 (2.6%)	11.15 (8.6%)	10.63 (7.8%)	4.41 (3.4%)	1.74 (1.3%)
Peat	-0.02 (-0.0%)	-0.08 (-0.1%)	-0.05 (-0.0%)	-0.06 (-0.1%)	-0.08 (-0.1%)	-0.07 (-0.1%)	-0.07 (-0.1%)	-0.03 (-0.0%)	-0.01 (-0.0%)	-0.00 (-0.0%)	0 (0%)
Non-ren. waste	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Electricity	2.15 (1.6%)	0.67 (0.5%)	-0.71 (-0.6%)	-0.68 (-0.6%)	-0.03 (-0.0%)	0.64 (0.5%)	-0.15 (-0.1%)	1.59 (1.2%)	0.25 (0.2%)	3.27 (2.5%)	5.06 (3.7%)
Total net imports	135.03 (100%)	146.90 (100%)	120.38 (100%)	114.91 (100%)	118.13 (100%)	120.78 (100%)	115.31 (100%)	129.41 (100%)	136.09 (100%)	131.45 (100%)	136.53 (100%)

3.3 Primary energy supply and requirement

SEAI uses the following standard definition of primary energy supply for each fuel and energy type in the energy balance:

$$\text{Primary energy supply} = \text{primary energy production} + \text{imports} - \text{exports} - \text{int. marine bunkers} + \text{stock change}$$

Primary energy production is discussed in section 3.1, while imports and exports are discussed in section 3.2. International marine bunkers refer to the fuel delivered at Ireland's ports to vessels that are engaged in international navigation; currently only fuel oil and gas oil are used for this purpose. Stock change is the change between opening and closing stock levels for a fuel held in Ireland during the year. A positive stock change indicates the net effect of the stock of a product being drawn down to enter the primary energy supply. A negative value indicates that the stock of the product has increased during the year. Stocks do not include reserves of fuels or products that are not yet extracted from natural deposits.

Another key measure of primary energy is primary energy requirement, defined as primary energy supply less non-energy use of the fuel or energy type:

$$\text{Primary energy requirement} = \text{primary energy supply} - \text{non-energy use}$$

For most fuel types included in the energy balance, there are no significant non-energy uses, and their primary energy supply and requirement are equal. In Ireland, the exceptions to this are bitumen, lubricating oil and white spirits, all oil products not used for energy purposes. Some natural gas was used for non-energy purposes up until 2002. Renewables used for materials, such as wood for construction or peat for gardening are not tracked in the energy balance as they are not used for energy purposes.

Figure 3.4 displays the breakdown of energy types in Ireland's primary energy requirement for 2024.

Figure 3.4: Share of primary energy requirement by energy type

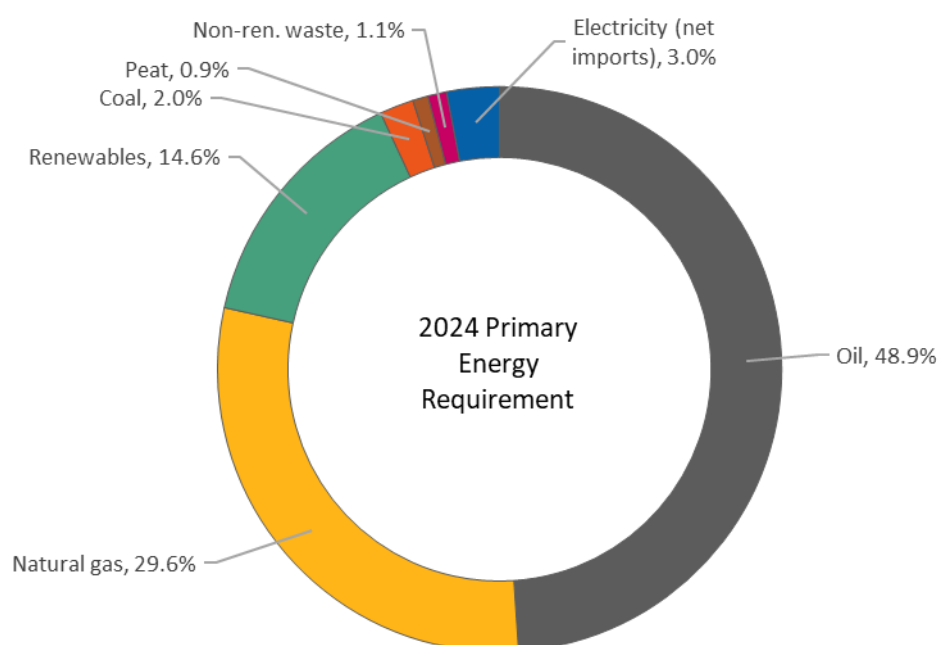


Figure 3.5 shows the trend in annual total primary energy requirement by energy type. Table 3.5 provides a detailed summary of the quantities and share of each energy type in Ireland's total primary energy requirement in comparison to previous years.

Total primary energy requirement peaked in 2008 before the economic downturn and reached a low in 2014 before growing to a relatively stable level from 2016 to 2019. A significant contraction to total primary energy requirement occurred in 2020 due to the impacts of the COVID-19 pandemic; this contraction was mostly confined to oil products, caused by a downturn in transport demand.

After growing in 2021 and 2022, total primary energy requirement decreased in 2023, due to declining use of natural gas. In 2024, it grew again, to near its pre-pandemic level.

In absolute terms, Ireland's 2024 total primary energy requirement (168.04 TWh) was below that from 20 years ago (176.20 TWh) and above that from 10 years ago (154.07 TWh), with intervening periods of growth and decline. Nevertheless, the mix of fuels and energy types in primary energy has evolved significantly during this time. The broad trend has been the growth of renewables and natural gas displacing oil, coal and peat. Despite the meaningful development of renewables, fossil fuels still dominate Ireland's primary energy supply.

Figure 3.5: Primary energy requirement by energy type

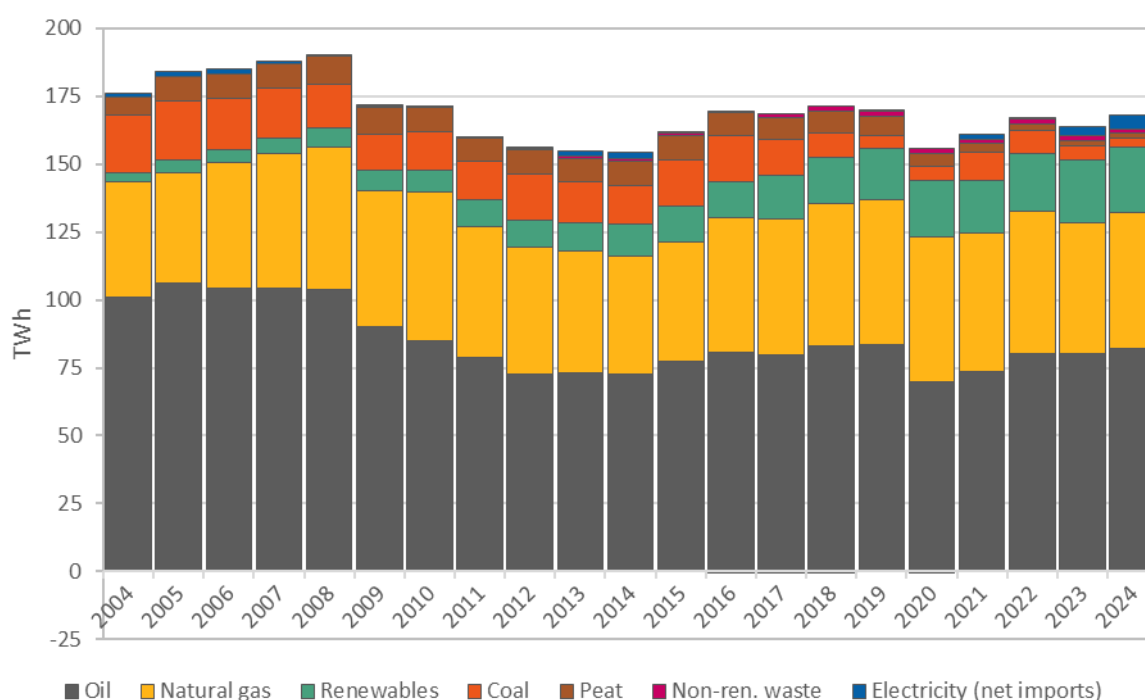


Table 3.5: Primary energy requirement by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	72.56 (47.1%)	77.38 (47.8%)	80.78 (47.8%)	79.79 (47.6%)	83.19 (48.6%)	83.48 (49.1%)	69.77 (44.8%)	73.71 (45.9%)	80.36 (48.2%)	80.06 (48.9%)	82.21 (48.9%)
Natural gas	43.40 (28.2%)	43.83 (27.1%)	49.44 (29.3%)	50.19 (30.0%)	52.11 (30.4%)	53.16 (31.3%)	53.22 (34.2%)	50.98 (31.7%)	52.00 (31.2%)	48.26 (29.5%)	49.70 (29.6%)
Renewables	11.76 (7.6%)	13.34 (8.2%)	13.31 (7.9%)	15.66 (9.3%)	17.31 (10.1%)	19.25 (11.3%)	21.10 (13.6%)	19.15 (11.9%)	21.54 (12.9%)	23.18 (14.1%)	24.49 (14.6%)
Coal	14.41 (9.4%)	16.89 (10.4%)	16.70 (9.9%)	13.19 (7.9%)	8.99 (5.3%)	4.38 (2.6%)	5.10 (3.3%)	10.51 (6.5%)	8.34 (5.0%)	5.02 (3.1%)	3.28 (2.0%)
Peat	9.03 (5.9%)	8.92 (5.5%)	8.54 (5.1%)	8.08 (4.8%)	7.98 (4.7%)	7.32 (4.3%)	4.86 (3.1%)	3.08 (1.9%)	2.59 (1.6%)	2.10 (1.3%)	1.49 (0.9%)
Non-ren. waste	0.77 (0.5%)	0.80 (0.5%)	0.78 (0.5%)	1.32 (0.8%)	1.69 (1.0%)	1.69 (1.0%)	1.71 (1.1%)	1.66 (1.0%)	1.73 (1.0%)	1.96 (1.2%)	1.81 (1.1%)
Electricity (net imports)	2.15 (1.4%)	0.67 (0.4%)	-0.71 (-0.4%)	-0.68 (-0.4%)	-0.03 (-0.0%)	0.64 (0.4%)	-0.15 (-0.1%)	1.59 (1.0%)	0.25 (0.2%)	3.27 (2.0%)	5.06 (3.0%)
Total	154.07 (100%)	161.83 (100%)	168.84 (100%)	167.55 (100%)	171.24 (100%)	169.93 (100%)	155.61 (100%)	160.69 (100%)	166.80 (100%)	163.86 (100%)	168.04 (100%)

3.4 Security of supply and import dependency

Energy security, in its simplest terms, means having uninterrupted access to reliable, affordable supplies of energy. Secure supplies of energy are essential for our economy and for maintaining safe and comfortable living conditions.

Energy imports dependency is one of the simplest and most widely used indicators of a country's energy security, with indigenous energy sources generally considered to be more secure than imported energy. While the overall energy imports dependency figure provides a useful context, a deeper understanding of energy security requires more detailed information on individual energy sources. This includes the countries from where each fuel is sourced, global market conditions, transportation and other infrastructure requirements. It also requires analysis of the current trends in energy use, and of the significant changes that will occur in energy use - both nationally and globally over the coming years. Energy security is considered in more detail in a separate SEAI publications [5], [6].

Energy imports dependency is defined as the ratio of net energy imports to gross available energy [7]:

$$\text{Energy import dependency} = \frac{\text{Imports} - \text{Exports}}{\text{Gross available energy}}$$

Or:

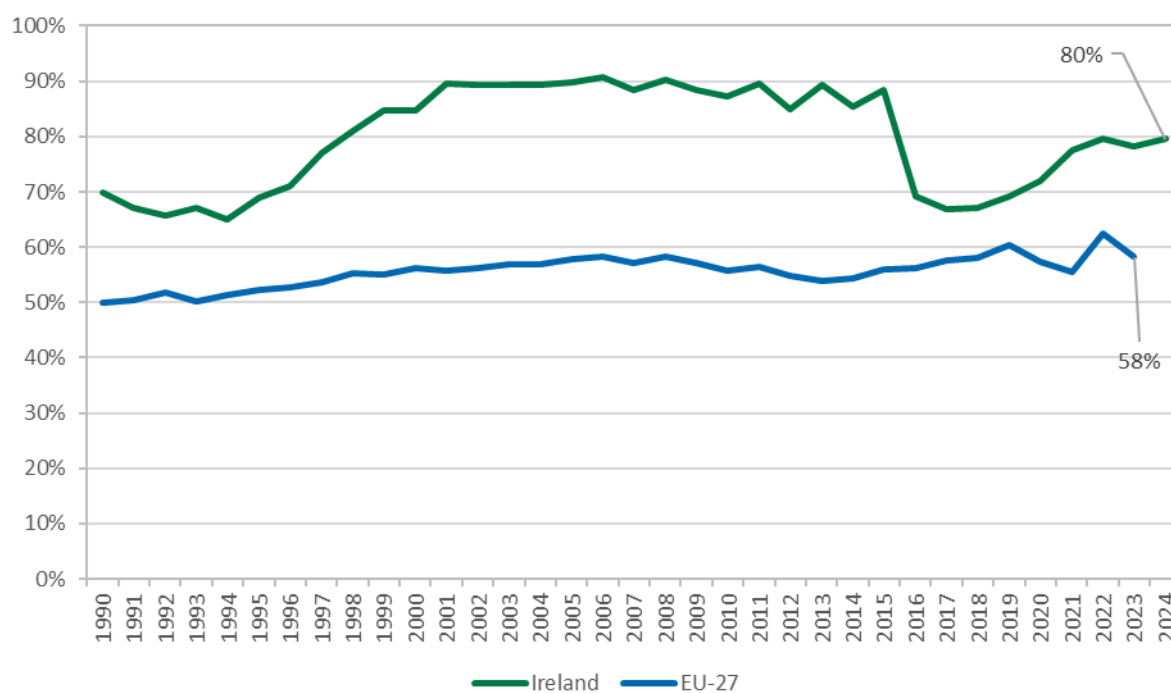
$$\text{Energy import dependency} = \frac{\text{Imports} - \text{Exports}}{\text{Primary energy requirement} + \text{Non energy use} + \text{Int. marine bunkers}}$$

Figure 3.6 illustrates the trend in import dependency since 1990, comparing it with that for the EU 27, and shows the dramatic change in Ireland's import dependency in 2016 resulting from the start of natural gas production from the Corrib gas field. Indigenous production accounted for 32% of Ireland's energy requirements in 1990. From the mid-1990s, import dependency grew significantly due to the increase in energy use, together with the decline in indigenous natural gas production at Kinsale since 1995, and decreasing peat production. Ireland's overall import dependency reached 91% in 2006. In 2016, import dependency fell sharply following the opening of the Corrib gas field but subsequently increased again, as annual production from the field decreased. Import dependency increased slightly

again from 78% in 2023 to 80% in 2024 due to increased primary energy requirement and decreased in primary energy production.

This overall trend reflects the fact that Ireland does not have significant indigenous fossil fuel resources and has only in recent years begun to harness significant quantities of renewables.

Figure 3.6: Import dependency of Ireland and the EU-27



Source: SEAI and Eurostat

4 Energy transformation

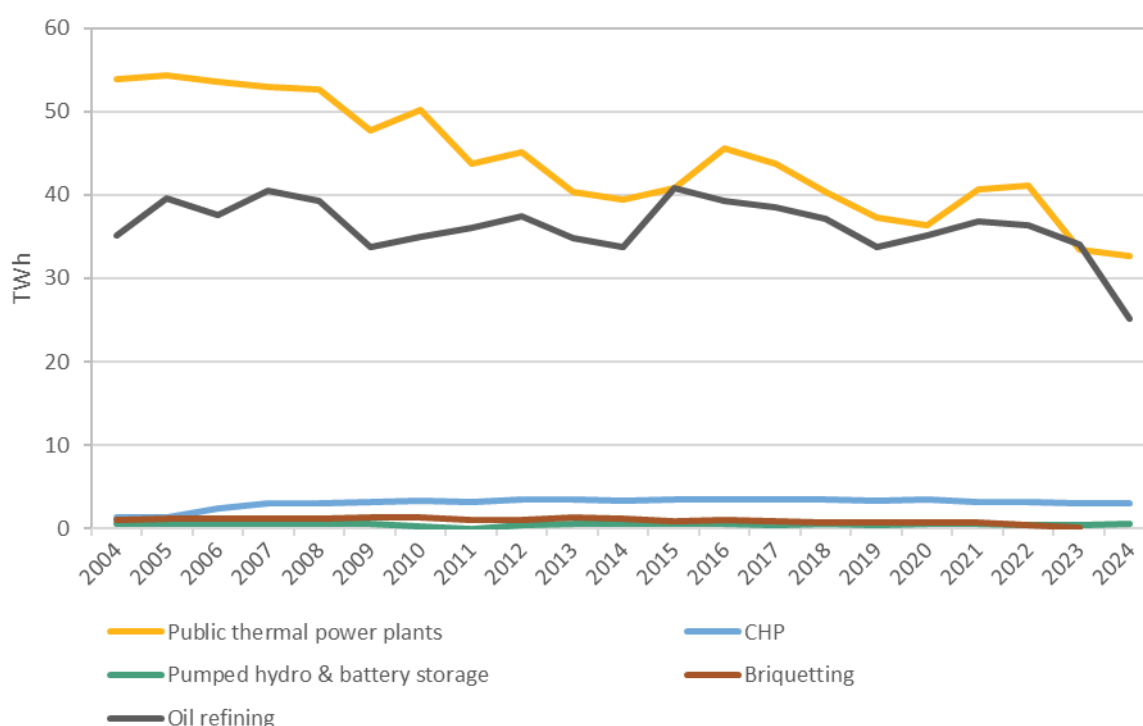
Energy transformation involves converting one fuel type or energy source into another, such as transforming crude oil into petrol and diesel in an oil refinery, or converting coal and gas into electricity in a thermal generation plant. Approximately half of all primary energy undergoes a transformation, of one kind or another, before it reaches an end user for final consumption.

Primary energy supply includes all inputs and losses in energy transformation processes, while final energy use only includes the outputs from those transformation processes. Transformation outputs are less than the primary supply inputs due to the energy required to make the transformations, and losses from those processes.

4.1 Overview of transformation

As shown in Figure 4.1, the two most significant energy transformation processes in Ireland are thermal electricity generation and oil refining. Oil refining has maintained a relatively stable long-term average input of around 37 TWh per year, with a decline in 2024 due to scheduled maintenance. Meanwhile, the transformation input of public thermal power plants has shown a long-term downward trend, as coal, oil, and peat-fired power are gradually replaced by gas, non-combustible renewables, and electricity imports.

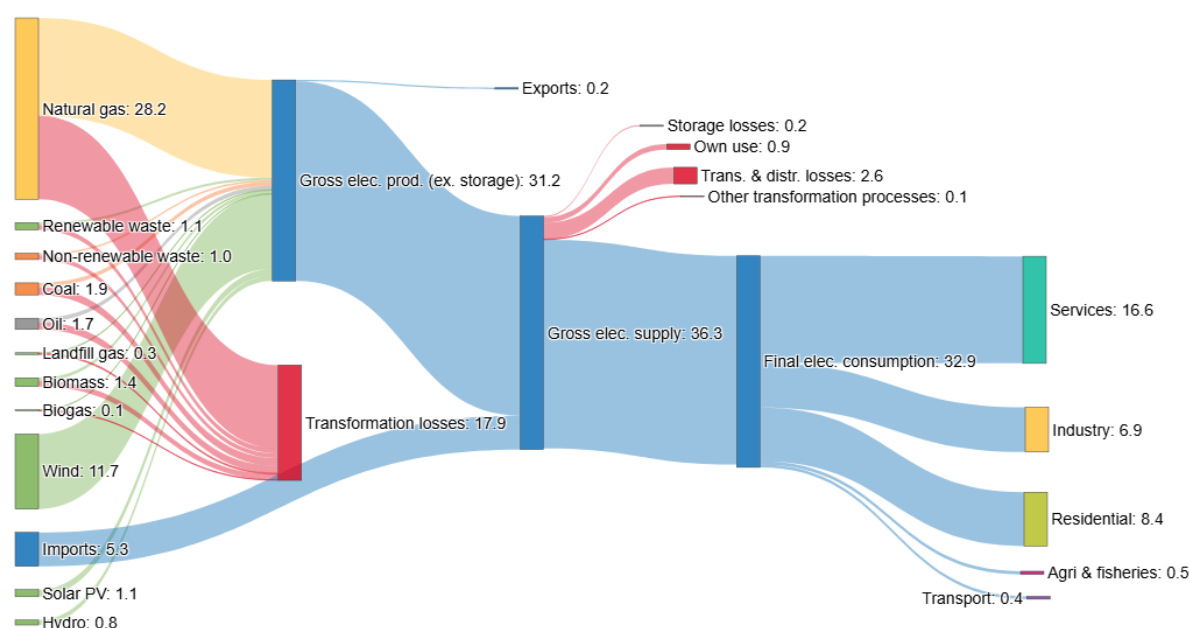
Figure 4.1: Energy inputs to transformation processes



4.2 Electricity generation and supply

Total energy inputs to electricity generation account for approximately one-third of Ireland's total primary energy supply. The Sankey diagram in Figure 4.2 shows the flow of energy from the inputs to electricity generation through to the final electricity used by the different sectors. As the diagram shows, a significant portion of the energy used to generate electricity is lost before the electricity reaches the end user, through a combination of transformation losses, own use of electricity by power plants, energy storage losses, and transmission and distribution losses.

Figure 4.2: Flow of energy in electricity generation and consumption (TWh)¹⁷



4.2.1 Energy input to electricity generation

Figure 4.3 shows the trends in energy input to electricity generation broken out by energy type.

Note that non-combustible renewable sources accounted for a higher share of generated electricity than of *energy input* to electricity generation. This is because the thermal generation of electricity from natural gas and coal has significant energy losses (see Figure 4.2), while electricity generation from non-combustible renewable sources (wind, hydro and solar) is considered to be 100% efficient as this is the point at which the measurable energy enters the system.

Table 4.1 shows the energy input to electricity generation by energy type with comparison to previous years.

¹⁷ GEP refers to *gross electricity production*.

Figure 4.3: Energy input to electricity generation by energy type

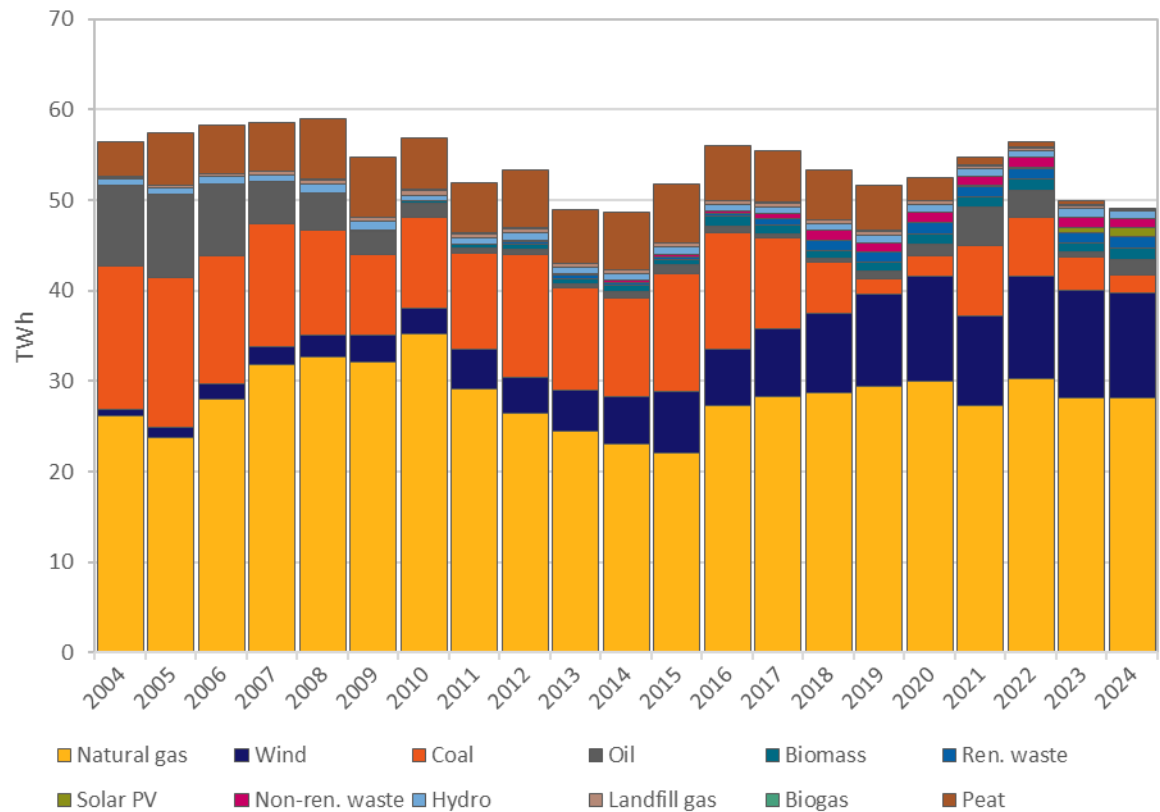


Table 4.1 provides the quantities and shares of energy types that go into electricity generation in Ireland, while Table 4.2 provides a breakdown of the energy input to electricity generation by source (renewable or non-renewable). Figure 4.4 shows the changes in energy input to electricity generation in the last year, by fuel type and energy source.

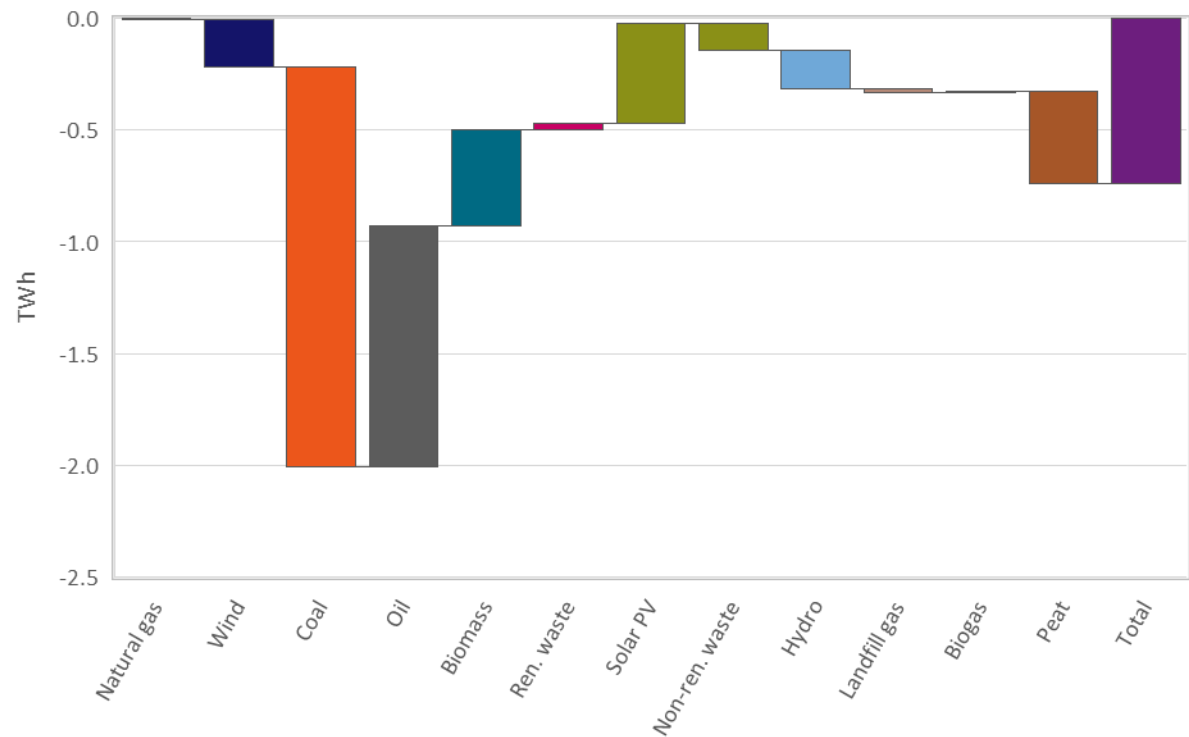
Table 4.1: Energy input to electricity generation by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	22.99 (47.2%)	22.12 (42.7%)	27.30 (48.7%)	28.25 (50.9%)	28.68 (53.8%)	29.38 (56.8%)	29.91 (57.0%)	27.29 (49.9%)	30.21 (53.5%)	28.16 (56.5%)	28.16 (57.3%)
Wind	5.23 (10.7%)	6.68 (12.9%)	6.25 (11.1%)	7.57 (13.6%)	8.78 (16.5%)	10.19 (19.7%)	11.74 (22.4%)	9.94 (18.2%)	11.40 (20.2%)	11.87 (23.8%)	11.65 (23.7%)
Coal	10.94 (22.4%)	13.09 (25.3%)	12.82 (22.9%)	10.09 (18.2%)	5.69 (10.7%)	1.72 (3.3%)	2.26 (4.3%)	7.82 (14.3%)	6.56 (11.6%)	3.66 (7.3%)	1.87 (3.8%)
Oil	0.70 (1.4%)	1.00 (1.9%)	0.80 (1.4%)	0.39 (0.7%)	0.40 (0.8%)	0.91 (1.8%)	1.25 (2.4%)	4.19 (7.7%)	2.97 (5.3%)	0.65 (1.3%)	1.73 (3.5%)
Biomass	0.68 (1.4%)	0.50 (1.0%)	1.02 (1.8%)	0.99 (1.8%)	0.88 (1.7%)	0.90 (1.7%)	1.14 (2.2%)	1.18 (2.2%)	1.26 (2.2%)	0.93 (1.9%)	1.36 (2.8%)
Ren. waste	0.30 (0.6%)	0.30 (0.6%)	0.30 (0.5%)	0.62 (1.1%)	1.16 (2.2%)	1.12 (2.2%)	1.18 (2.3%)	1.13 (2.1%)	1.14 (2.0%)	1.11 (2.2%)	1.15 (2.3%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.03 (0.1%)	0.06 (0.1%)	0.08 (0.2%)	0.15 (0.3%)	0.65 (1.3%)	1.09 (2.2%)
Non-ren. waste	0.29 (0.6%)	0.29 (0.6%)	0.29 (0.5%)	0.66 (1.2%)	1.06 (2.0%)	1.03 (2.0%)	1.08 (2.1%)	1.04 (1.9%)	1.05 (1.9%)	1.09 (2.2%)	0.97 (2.0%)
Hydro	0.71 (1.5%)	0.81 (1.6%)	0.68 (1.2%)	0.69 (1.2%)	0.69 (1.3%)	0.89 (1.7%)	0.93 (1.8%)	0.75 (1.4%)	0.70 (1.2%)	0.94 (1.9%)	0.77 (1.6%)
Landfill gas	0.45 (0.9%)	0.49 (0.9%)	0.46 (0.8%)	0.45 (0.8%)	0.39 (0.7%)	0.37 (0.7%)	0.34 (0.6%)	0.34 (0.6%)	0.30 (0.5%)	0.28 (0.6%)	0.27 (0.5%)
Biogas	0.06 (0.1%)	0.06 (0.1%)	0.08 (0.1%)	0.08 (0.1%)	0.08 (0.2%)	0.10 (0.2%)	0.10 (0.2%)	0.11 (0.2%)	0.12 (0.2%)	0.12 (0.2%)	0.13 (0.3%)
Peat	6.39 (13.1%)	6.44 (12.4%)	6.07 (10.8%)	5.68 (10.2%)	5.50 (10.3%)	5.05 (9.8%)	2.49 (4.7%)	0.83 (1.5%)	0.64 (1.1%)	0.42 (0.8%)	0 (0%)
Total	48.73 (100%)	51.78 (100%)	56.07 (100%)	55.48 (100%)	53.34 (100%)	51.69 (100%)	52.49 (100%)	54.71 (100%)	56.50 (100%)	49.88 (100%)	49.14 (100%)

Table 4.2: Energy input to electricity generation from renewables and non-renewables (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Renewable	7.42 (15.2%)	8.83 (17.1%)	8.80 (15.7%)	10.41 (18.8%)	12.01 (22.5%)	13.59 (26.3%)	15.50 (29.5%)	13.54 (24.8%)	15.07 (26.7%)	15.90 (31.9%)	16.41 (33.4%)
Non-renewable	41.31 (84.8%)	42.95 (82.9%)	47.28 (84.3%)	45.06 (81.2%)	41.33 (77.5%)	38.10 (73.7%)	36.99 (70.5%)	41.17 (75.2%)	41.43 (73.3%)	33.98 (68.1%)	32.73 (66.6%)
Total	48.73 (100%)	51.78 (100%)	56.07 (100%)	55.48 (100%)	53.34 (100%)	51.69 (100%)	52.49 (100%)	54.71 (100%)	56.50 (100%)	49.88 (100%)	49.14 (100%)

Figure 4.4: Change in input energy type to electricity generation in 2024 compared with 2023



4.2.2 Gross electricity production from generation

Electricity output from generation is recorded as *gross electricity production* in the National Energy Balance. As defined by the IEA and Eurostat, gross electricity production is the amount of electrical energy measured at the output terminals of the main generator or inverter and includes energy which is used by plant auxiliaries and transformer losses within the site or station [8].

Electricity output from grid-connected energy storage, *i.e.* pumped hydroelectric or batteries, is not included in the data presented in this section. See section 4.2.5 for details on energy inputs and outputs for grid-connected energy storage.

Figure 4.5 and Table 4.3 show the gross electricity production from each energy sub-type; this includes electricity from all electricity producers, from residential rooftop solar PV to utility scale thermal power plants and wind farms.

Figure 4.5: Gross electricity production from all producers, excluding storage, by energy sub-type

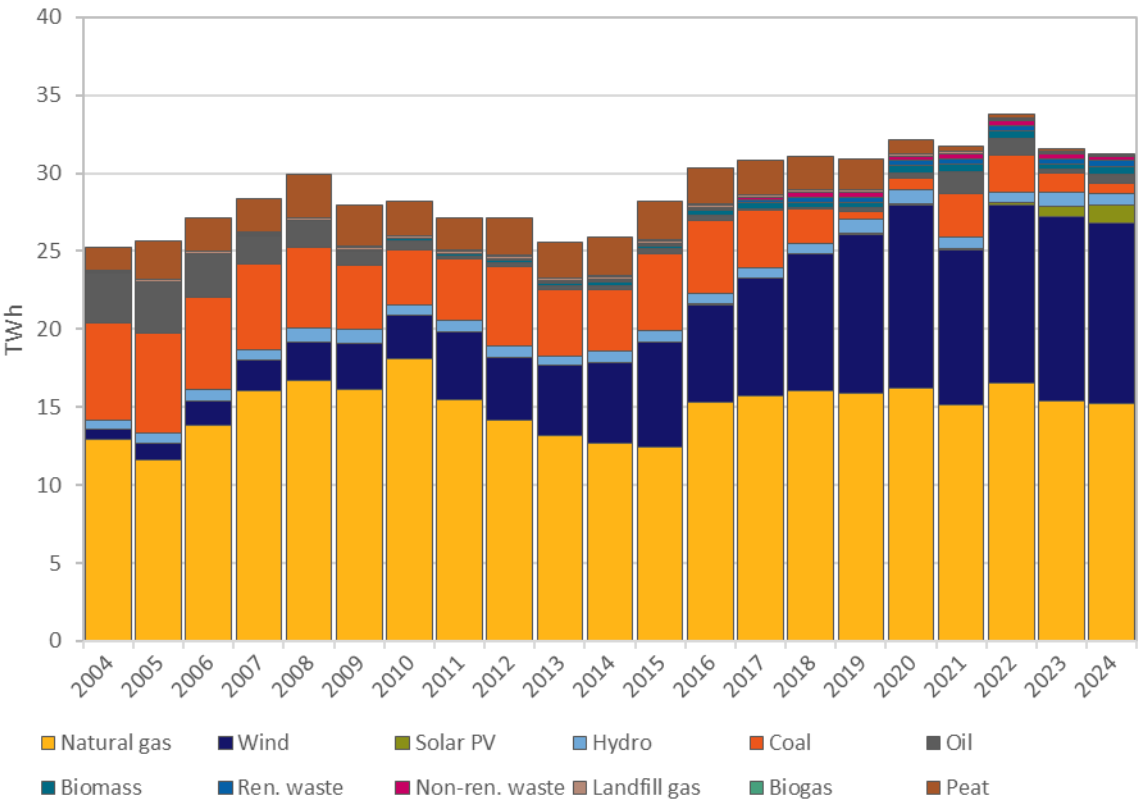


Table 4.3: Gross electricity production from all producers, excluding storage, by energy sub-type (share in brackets)

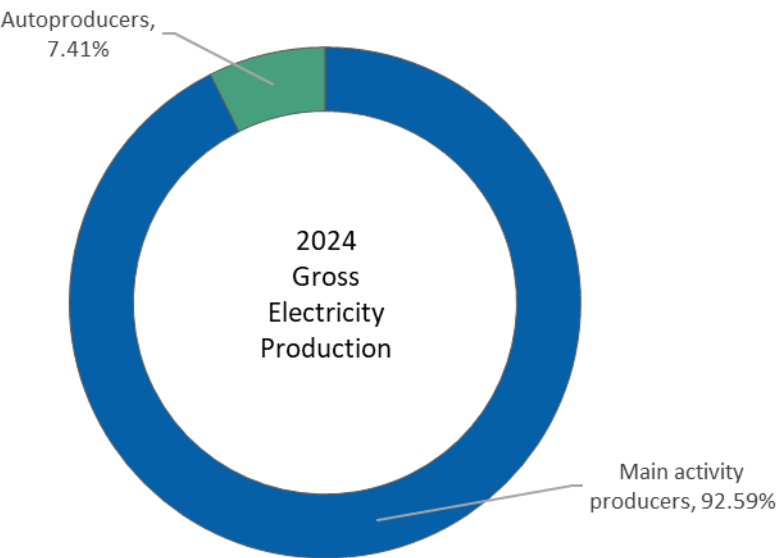
Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	12.64 (48.8%)	12.44 (44.1%)	15.33 (50.6%)	15.68 (50.9%)	16.01 (51.6%)	15.91 (51.5%)	16.24 (50.5%)	15.16 (47.7%)	16.53 (48.9%)	15.35 (48.7%)	15.18 (48.6%)
Wind	5.23 (20.2%)	6.68 (23.7%)	6.25 (20.6%)	7.57 (24.6%)	8.78 (28.3%)	10.19 (33.0%)	11.74 (36.5%)	9.94 (31.3%)	11.40 (33.7%)	11.87 (37.6%)	11.65 (37.3%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.02 (0.1%)	0.03 (0.1%)	0.06 (0.2%)	0.08 (0.3%)	0.15 (0.4%)	0.65 (2.0%)	1.09 (3.5%)
Hydro	0.71 (2.7%)	0.81 (2.9%)	0.68 (2.2%)	0.69 (2.2%)	0.69 (2.2%)	0.89 (2.9%)	0.93 (2.9%)	0.75 (2.4%)	0.70 (2.1%)	0.94 (3.0%)	0.77 (2.5%)
Coal	3.96 (15.3%)	4.87 (17.3%)	4.70 (15.5%)	3.64 (11.8%)	2.15 (6.9%)	0.51 (1.6%)	0.67 (2.1%)	2.72 (8.6%)	2.37 (7.0%)	1.21 (3.8%)	0.63 (2.0%)
Oil	0.26 (1.0%)	0.33 (1.2%)	0.29 (0.9%)	0.14 (0.5%)	0.14 (0.4%)	0.28 (0.9%)	0.38 (1.2%)	1.45 (4.6%)	1.07 (3.2%)	0.23 (0.7%)	0.60 (1.9%)
Biomass	0.26 (1.0%)	0.20 (0.7%)	0.40 (1.3%)	0.38 (1.2%)	0.33 (1.1%)	0.35 (1.1%)	0.43 (1.3%)	0.47 (1.5%)	0.51 (1.5%)	0.35 (1.1%)	0.51 (1.6%)
Ren. waste	0.07 (0.3%)	0.08 (0.3%)	0.08 (0.2%)	0.15 (0.5%)	0.33 (1.1%)	0.32 (1.0%)	0.33 (1.0%)	0.35 (1.1%)	0.35 (1.0%)	0.33 (1.1%)	0.34 (1.1%)
Non-ren. waste	0.07 (0.3%)	0.07 (0.3%)	0.07 (0.2%)	0.16 (0.5%)	0.30 (1.0%)	0.29 (1.0%)	0.30 (0.9%)	0.32 (1.0%)	0.32 (0.9%)	0.33 (1.0%)	0.29 (0.9%)
Landfill gas	0.17 (0.6%)	0.18 (0.6%)	0.16 (0.5%)	0.16 (0.5%)	0.14 (0.4%)	0.13 (0.4%)	0.12 (0.4%)	0.12 (0.4%)	0.10 (0.3%)	0.10 (0.3%)	0.09 (0.3%)
Biogas	0.04 (0.1%)	0.03 (0.1%)	0.04 (0.1%)	0.04 (0.1%)	0.04 (0.1%)	0.06 (0.2%)	0.05 (0.2%)	0.05 (0.2%)	0.06 (0.2%)	0.05 (0.2%)	0.06 (0.2%)
Peat	2.50 (9.6%)	2.52 (8.9%)	2.32 (7.6%)	2.16 (7.0%)	2.09 (6.7%)	1.93 (6.2%)	0.92 (2.9%)	0.32 (1.0%)	0.25 (0.7%)	0.15 (0.5%)	0 (0%)
Total	25.89 (100%)	28.21 (100%)	30.32 (100%)	30.80 (100%)	31.04 (100%)	30.88 (100%)	32.17 (100%)	31.74 (100%)	33.80 (100%)	31.55 (100%)	31.22 (100%)

Figure 4.6 shows the split in gross electricity production between two categories of producers:

- **main activity producers** - entities whose primary activity is the production of electricity for sale to other parties; this category includes public thermal power plants, wind farms, and solar farms.
- **autoproducers** – entities that generate electricity either wholly or partly for their own use which supports their main activity; this category includes rooftop solar PV on homes or businesses or CHP units at manufacturing sites.

The following sub-sections, 4.2.2.1 and 4.2.2.2, provide the gross electricity production from main activity producers and auto producers, respectively.

Figure 4.6: Share of gross electricity production by producer category



4.2.2.1 Main activity producers

Figure 4.7 and Table 4.4 present the gross electricity production from main activity producers by energy sub-type. The production figures for combustible fuels (natural gas, coal, oil, biomass, renewable waste, non-renewable waste, landfill gas and peat) correspond to the output from *public thermal power plants* as recorded in the National Energy Balance, emissions from which are included in *public electricity and heat production* (IPCC category 1.A.1.a) in the EPA's National GHG Emissions Inventories and in the *electricity CAP* sector.

Figure 4.7: Main activity producer - gross electricity production by energy type, excluding storage

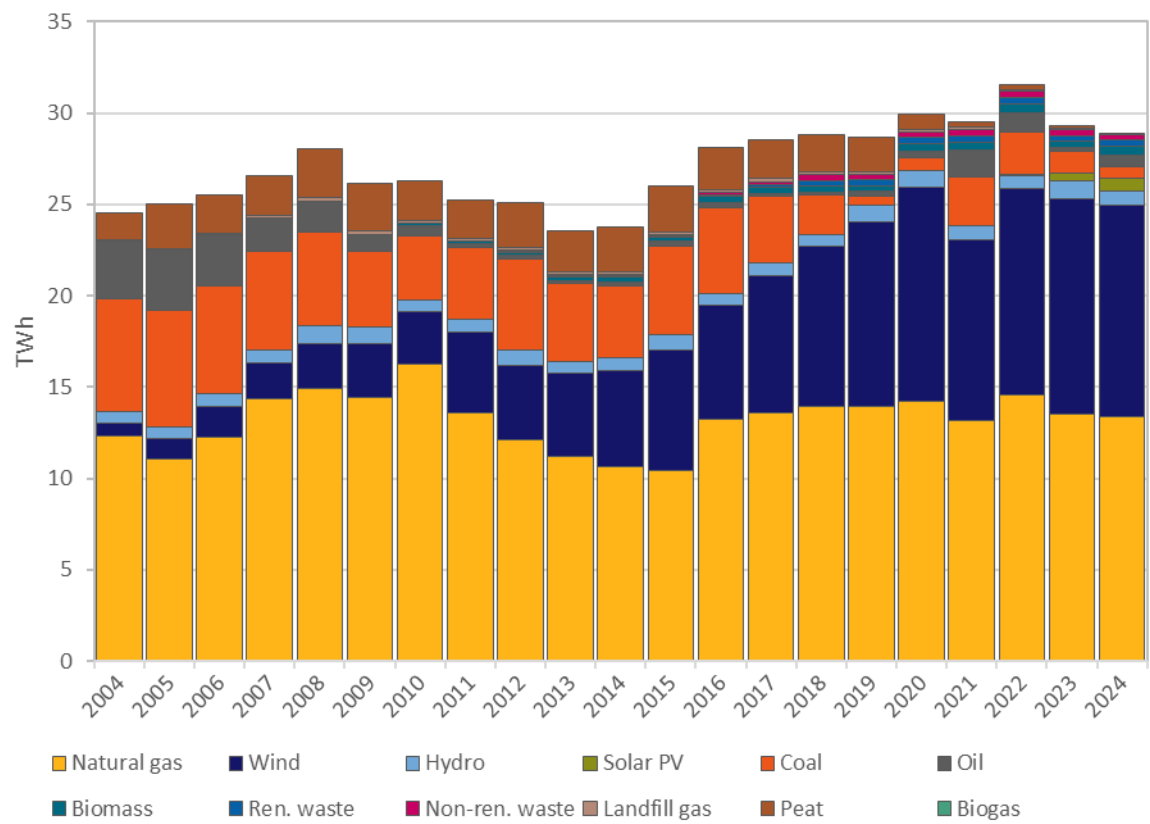


Table 4.4: Main activity producer - gross electricity production by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	10.67 (44.8%)	10.40 (40.0%)	13.24 (47.2%)	13.58 (47.6%)	13.94 (48.4%)	13.92 (48.5%)	14.25 (47.5%)	13.19 (44.7%)	14.56 (46.2%)	13.53 (46.1%)	13.38 (46.3%)
Wind	5.20 (21.9%)	6.65 (25.5%)	6.22 (22.1%)	7.53 (26.4%)	8.74 (30.3%)	10.13 (35.3%)	11.69 (39.0%)	9.88 (33.5%)	11.33 (35.9%)	11.80 (40.2%)	11.59 (40.1%)
Hydro	0.71 (3.0%)	0.81 (3.1%)	0.68 (2.4%)	0.69 (2.4%)	0.69 (2.4%)	0.89 (3.1%)	0.93 (3.1%)	0.75 (2.5%)	0.70 (2.2%)	0.94 (3.2%)	0.77 (2.7%)
Solar PV	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.02 (0.1%)	0.41 (1.4%)	0.73 (2.5%)
Coal	3.96 (16.6%)	4.87 (18.7%)	4.70 (16.7%)	3.64 (12.8%)	2.15 (7.5%)	0.51 (1.8%)	0.67 (2.2%)	2.72 (9.2%)	2.37 (7.5%)	1.21 (4.1%)	0.63 (2.2%)
Oil	0.23 (1.0%)	0.28 (1.1%)	0.26 (0.9%)	0.14 (0.5%)	0.14 (0.5%)	0.28 (1.0%)	0.38 (1.3%)	1.45 (4.9%)	1.07 (3.4%)	0.22 (0.8%)	0.60 (2.1%)
Biomass	0.25 (1.1%)	0.18 (0.7%)	0.38 (1.4%)	0.37 (1.3%)	0.32 (1.1%)	0.33 (1.1%)	0.41 (1.4%)	0.45 (1.5%)	0.48 (1.5%)	0.32 (1.1%)	0.49 (1.7%)
Ren. waste	0.07 (0.3%)	0.08 (0.3%)	0.08 (0.3%)	0.15 (0.5%)	0.33 (1.1%)	0.32 (1.1%)	0.33 (1.1%)	0.35 (1.2%)	0.35 (1.1%)	0.33 (1.1%)	0.34 (1.2%)
Non-ren. waste	0.07 (0.3%)	0.07 (0.3%)	0.07 (0.3%)	0.16 (0.6%)	0.30 (1.0%)	0.29 (1.0%)	0.30 (1.0%)	0.32 (1.1%)	0.32 (1.0%)	0.33 (1.1%)	0.29 (1.0%)
Landfill gas	0.17 (0.7%)	0.18 (0.7%)	0.16 (0.6%)	0.16 (0.6%)	0.14 (0.5%)	0.13 (0.5%)	0.12 (0.4%)	0.12 (0.4%)	0.10 (0.3%)	0.10 (0.3%)	0.09 (0.3%)
Peat	2.47 (10.4%)	2.50 (9.6%)	2.29 (8.2%)	2.14 (7.5%)	2.07 (7.2%)	1.91 (6.7%)	0.90 (3.0%)	0.31 (1.0%)	0.23 (0.7%)	0.14 (0.5%)	0 (0%)
Biogas	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	23.79 (100%)	26.01 (100%)	28.09 (100%)	28.56 (100%)	28.83 (100%)	28.71 (100%)	29.97 (100%)	29.53 (100%)	31.53 (100%)	29.35 (100%)	28.91 (100%)

4.2.2.2 Autoproducers

Figure 4.8 and Table 4.5 present the gross electricity production from autoproducers by energy sub-type. The production figures for combustible fuels (natural gas, biogas, biomass, oil, coal and peat) correspond to the output from *combined heat and power* as recorded in the National Energy Balance. In the autoproducer category, coal and peat are no longer used to generate electricity, with coal was last used in 2005 and peat in 2023.

Figure 4.8: Autoproducers - gross electricity production by energy type

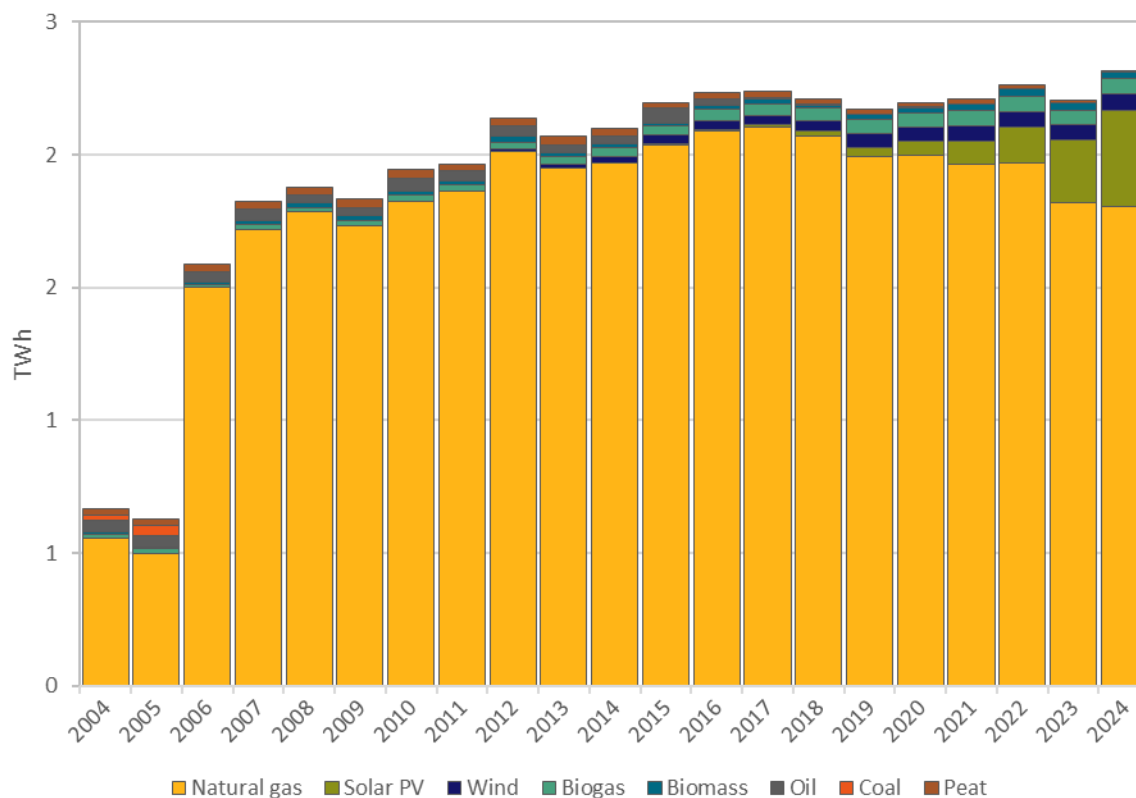


Table 4.5: Autoproducers - gross electricity production by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	1.97 (93.8%)	2.04 (92.8%)	2.09 (93.5%)	2.10 (93.9%)	2.07 (93.7%)	1.99 (91.8%)	2.00 (90.8%)	1.97 (89.0%)	1.97 (87.0%)	1.82 (82.6%)	1.81 (78.0%)
Solar PV	0.00 (0.1%)	0.00 (0.1%)	0.00 (0.2%)	0.01 (0.4%)	0.02 (0.8%)	0.03 (1.6%)	0.06 (2.5%)	0.08 (3.8%)	0.13 (5.8%)	0.23 (10.6%)	0.36 (15.7%)
Wind	0.02 (1.2%)	0.04 (1.7%)	0.03 (1.4%)	0.04 (1.6%)	0.04 (1.8%)	0.05 (2.4%)	0.05 (2.4%)	0.06 (2.7%)	0.06 (2.7%)	0.06 (2.8%)	0.06 (2.7%)
Biogas	0.04 (1.7%)	0.03 (1.3%)	0.04 (2.0%)	0.04 (2.0%)	0.04 (2.0%)	0.06 (2.6%)	0.05 (2.3%)	0.05 (2.4%)	0.06 (2.6%)	0.05 (2.4%)	0.06 (2.4%)
Biomass	0.01 (0.7%)	0.01 (0.6%)	0.02 (0.7%)	0.02 (0.7%)	0.01 (0.6%)	0.02 (0.8%)	0.02 (1.0%)	0.02 (1.1%)	0.03 (1.1%)	0.03 (1.2%)	0.03 (1.1%)
Oil	0.03 (1.4%)	0.06 (2.5%)	0.02 (1.1%)	0.01 (0.2%)	0.00 (0.2%)	0.00 (0.2%)	0.00 (0.1%)	0.00 (0.1%)	0.00 (0.1%)	0.00 (0.2%)	0.00 (0.1%)
Coal	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Peat	0.03 (1.2%)	0.02 (1.0%)	0.02 (1.1%)	0.02 (1.1%)	0.02 (0.9%)	0.02 (0.7%)	0.02 (0.8%)	0.02 (0.8%)	0.01 (0.6%)	0.01 (0.3%)	0 (0%)
Total	2.10 (100%)	2.20 (100%)	2.24 (100%)	2.24 (100%)	2.21 (100%)	2.17 (100%)	2.20 (100%)	2.21 (100%)	2.26 (100%)	2.20 (100%)	2.31 (100%)

4.2.3 Electricity imports and exports

Figure 4.9 and Table 4.6 show the annual quantities of electricity imported and export; net imports (*i.e.* imports minus exports) are also shown. As with all energy products, all physical flows of electricity across the Ireland's borders are considered imports or exports in the national energy statistics; this includes flows of electricity between Ireland and Northern Ireland, which are within the integrated single electricity market (I-SEM), as well as interconnector flows between Ireland and Great Britain, which are between I-SEM and the GB electricity market.

Figure 4.9: Electricity imports and exports

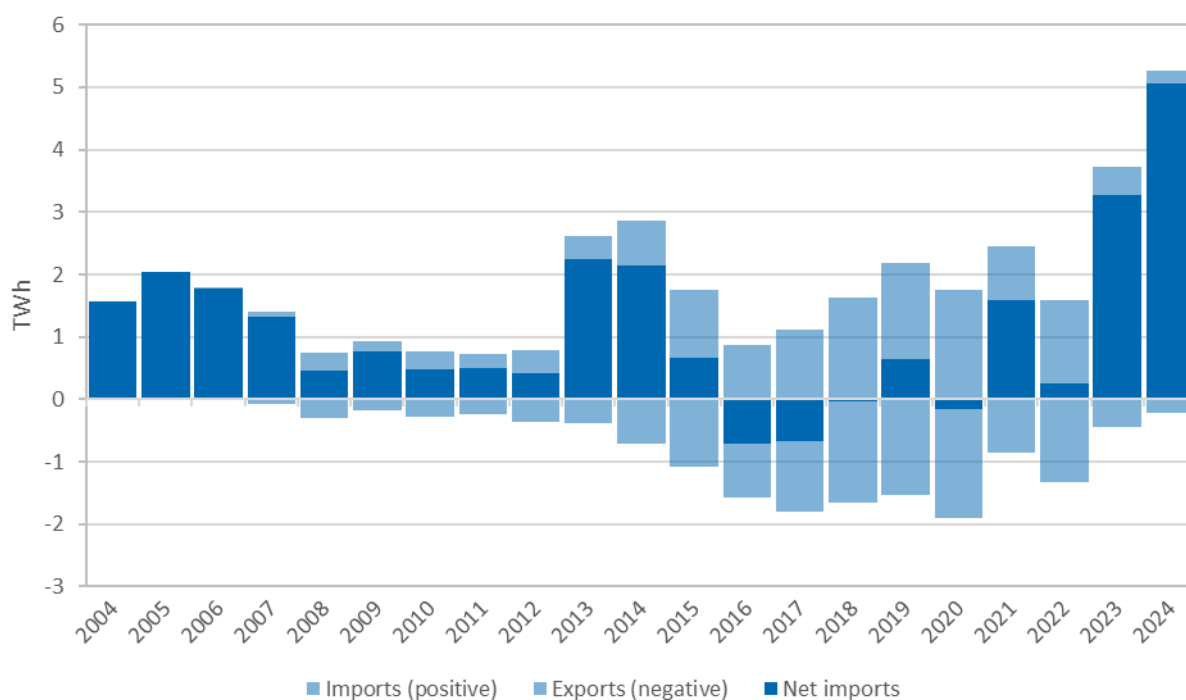


Table 4.6: Electricity imports and exports

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Imports (positive)	2.85	1.75	0.87	1.12	1.62	2.18	1.76	2.45	1.58	3.72	5.27
Exports (negative)	-0.70	-1.08	-1.58	-1.79	-1.65	-1.54	-1.91	-0.86	-1.33	-0.45	-0.21
Net imports	2.15	0.67	-0.71	-0.68	-0.03	0.64	-0.15	1.59	0.25	3.27	5.06

4.2.4 Gross electricity supply

This subsection presents the gross quantity of electricity supplied annually in Ireland, prior to losses, own use or energy storage.

The follow definition of gross electricity supply is followed:

Gross electricity supply = gross electricity production from generation (excluding storage) + net electricity imports

Figure 4.10 and Table 4.7 detail the *gross electricity supply* from each energy type. It includes all sources of electricity, from residential rooftop solar PV to utility scale thermal power plants, wind farms and net imports. It includes all electricity that enters the national grid, as well as electricity that is generated and consumed on the same site without entering the grid (referred to as auto-production, auto-consumption or self-consumption).

Figure 4.10: Gross electricity supply by energy type

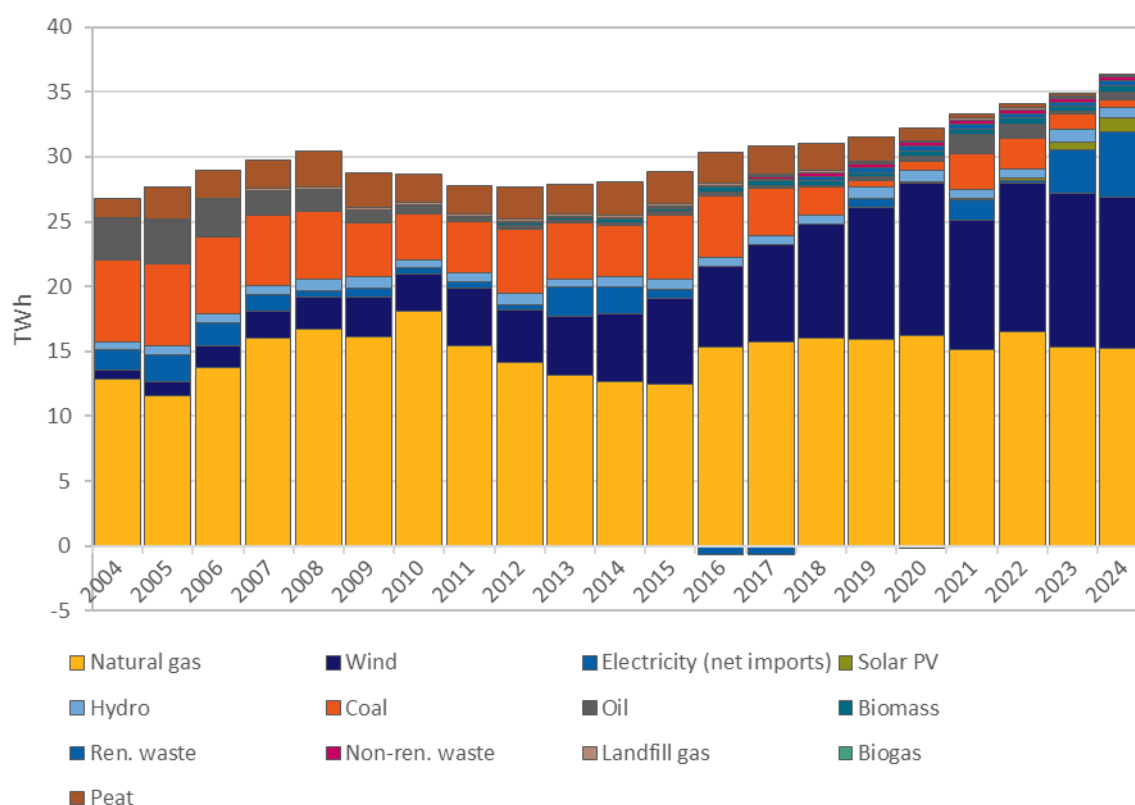


Table 4.7: Gross electricity supply by energy type (share in brackets)

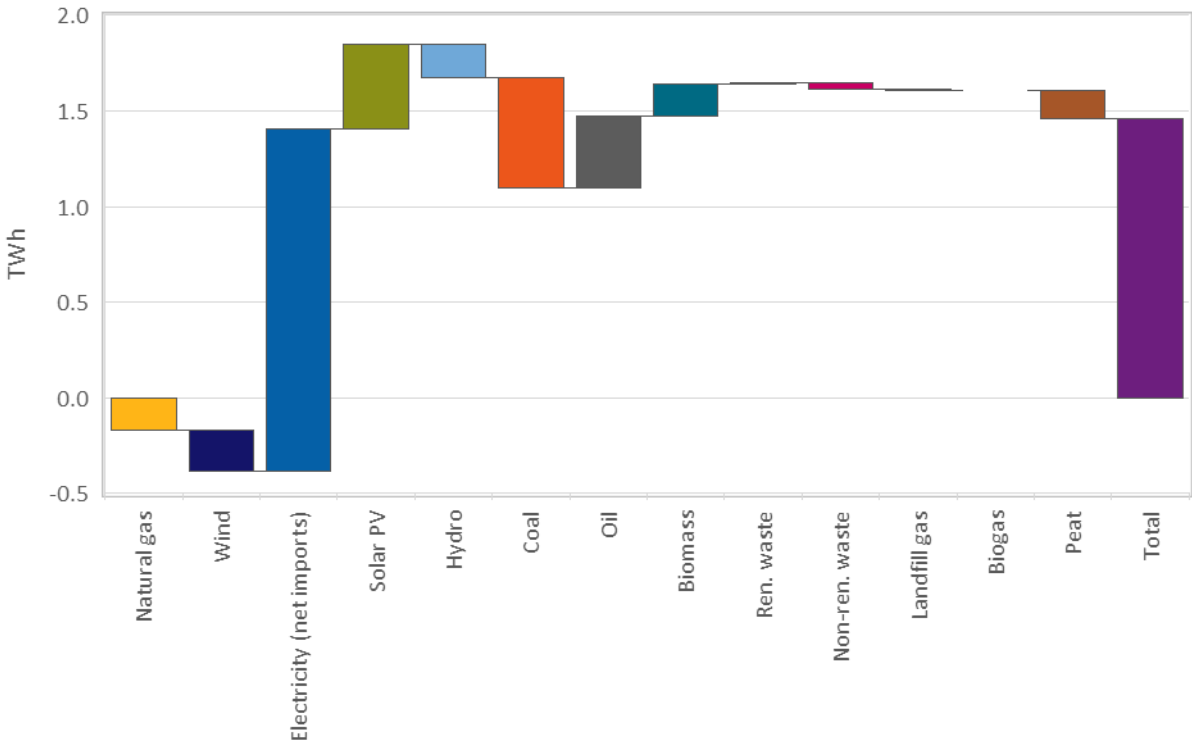
Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	12.64 (45.1%)	12.44 (43.1%)	15.33 (51.8%)	15.68 (52.1%)	16.01 (51.6%)	15.91 (50.5%)	16.24 (50.7%)	15.16 (45.5%)	16.53 (48.6%)	15.35 (44.1%)	15.18 (41.8%)
Wind	5.23 (18.6%)	6.68 (23.1%)	6.25 (21.1%)	7.57 (25.1%)	8.78 (28.3%)	10.19 (32.3%)	11.74 (36.7%)	9.94 (29.8%)	11.40 (33.5%)	11.87 (34.1%)	11.65 (32.1%)
Electricity (net imports)	2.15 (7.7%)	0.67 (2.3%)	-0.71 (-2.4%)	-0.68 (-2.3%)	-0.03 (-0.1%)	0.64 (2.0%)	-0.15 (-0.5%)	1.59 (4.8%)	0.25 (0.7%)	3.27 (9.4%)	5.06 (13.9%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.02 (0.1%)	0.03 (0.1%)	0.06 (0.2%)	0.08 (0.3%)	0.15 (0.4%)	0.65 (1.9%)	1.09 (3.0%)
Hydro	0.71 (2.5%)	0.81 (2.8%)	0.68 (2.3%)	0.69 (2.3%)	0.69 (2.2%)	0.89 (2.8%)	0.93 (2.9%)	0.75 (2.2%)	0.70 (2.1%)	0.94 (2.7%)	0.77 (2.1%)
Coal	3.96 (14.1%)	4.87 (16.9%)	4.70 (15.9%)	3.64 (12.1%)	2.15 (6.9%)	0.51 (1.6%)	0.67 (2.1%)	2.72 (8.2%)	2.37 (6.9%)	1.21 (3.5%)	0.63 (1.7%)
Oil	0.26 (0.9%)	0.33 (1.1%)	0.29 (1.0%)	0.14 (0.5%)	0.14 (0.4%)	0.28 (0.9%)	0.38 (1.2%)	1.45 (4.4%)	1.07 (3.1%)	0.23 (0.6%)	0.60 (1.7%)
Biomass	0.26 (0.9%)	0.20 (0.7%)	0.40 (1.3%)	0.38 (1.3%)	0.33 (1.1%)	0.35 (1.1%)	0.43 (1.3%)	0.47 (1.4%)	0.51 (1.5%)	0.35 (1.0%)	0.51 (1.4%)
Ren. waste	0.07 (0.3%)	0.08 (0.3%)	0.08 (0.3%)	0.15 (0.5%)	0.33 (1.1%)	0.32 (1.0%)	0.33 (1.0%)	0.35 (1.1%)	0.35 (1.0%)	0.33 (1.0%)	0.34 (0.9%)
Non-ren. waste	0.07 (0.2%)	0.07 (0.3%)	0.07 (0.2%)	0.16 (0.5%)	0.30 (1.0%)	0.29 (0.9%)	0.30 (0.9%)	0.32 (1.0%)	0.32 (0.9%)	0.33 (0.9%)	0.29 (0.8%)
Landfill gas	0.17 (0.6%)	0.18 (0.6%)	0.16 (0.6%)	0.16 (0.5%)	0.14 (0.4%)	0.13 (0.4%)	0.12 (0.4%)	0.12 (0.4%)	0.10 (0.3%)	0.10 (0.3%)	0.09 (0.2%)
Biogas	0.04 (0.1%)	0.03 (0.1%)	0.04 (0.1%)	0.04 (0.1%)	0.04 (0.1%)	0.06 (0.2%)	0.05 (0.2%)	0.05 (0.2%)	0.06 (0.2%)	0.05 (0.2%)	0.06 (0.2%)
Peat	2.50 (8.9%)	2.52 (8.7%)	2.32 (7.8%)	2.16 (7.2%)	2.09 (6.7%)	1.93 (6.1%)	0.92 (2.9%)	0.32 (1.0%)	0.25 (0.7%)	0.15 (0.4%)	0 (0%)
Total	28.04 (100%)	28.88 (100%)	29.61 (100%)	30.12 (100%)	31.01 (100%)	31.52 (100%)	32.02 (100%)	33.33 (100%)	34.05 (100%)	34.83 (100%)	36.28 (100%)

Table 4.8 shows gross electricity production (including net imports) from renewables, non-renewables and net imports. Figure 4.11 shows the changes in gross electricity supply from 2023 to 2024.

Table 4.8: Gross electricity supply from renewables, non-renewables and net imports (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Renewable	6.48 (23.1%)	7.97 (27.6%)	7.61 (25.7%)	9.01 (29.9%)	10.34 (33.4%)	11.96 (37.9%)	13.65 (42.6%)	11.77 (35.3%)	13.26 (38.9%)	14.29 (41.0%)	14.51 (40.0%)
Non-renewable	19.42 (69.2%)	20.24 (70.1%)	22.71 (76.7%)	21.79 (72.4%)	20.70 (66.7%)	18.92 (60.0%)	18.52 (57.8%)	19.97 (59.9%)	20.54 (60.3%)	17.27 (49.6%)	16.71 (46.1%)
Electricity (net imports)	2.15 (7.7%)	0.67 (2.3%)	-0.71 (-2.4%)	-0.68 (-2.3%)	-0.03 (-0.1%)	0.64 (2.0%)	-0.15 (-0.5%)	1.59 (4.8%)	0.25 (0.7%)	3.27 (9.4%)	5.06 (13.9%)
Total	28.04 (100%)	28.88 (100%)	29.61 (100%)	30.12 (100%)	31.01 (100%)	31.52 (100%)	32.02 (100%)	33.33 (100%)	34.05 (100%)	34.83 (100%)	36.28 (100%)

Figure 4.11: Change in gross electricity supply by energy type in 2024 compared with 2023



4.2.5 Energy storage

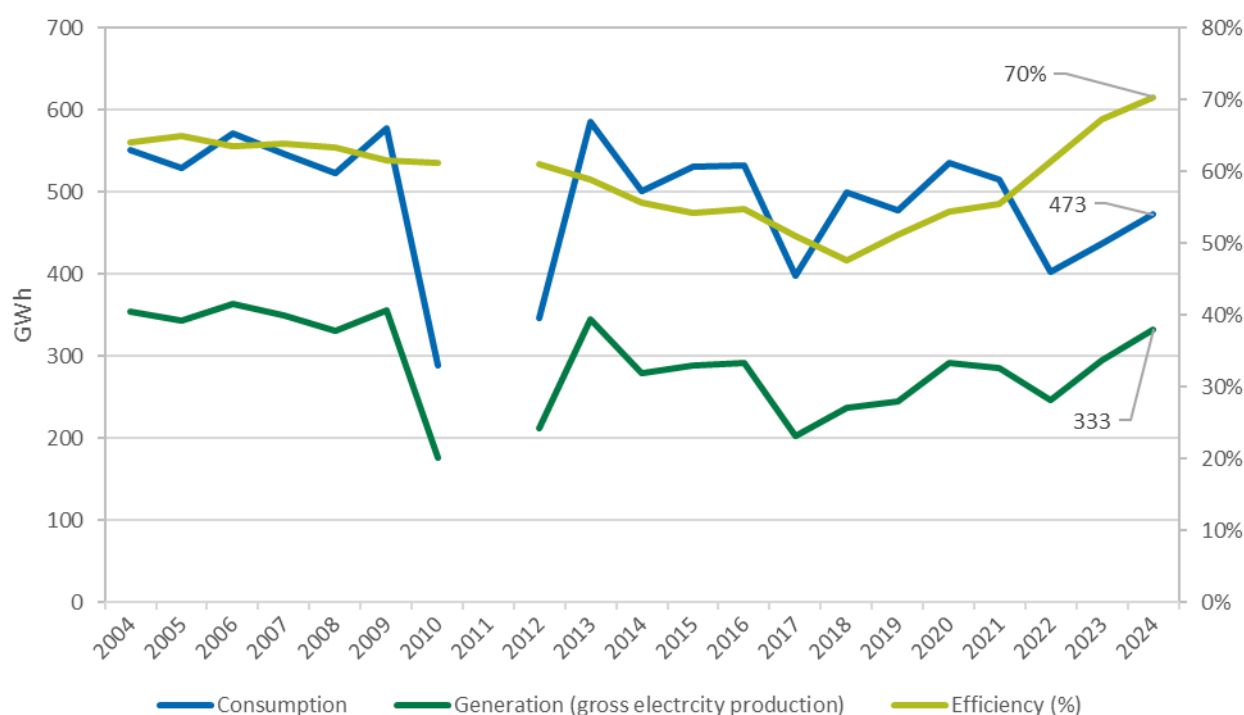
4.2.5.1 Pumped hydroelectric storage

Pumped hydroelectricity storage is the process of using electricity to pump water to an uphill reservoir, and later releasing the water from the reservoir through a turbine to generate electricity. A pumped storage facility acts like a battery to store relatively large amounts of energy. There is one pumped hydroelectric station in Ireland at Turlough Hill in County Wicklow, with a total capacity of 292 MW. It can operate for roughly 6 hours at maximum generating capacity.

The electricity generated from pumped hydro storage is not considered hydroelectricity for statistical purposes and is not counted as renewable energy. Pumped storage facilities act to store energy previously generated by other sources. Although it is not a renewable source, pumped hydro storage is useful for integrating and smoothing variable non-synchronous renewable electricity sources, such as wind, onto the electricity system.

Figure 4.12 shows the annual consumption and generation of electricity, along with associated overall efficiency, at Turlough Hill. The facility underwent significant overhaul work from 2010 to 2012, producing no electricity in 2011.

Figure 4.12: Pumped hydroelectric storage (Turlough Hill)



4.2.5.2 Battery storage

Figure 4.13 and Table 4.9 show the generation capacity of grid-connected battery storage at the end of each year, while Figure 4.14 and Table 4.10 show the storage capacity. Generation capacity is the maximum electrical power that the unit(s) can deliver, defined as maximum export capacity (MEC) to the grid. Storage capacity is the maximum energy the unit(s) can store when fully charged.

The data presented here only includes battery storage capable of exporting power to the grid with an MEC of at least 0.2 MW. Consequently, batteries installed in homes or businesses for the purposes of storing energy for own use are not included in this dataset.

Approximately 21 battery storage units had been connected to Ireland's transmission and distribution networks by the end of 2024. Table 4.11 shows the annual electrical energy consumed and generated by these units; 2022 is the first year for which data is available.

Figure 4.13: Battery generation capacity [MW] at year end

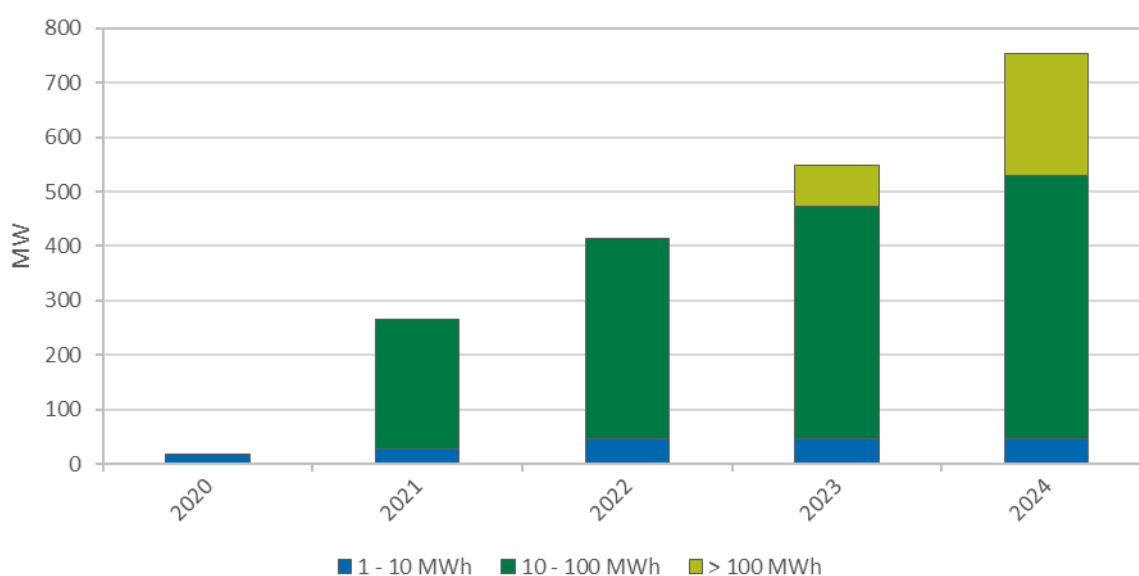
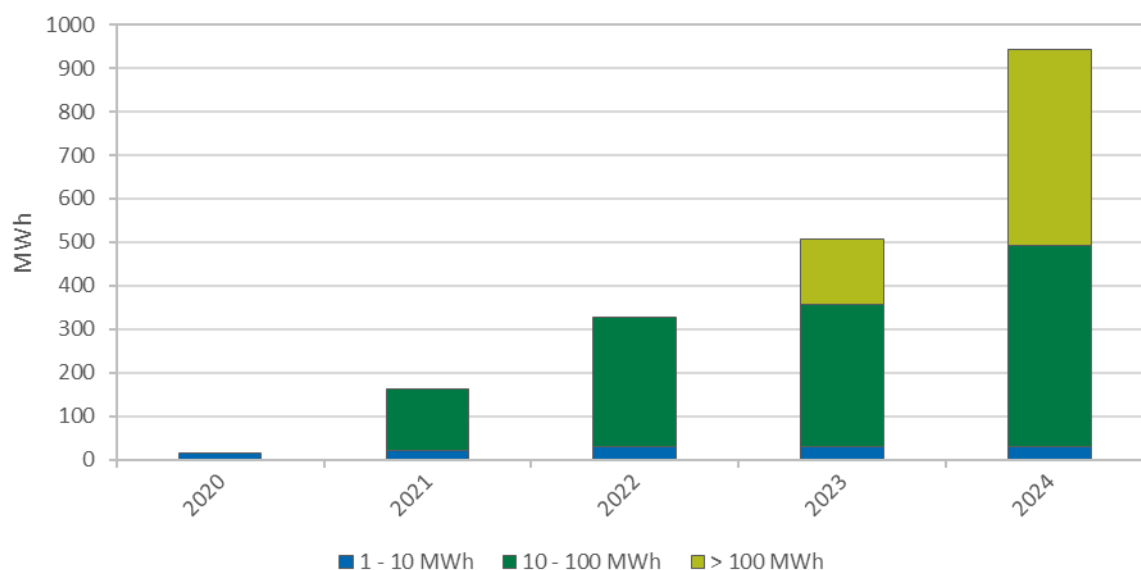


Table 4.9: Battery generation capacity at year end

Generation capacity [MW]	2020	2021	2022	2023	2024
1 - 10 MWh	19.50	28.50	47.50	47.50	47.50
10 - 100 MWh	0	237.40	366.40	426.40	481.40
> 100 MWh	0	0	0	75.00	225.00
Total	19.50	265.90	413.90	548.90	753.90

Figure 4.14: Battery storage – energy storage capacity (MWh) at year end**Table 4.10: Battery storage capacity at year end**

Storage capacity [MWh]	2020	2021	2022	2023	2024
1 - 10 MWh	15.24	19.74	30.46	30.46	30.46
10 - 100 MWh	0	144.20	297.45	327.45	462.45
> 100 MWh	0	0	0	150.00	450.00
Total	15.24	163.94	327.91	507.91	942.91

Table 4.11: Battery storage consumption and generation

Energy [GWh]	2022	2023	2024
Consumption	24.24	35.14	70.14
Generation	3.69	7.17	26.86

4.2.6 Combined heat and power generation

Combined heat and power (CHP) is the simultaneous generation of usable heat and electricity in a single process. In conventional electricity generation, much of the input energy is lost as waste heat. The efficiency of a CHP plant can be 20-25% higher than the combined efficiency of heat-only boilers and conventional power stations. Also, if embedded in the network close to the point of electrical consumption, CHP can avoid some of the transmission losses incurred by centralised generation. In the right circumstances, despite higher capital outlay, CHP can be an economical means of improving the efficiency of energy use and reducing emissions. Table 4.5 shows the operational electrical capacity (MWe) over the last ten years, broken down by fuel.

All of the CHP plants currently operating in Ireland are categorised as autoproducers, that is they generate electricity to support their own primary activities, and so their gross electricity production is included within the gross electricity production of autoproducers in sub-section 4.2.2.2.

Table 4.12: Operational CHP electrical capacity by fuel (share in brackets)

Capacity (MWe)	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural Gas	285.4 (91.8%)	287.7 (92.1%)	285.5 (91.5%)	292.1 (91.7%)	298.3 (93.5%)	300.3 (93.2%)	301.2 (93.1%)	303.7 (93.2%)	305.5 (93.2%)	301.8 (93.2%)	306.1 (94.0%)
Solid Fuels	5.2 (1.7%)	5.2 (1.7%)	5.2 (1.7%)	5.2 (1.6%)	5.2 (1.6%)	2.6 (0.8%)	2.6 (0.8%)	2.6 (0.8%)	2.6 (0.8%)	2.6 (0.8%)	0 (0%)
Biomass	5.4 (1.7%)	5.4 (1.7%)	5.4 (1.7%)	5.4 (1.7%)	5.4 (1.7%)	6.6 (2.0%)	6.6 (2.0%)	6.6 (2.0%)	6.6 (2.0%)	6.6 (2.0%)	6.6 (2.0%)
Oil	8.8 (2.8%)	7.9 (2.5%)	7.5 (2.4%)	7.6 (2.4%)	1.0 (0.3%)	1.0 (0.3%)	1.1 (0.3%)	1.1 (0.3%)	1.1 (0.3%)	0.9 (0.3%)	0.9 (0.3%)
Biogas	6.3 (2.0%)	6.3 (2.0%)	8.4 (2.7%)	8.4 (2.6%)	9.3 (2.9%)	11.9 (3.7%)	11.9 (3.7%)	11.9 (3.6%)	11.9 (3.6%)	11.9 (3.7%)	11.9 (3.7%)
Total	311.0 (100%)	312.5 (100%)	312.1 (100%)	318.7 (100%)	319.2 (100%)	322.3 (100%)	323.4 (100%)	325.8 (100%)	327.7 (100%)	323.9 (100%)	325.5 (100%)

Figure 4.9 illustrates the fuel consumption, along with electricity and heat outputs, from CHP in Ireland over the last 21 years. The step-change increase observed in 2006 is due to the Aughinish Alumina CHP plant coming online.

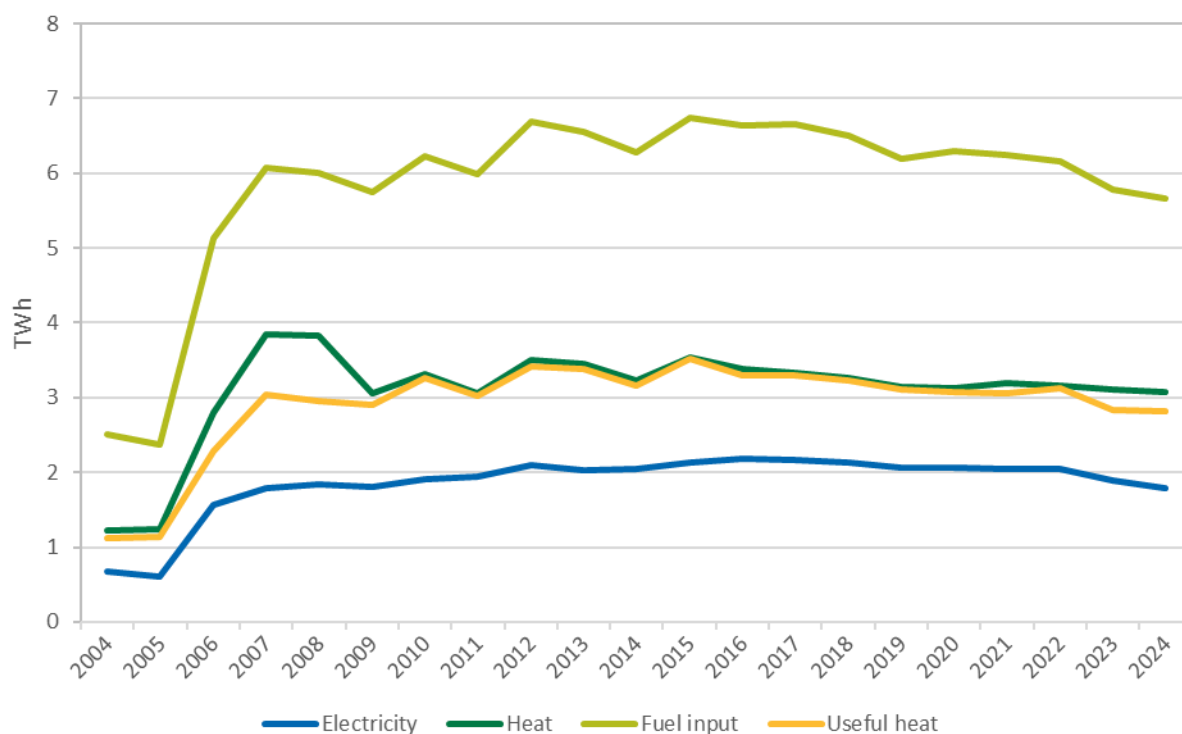
Figure 4.15: CHP fuel input and thermal/electricity output

Figure 4.10 focuses on CHP generated electricity in Ireland as a proportion of gross electricity consumption (that is electricity generation plus net imports). Table 4.6 provides data on the number of CHP units that exported electricity to the grid in the last two years and the total quantity of electricity exported.

Figure 4.16: CHP electricity as percentage of total electricity generation

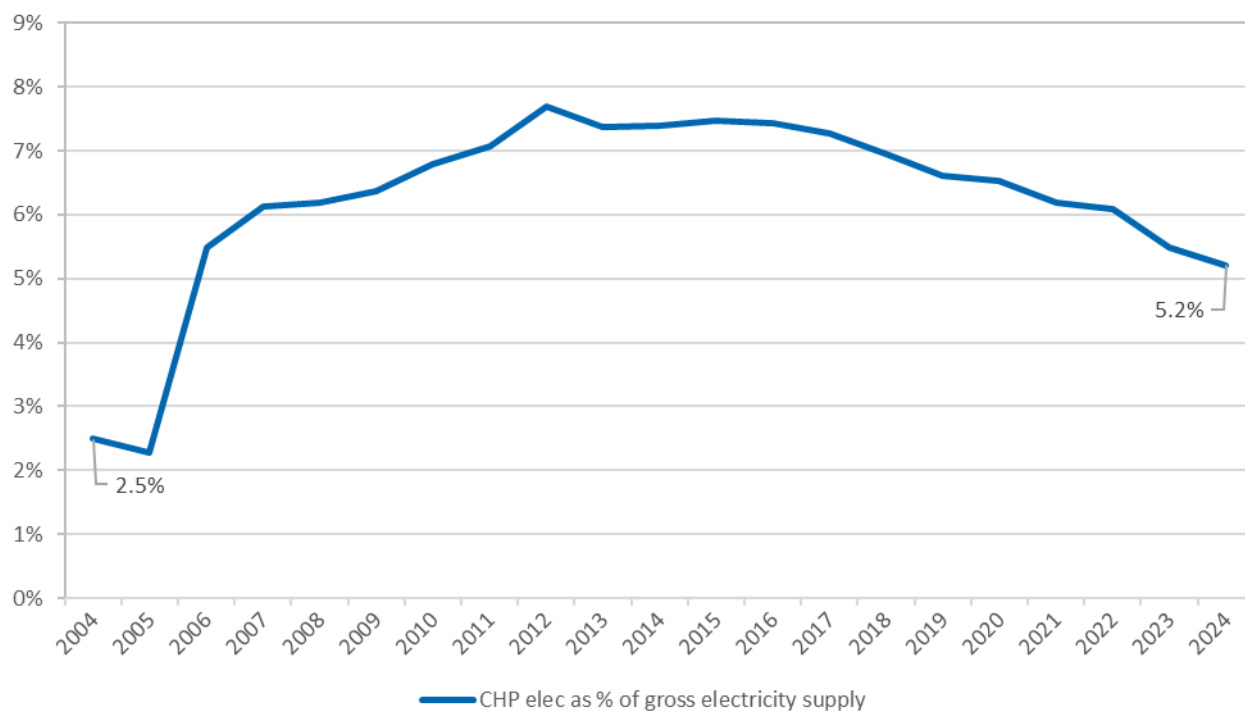


Table 4.13: Electricity exported to the grid from CHP

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
No. of CHP units exporting to grid	18	20	16	17	17	17	16	16	18	18	19
Electricity exported to grid [GWh]	1,372	1,417	1,375	1,384	1,395	1,337	1,378	1,317	1,309	1,194	1,087

4.2.7 Solar, wind and hydro capacity

Figure 4.17 and Table 4.14 show the installed capacity of rooftop and grid-scale solar PV in Ireland at the end of each year. *Rooftop* refers to all solar PV installations with a maximum export capacity (MEC) 0.2 MW, as well as some generators above this threshold that do not export power to the grid (parallel generators). All solar PV generators that are above 0.2 MW and export power to the grid are included in *grid-scale* capacity.

Figure 4.18 shows the location of grid-scale solar PV generators (*i.e.* solar farms) connected in Ireland.

Figure 4.17: Installed solar PV capacity (AC) at year end

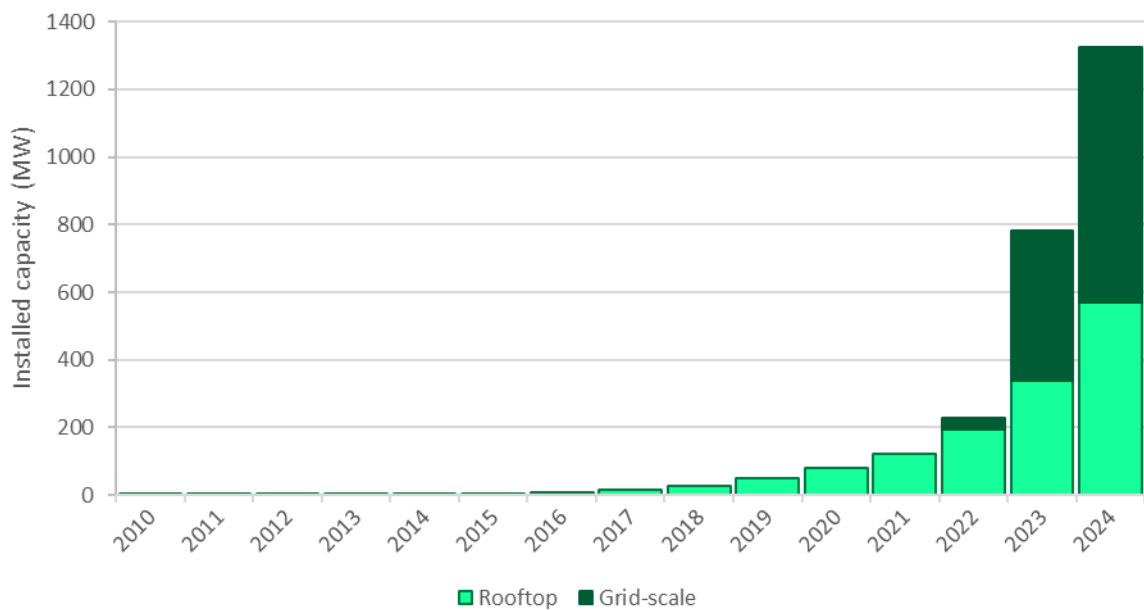


Table 4.14: Installed solar PV capacity (AC) at year end

Capacity [MW]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Rooftop	1.64	2.98	6.89	14.48	26.41	49.19	80.18	123.15	192.63	338.56	570.08
Grid-scale	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.19	445.23	753.32
Total	1.64	2.98	6.89	14.48	26.41	49.19	80.18	123.15	226.82	783.79	1,323.40

Figure 4.18: Map of grid-scale solar PV (solar farms) connected in Ireland

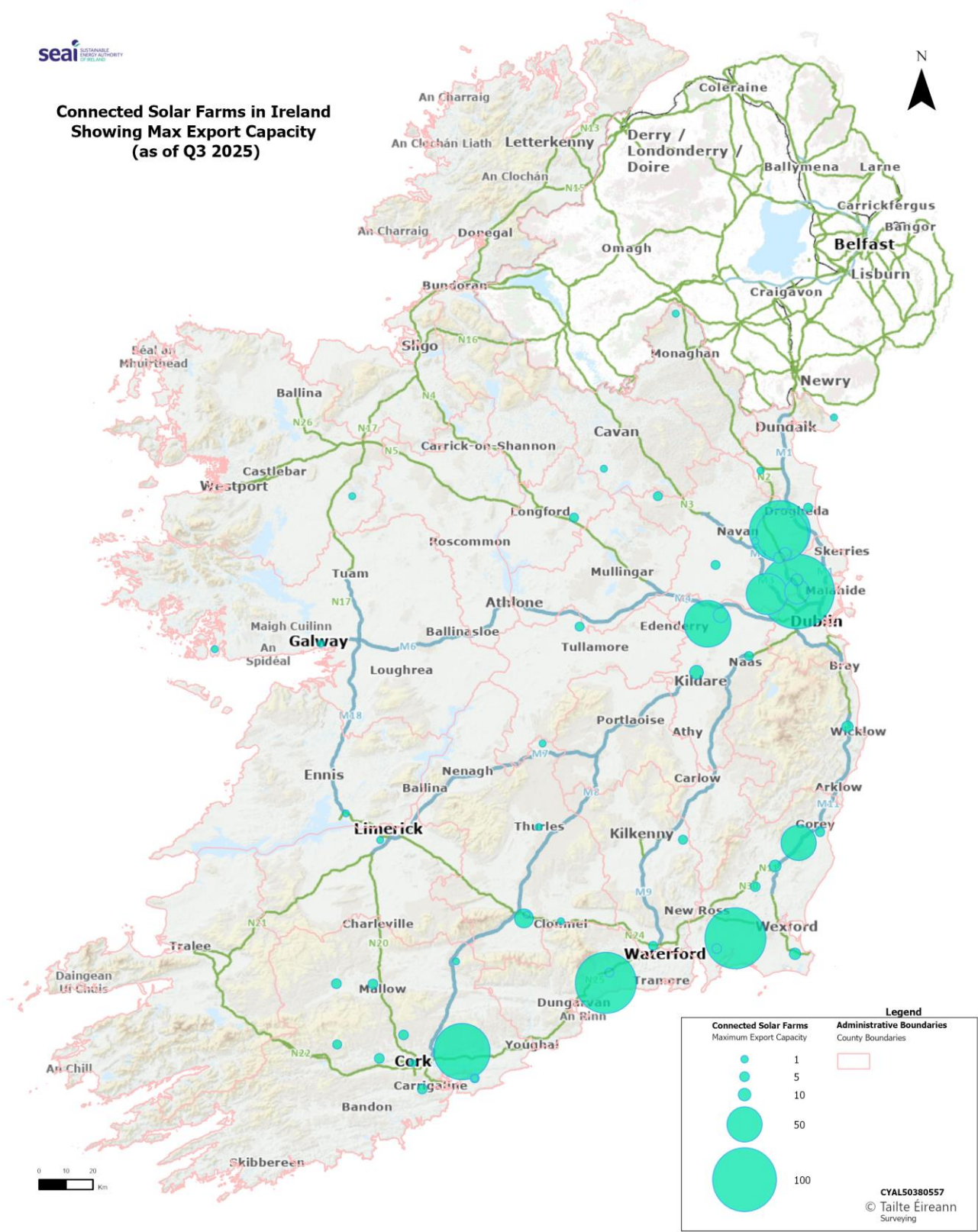
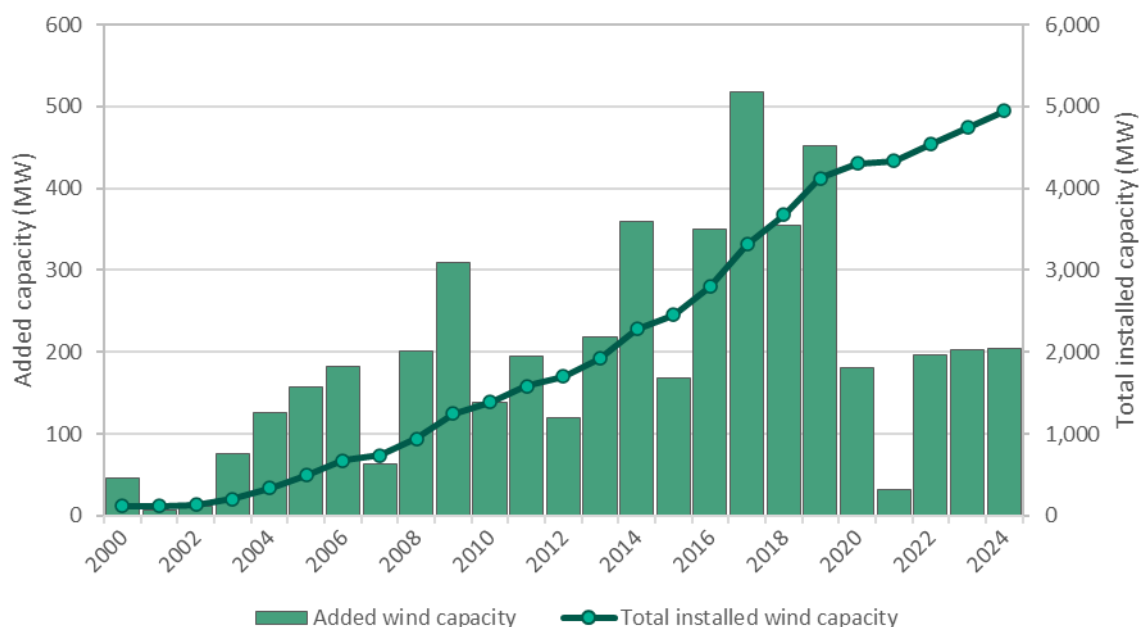


Figure 4.19 shows the annual growth in installed wind generation capacity and overall cumulative capacity since 2000 [9]. 204 MW of wind capacity was added during 2024 and the total installed capacity

at the end of the year was 4,912 MW. Note that these capacity figures include both transmission-connected and distribution-connected windfarms, along with a small number of autoproducers (sites that generate electricity for their own consumption). The location of grid-connected wind farms in Ireland can be seen in Figure 4.20.

Figure 4.19: Installed wind generation capacity

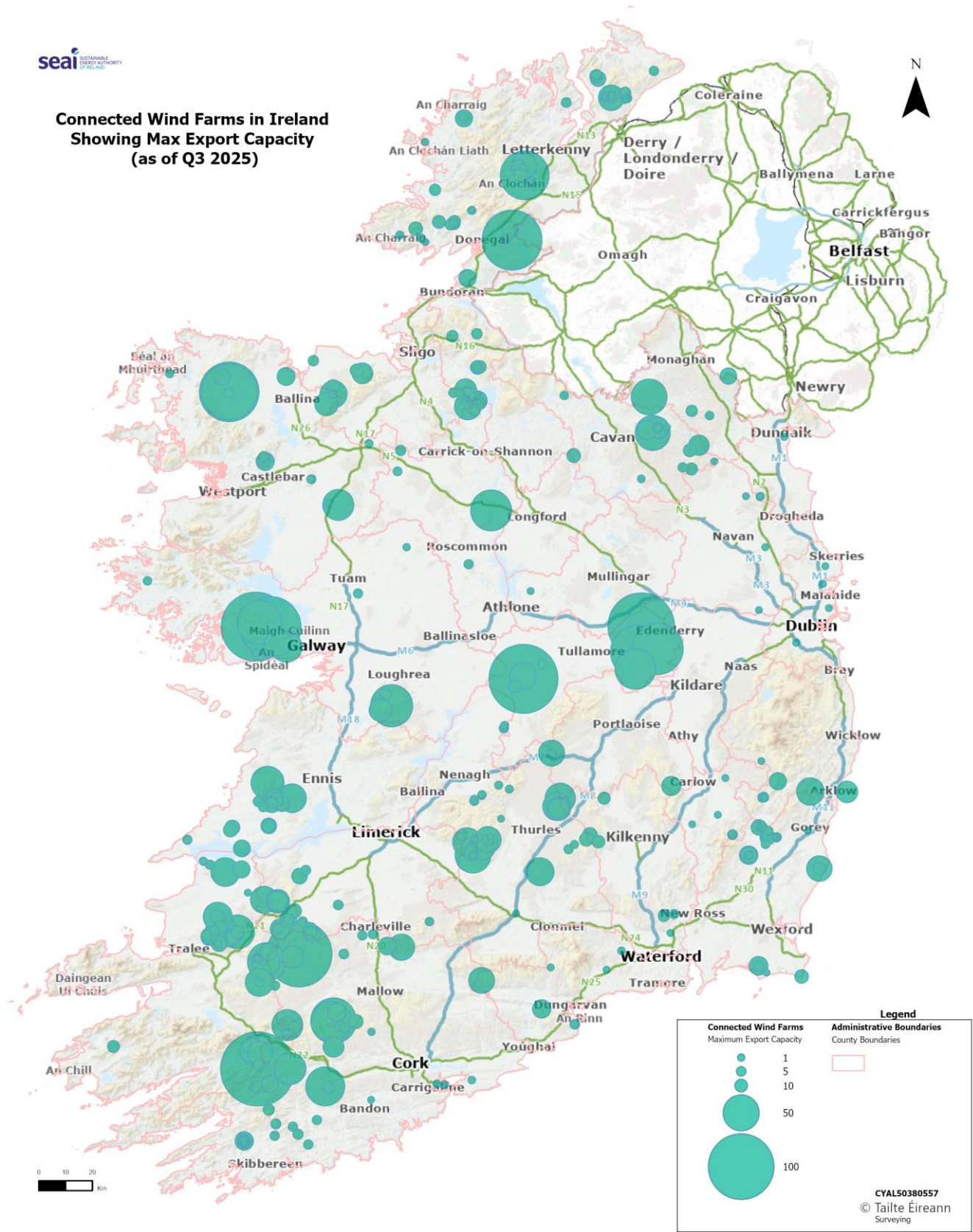


Source: EirGrid, ESB Networks and SEAI

Table 4.15: Wind capacity

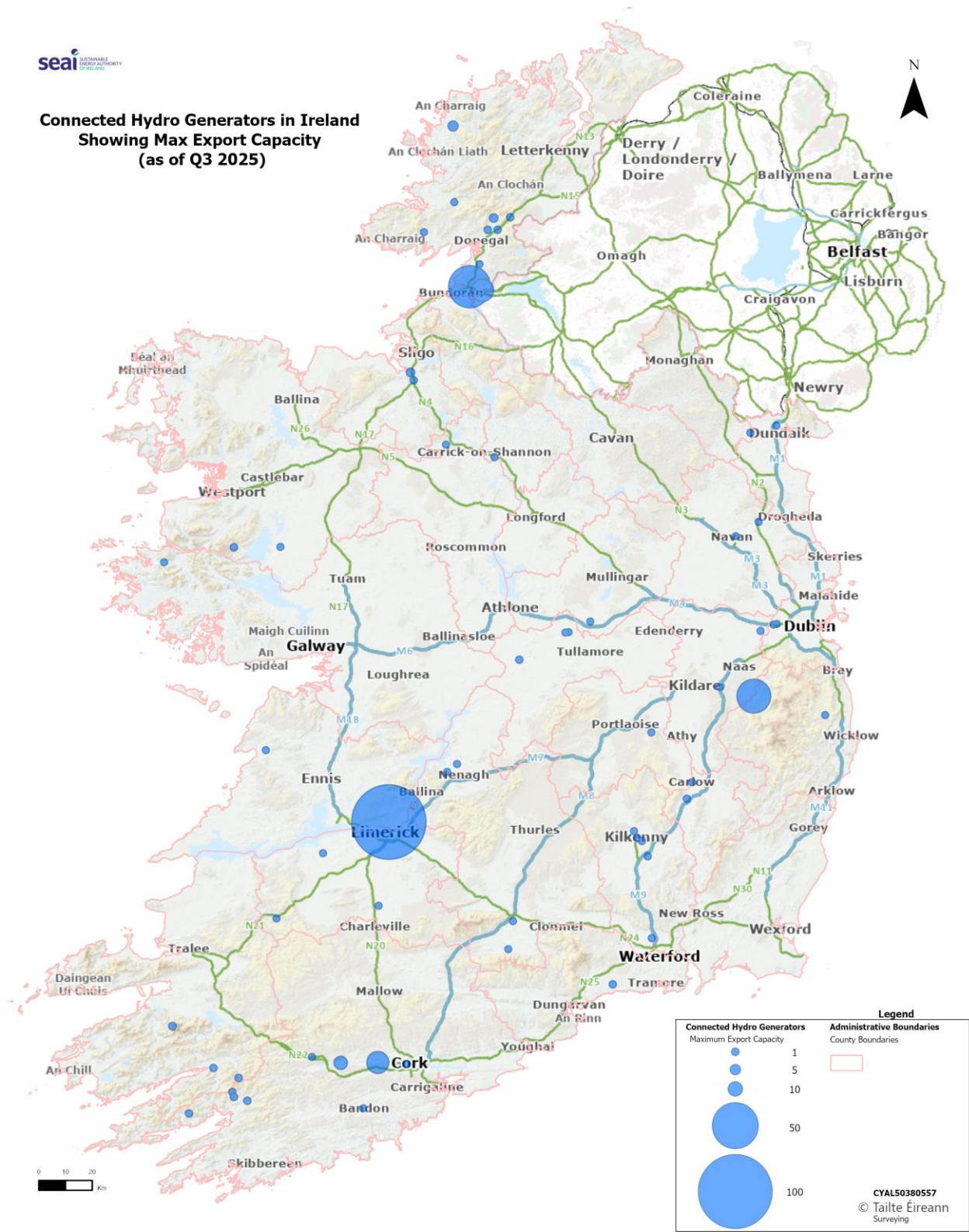
Capacity [MW]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Added wind capacity	360	168	351	518	355	453	180	32	197	203	204
Main activity capacity	2,267	2,435	2,785	3,299	3,651	4,100	4,280	4,308	4,505	4,708	4,912
Autoproducer capacity	16	16	16	20	23	26	26	31	31	31	31
Total installed wind capacity	2,283	2,451	2,802	3,319	3,674	4,126	4,307	4,339	4,536	4,739	4,943

Figure 4.20: Map of grid-connected wind farms in Ireland



Unlike solar and wind generation, the installed capacity of hydroelectric generation has been consistent for over 10 years at approximately 237 MW (excluding pumped storage). The location of hydro generators can be seen in Figure 4.21.

Figure 4.21: Map of grid-connected hydroelectric generators in Ireland (pumped hydro not shown)



The output from wind and hydro generation is affected by the amount of the resource (wind and rainfall) in a particular year. It is also affected by the extent of outages of the plant for reasons such as faults, maintenance and curtailment. An indication of how these factors affect the output of wind and hydro can be obtained by examining the capacity factors for these generation types. The capacity factor is the ratio of average electricity produced to the theoretical maximum possible if the installed capacity was generating at a maximum for a full year. Capacity factors for wind and hydro for the last 11 years are presented in Table 4.16

The rates of capacity increase each year can have a significant impact on the capacity factor in periods of large annual capacity increases. If significant capacity is added late in the year, this artificially reduces the capacity factor for the year. To mitigate this, the wind capacity factor is calculated using the average of the installed capacity at the beginning and end of the year.

Table 4.16: Annual capacity factor for wind and hydro

Capacity factor [%]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Wind (based on average in-year installed capacity)	29.1%	28.4%	32.2%	27.2%	28.2%	28.7%	29.8%	31.8%	26.2%	29.3%	29.2%
Hydro (based on year-end capacity)	28.9%	34.1%	38.8%	32.8%	33.3%	33.4%	42.7%	44.9%	36.1%	33.8%	45.4%

4.3 Oil refining

Ireland has one oil refinery at Whitegate in Cork, that is currently operated by Irving Oil. Whereas electricity generation has a variety of fuel inputs and just one output (electricity), oil refining has one major fuel input (crude oil), but multiple fuel outputs (petrol, diesel and kerosene, *etc.*). Figure 4.11 and Table 4.7 show the outputs from oil refining, primarily being fuel oil, diesel (gas oil) and petrol (gasoline).

Due to the highly international nature of the oil market, refinery outputs are not heavily influenced by local demand in the Irish market, but on the configuration and capabilities of the refinery. A significant portion of the refinery's output is exported directly, while the majority of oil products used for final energy in Ireland are imported as finished products. Nonetheless, the refinery is an important piece of infrastructure regarding energy security.

Figure 4.22: Outputs from oil refining

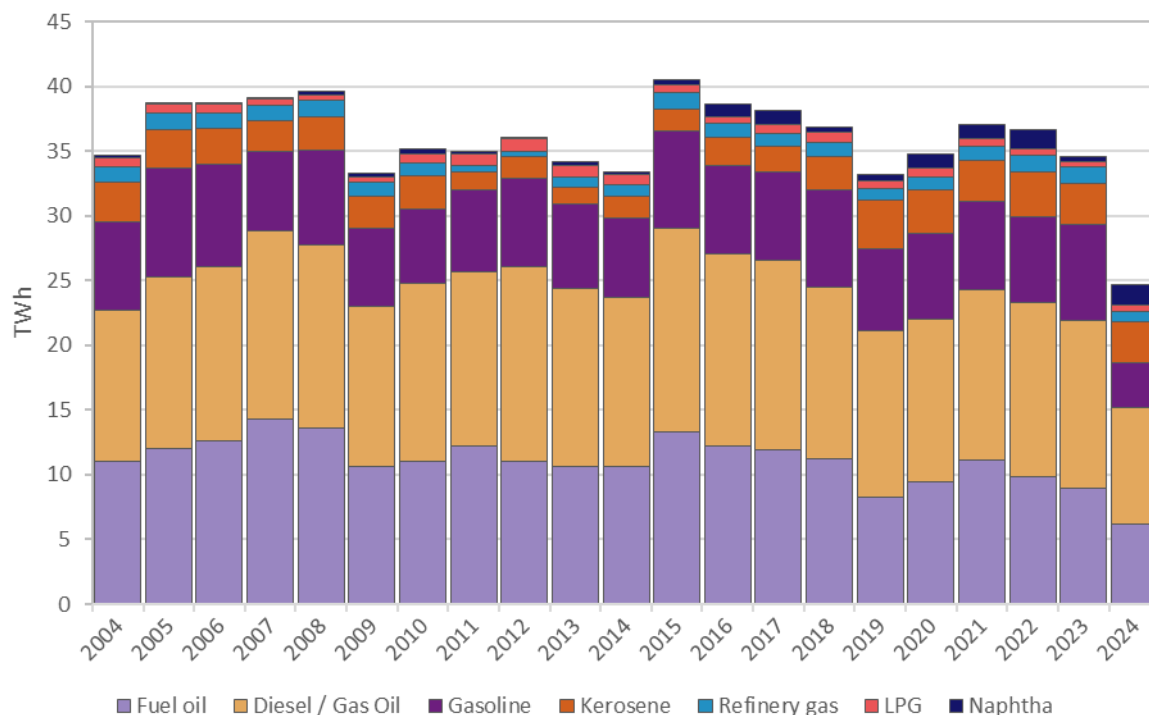


Table 4.17: Outputs from oil refining (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Fuel oil	10.68 (32.0%)	13.28 (32.8%)	12.17 (31.5%)	11.93 (31.3%)	11.18 (30.3%)	8.30 (25.0%)	9.44 (27.1%)	11.12 (30.0%)	9.85 (26.9%)	9.00 (26.0%)	6.16 (25.0%)
Diesel / Gas Oil	13.06 (39.1%)	15.76 (38.9%)	14.86 (38.5%)	14.62 (38.3%)	13.34 (36.2%)	12.77 (38.4%)	12.54 (36.0%)	13.20 (35.7%)	13.44 (36.6%)	12.95 (37.4%)	9.02 (36.5%)
Gasoline	6.09 (18.2%)	7.49 (18.5%)	6.86 (17.8%)	6.83 (17.9%)	7.45 (20.2%)	6.42 (19.3%)	6.65 (19.1%)	6.80 (18.4%)	6.62 (18.1%)	7.38 (21.3%)	3.51 (14.2%)
Kerosene	1.68 (5.0%)	1.75 (4.3%)	2.15 (5.6%)	1.94 (5.1%)	2.64 (7.2%)	3.76 (11.3%)	3.41 (9.8%)	3.13 (8.4%)	3.43 (9.4%)	3.17 (9.2%)	3.09 (12.5%)
Jet kerosene	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Refinery gas	0.86 (2.6%)	1.24 (3.1%)	1.07 (2.8%)	1.05 (2.7%)	1.05 (2.8%)	0.88 (2.6%)	0.99 (2.8%)	1.13 (3.1%)	1.37 (3.7%)	1.24 (3.6%)	0.79 (3.2%)
LPG	0.78 (2.3%)	0.61 (1.5%)	0.56 (1.4%)	0.64 (1.7%)	0.75 (2.0%)	0.59 (1.8%)	0.68 (2.0%)	0.55 (1.5%)	0.49 (1.3%)	0.48 (1.4%)	0.51 (2.1%)
Naphtha	0.24 (0.7%)	0.35 (0.9%)	0.93 (2.4%)	1.15 (3.0%)	0.43 (1.2%)	0.51 (1.5%)	1.09 (3.1%)	1.09 (2.9%)	1.47 (4.0%)	0.37 (1.1%)	1.59 (6.5%)
Total	33.39 (100%)	40.49 (100%)	38.59 (100%)	38.16 (100%)	36.83 (100%)	33.23 (100%)	34.80 (100%)	37.03 (100%)	36.67 (100%)	34.58 (100%)	24.67 (100%)

4.4 Other transformation processes

Several other energy transformation processes operate in the Irish energy sector, though all are relatively small compared to electricity generation, CHP and oil refining. The following subsections provide data on biomethane injection into the gas network and peat briquetting.

4.4.1 Biomethane in natural gas network

The first biomethane injection point in Ireland's gas network was installed in 2019, with the first statistically significant quantities of biomethane recorded in 2020. Gas Networks Ireland record the quantities of biomethane injected into the network and provide annual data to SEAI.

Unblended biomethane is recorded as *biogas* in the National Energy Balance. When biomethane is injected into the natural gas network, it is recorded as a transfer from *biogas* to *natural gas* within the balance.

Figure 4.23: Biomethane injected into the natural gas network

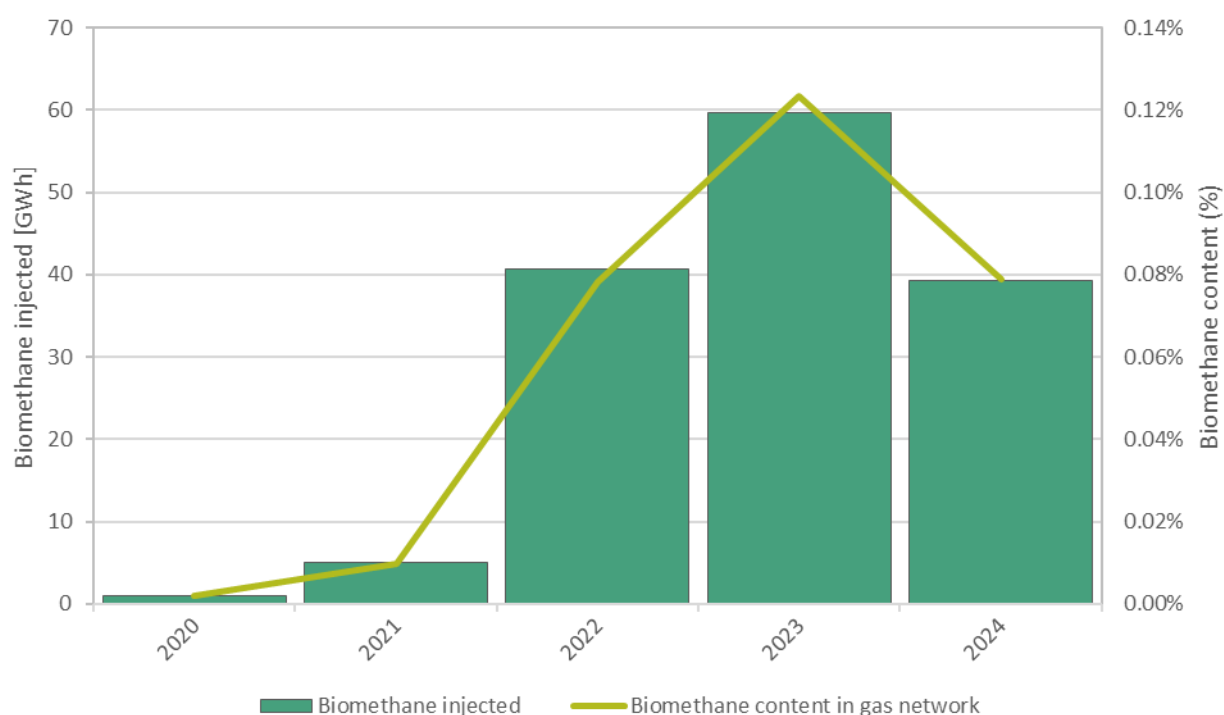


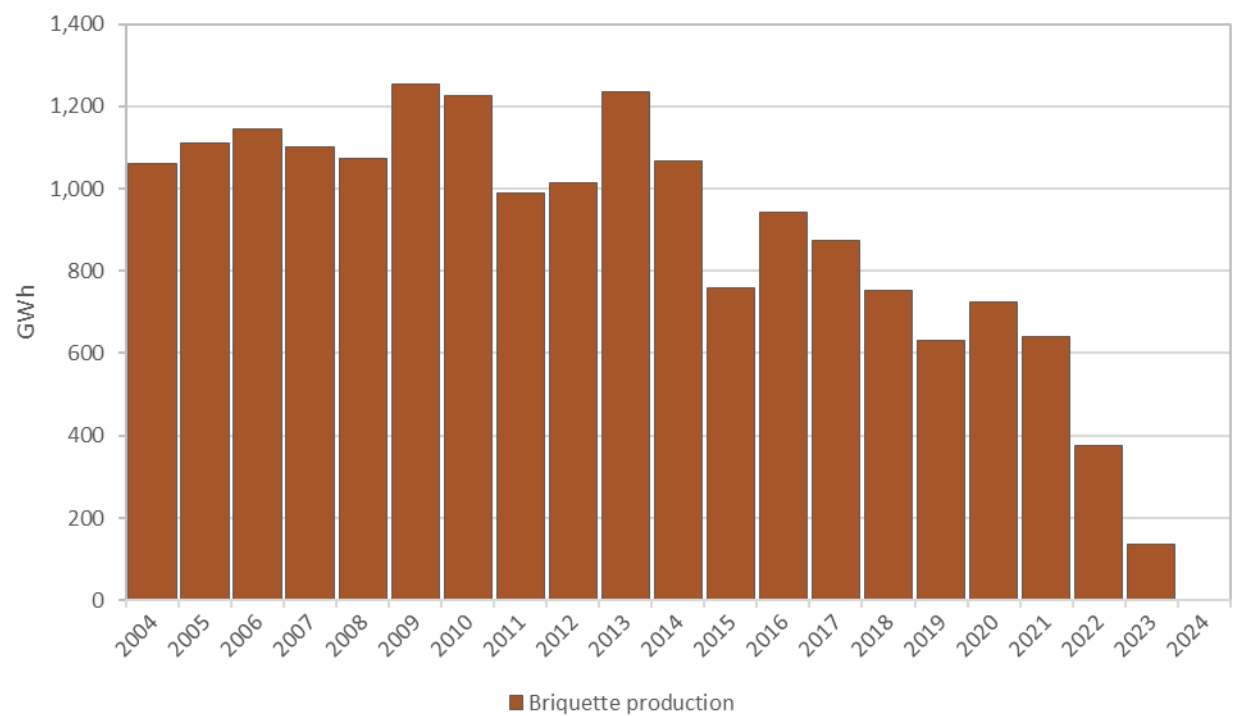
Table 4.18: Biomethane injected into the natural gas network

Biomethane in natural gas network [GWh]	2020	2021	2022	2023	2024
Biomethane injected	1	5	41	60	39
Natural gas (primary energy requirement)	53,219	50,983	52,000	48,263	49,704
Biomethane content in gas network	0.00%	0.01%	0.08%	0.12%	0.08%

4.4.2 Peat briquetting

Peat briquetting converts milled peat into briquettes for residential use. As shown in Figure 4.13, production of peat briquettes in Ireland ended in 2023 with the closure of Bord na Móna's Derrinlough facility in Co Offaly.

Figure 4.24: Annual production of peat briquettes



5 Final energy

The term final energy describes the energy that is directly consumed by an end-user. It covers energy delivered for manufacturing, transport of goods and people, and the day-to-day energy requirements of living, such as heating and cooking. SEAI analyses final energy consumption by fuel and by sector.

Final energy use excludes energy lost in the transformation or transmission of primary energy supply, because this energy is not available to the end user. Multiple primary energy sources may be aggregated and transformed into a single final energy type for an end user. For example, when an end user consumes electricity (as final energy), that electricity originated from a blend of wind, natural gas, *etc.*, (that is, primary energy sources). Similarly, final energy use covers the energy in petrol and diesel consumed by end users, but not the energy that was needed to convert crude oil into that petrol and diesel in a refinery.

Final energy use is important, because individuals and businesses have direct control over how they decide to consume it – petrol *vs.* diesel *vs.* electric vehicles for surface transport, gas *vs.* oil *vs.* heat-pumps for heating, *etc.*

5.1 Final energy consumption by energy type

In evaluating final energy use or consumption in Ireland, SEAI collects data on each fuel, or energy type, from various sources:

- Aggregated electricity consumption data collected by the meter registration system operator (MRSO, a function within ESB Networks).
- Aggregated gas consumption data collected by Gas Networks Ireland (GNI).
- Oil supply data reported to the DCEE for the payment of the National Oil Reserves Agency (NORA) levy.
- Biofuel data collected by NORA for the Renewable Transport Fuel Obligation (RTFO).
- Fuel consumption data reported by permit holders to the EPA under EU Emissions Trading System (ETS).
- Aggregated carbon tax receipts published by the Central Statistics Office (CSO).
- SEAI surveys of solid fuel suppliers.
- SEAI surveys of biomass/wood suppliers.
- Other data requests made by SEAI.
- SEAI data on the number and size of heat pumps and solar thermal collectors in domestic and non-domestic buildings.
- Aggregated data from SEAI's domestic and non-domestic Building Energy Rating (BER) database on the number and size of heat pumps and solar thermal collectors

Figure 5.1 shows the current percentage breakdown of Ireland's final energy use by energy type. Oil products account for more than half of Ireland's final energy use, followed by electricity, gas, renewables, coal and peat, in that order.

Oil makes up a significant share of final energy in each sector. Ireland is almost completely dependent on oil for the servicing of its transport sector, and that sector is Ireland's largest end use of energy.

Electricity from renewable sources (e.g. wind, solar PV, renewable waste) is included in the *electricity* energy type for final energy use, not *renewables*. Final consumption of electricity includes grid-supplied electricity as well as electricity that is generated and consumed on the same site without entering the grid (referred to as auto-production or auto-consumption).

The *renewables* energy type in final energy comprises the combustion of biomass and biogas for heat, the combustion of biofuels in transport, solar thermal and ambient heat (*via* heat pumps). Similarly, consumption of oil, gas, coal and peat reported here is the direct consumption of these fuels and excludes fuels consumed for generation of electricity, non-energy use, and those converted into other fuels as discussed in section 4.

Figure 5.1: Final energy consumption by energy type

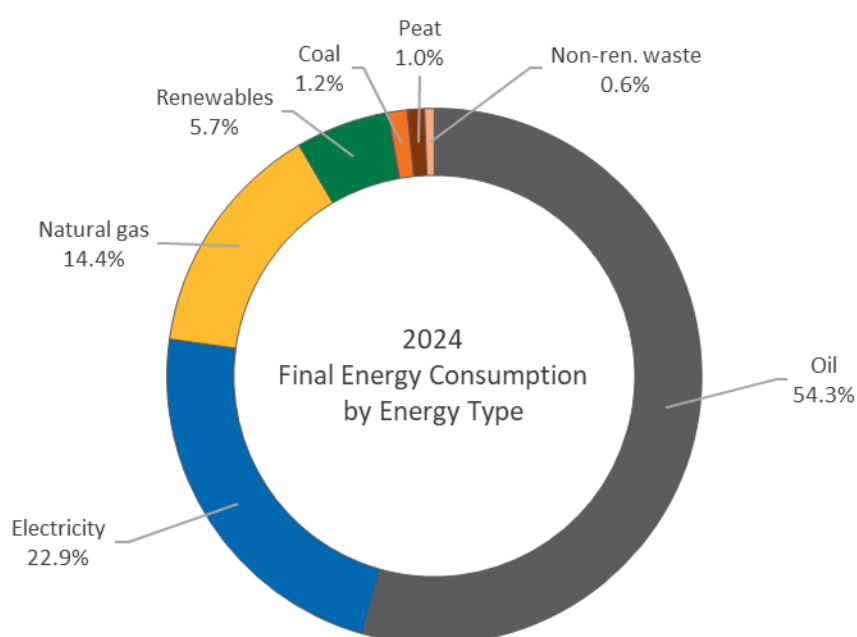


Figure 5.2 shows the annual trends in final energy use by energy type for the most recent 21-year period. Table 5.1 provides numerical details on the absolute values and relative shares in the final energy use by energy type shown in Figure 5.2. Although Ireland remains heavily reliant on oil, its consumption has decreased, in absolute and percentage terms, over the period presented. Decreases were precipitated by two events: the economic downturn from 2008 to 2012 and the COVID-19 impact in 2020 and 2021. Outside of these events, oil consumption has increased each year, albeit only very slightly in 2024. Other general trends over the period shown have been decreases in the consumption of coal and peat, and increases in the consumption of gas, electricity and renewables.

Table 5.2 presents the breakdown of final energy consumption by sub-type (*i.e.* individual energy products). Diesel/gas oil remains the largest single product, accounting for roughly 30% of total consumption. This highlights the transport sector's heavy dependence on diesel/gas oil across road, rail, and navigation, as well as its widespread use in other sectors for machinery (*e.g.* construction and agriculture) and heating. Similarly, biodiesel is the leading renewable energy product, representing 2.5% of final energy consumption in 2024. While most biodiesel is used in transport, it is increasingly adopted in other sectors as a substitute for gas oil.

Figure 5.2: Total final consumption by energy type

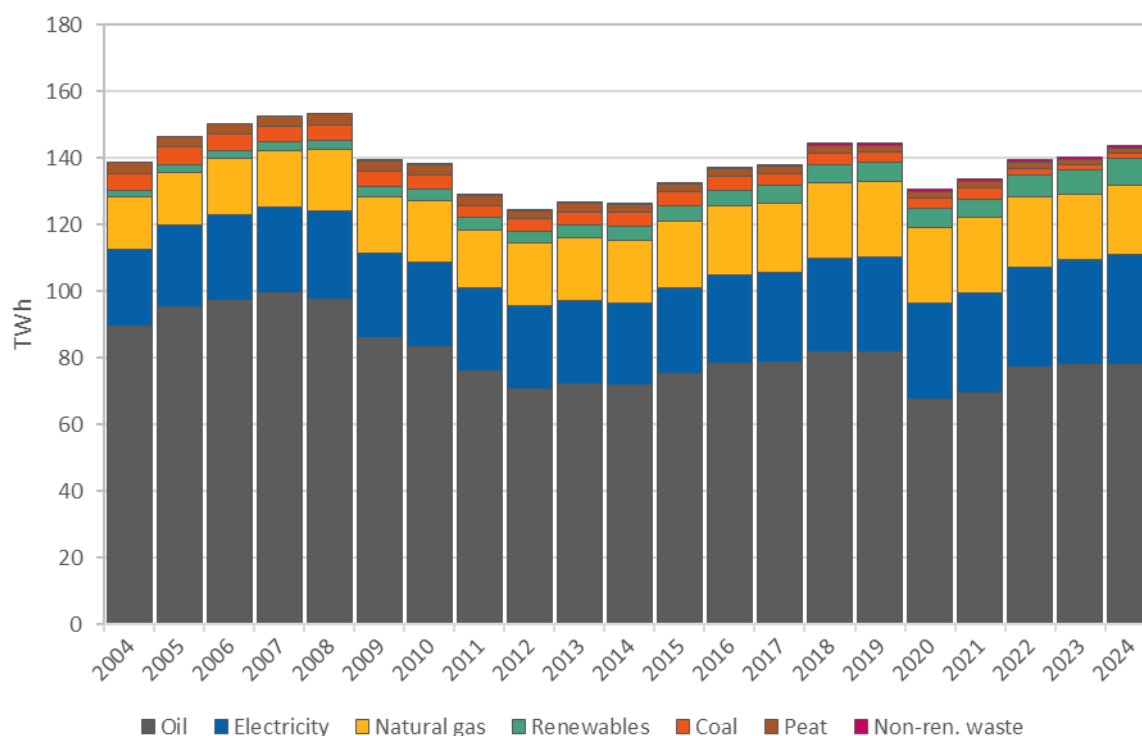


Table 5.1: Final energy consumption by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	71.63 (56.7%)	75.40 (56.9%)	78.34 (57.2%)	78.82 (57.1%)	81.93 (56.7%)	81.67 (56.6%)	67.75 (51.9%)	69.50 (52.0%)	77.02 (55.2%)	78.10 (55.6%)	78.12 (54.3%)
Electricity	24.81 (19.6%)	25.79 (19.5%)	26.38 (19.2%)	26.63 (19.3%)	27.87 (19.3%)	28.40 (19.7%)	28.62 (21.9%)	29.92 (22.4%)	30.32 (21.7%)	31.57 (22.5%)	32.86 (22.9%)
Natural gas	18.80 (14.9%)	19.92 (15.0%)	20.81 (15.2%)	21.13 (15.3%)	22.76 (15.8%)	22.90 (15.9%)	22.75 (17.4%)	22.65 (17.0%)	21.01 (15.1%)	19.27 (13.7%)	20.63 (14.4%)
Renewables	4.42 (3.5%)	4.66 (3.5%)	4.69 (3.4%)	5.32 (3.9%)	5.34 (3.7%)	5.64 (3.9%)	5.61 (4.3%)	5.66 (4.2%)	6.42 (4.6%)	7.32 (5.2%)	8.15 (5.7%)
Coal	3.88 (3.1%)	3.95 (3.0%)	4.08 (3.0%)	3.33 (2.4%)	3.59 (2.5%)	2.97 (2.1%)	3.06 (2.3%)	3.09 (2.3%)	2.15 (1.5%)	1.67 (1.2%)	1.66 (1.2%)
Peat	2.33 (1.8%)	2.34 (1.8%)	2.30 (1.7%)	2.20 (1.6%)	2.30 (1.6%)	2.13 (1.5%)	2.20 (1.7%)	2.10 (1.6%)	1.87 (1.3%)	1.63 (1.2%)	1.49 (1.0%)
Non-ren. waste	0.48 (0.4%)	0.51 (0.4%)	0.49 (0.4%)	0.66 (0.5%)	0.64 (0.4%)	0.66 (0.5%)	0.62 (0.5%)	0.63 (0.5%)	0.68 (0.5%)	0.87 (0.6%)	0.84 (0.6%)
Total	126.36 (100%)	132.58 (100%)	137.07 (100%)	138.09 (100%)	144.42 (100%)	144.38 (100%)	130.61 (100%)	133.54 (100%)	139.46 (100%)	140.42 (100%)	143.74 (100%)

Table 5.2: Final energy consumption disaggregated by energy sub-type (share in brackets)

Energy [TWh]		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	Diesel / Gas oil	37.03 (29.3%)	39.20 (29.6%)	42.06 (30.7%)	42.54 (30.8%)	44.47 (30.8%)	44.95 (31.1%)	40.04 (30.7%)	42.13 (31.5%)	44.01 (31.6%)	43.11 (30.7%)	42.71 (29.7%)
	Jet Kerosene	8.70 (6.9%)	9.84 (7.4%)	10.10 (7.4%)	11.88 (8.6%)	12.83 (8.9%)	12.98 (9.0%)	4.63 (3.5%)	5.18 (3.9%)	11.84 (8.5%)	13.26 (9.4%)	13.03 (9.1%)
	Kerosene	8.65 (6.8%)	10.01 (7.5%)	10.53 (7.7%)	9.99 (7.2%)	10.87 (7.5%)	10.59 (7.3%)	12.44 (9.5%)	10.98 (8.2%)	9.17 (6.6%)	9.13 (6.5%)	9.91 (6.9%)
	Gasoline	13.18 (10.4%)	12.50 (9.4%)	11.66 (8.5%)	10.52 (7.6%)	9.59 (6.6%)	9.08 (6.3%)	6.73 (5.2%)	7.13 (5.3%)	8.12 (5.8%)	8.65 (6.2%)	8.90 (6.2%)
	LPG	1.75 (1.4%)	1.78 (1.3%)	1.90 (1.4%)	1.94 (1.4%)	2.11 (1.5%)	2.10 (1.5%)	2.02 (1.5%)	2.07 (1.6%)	2.13 (1.5%)	2.41 (1.7%)	2.55 (1.8%)
	Pet. coke	1.41 (1.1%)	1.56 (1.2%)	1.66 (1.2%)	1.60 (1.2%)	1.70 (1.2%)	1.63 (1.1%)	1.54 (1.2%)	1.75 (1.3%)	1.51 (1.1%)	1.34 (1.0%)	0.91 (0.6%)
	Fuel oil	0.91 (0.7%)	0.51 (0.4%)	0.42 (0.3%)	0.37 (0.3%)	0.36 (0.2%)	0.35 (0.2%)	0.34 (0.3%)	0.25 (0.2%)	0.25 (0.2%)	0.20 (0.1%)	0.09 (0.1%)
Electricity		24.81 (19.6%)	25.79 (19.5%)	26.38 (19.2%)	26.63 (19.3%)	27.87 (19.3%)	28.40 (19.7%)	28.62 (21.9%)	29.92 (22.4%)	30.32 (21.7%)	31.57 (22.5%)	32.86 (22.9%)
Natural gas		18.80 (14.9%)	19.92 (15.0%)	20.81 (15.2%)	21.13 (15.3%)	22.76 (15.8%)	22.90 (15.9%)	22.75 (17.4%)	22.65 (17.0%)	21.01 (15.1%)	19.27 (13.7%)	20.63 (14.4%)
Renewables	Biodiesel	1.04 (0.8%)	1.14 (0.9%)	1.00 (0.7%)	1.52 (1.1%)	1.48 (1.0%)	1.90 (1.3%)	1.82 (1.4%)	1.87 (1.4%)	2.39 (1.7%)	3.18 (2.3%)	3.56 (2.5%)
	Biomass	2.28 (1.8%)	2.26 (1.7%)	2.24 (1.6%)	2.31 (1.7%)	2.37 (1.6%)	2.20 (1.5%)	2.17 (1.7%)	2.04 (1.5%)	1.98 (1.4%)	1.75 (1.2%)	1.80 (1.3%)
	Ambient heat	0.27 (0.2%)	0.31 (0.2%)	0.38 (0.3%)	0.41 (0.3%)	0.43 (0.3%)	0.50 (0.3%)	0.58 (0.4%)	0.67 (0.5%)	0.83 (0.6%)	1.02 (0.7%)	1.21 (0.8%)
	Ren. waste	0.30 (0.2%)	0.36 (0.3%)	0.44 (0.3%)	0.47 (0.3%)	0.47 (0.3%)	0.47 (0.3%)	0.50 (0.4%)	0.54 (0.4%)	0.63 (0.4%)	0.68 (0.5%)	0.67 (0.5%)
	Bioethanol	0.31 (0.2%)	0.35 (0.3%)	0.38 (0.3%)	0.34 (0.2%)	0.32 (0.2%)	0.30 (0.2%)	0.23 (0.2%)	0.24 (0.2%)	0.27 (0.2%)	0.38 (0.3%)	0.57 (0.4%)
	Biogas	0.09 (0.1%)	0.10 (0.1%)	0.11 (0.1%)	0.11 (0.1%)	0.11 (0.1%)	0.12 (0.1%)	0.14 (0.1%)	0.14 (0.1%)	0.16 (0.1%)	0.15 (0.1%)	0.17 (0.1%)
	Solar Th.	0.12 (0.1%)	0.13 (0.1%)	0.14 (0.1%)	0.15 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)
Coal	Anthracite & ovoids	1.06 (0.8%)	0.87 (0.7%)	1.03 (0.8%)	0.99 (0.7%)	1.04 (0.7%)	1.03 (0.7%)	1.03 (0.8%)	1.06 (0.8%)	1.03 (0.7%)	1.05 (0.8%)	1.16 (0.8%)
	Bit. coal	2.67 (2.1%)	2.88 (2.2%)	2.85 (2.1%)	2.17 (1.6%)	2.41 (1.7%)	1.84 (1.3%)	1.94 (1.5%)	1.89 (1.4%)	1.03 (0.7%)	0.53 (0.4%)	0.45 (0.3%)
	Lignite / BC Briq.	0.15 (0.1%)	0.21 (0.2%)	0.19 (0.1%)	0.17 (0.1%)	0.14 (0.1%)	0.11 (0.1%)	0.08 (0.1%)	0.14 (0.1%)	0.09 (0.1%)	0.09 (0.1%)	0.05 (0.0%)
Peat	Sod peat	1.49 (1.2%)	1.49 (1.1%)	1.49 (1.1%)	1.49 (1.1%)	1.49 (1.0%)	1.49 (1.0%)	1.49 (1.1%)	1.49 (1.1%)	1.49 (1.1%)	1.49 (1.1%)	1.49 (1.0%)
	Milled peat	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	Briquettes	0.84 (0.7%)	0.85 (0.6%)	0.80 (0.6%)	0.71 (0.5%)	0.80 (0.6%)	0.65 (0.4%)	0.72 (0.5%)	0.61 (0.5%)	0.38 (0.3%)	0.14 (0.1%)	0 (0%)
Non-ren. waste		0.48 (0.4%)	0.51 (0.4%)	0.49 (0.4%)	0.66 (0.5%)	0.64 (0.4%)	0.66 (0.5%)	0.62 (0.5%)	0.63 (0.5%)	0.68 (0.5%)	0.87 (0.6%)	0.84 (0.6%)
Total		126.36 (100%)	132.58 (100%)	137.07 (100%)	138.09 (100%)	144.42 (100%)	144.38 (100%)	130.61 (100%)	133.54 (100%)	139.46 (100%)	140.42 (100%)	143.74 (100%)

5.2 Final energy consumption by sector

In calculating the final energy use by sector, SEAI uses the data on final energy use of each fuel type as described in section 5.1 and apportions it to each sector using data from, among other things:

- CSO's most recent annual Business Energy Use Survey.
- SEAI solid fuel and biomass/wood supplier surveys indicating sales to industry, commercial and residential sectors.
- Aggregated metered electricity consumption in each sector.
- Aggregated metered gas consumption in each sector.
- SEAI's Public Sector Monitoring and Reporting Programme.

Figure 5.3 shows the share of each sector in final energy consumption. The transport sector consumes more than 40% of total final energy, more than the next two sectors (residential and industry) combined. The services sector contains both commercial and public services, a further breakdown is given in section 5.2.4.

Figure 5.3: Share final energy consumption by sector

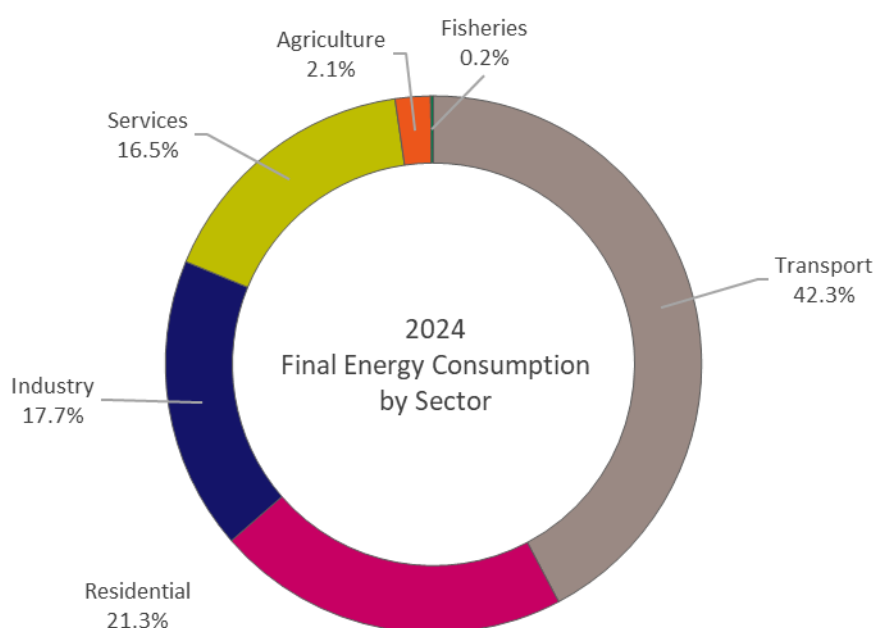


Figure 5.4 shows the annual final energy demand broken out by sector. The broad reduction in final energy use across all sectors from 2008 to 2012 is attributed to the international economic downturn, with the industry, transport and services sectors returning to growth after 2012, and growth in the residential sector delayed until 2014.

The reduction in 2020 final energy use was mostly due to the COVID-19 restrictions and was almost entirely limited to the transport sector. Prior to 2020, final energy demand for transport had risen every year since 2012 as activity growth outstripped efficiency improvements. Transport remains the sector with greatest final energy use, followed in order by the residential sector, industry and services.

Figure 5.4: Final energy consumption by sector

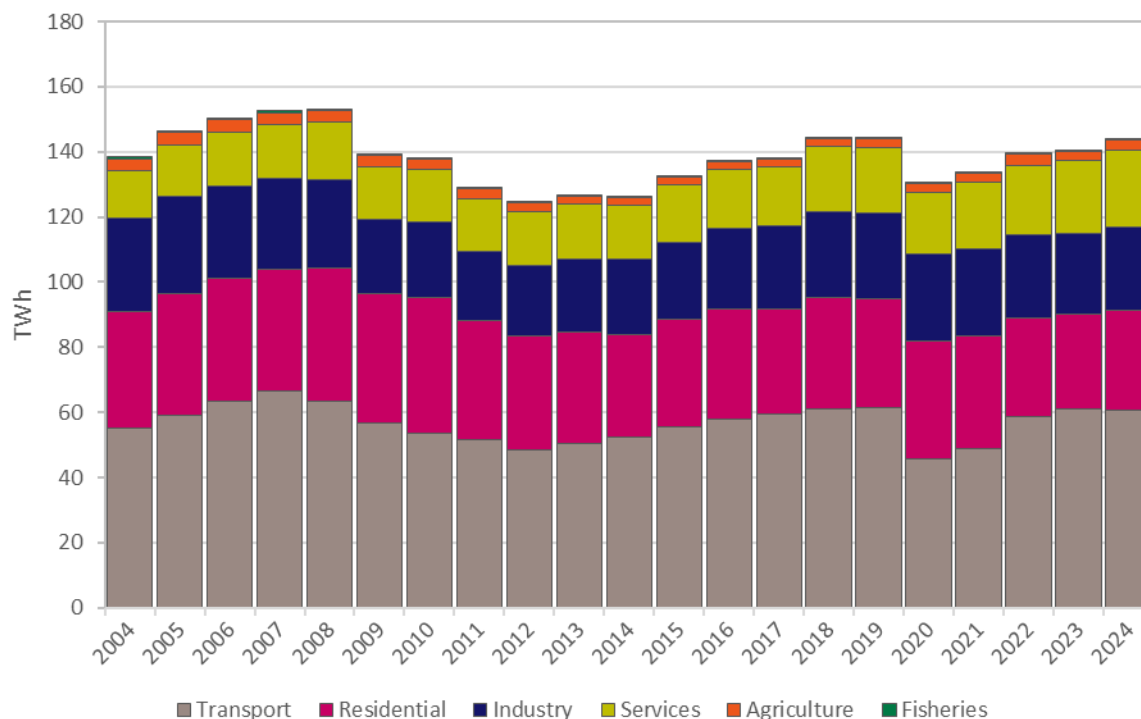


Table 5.2 provides absolute values and relative shares in the final energy consumption by sector.

Table 5.3: Final energy consumption by sector (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Transport	52.58 (41.6%)	55.67 (42.0%)	57.79 (42.2%)	59.38 (43.0%)	60.87 (42.1%)	61.36 (42.5%)	45.53 (34.9%)	48.75 (36.5%)	58.56 (42.0%)	60.89 (43.4%)	60.76 (42.3%)
Residential	31.10 (24.6%)	32.90 (24.8%)	33.99 (24.8%)	32.24 (23.3%)	34.37 (23.8%)	33.42 (23.2%)	36.45 (27.9%)	34.80 (26.1%)	30.47 (21.8%)	29.12 (20.7%)	30.66 (21.3%)
Industry	23.31 (18.4%)	23.75 (17.9%)	24.68 (18.0%)	25.84 (18.7%)	26.46 (18.3%)	26.43 (18.3%)	26.52 (20.3%)	26.47 (19.8%)	25.34 (18.2%)	24.84 (17.7%)	25.38 (17.7%)
Services	16.70 (13.2%)	17.70 (13.4%)	17.97 (13.1%)	17.89 (13.0%)	19.81 (13.7%)	20.22 (14.0%)	19.06 (14.6%)	20.51 (15.4%)	21.39 (15.3%)	22.34 (15.9%)	23.65 (16.5%)
Agriculture	2.39 (1.9%)	2.32 (1.8%)	2.41 (1.8%)	2.48 (1.8%)	2.60 (1.8%)	2.66 (1.8%)	2.82 (2.2%)	2.78 (2.1%)	3.49 (2.5%)	3.00 (2.1%)	3.05 (2.1%)
Fisheries	0.28 (0.2%)	0.24 (0.2%)	0.22 (0.2%)	0.27 (0.2%)	0.32 (0.2%)	0.27 (0.2%)	0.22 (0.2%)	0.22 (0.2%)	0.20 (0.1%)	0.24 (0.2%)	0.24 (0.2%)
Total	126.36 (100%)	132.58 (100%)	137.07 (100%)	138.09 (100%)	144.42 (100%)	144.38 (100%)	130.61 (100%)	133.54 (100%)	139.46 (100%)	140.42 (100%)	143.74 (100%)

5.2.1 Final energy consumption in industry

In determining the sectoral breakdown of energy consumption, SEAI uses a blend of data sources, including the BEUS from the CSO. In June 2025, the CSO published the latest version of the BEUS, which is based on energy use in 2023 [10]. SEAI uses this most recent BEUS release in estimating the breakdown of 2023 energy consumption across the industry sector.

Figure 5.5 shows the current share of each energy type in the final energy consumed in industry. Natural gas forms the largest share, at more than 40%, followed by electricity and oil. Renewables accounts for around 9%.

Figure 5.5: Shares of energy types in industry final energy

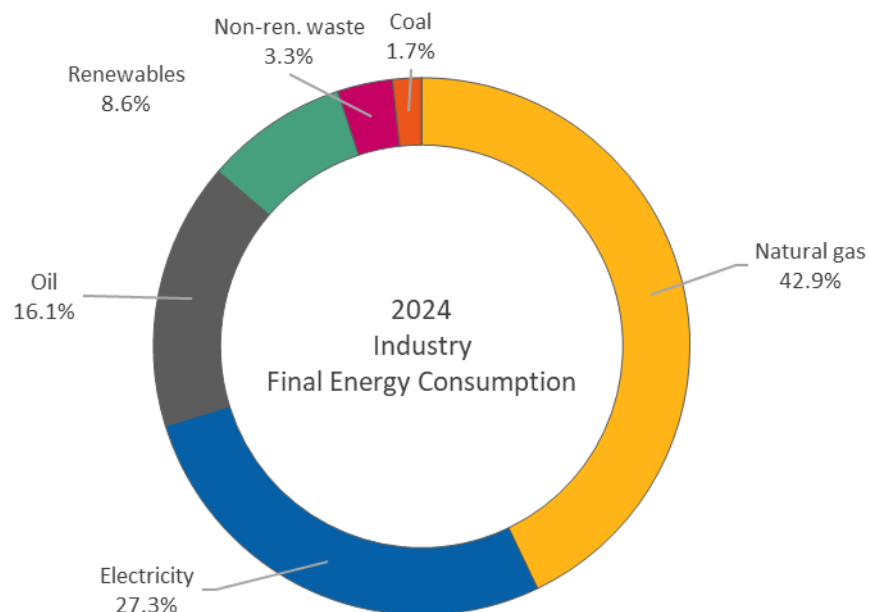


Figure 5.6 shows that natural gas, wastes and renewables have all increased their shares of industrial energy use over the period, while the shares of oil and coal have decreased. The share of electricity has declined slightly faster than overall energy consumption within the industry sector. The increase evident in renewables, is mainly due to the use of biomass in the wood-processing industry, and the use of the renewable portion of wastes in cement manufacturing, while biodiesel has grown since 2023.

There was also significant fuel switching from coal and oil to natural gas during this period. Because gas is less carbon intensive than oil or coal, this fuel switching, along with increased use of renewable energy, has resulted in lower average emissions per unit of energy used in industry during this period. Peat has not been used in the sector since 2018. Natural gas consumption increased in 2024 but was still around 10% below its 2020 peak.

Figure 5.6: Final energy in industry by energy type

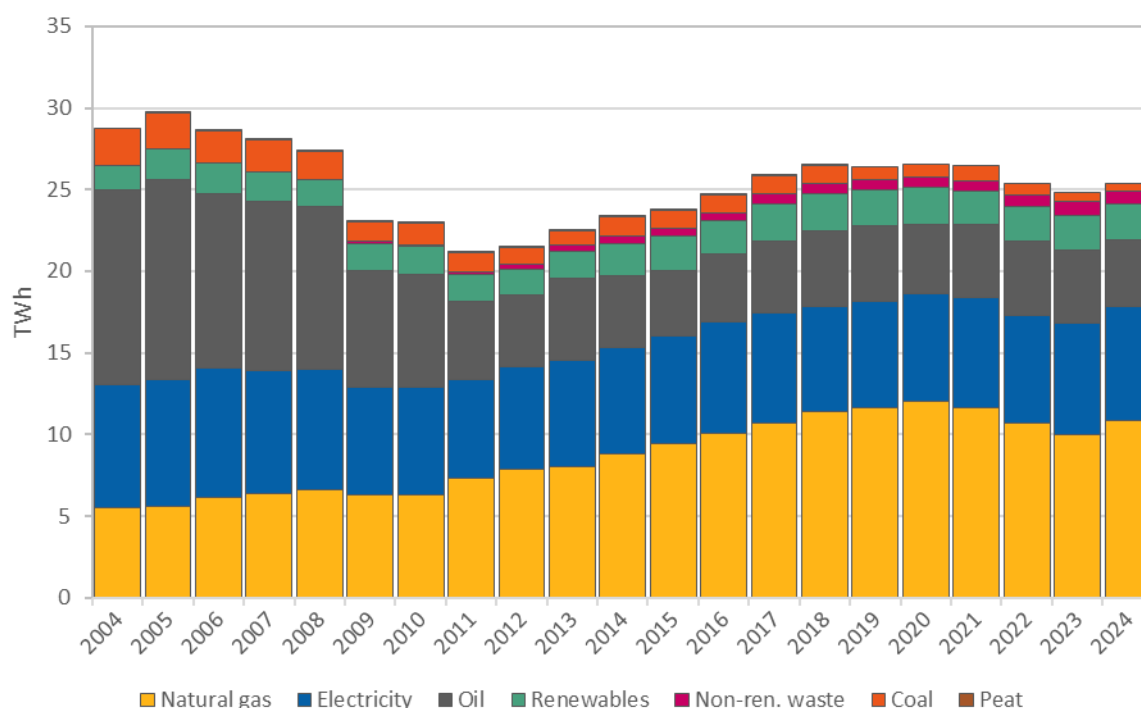


Table 5.3 shows the quantities and relative shares of energy consumption in industry by energy type, while Table 5.5 provides further disaggregation by energy sub-type.

Table 5.4: Final energy in industry by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	8.79 (37.7%)	9.41 (39.6%)	10.07 (40.8%)	10.65 (41.2%)	11.37 (43.0%)	11.60 (43.9%)	12.05 (45.4%)	11.64 (44.0%)	10.70 (42.2%)	10.02 (40.3%)	10.89 (42.9%)
Electricity	6.54 (28.0%)	6.62 (27.9%)	6.85 (27.8%)	6.82 (26.4%)	6.47 (24.4%)	6.56 (24.8%)	6.54 (24.6%)	6.73 (25.4%)	6.58 (26.0%)	6.80 (27.4%)	6.93 (27.3%)
Oil	4.36 (18.7%)	4.04 (17.0%)	4.14 (16.8%)	4.40 (17.0%)	4.60 (17.4%)	4.59 (17.4%)	4.31 (16.2%)	4.46 (16.8%)	4.54 (17.9%)	4.43 (17.8%)	4.09 (16.1%)
Renewables	2.00 (8.6%)	2.08 (8.8%)	2.05 (8.3%)	2.25 (8.7%)	2.32 (8.8%)	2.22 (8.4%)	2.22 (8.4%)	2.09 (7.9%)	2.14 (8.5%)	2.19 (8.8%)	2.19 (8.6%)
Non-ren. waste	0.48 (2.1%)	0.51 (2.2%)	0.49 (2.0%)	0.66 (2.6%)	0.64 (2.4%)	0.66 (2.5%)	0.62 (2.4%)	0.63 (2.4%)	0.68 (2.7%)	0.87 (3.5%)	0.84 (3.3%)
Coal	1.13 (4.8%)	1.08 (4.5%)	1.08 (4.4%)	1.05 (4.0%)	1.06 (4.0%)	0.80 (3.0%)	0.78 (3.0%)	0.93 (3.5%)	0.71 (2.8%)	0.53 (2.1%)	0.44 (1.7%)
Peat	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	23.31 (100%)	23.75 (100%)	24.68 (100%)	25.84 (100%)	26.46 (100%)	26.43 (100%)	26.52 (100%)	26.47 (100%)	25.34 (100%)	24.84 (100%)	25.38 (100%)

Table 5.5: Final energy consumption in industry disaggregated by energy sub-type (share in brackets)

Energy [TWh]		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas		8.79 (37.7%)	9.41 (39.6%)	10.07 (40.8%)	10.65 (41.2%)	11.37 (43.0%)	11.60 (43.9%)	12.05 (45.4%)	11.64 (44.0%)	10.70 (42.2%)	10.02 (40.3%)	10.89 (42.9%)
Electricity		6.54 (28.0%)	6.62 (27.9%)	6.85 (27.8%)	6.82 (26.4%)	6.47 (24.4%)	6.56 (24.8%)	6.54 (24.6%)	6.73 (25.4%)	6.58 (26.0%)	6.80 (27.4%)	6.93 (27.3%)
Oil	Diesel / Gas oil	1.55 (6.7%)	1.49 (6.3%)	1.48 (6.0%)	1.78 (6.9%)	1.85 (7.0%)	1.85 (7.0%)	1.71 (6.5%)	1.69 (6.4%)	1.92 (7.6%)	1.85 (7.5%)	1.95 (7.7%)
	LPG	0.54 (2.3%)	0.53 (2.3%)	0.64 (2.6%)	0.66 (2.5%)	0.68 (2.6%)	0.73 (2.8%)	0.72 (2.7%)	0.79 (3.0%)	0.88 (3.5%)	1.01 (4.1%)	1.08 (4.3%)
	Pet. coke	1.32 (5.6%)	1.45 (6.1%)	1.55 (6.3%)	1.53 (5.9%)	1.64 (6.2%)	1.59 (6.0%)	1.46 (5.5%)	1.67 (6.3%)	1.43 (5.6%)	1.31 (5.3%)	0.90 (3.5%)
	Fuel oil	0.87 (3.7%)	0.48 (2.0%)	0.40 (1.6%)	0.35 (1.4%)	0.35 (1.3%)	0.34 (1.3%)	0.34 (1.3%)	0.25 (0.9%)	0.25 (1.0%)	0.19 (0.8%)	0.09 (0.4%)
	Kerosene	0.08 (0.3%)	0.09 (0.4%)	0.07 (0.3%)	0.07 (0.3%)	0.07 (0.3%)	0.07 (0.3%)	0.08 (0.3%)	0.06 (0.2%)	0.07 (0.3%)	0.06 (0.3%)	0.07 (0.3%)
	Naphtha	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Renewables	Biomass	1.65 (7.1%)	1.68 (7.1%)	1.56 (6.3%)	1.73 (6.7%)	1.80 (6.8%)	1.71 (6.5%)	1.65 (6.2%)	1.48 (5.6%)	1.44 (5.7%)	1.25 (5.0%)	1.22 (4.8%)
	Ren. waste	0.30 (1.3%)	0.36 (1.5%)	0.44 (1.8%)	0.47 (1.8%)	0.47 (1.8%)	0.47 (1.8%)	0.50 (1.9%)	0.54 (2.0%)	0.63 (2.5%)	0.68 (2.7%)	0.67 (2.7%)
	Biodiesel	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.19 (0.8%)	0.21 (0.8%)
	Biogas	0.04 (0.2%)	0.05 (0.2%)	0.05 (0.2%)	0.04 (0.2%)	0.05 (0.2%)	0.05 (0.2%)	0.07 (0.3%)	0.07 (0.3%)	0.08 (0.3%)	0.06 (0.3%)	0.09 (0.3%)
Non-ren. waste		0.48 (2.1%)	0.51 (2.2%)	0.49 (2.0%)	0.66 (2.6%)	0.64 (2.4%)	0.66 (2.5%)	0.62 (2.4%)	0.63 (2.4%)	0.68 (2.7%)	0.87 (3.5%)	0.84 (3.3%)
Coal	Bit. coal	1.13 (4.8%)	1.08 (4.5%)	1.08 (4.4%)	1.05 (4.0%)	1.06 (4.0%)	0.80 (3.0%)	0.78 (3.0%)	0.93 (3.5%)	0.71 (2.8%)	0.53 (2.1%)	0.44 (1.7%)
Peat	Milled peat	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	Briquettes	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total		23.31 (100%)	23.75 (100%)	24.68 (100%)	25.84 (100%)	26.46 (100%)	26.43 (100%)	26.52 (100%)	26.47 (100%)	25.34 (100%)	24.84 (100%)	25.38 (100%)

5.2.2 Final energy consumption in transport

Fuel consumption in transport is often closely aligned to the mode of transport used: jet kerosene is used for air transport (both international and domestic); petrol and bioethanol are almost exclusively used for road transport; while diesel and biodiesel are used for road transport, domestic navigation¹⁸, and rail.

Figure 5.7 shows the share of each energy type in transport energy consumption. Figure 5.7 shows the share of energy types in transport in 2024. Figure 5.8 and Table 5.4 show the trends in transport's final energy use, split by energy type over the last 21 years. Blended fuels (fossil fuels and biofuels) are reported separately as their renewable and non-renewable constituent components, despite being consumed as blended products.

The most important point to note is that transport remains heavily dependent on fossil fuels, particularly oil products which account for around 92.6% of transport consumption. This lack of fuel diversity is unique amongst the sectors of final consumption. Renewables made up around 6.3% share of transport energy use in 2024. Electricity also remains a relatively small share of transport energy use despite growing from around 0.1% in 2019 to 0.7% in 2024, which is split between electric rail (Dublin Area Rapid Transit (DART) and Luas) and electric vehicles (mostly private cars). This has meant that there has been very little decarbonisation of the transport fuel mix to date, with transport greenhouse gas emissions remaining tightly coupled to energy use.

There was a clear shift from petrol to diesel over the time period, due to the switch to diesel private cars accelerated by the changes to the private car tax system from 2008 onwards.

In accordance with the principle of recording physical flows of energy in the national energy statistics, and reporting guidance from Eurostat under the Energy Statistics Regulation, all gas delivered to road vehicles from the gas network is recorded as *natural gas*, regardless of any mass balancing of grid-injected biomethane via sustainability certificates. Biomethane used directly road transport without injection into the gas network (*i.e.* the physical flow of biomethane). Under the Renewable Energy Directive, biomethane from the gas network can be counted in the transport sector via mass balancing and transfer of sustainability certificates, see section 8.2.6.

Biopropane (also referred to as bioLPG) is a biofuel (*i.e.* renewable liquid transport fuel) that is used in place of LPG. Biopropane is included in the biodiesel figures in the transport sector in the national energy statistics.

¹⁸ Under the EU's Energy Statistics Regulation, and in accordance with guidance issued by Eurostat and IEA, fuel consumed in ships engaged in international navigation (based on port of departure and arrival) is considered *international marine bunkers* and is not included within the transport sector. *International marine bunkers* is included within the primary energy section of the National Energy Balance, similar to imports and exports.

Figure 5.7: Shares of energy sub-types in transport final energy

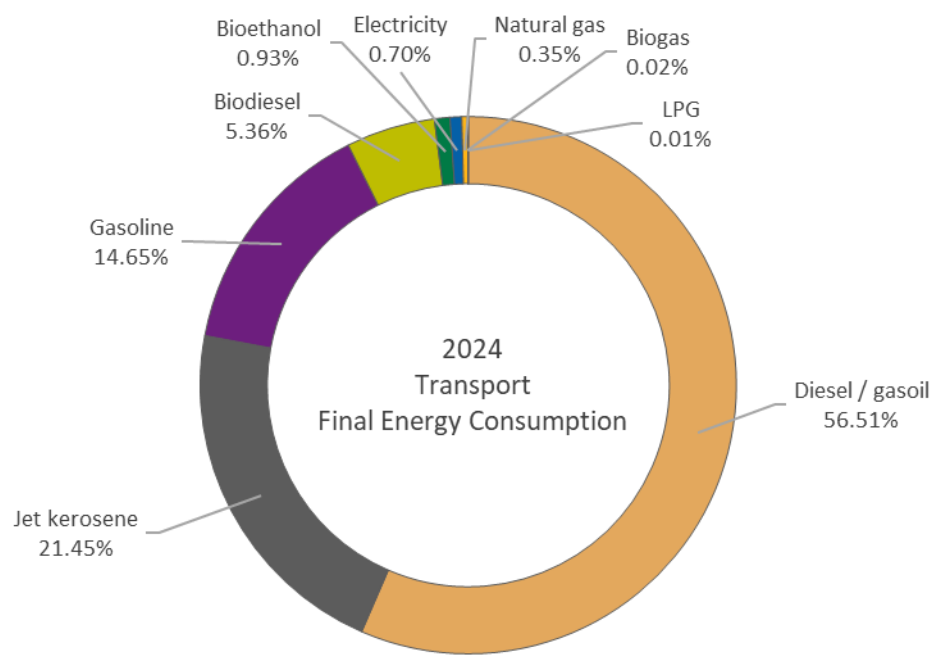


Figure 5.8: Final energy in transport sector by energy sub-type

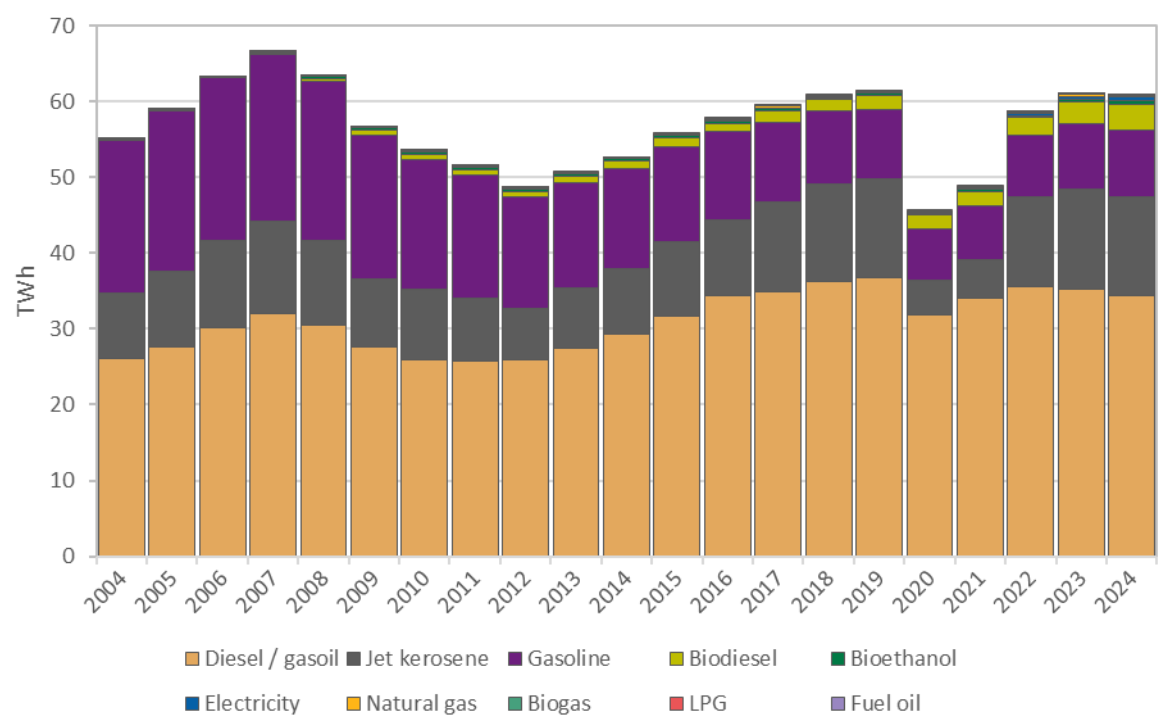


Table 5.6: Final energy in transport sector by energy sub-types (share in brackets)

Energy [TWh]		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	Diesel / gasoil	29.25 (55.6%)	31.72 (57.0%)	34.32 (59.4%)	34.80 (58.6%)	36.31 (59.6%)	36.80 (60.0%)	31.83 (69.9%)	33.98 (69.7%)	35.50 (60.6%)	35.16 (57.7%)	34.34 (56.5%)
	Jet kerosene	8.70 (16.5%)	9.84 (17.7%)	10.10 (17.5%)	11.88 (20.0%)	12.83 (21.1%)	12.98 (21.1%)	4.63 (10.2%)	5.18 (10.6%)	11.84 (20.2%)	13.26 (21.8%)	13.03 (21.5%)
	Gasoline	13.18 (25.1%)	12.50 (22.4%)	11.66 (20.2%)	10.52 (17.7%)	9.59 (15.7%)	9.08 (14.8%)	6.73 (14.8%)	7.13 (14.6%)	8.12 (13.9%)	8.65 (14.2%)	8.90 (14.7%)
	LPG	0.02 (0.0%)	0.03 (0.1%)	0.03 (0.1%)	0.03 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)
	Fuel oil	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Renewables	Biodiesel	1.04 (2.0%)	1.14 (2.1%)	1.00 (1.7%)	1.52 (2.6%)	1.48 (2.4%)	1.90 (3.1%)	1.82 (4.0%)	1.87 (3.8%)	2.39 (4.1%)	2.89 (4.8%)	3.26 (5.4%)
	Bioethanol	0.31 (0.6%)	0.35 (0.6%)	0.38 (0.7%)	0.34 (0.6%)	0.32 (0.5%)	0.30 (0.5%)	0.23 (0.5%)	0.24 (0.5%)	0.27 (0.5%)	0.38 (0.6%)	0.57 (0.9%)
	Biogas	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.01 (0.0%)	0.00 (0.0%)	0.01 (0.0%)
Electricity		0.04 (0.1%)	0.04 (0.1%)	0.05 (0.1%)	0.05 (0.1%)	0.07 (0.1%)	0.09 (0.1%)	0.10 (0.2%)	0.15 (0.3%)	0.22 (0.4%)	0.33 (0.5%)	0.43 (0.7%)
Natural gas		0.03 (0.1%)	0.05 (0.1%)	0.25 (0.4%)	0.24 (0.4%)	0.26 (0.4%)	0.20 (0.3%)	0.18 (0.4%)	0.19 (0.4%)	0.20 (0.3%)	0.20 (0.3%)	0.21 (0.3%)
Total		52.58 (100%)	55.67 (100%)	57.79 (100%)	59.38 (100%)	60.87 (100%)	61.36 (100%)	45.53 (100%)	48.75 (100%)	58.56 (100%)	60.89 (100%)	60.76 (100%)

Figure 5.9 shows the current share of each subsector in transport by final energy. Road transport accounts for just over 75% of energy consumption in the transport sector, followed by international aviation at around 21%. Navigation (domestic), rail, pipeline and domestic aviation account for the remaining 3%.

SEAI collects data on the quantity of each fuel sold for consumption in road transport from the national Oil Levy Application (OLA) operated by the Department of Climate, Energy and the Environment. SEAI uses various methods to further disaggregate road transport energy consumption into:

- **Private car** – vehicles in the *private* taxation class
- **Freight** – vehicles in the *goods* taxation class with an unladen weight equal to or greater than 2,033 kg.
- **Light goods vehicle** – vehicles in the *goods* taxation class with an unladen weight less than 2,033 kg.
- **Public passenger** – vehicles in the *large PSV* and *small PSV* taxation classes. This includes public and private buses, coaches, taxis and hackneys.
- **Fuel tourism** – defined as fuel that is bought within the state by motorists and hauliers but consumed outside the State.
- **Road unspecified** – the balance of recorded road transport fuel consumption that is not attributable to the items above. This includes fuel consumed by vehicles for which insufficient data is available. This includes vehicles taxed as *motorcycles*, *vintage/veteran*, *hearses*, *motor caravan*, *youth community bus* or *school bus*. It also includes state-owned vehicles, diplomatic vehicles, emergency vehicles, and vehicles exempt under the Disabled Drivers and Disabled Passengers (Tax Concessions) Regulations. Fuel consumption by vehicles not under taxation (e.g. foreign registered vehicles) and not reflected in the fuel tourism end use would also be captured here. The unspecified figure also, by default, accounts for any discrepancy between the estimated and the real-world energy demand of the individual road transport modes above.

Figure 5.9: Shares of subsectors in transport final energy

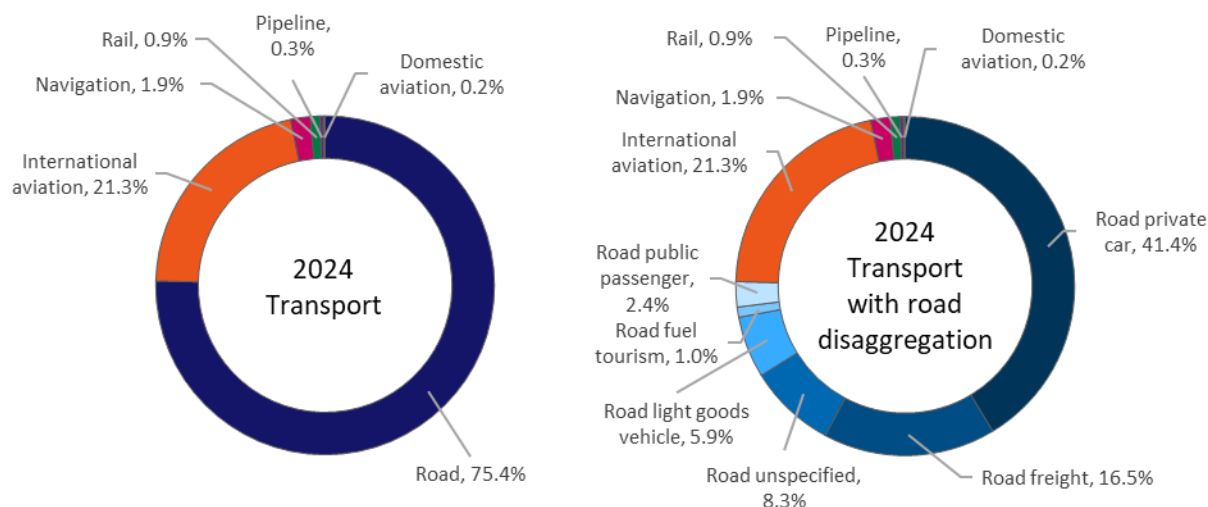


Figure 5.10 shows the trend for energy use of transport by subsector over the period. The energy use of light goods vehicles (LGV) is estimated from 2008 onwards. Prior to 2008, the energy use of this subsector was included in the 'unspecified' category.

Private car energy use dominates and accounted for 41.4% of transport energy use. Private car energy use declined briefly during the recession in 2009 and 2010 but returned to growth in 2011. It peaked in 2015 and remained relatively flat until 2019, before the sharp drop in 2020. In 2024 consumption by private cars exceeded pre-pandemic levels (2019).

Aviation energy use is notable in that it usually makes up a large share of transport energy use in Ireland, particularly since 2000, but can be severely affected by external factors, such as recessions or the COVID-19 pandemic. International aviation energy consumption fell by 44% between 2007 and 2012 during the recession. It returned to strong growth after 2012, reaching an all-time-high in 2019, 6.8% above the previous 2007 peak, before the dramatic fall in 2020. In 2022, consumption in international aviation had returned to around the same level as 2017, followed by a further increase to a new record high in 2023. International aviation decreased by 1.8% in 2024.

Road freight energy use also saw a large reduction during the recession, falling by 49% between 2007 and 2013. It has increased since, but remains below the 2007 peak. These changes are due to changes in the amount of goods transported, as discussed further in section 10.3.4.

Figure 5.10: Final energy in transport by subsector

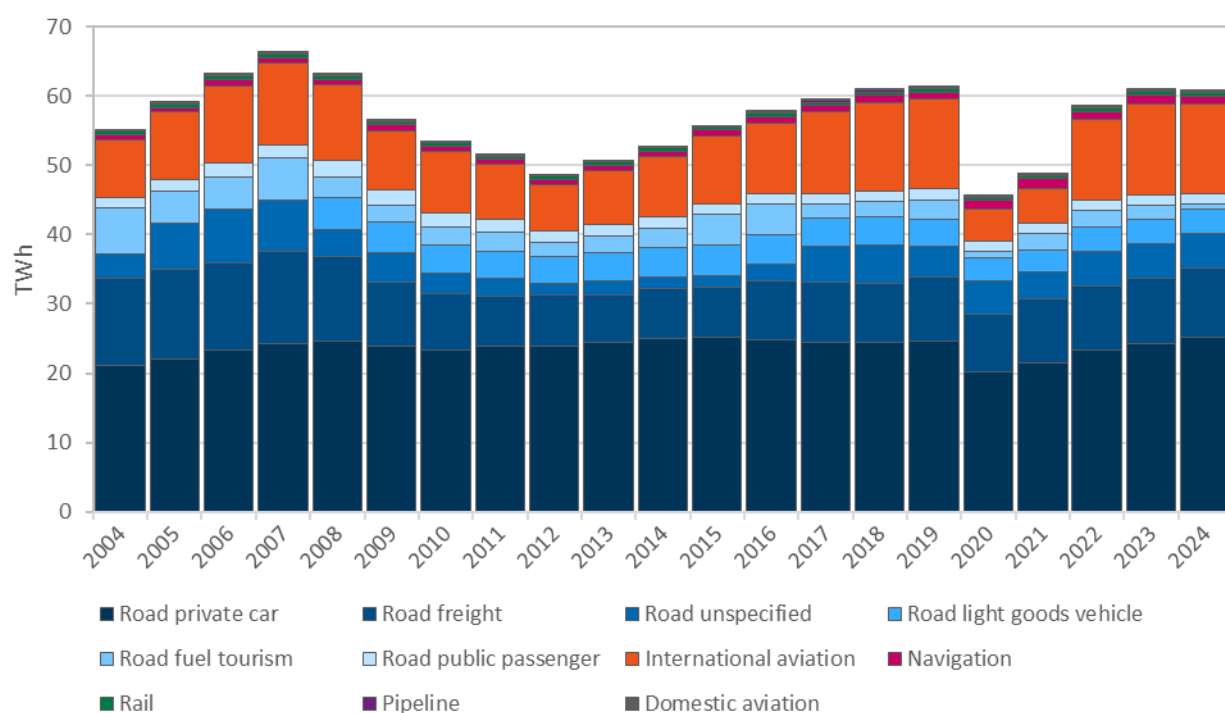


Table 5.7: Transport final energy by subsector (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Road private car	25.08 (47.7%)	25.19 (45.3%)	24.81 (42.9%)	24.36 (41.0%)	24.36 (40.0%)	24.64 (40.1%)	20.16 (44.3%)	21.43 (43.9%)	23.36 (39.9%)	24.35 (40.0%)	25.13 (41.4%)
Road freight	7.23 (13.7%)	7.28 (13.1%)	8.55 (14.8%)	8.70 (14.6%)	8.53 (14.0%)	9.18 (15.0%)	8.43 (18.5%)	9.25 (19.0%)	9.19 (15.7%)	9.28 (15.2%)	10.01 (16.5%)
Road unspecified	1.52 (2.9%)	1.55 (2.8%)	2.41 (4.2%)	5.31 (8.9%)	5.68 (9.3%)	4.51 (7.4%)	4.68 (10.3%)	3.88 (8.0%)	5.07 (8.7%)	5.08 (8.3%)	5.07 (8.3%)
Road light goods vehicle	4.33 (8.2%)	4.39 (7.9%)	4.18 (7.2%)	4.10 (6.9%)	3.97 (6.5%)	3.82 (6.2%)	3.45 (7.6%)	3.20 (6.6%)	3.52 (6.0%)	3.46 (5.7%)	3.56 (5.9%)
Road fuel tourism	2.82 (5.4%)	4.51 (8.1%)	4.47 (7.7%)	1.92 (3.2%)	2.17 (3.6%)	2.89 (4.7%)	0.94 (2.1%)	2.41 (4.9%)	2.29 (3.9%)	2.03 (3.3%)	0.59 (1.0%)
Road public passenger	1.58 (3.0%)	1.55 (2.8%)	1.54 (2.7%)	1.51 (2.5%)	1.59 (2.6%)	1.59 (2.6%)	1.38 (3.0%)	1.40 (2.9%)	1.46 (2.5%)	1.49 (2.5%)	1.46 (2.4%)
International aviation	8.65 (16.5%)	9.79 (17.6%)	10.04 (17.4%)	11.82 (19.9%)	12.77 (21.0%)	12.91 (21.0%)	4.58 (10.1%)	5.12 (10.5%)	11.76 (20.1%)	13.18 (21.7%)	12.94 (21.3%)
Navigation	0.84 (1.6%)	0.83 (1.5%)	1.00 (1.7%)	0.88 (1.5%)	0.98 (1.6%)	1.04 (1.7%)	1.27 (2.8%)	1.36 (2.8%)	1.16 (2.0%)	1.22 (2.0%)	1.14 (1.9%)
Rail	0.45 (0.9%)	0.46 (0.8%)	0.47 (0.8%)	0.48 (0.8%)	0.49 (0.8%)	0.51 (0.8%)	0.42 (0.9%)	0.45 (0.9%)	0.49 (0.8%)	0.54 (0.9%)	0.57 (0.9%)
Pipeline	0.03 (0.1%)	0.05 (0.1%)	0.25 (0.4%)	0.24 (0.4%)	0.26 (0.4%)	0.20 (0.3%)	0.18 (0.4%)	0.18 (0.4%)	0.19 (0.3%)	0.17 (0.3%)	0.19 (0.3%)
Domestic aviation	0.06 (0.1%)	0.06 (0.1%)	0.07 (0.1%)	0.07 (0.1%)	0.06 (0.1%)	0.07 (0.1%)	0.05 (0.1%)	0.08 (0.2%)	0.08 (0.1%)	0.09 (0.1%)	0.10 (0.2%)
Total	52.58 (100%)	55.67 (100%)	57.79 (100%)	59.38 (100%)	60.87 (100%)	61.36 (100%)	45.53 (100%)	48.75 (100%)	58.56 (100%)	60.89 (100%)	60.76 (100%)

5.2.2.1 Breakdown of energy sub-types and use in transport

This subsection provides additional breakdowns on the quantities of each energy sub-type used within each subsector of transport. Figure 5.11 and Table 5.8 provide the annual quantities of each energy sub-type within road transport.

Figure 5.11: Road transport – final energy by energy sub-type

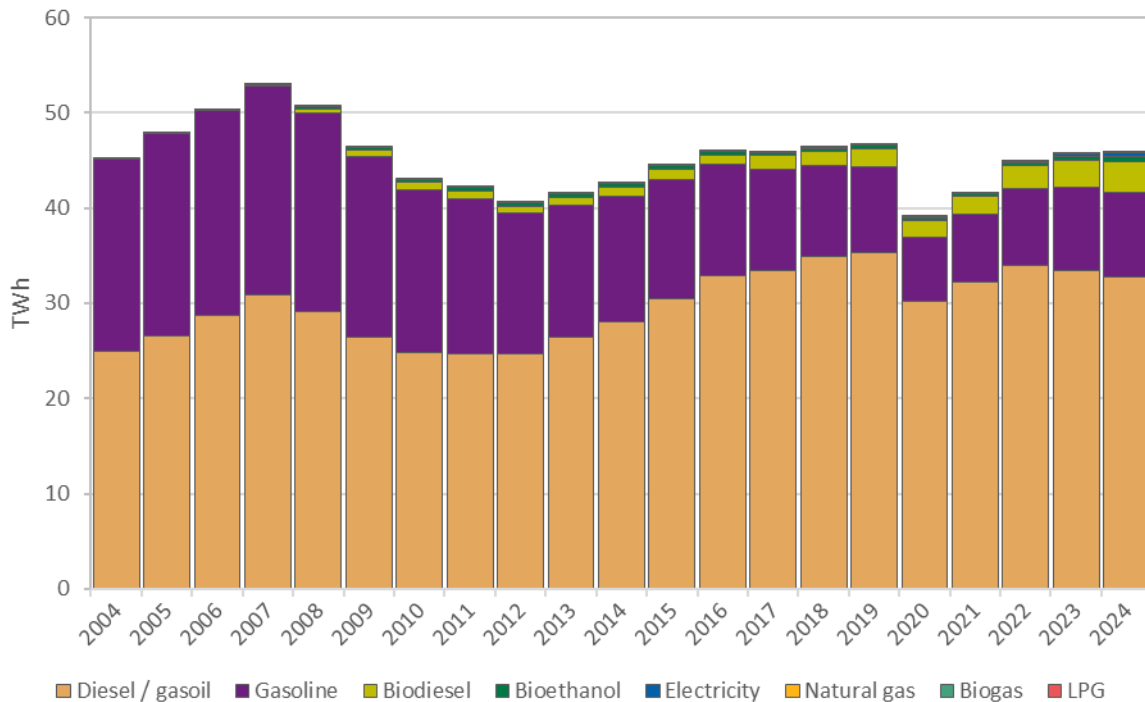


Table 5.8: Road transport – final energy by energy sub-type

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Diesel / gasoil	28.00 (65.8%)	30.47 (68.5%)	32.90 (71.6%)	33.48 (73.0%)	34.89 (75.3%)	35.30 (75.7%)	30.19 (77.4%)	32.22 (77.5%)	33.91 (75.5%)	33.47 (73.3%)	32.71 (71.4%)
Gasoline	13.17 (31.0%)	12.49 (28.1%)	11.65 (25.4%)	10.51 (22.9%)	9.58 (20.7%)	9.07 (19.5%)	6.72 (17.2%)	7.12 (17.1%)	8.11 (18.1%)	8.64 (18.9%)	8.90 (19.4%)
Biodiesel	1.04 (2.4%)	1.14 (2.6%)	1.00 (2.2%)	1.52 (3.3%)	1.48 (3.2%)	1.90 (4.1%)	1.82 (4.7%)	1.87 (4.5%)	2.38 (5.3%)	2.87 (6.3%)	3.23 (7.1%)
Bioethanol	0.31 (0.7%)	0.35 (0.8%)	0.38 (0.8%)	0.34 (0.8%)	0.32 (0.7%)	0.30 (0.7%)	0.23 (0.6%)	0.24 (0.6%)	0.27 (0.6%)	0.38 (0.8%)	0.57 (1.2%)
Electricity	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.03 (0.1%)	0.05 (0.1%)	0.10 (0.2%)	0.17 (0.4%)	0.28 (0.6%)	0.38 (0.8%)
Natural gas	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.03 (0.1%)
Biogas	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.01 (0.0%)	0.00 (0.0%)	0.01 (0.0%)
LPG	0.02 (0.1%)	0.03 (0.1%)	0.03 (0.1%)	0.03 (0.1%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)
Total	42.55 (100%)	44.48 (100%)	45.97 (100%)	45.90 (100%)	46.30 (100%)	46.63 (100%)	39.03 (100%)	41.58 (100%)	44.89 (100%)	45.69 (100%)	45.83 (100%)

Figure 5.12 and Table 5.9 provide the annual quantities of each energy sub-type within international and domestic aviation.

Figure 5.12: International and domestic aviation – final energy by energy sub-type

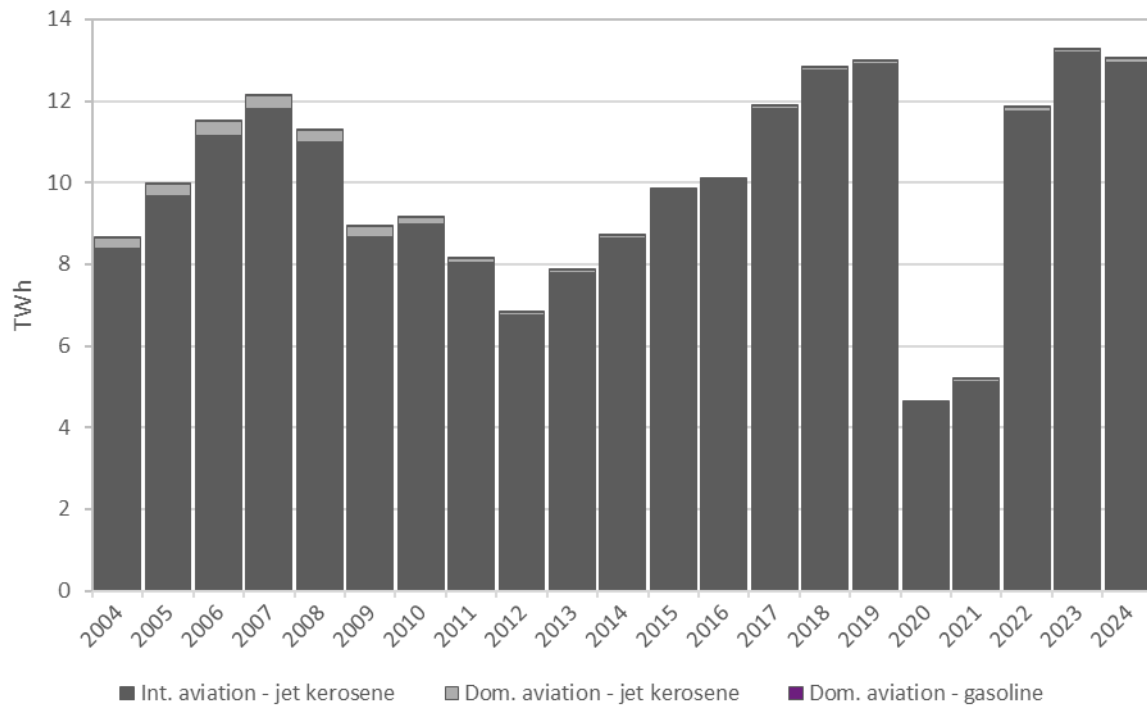


Table 5.9: International and domestic aviation – final energy by energy sub-type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Int. aviation - jet kerosene	8.65 (99.3%)	9.79 (99.4%)	10.04 (99.4%)	11.82 (99.4%)	12.77 (99.5%)	12.91 (99.5%)	4.58 (98.9%)	5.12 (98.5%)	11.76 (99.3%)	13.18 (99.3%)	12.94 (99.2%)
Dom. aviation - jet kerosene	0.05 (0.6%)	0.05 (0.5%)	0.06 (0.6%)	0.06 (0.5%)	0.06 (0.4%)	0.06 (0.5%)	0.04 (0.9%)	0.07 (1.3%)	0.08 (0.6%)	0.08 (0.6%)	0.09 (0.7%)
Dom. aviation - gasoline	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.0%)
Total	8.71 (100%)	9.85 (100%)	10.11 (100%)	11.88 (100%)	12.83 (100%)	12.98 (100%)	4.64 (100%)	5.19 (100%)	11.84 (100%)	13.27 (100%)	13.04 (100%)

Figure 5.13 and Table 5.10 provide the annual quantities of each energy sub-type within domestic navigation, rail and pipeline transport.

Figure 5.13: Navigation, rail and pipeline – final energy by energy sub-type

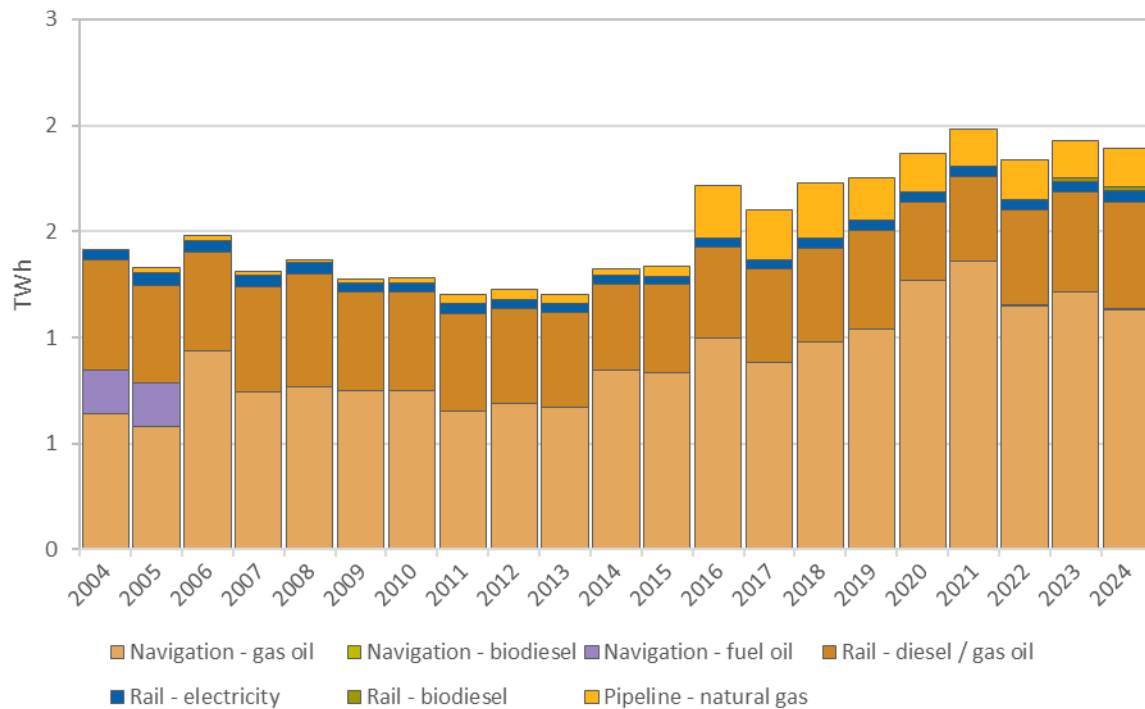


Table 5.10: Navigation, rail and pipeline – final energy by energy sub-type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Navigation - gas oil	0.84 (63.7%)	0.83 (62.3%)	1.00 (58.3%)	0.88 (55.1%)	0.98 (56.4%)	1.04 (59.3%)	1.27 (68.2%)	1.36 (68.5%)	1.15 (62.5%)	1.22 (63.1%)	1.13 (59.5%)
Navigation - biodiesel	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.01 (0.5%)	0 (0%)	0.01 (0.5%)
Navigation - fuel oil	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Rail - diesel / gas oil	0.41 (30.9%)	0.42 (31.2%)	0.42 (24.7%)	0.44 (27.4%)	0.44 (25.6%)	0.46 (26.4%)	0.37 (19.8%)	0.40 (20.1%)	0.45 (24.3%)	0.47 (24.3%)	0.50 (26.4%)
Rail - electricity	0.04 (3.0%)	0.04 (3.1%)	0.04 (2.6%)	0.05 (2.8%)	0.05 (2.9%)	0.05 (3.0%)	0.05 (2.6%)	0.05 (2.5%)	0.05 (2.7%)	0.05 (2.7%)	0.05 (2.8%)
Rail - biodiesel	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.02 (1.0%)	0.02 (1.0%)
Pipeline - natural gas	0.03 (2.5%)	0.05 (3.4%)	0.25 (14.4%)	0.24 (14.7%)	0.26 (15.1%)	0.20 (11.3%)	0.18 (9.5%)	0.18 (8.9%)	0.19 (10.1%)	0.17 (9.0%)	0.19 (9.8%)
Total	1.32 (100%)	1.33 (100%)	1.72 (100%)	1.60 (100%)	1.73 (100%)	1.75 (100%)	1.86 (100%)	1.98 (100%)	1.84 (100%)	1.93 (100%)	1.89 (100%)

5.2.3 Final energy consumption in residential sector

Figure 5.11 shows the most recent breakdown of energy consumption in the residential sector by energy type, while Figure 5.12 shows the annual quantities of each energy type consumed over the last 21 years. The shares of each have been relatively stable, with a gradual increase in the share of electricity, gas, and renewables, and a continuing, though gradual, decline in coal, peat and oil use.

Oil remains the dominant fuel in the residential sector at just over 41%. Electricity is the second largest source, with natural gas having the third largest share. Use of renewables directly in households has grown each year since 2017.

Table 5.11 provides the quantities and shares of each energy type in the residential sector over the last 11 years, while Table 5.12 presents further a disaggregation by energy sub-type. It is notable that electricity consumption over the period was at its highest in 2021. Oil (specifically kerosene), natural gas and electricity all decreased in 2022 and 2023, before growing again in 2024.

Figure 5.14: Shares of energy types in residential final energy

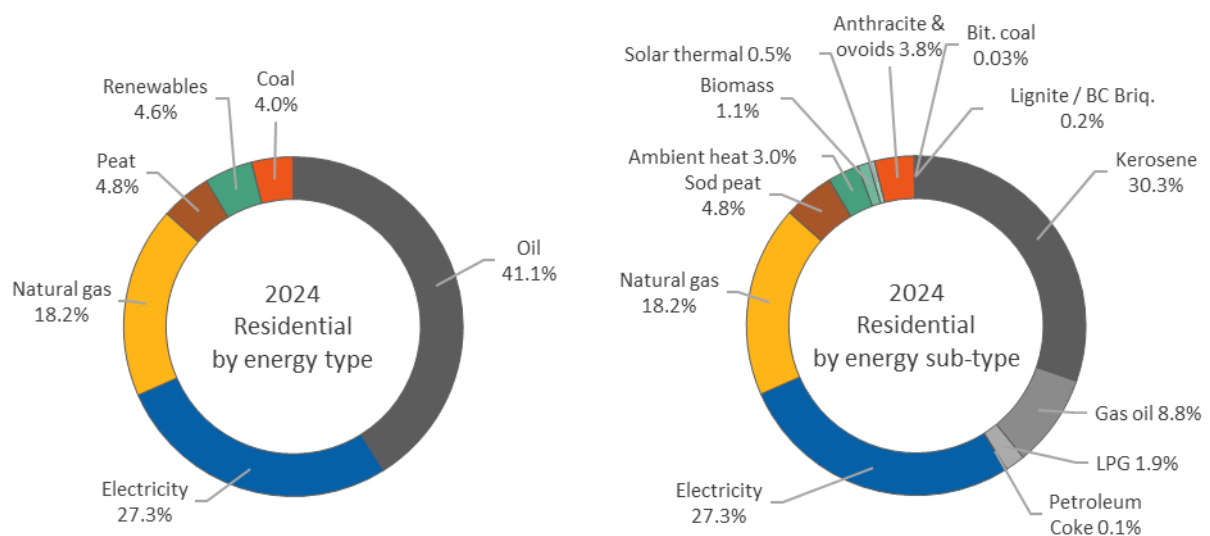


Figure 5.15: Final energy in residential by energy type

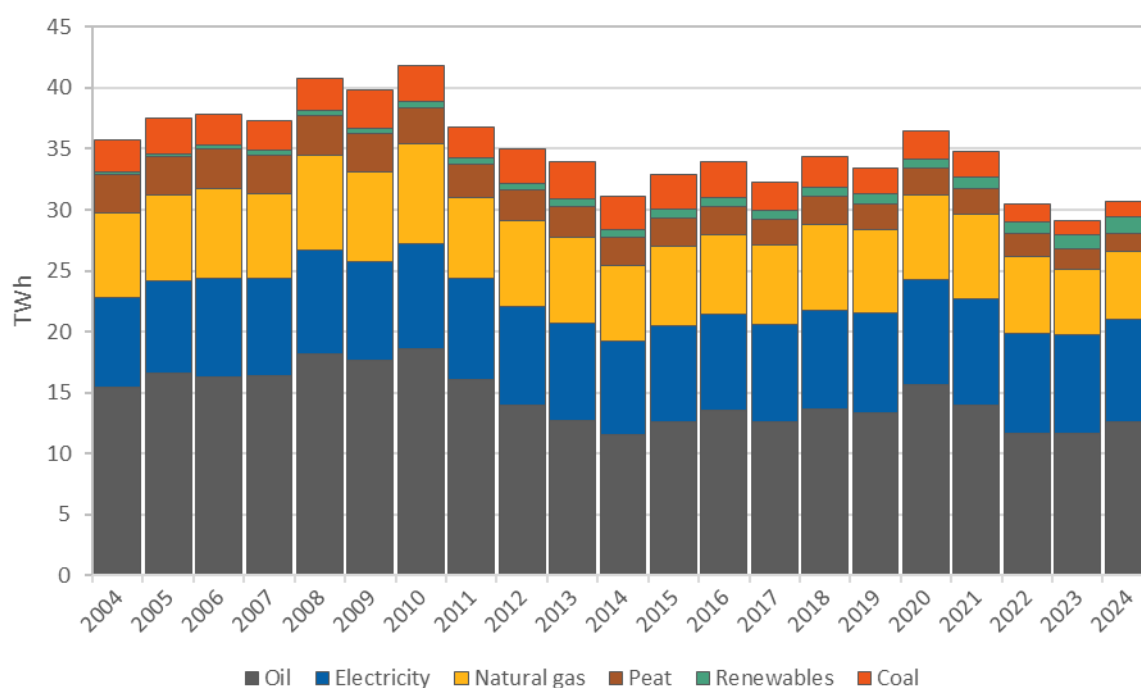


Table 5.11: Final energy in residential sector by energy type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	11.50 (37.0%)	12.63 (38.4%)	13.52 (39.8%)	12.65 (39.2%)	13.62 (39.6%)	13.39 (40.1%)	15.61 (42.8%)	13.99 (40.2%)	11.66 (38.3%)	11.65 (40.0%)	12.60 (41.1%)
Electricity	7.70 (24.8%)	7.88 (24.0%)	7.87 (23.2%)	7.95 (24.7%)	8.16 (23.8%)	8.11 (24.3%)	8.68 (23.8%)	8.74 (25.1%)	8.21 (26.9%)	8.10 (27.8%)	8.37 (27.3%)
Natural gas	6.23 (20.0%)	6.46 (19.6%)	6.55 (19.3%)	6.46 (20.0%)	7.03 (20.4%)	6.88 (20.6%)	6.86 (18.8%)	6.92 (19.9%)	6.28 (20.6%)	5.41 (18.6%)	5.58 (18.2%)
Peat	2.33 (7.5%)	2.33 (7.1%)	2.29 (6.7%)	2.19 (6.8%)	2.29 (6.6%)	2.13 (6.4%)	2.20 (6.0%)	2.10 (6.0%)	1.87 (6.1%)	1.63 (5.6%)	1.49 (4.8%)
Renewables	0.59 (1.9%)	0.72 (2.2%)	0.77 (2.3%)	0.72 (2.2%)	0.76 (2.2%)	0.75 (2.3%)	0.82 (2.2%)	0.90 (2.6%)	1.01 (3.3%)	1.19 (4.1%)	1.41 (4.6%)
Coal	2.75 (8.8%)	2.87 (8.7%)	2.99 (8.8%)	2.28 (7.1%)	2.52 (7.3%)	2.16 (6.5%)	2.27 (6.2%)	2.16 (6.2%)	1.44 (4.7%)	1.14 (3.9%)	1.22 (4.0%)
Non-ren. waste	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	31.10 (100%)	32.90 (100%)	33.99 (100%)	32.24 (100%)	34.37 (100%)	33.42 (100%)	36.45 (100%)	34.80 (100%)	30.47 (100%)	29.12 (100%)	30.66 (100%)

Table 5.12: Final energy in residential sector disaggregated by energy sub-type (share in brackets)

Energy [TWh]		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	Kerosene	7.99 (25.7%)	9.31 (28.3%)	9.84 (29.0%)	9.32 (28.9%)	10.16 (29.5%)	9.90 (29.6%)	11.83 (32.5%)	10.33 (29.7%)	8.56 (28.1%)	8.55 (29.4%)	9.28 (30.3%)
	Gas oil	2.99 (9.6%)	2.79 (8.5%)	3.00 (8.8%)	2.70 (8.4%)	2.77 (8.1%)	2.87 (8.6%)	3.06 (8.4%)	3.02 (8.7%)	2.50 (8.2%)	2.50 (8.6%)	2.71 (8.8%)
	LPG	0.42 (1.3%)	0.42 (1.3%)	0.56 (1.7%)	0.56 (1.7%)	0.63 (1.8%)	0.58 (1.7%)	0.64 (1.8%)	0.56 (1.6%)	0.53 (1.7%)	0.58 (2.0%)	0.59 (1.9%)
	Petroleum Coke	0.10 (0.3%)	0.11 (0.3%)	0.11 (0.3%)	0.07 (0.2%)	0.06 (0.2%)	0.04 (0.1%)	0.09 (0.2%)	0.08 (0.2%)	0.08 (0.3%)	0.02 (0.1%)	0.02 (0.1%)
Electricity		7.70 (24.8%)	7.88 (24.0%)	7.87 (23.2%)	7.95 (24.7%)	8.16 (23.8%)	8.11 (24.3%)	8.68 (23.8%)	8.74 (25.1%)	8.21 (26.9%)	8.10 (27.8%)	8.37 (27.3%)
Natural gas		6.23 (20.0%)	6.46 (19.6%)	6.55 (19.3%)	6.46 (20.0%)	7.03 (20.4%)	6.88 (20.6%)	6.86 (18.8%)	6.92 (19.9%)	6.28 (20.6%)	5.41 (18.6%)	5.58 (18.2%)
Peat	Sod peat	1.49 (4.8%)	1.49 (4.5%)	1.49 (4.4%)	1.49 (4.6%)	1.49 (4.3%)	1.49 (4.4%)	1.49 (4.1%)	1.49 (4.3%)	1.49 (4.9%)	1.49 (5.1%)	1.49 (4.8%)
	Briquettes	0.84 (2.7%)	0.85 (2.6%)	0.80 (2.4%)	0.71 (2.2%)	0.80 (2.3%)	0.65 (1.9%)	0.72 (2.0%)	0.61 (1.8%)	0.38 (1.3%)	0.14 (0.5%)	0 (0%)
Renewables	Ambient heat	0.17 (0.5%)	0.20 (0.6%)	0.23 (0.7%)	0.24 (0.7%)	0.25 (0.7%)	0.29 (0.9%)	0.34 (0.9%)	0.41 (1.2%)	0.56 (1.8%)	0.73 (2.5%)	0.91 (3.0%)
	Biomass	0.30 (1.0%)	0.39 (1.2%)	0.40 (1.2%)	0.33 (1.0%)	0.35 (1.0%)	0.31 (0.9%)	0.32 (0.9%)	0.33 (0.9%)	0.29 (1.0%)	0.29 (1.0%)	0.33 (1.1%)
	Solar thermal	0.12 (0.4%)	0.13 (0.4%)	0.14 (0.4%)	0.15 (0.5%)	0.16 (0.5%)	0.16 (0.5%)	0.16 (0.4%)	0.16 (0.5%)	0.16 (0.5%)	0.16 (0.6%)	0.16 (0.5%)
Coal	Anthracite & ovoids	1.06 (3.4%)	0.87 (2.6%)	1.03 (3.0%)	0.99 (3.1%)	1.04 (3.0%)	1.03 (3.1%)	1.03 (2.8%)	1.06 (3.0%)	1.03 (3.4%)	1.05 (3.6%)	1.16 (3.8%)
	Lignite / BC Briq.	0.15 (0.5%)	0.21 (0.6%)	0.19 (0.6%)	0.17 (0.5%)	0.14 (0.4%)	0.11 (0.3%)	0.08 (0.2%)	0.14 (0.4%)	0.09 (0.3%)	0.09 (0.3%)	0.05 (0.2%)
	Bit. coal	1.54 (5.0%)	1.80 (5.5%)	1.77 (5.2%)	1.12 (3.5%)	1.34 (3.9%)	1.03 (3.1%)	1.16 (3.2%)	0.96 (2.7%)	0.32 (1.0%)	0.00 (0.0%)	0.01 (0.0%)
Total		31.10 (100%)	32.90 (100%)	33.99 (100%)	32.24 (100%)	34.37 (100%)	33.42 (100%)	36.45 (100%)	34.80 (100%)	30.47 (100%)	29.12 (100%)	30.66 (100%)

5.2.4 Final energy consumption in services

In determining the sectoral breakdown of commercial and public service energy use, SEAI uses a blend of data sources, including the Business Energy Use Survey (BEUS) from the CSO [10]. In June 2025, the CSO published the latest version of the BEUS, which is based on energy use in 2023. SEAI uses this most recent BEUS release in apportioning 2024 energy consumption across the commercial and public service sectors.

Figure 5.16: Shares of energy types in commercial and public services final energy

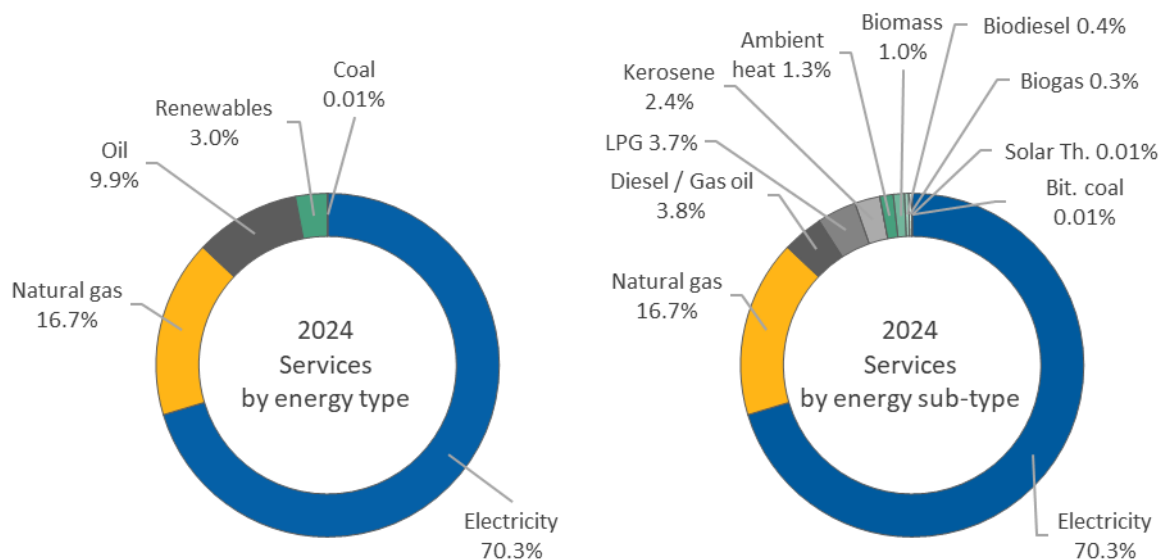


Figure 5.17: Final energy in commercial and public services by energy type

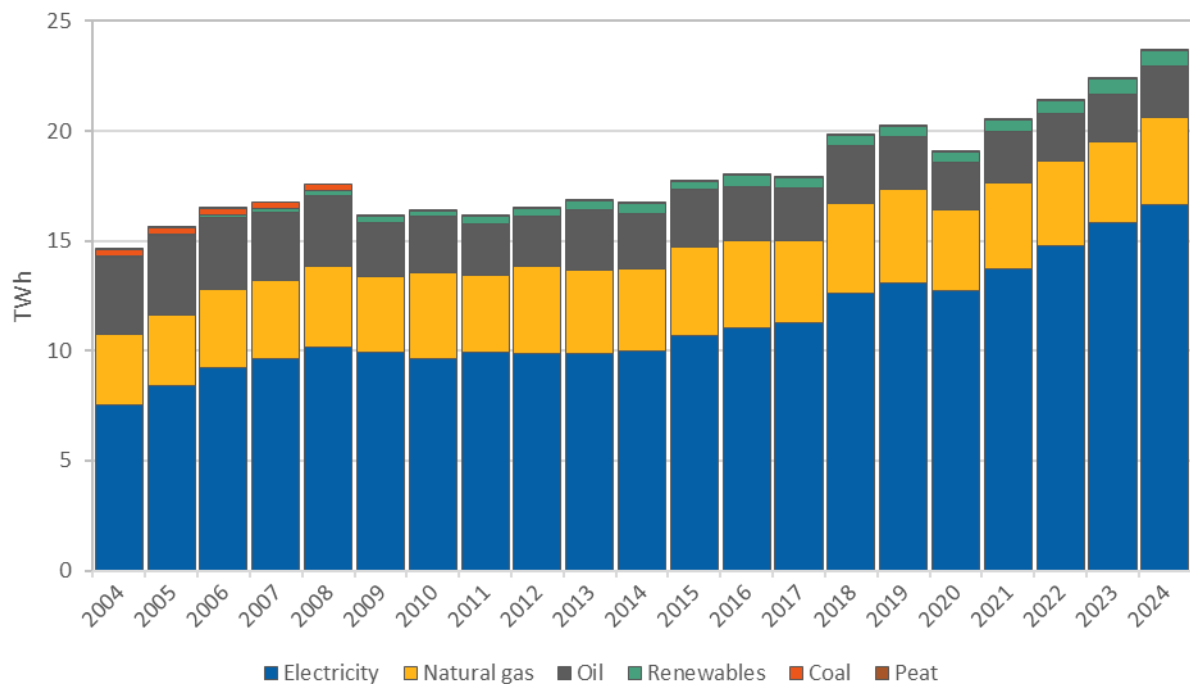


Figure 5.14 shows the changes in the fuel mix in the services sector over the period. The range of fuels used in this sector is small – essentially oil, gas and electricity. Oil and gas are used predominantly for

space heating, but also for water heating, cooking and, in some subsectors, laundry. Electricity is used in buildings for heating, air conditioning, water heating, lighting, lifts and escalators, automatic doors, and ICT. Electricity in services is also used for public lighting, off-road electric vehicles, and water and sanitation services.

Electricity use in services is driven by the changing structure of this sector and the general increase in use of ICT, electric heating and air conditioning. Data centres are also included under commercial services.

Small quantities were consumed in services until 2006; however, consumption of peat is no longer recorded as statistically significant. Similarly, the consumption of coal in the sector is now very low, at approximately 3 GWh per annum.

The quantities and shares of energy types consumed in commercial and public services are shown in Table 5.7. Table 5.14 provides further disaggregation of consumption in the services sector by energy sub-type. The following energy products are included in Table 5.14 for completeness but have not been consumed in statistically significant quantities since 2008 (at the latest): anthracite & ovoids, lignite / brown coal briquettes, milled peat, sod peat, and peat briquettes.

Table 5.13: Final energy in commercial and public services (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Electricity	9.97 (59.7%)	10.69 (60.4%)	11.05 (61.5%)	11.24 (62.9%)	12.62 (63.7%)	13.10 (64.8%)	12.73 (66.8%)	13.74 (67.0%)	14.77 (69.1%)	15.83 (70.9%)	16.63 (70.3%)
Natural gas	3.74 (22.4%)	4.01 (22.6%)	3.94 (21.9%)	3.77 (21.1%)	4.10 (20.7%)	4.22 (20.8%)	3.66 (19.2%)	3.90 (19.0%)	3.82 (17.9%)	3.64 (16.3%)	3.95 (16.7%)
Oil	2.50 (15.0%)	2.63 (14.9%)	2.49 (13.8%)	2.38 (13.3%)	2.61 (13.2%)	2.43 (12.0%)	2.15 (11.3%)	2.31 (11.3%)	2.20 (10.3%)	2.20 (9.8%)	2.35 (9.9%)
Renewables	0.48 (2.9%)	0.37 (2.1%)	0.49 (2.7%)	0.49 (2.7%)	0.47 (2.4%)	0.47 (2.3%)	0.51 (2.7%)	0.56 (2.7%)	0.60 (2.8%)	0.66 (3.0%)	0.71 (3.0%)
Coal	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Peat	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Non-ren. waste	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	16.70 (100%)	17.70 (100%)	17.97 (100%)	17.89 (100%)	19.81 (100%)	20.22 (100%)	19.06 (100%)	20.51 (100%)	21.39 (100%)	22.34 (100%)	23.65 (100%)

Table 5.14: Final energy in commercial and public services disaggregated by energy sub-type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Electricity	9.97 (59.7%)	10.69 (60.4%)	11.05 (61.5%)	11.24 (62.9%)	12.62 (63.7%)	13.10 (64.8%)	12.73 (66.8%)	13.74 (67.0%)	14.77 (69.1%)	15.83 (70.9%)	16.63 (70.3%)
Natural gas	3.74 (22.4%)	4.01 (22.6%)	3.94 (21.9%)	3.77 (21.1%)	4.10 (20.7%)	4.22 (20.8%)	3.66 (19.2%)	3.90 (19.0%)	3.82 (17.9%)	3.64 (16.3%)	3.95 (16.7%)
Oil	Diesel / Gas oil	1.12 (6.7%)	1.19 (6.7%)	1.18 (6.6%)	1.06 (5.9%)	1.19 (6.0%)	1.05 (5.2%)	0.97 (5.1%)	1.01 (4.9%)	0.94 (4.4%)	0.86 (3.9%)
	LPG	0.76 (4.5%)	0.79 (4.4%)	0.67 (3.7%)	0.70 (3.9%)	0.78 (3.9%)	0.77 (3.8%)	0.64 (3.4%)	0.71 (3.5%)	0.71 (3.3%)	0.81 (3.6%)
	Kerosene	0.58 (3.5%)	0.62 (3.5%)	0.62 (3.4%)	0.60 (3.3%)	0.64 (3.2%)	0.62 (3.0%)	0.54 (2.8%)	0.59 (2.9%)	0.54 (2.5%)	0.52 (2.3%)
	Fuel oil	0.04 (0.2%)	0.04 (0.2%)	0.02 (0.1%)	0.01 (0.1%)	0.01 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
	Pet. coke	0.00 (0.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Renewables	Ambient heat	0.10 (0.6%)	0.12 (0.7%)	0.14 (0.8%)	0.17 (1.0%)	0.18 (0.9%)	0.21 (1.0%)	0.24 (1.3%)	0.26 (1.2%)	0.27 (1.3%)	0.30 (1.3%)
	Biomass	0.32 (1.9%)	0.19 (1.1%)	0.29 (1.6%)	0.24 (1.4%)	0.22 (1.1%)	0.18 (0.9%)	0.20 (1.1%)	0.23 (1.1%)	0.25 (1.1%)	0.21 (0.9%)
	Biodiesel	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0.4%)	0.09 (0.4%)
	Biogas	0.05 (0.3%)	0.06 (0.3%)	0.06 (0.3%)	0.07 (0.4%)	0.07 (0.3%)	0.07 (0.4%)	0.07 (0.4%)	0.07 (0.3%)	0.08 (0.4%)	0.08 (0.3%)
	Solar Th.	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Coal	Bit. coal	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
	Anthracite & ovoids	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	Lignite / BC Briq.	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Peat	Milled peat	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	Sod peat	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	Briquettes	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	16.70 (100%)	17.70 (100%)	17.97 (100%)	17.89 (100%)	19.81 (100%)	20.22 (100%)	19.06 (100%)	20.51 (100%)	21.39 (100%)	22.34 (100%)	23.65 (100%)

Figure 5.15 shows the current breakdown of final energy consumption by subsector in commercial services alone, while Figure 5.16 shows the timeseries of the annual final energy consumption in commercial services. Figure 5.17 shows the current share of energy types in public services alone and Figure 5.18 shows the corresponding timeseries of annual final energy consumption in public services.

Figure 5.18: Shares of final energy in commercial services by subsector

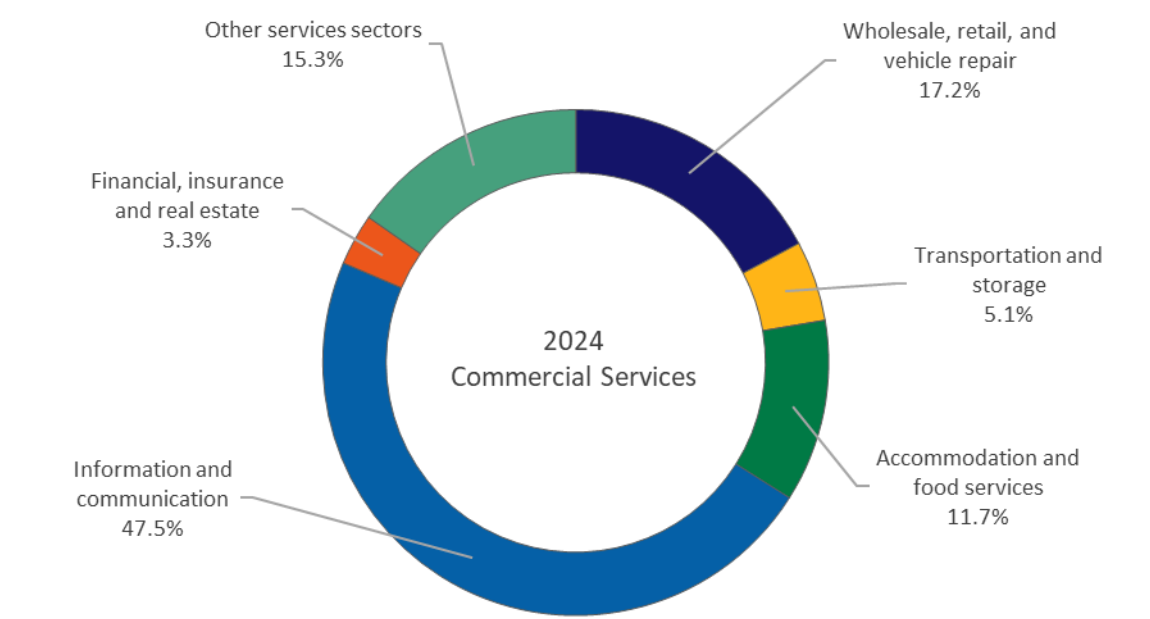


Figure 5.19: Final energy in commercial services by subsector (breakdown not available prior to 2009)

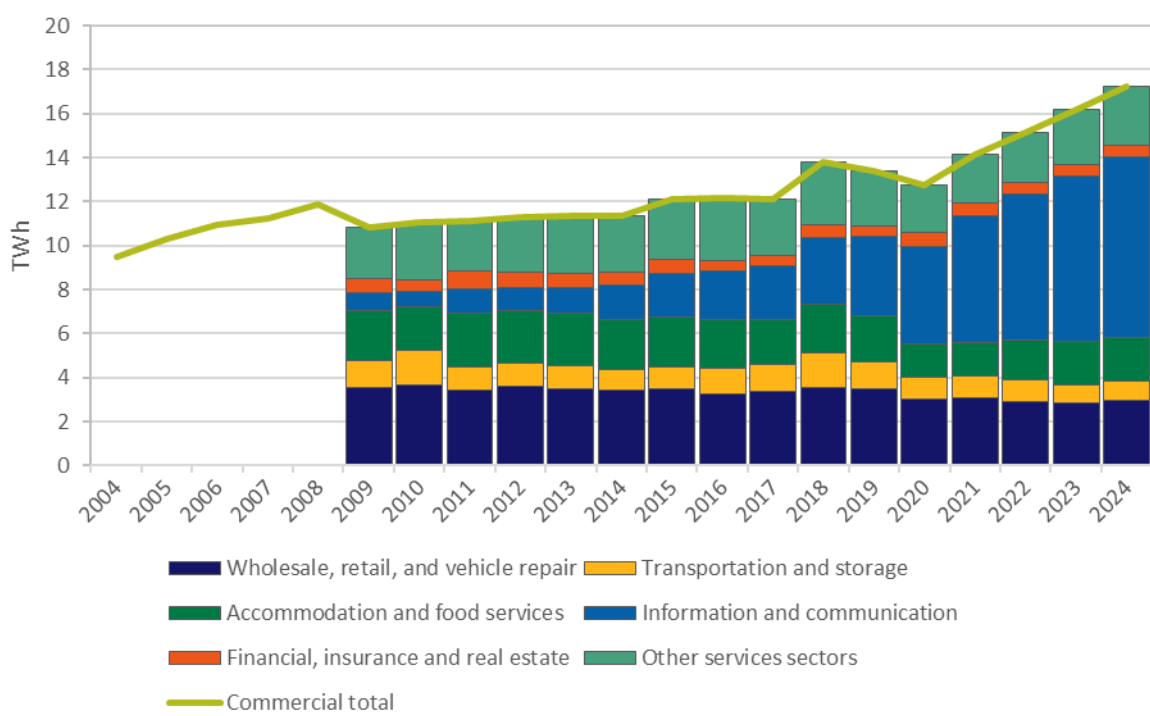


Figure 5.20: Shares of final energy in public services by subsector

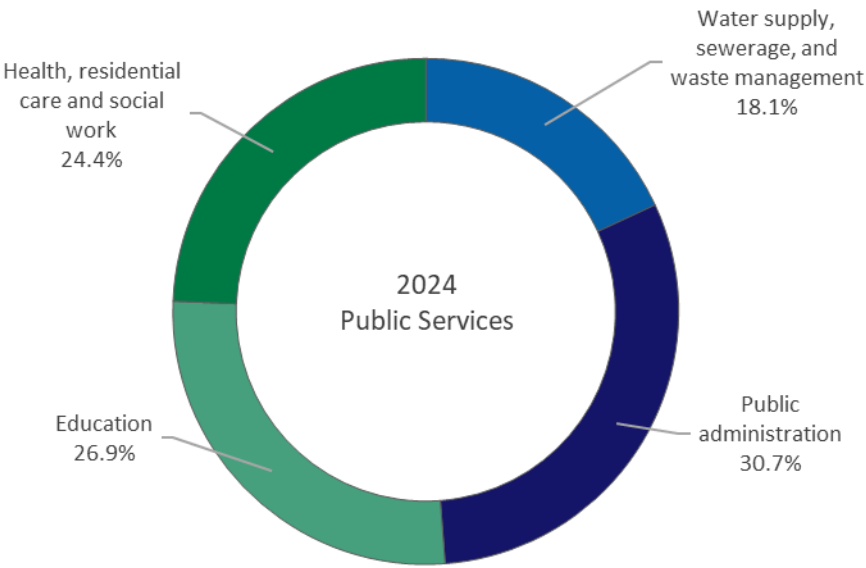
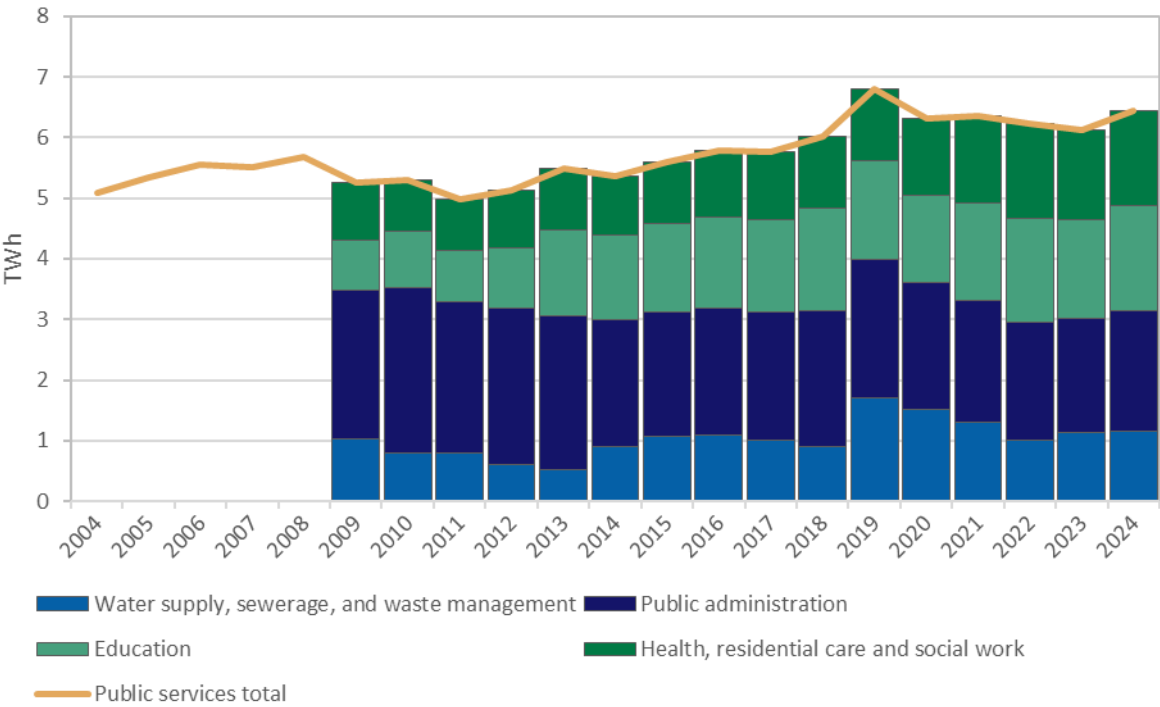


Figure 5.21: Final energy in public services by subsector (breakdown not available prior to 2009)



5.2.5 Final energy consumption in agriculture and fisheries

Figure 5.22 and Table 5.15 show the final energy consumption in the agricultural sector (including forestry). Gas oil and electricity are the only energy products recorded in statistically significant quantities in the sector.

Figure 5.22: Final energy consumption in agriculture by energy type

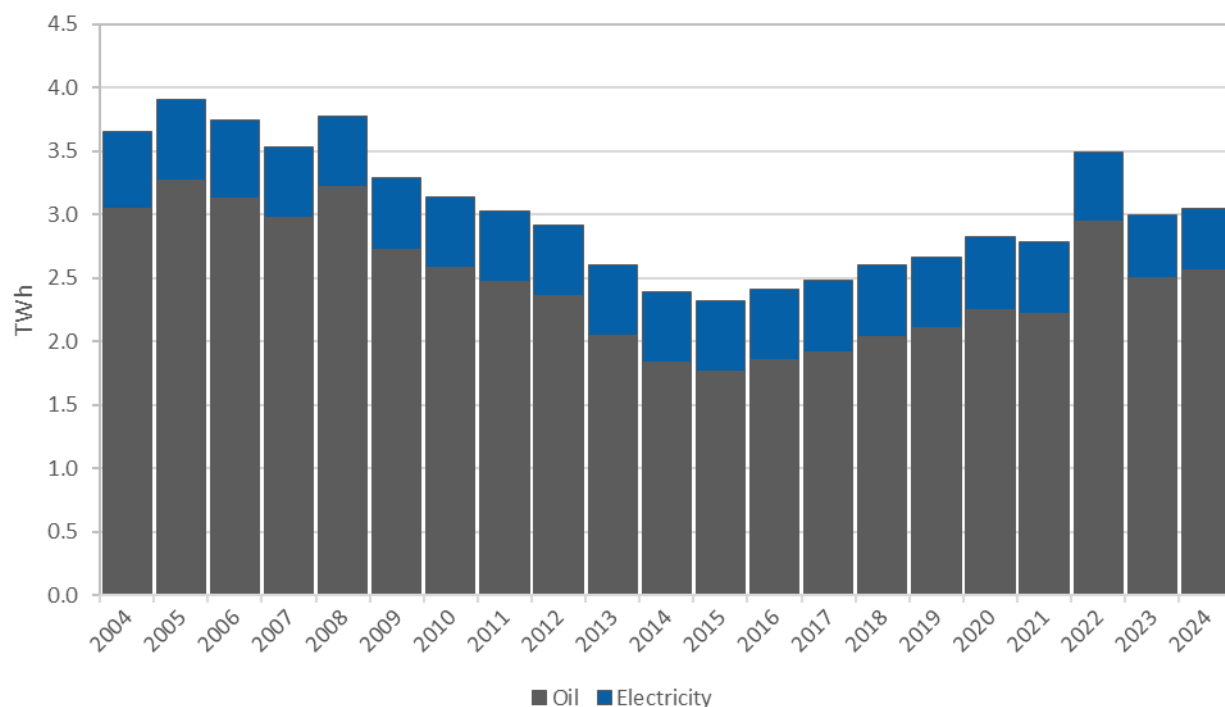


Table 5.15: Final energy in agriculture by energy sub-type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Gas oil	1.83 (76.6%)	1.76 (76.0%)	1.85 (76.8%)	1.92 (77.5%)	2.04 (78.5%)	2.11 (79.3%)	2.25 (79.7%)	2.22 (79.9%)	2.95 (84.5%)	2.50 (83.5%)	2.56 (83.9%)
Electricity	0.56 (23.4%)	0.56 (24.0%)	0.56 (23.2%)	0.56 (22.5%)	0.56 (21.5%)	0.55 (20.7%)	0.57 (20.3%)	0.56 (20.1%)	0.54 (15.5%)	0.49 (16.5%)	0.49 (16.1%)
Total	2.39 (100%)	2.32 (100%)	2.41 (100%)	2.48 (100%)	2.60 (100%)	2.66 (100%)	2.82 (100%)	2.78 (100%)	3.49 (100%)	3.00 (100%)	3.05 (100%)

Figure 5.23 and Table 5.16 show the final energy consumption in the fisheries sector. Gas oil is the only energy product recorded in statistically significant quantities in the sector.

Figure 5.23: Final energy in fisheries by energy type

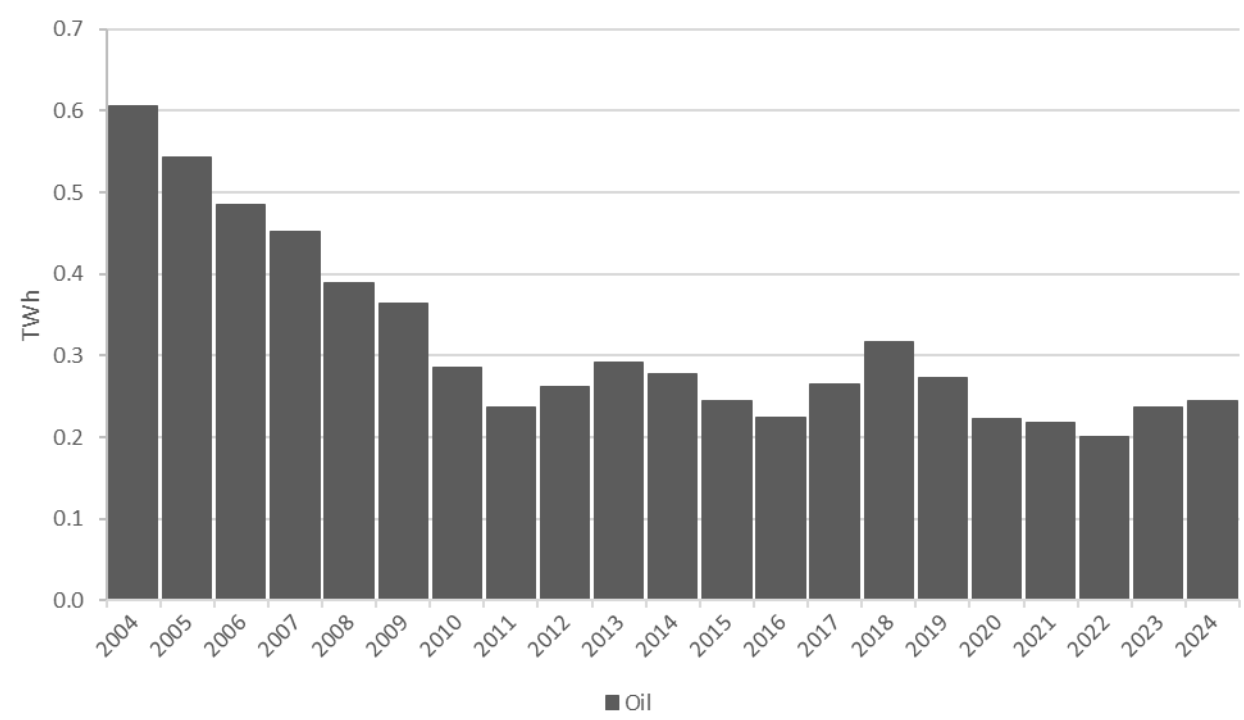


Table 5.16: Final energy in fisheries by energy sub-type (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Gas oil	0.28 (100%)	0.24 (100%)	0.22 (100%)	0.27 (100%)	0.32 (100%)	0.27 (100%)	0.22 (100%)	0.22 (100%)	0.20 (100%)	0.24 (100%)	0.24 (100%)
Total	0.28 (100%)	0.24 (100%)	0.22 (100%)	0.27 (100%)	0.32 (100%)	0.27 (100%)	0.22 (100%)	0.22 (100%)	0.20 (100%)	0.24 (100%)	0.24 (100%)

6 Energy modes

It is useful to split energy supply or use into the three modes of electricity, transport and heat. These represent distinct energy services and markets, and also map onto national and European renewable energy targets. To avoid double counting across modes, any heat and transport energy provided by electricity (for example electric heaters and electric vehicles (EVs)) is counted in the electricity mode only, not the heat or transport modes. This ensures that summing across the three modes gives a consistent total energy use.

6.1 Final energy consumption by mode

Figure 6.1 shows the current split in final energy use between electricity, transport and heat. Figure 6.2 shows the historical trend of the three modes in terms of final energy use. Table 6.1 shows the quantities and shares of final energy use in each mode and changes relative to previous years.

The transport and heat modes historically account for approximately 40% of final energy use each, with the electricity mode accounting for the remaining 20%. However, electricity consumption has increased steadily over this period at a faster rate than total energy consumption. The heat mode tends to show the greatest year-to-year fluctuations, due to its sensitivity to weather effects. Section 9.2 provides more detail on weather effects on heating energy. Final energy use in transport decreased during the 2008-2012 recession, and again during the COVID-19 restrictions in 2020 and early 2021. Outside of these events, final energy use in the transport mode has increased each year, falling slightly in 2024.

Figure 6.1: Current split of final energy by mode of application

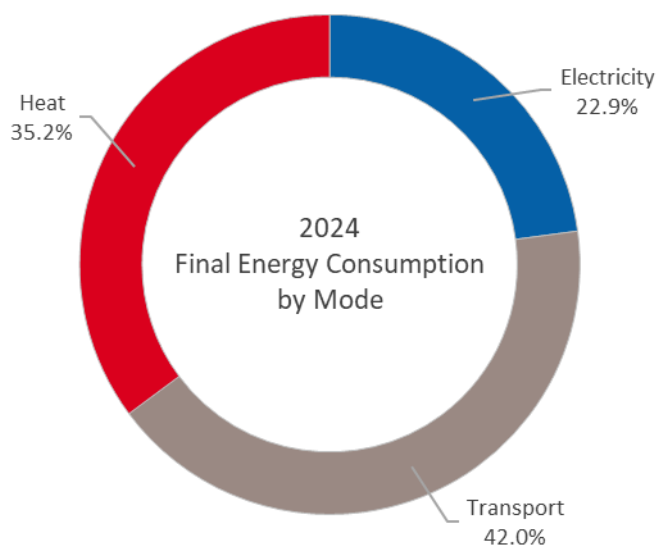


Figure 6.2: Final energy in electricity, transport and heat modes

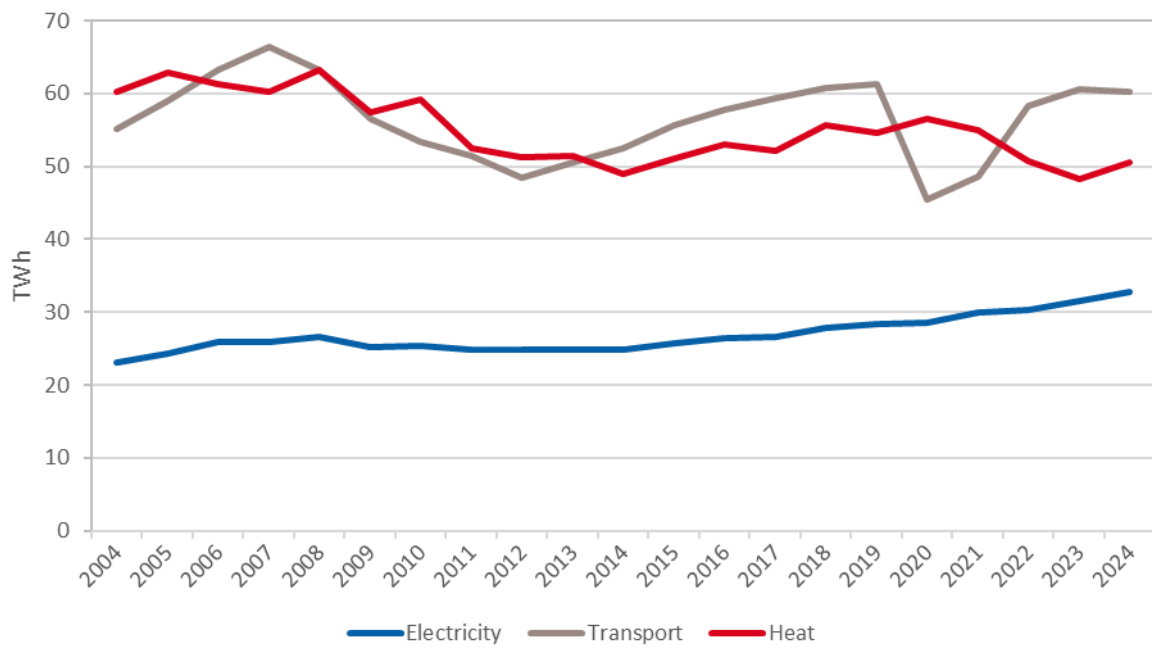


Table 6.1: Final energy in electricity, transport and heat compared (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Electricity	24.81 (19.6%)	25.79 (19.5%)	26.38 (19.2%)	26.63 (19.3%)	27.87 (19.3%)	28.40 (19.7%)	28.62 (21.9%)	29.92 (22.4%)	30.32 (21.7%)	31.57 (22.5%)	32.86 (22.9%)
Transport	52.54 (41.6%)	55.62 (42.0%)	57.74 (42.1%)	59.33 (43.0%)	60.80 (42.1%)	61.28 (42.4%)	45.43 (34.8%)	48.60 (36.4%)	58.34 (41.8%)	60.56 (43.1%)	60.33 (42.0%)
Heat	49.00 (38.8%)	51.17 (38.6%)	52.95 (38.6%)	52.14 (37.8%)	55.75 (38.6%)	54.69 (37.9%)	56.56 (43.3%)	55.02 (41.2%)	50.80 (36.4%)	48.30 (34.4%)	50.55 (35.2%)
Total	126.36 (100%)	132.58 (100%)	137.07 (100%)	138.09 (100%)	144.42 (100%)	144.38 (100%)	130.61 (100%)	133.54 (100%)	139.46 (100%)	140.42 (100%)	143.74 (100%)

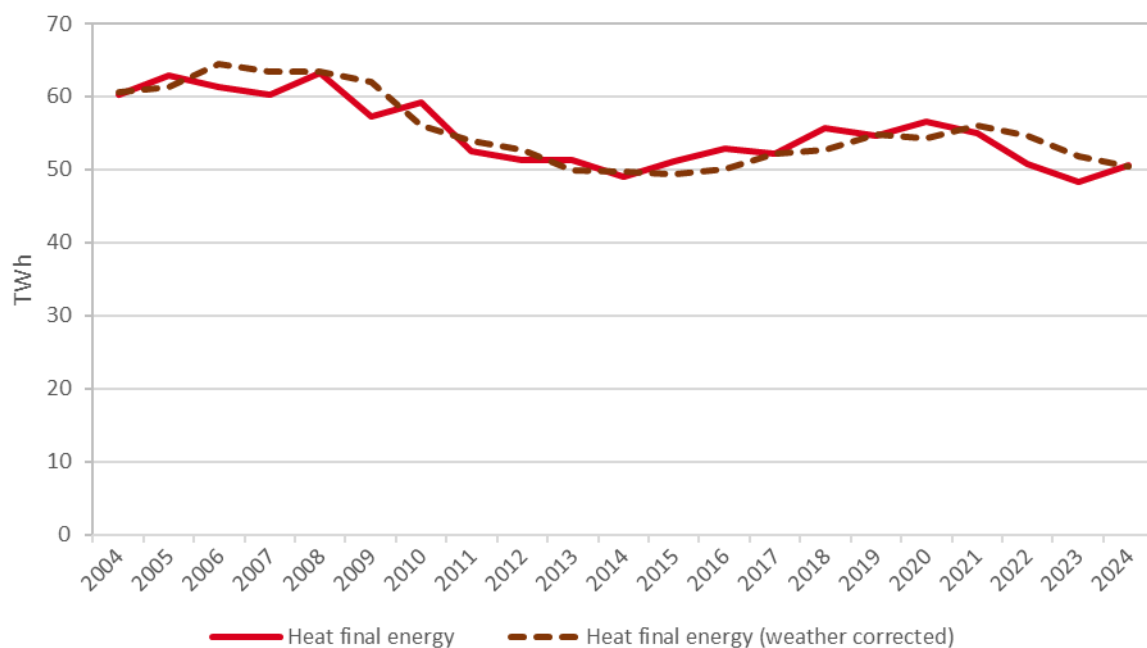
6.2 Heat mode

The heat mode captures all final energy consumption outside of transport and not in the form of electricity. It captures space, water and process heating not provided by electricity. It also includes fuel use in machinery outside of the transport sector: construction, mining, agriculture, forestry, fishing *etc.* The heat mode does not include fuel use outside of final energy consumption: fuel input to energy transformation or own use by energy sector (electricity generation, oil refining, gas refining *etc.*).

Figure 6.3 shows the historical final energy use in the heat mode with and without a weather correction. SEAI uses the concept of degree days for weather normalisation across warmer and colder years, which is the established standard recommended by Eurostat [11]. Further details are provided in section 9.2.

Weather-corrected heat demand reached a broad minimum in the period of 2012 to 2015 (averaging 49.5 TWh), likely due to a combination of impacts from the economic recession, a period of (then) record high oil prices, and efficiency improvements in the heating and insulation of buildings. After 2015, reduced international oil prices coupled with the recovery of the Irish economy acted to increase heat demand. Heat demand decreased since 2020, coinciding with the end of COVID restrictions in 2021 followed by high energy prices and milder weather in 2022 and 2023. Heat demand increased in 2024.

Figure 6.3: Final energy use for heat, actual and weather corrected



6.2.1 Final energy in heat mode by sector

As shown in Figure 6.4 and detailed in Table 6.2, the residential sector is the largest end user of final energy in the heat mode, followed by industry, services, and agricultural and fisheries sectors.

Household demand for heat is strongly affected by weather, as evidenced by the historic peak in 2008, a year that had periods of extremely cold weather.

Figure 6.4: Final energy in heat mode by sector

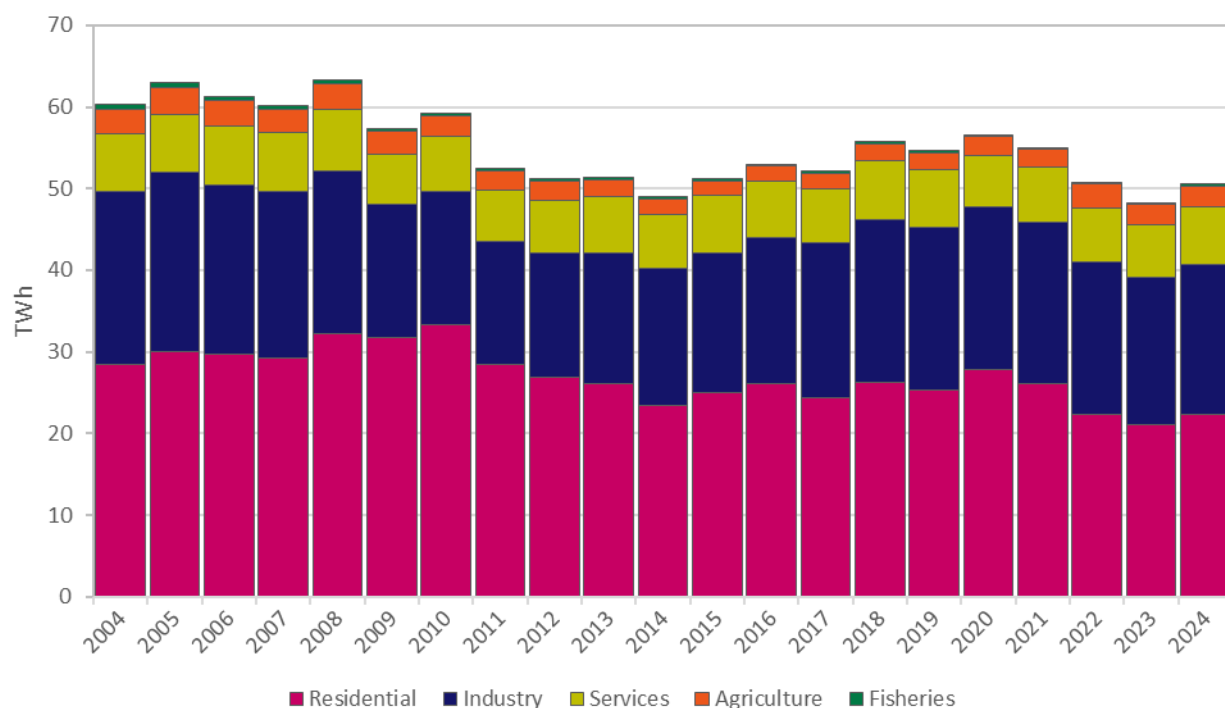


Table 6.2: Final energy in heat mode by sector (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Residential	23.40 (47.8%)	25.02 (48.9%)	26.12 (49.3%)	24.29 (46.6%)	26.21 (47.0%)	25.32 (46.3%)	27.77 (49.1%)	26.06 (47.4%)	22.26 (43.8%)	21.02 (43.5%)	22.29 (44.1%)
Industry	16.77 (34.2%)	17.13 (33.5%)	17.83 (33.7%)	19.02 (36.5%)	20.00 (35.9%)	19.87 (36.3%)	19.99 (35.3%)	19.74 (35.9%)	18.77 (36.9%)	18.03 (37.3%)	18.45 (36.5%)
Services	6.72 (13.7%)	7.01 (13.7%)	6.93 (13.1%)	6.65 (12.7%)	7.19 (12.9%)	7.12 (13.0%)	6.33 (11.2%)	6.77 (12.3%)	6.62 (13.0%)	6.50 (13.5%)	7.02 (13.9%)
Agriculture	1.83 (3.7%)	1.76 (3.4%)	1.85 (3.5%)	1.92 (3.7%)	2.04 (3.7%)	2.11 (3.9%)	2.25 (4.0%)	2.22 (4.0%)	2.95 (5.8%)	2.50 (5.2%)	2.56 (5.1%)
Fisheries	0.28 (0.6%)	0.24 (0.5%)	0.22 (0.4%)	0.27 (0.5%)	0.32 (0.6%)	0.27 (0.5%)	0.22 (0.4%)	0.22 (0.4%)	0.20 (0.4%)	0.24 (0.5%)	0.24 (0.5%)
Total	49.00 (100%)	51.17 (100%)	52.95 (100%)	52.14 (100%)	55.75 (100%)	54.69 (100%)	56.56 (100%)	55.02 (100%)	50.80 (100%)	48.30 (100%)	50.55 (100%)

6.2.2 Final energy in heat mode by fuel

Figure 6.5 and Table 6.3 detail the fuels and energy sources in the final energy heat mode. Oil remains the largest fuel type for the delivery of heat, followed closely by natural gas. The last two decades have seen Ireland shift from an oil dominance in the heat mode to near parity between oil and gas for heat supply. This was driven by Ireland's expanding gas network and an industrial transition from oil to gas. Since 2015, both oil and natural gas have been approximately equal sources of final energy for the heat mode, with oil always slightly leading.

Since 2023 renewables make up more of Ireland's final energy demand in the heat mode than coal and peat combined. These are followed by non-renewable waste.

Figure 6.5: Final energy in heat mode by fuel

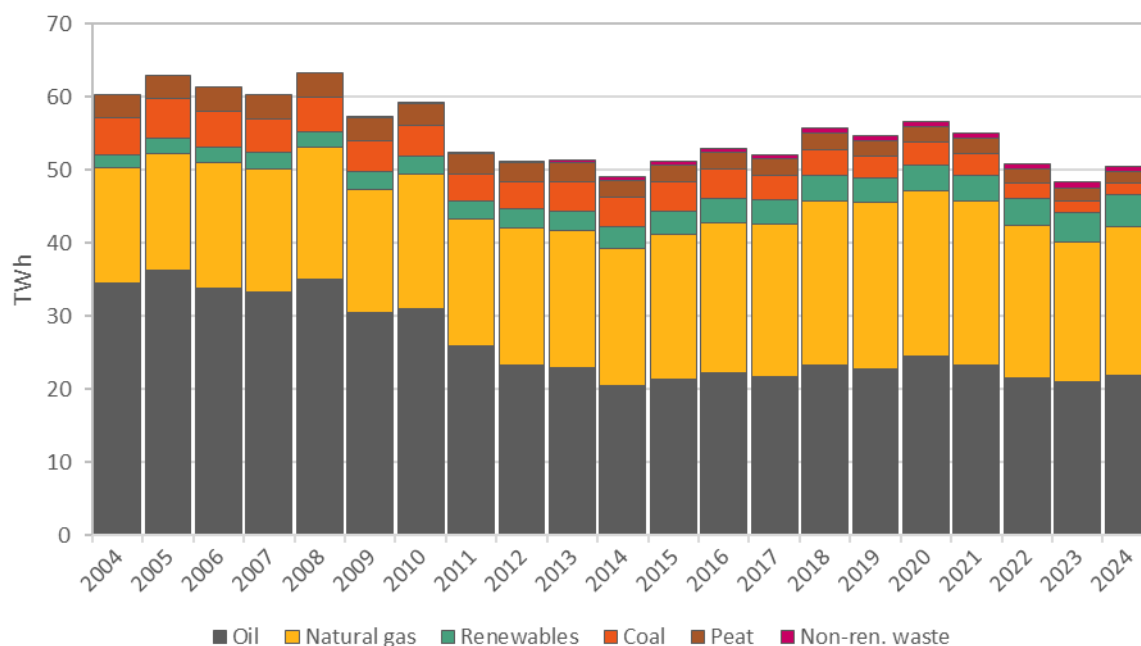


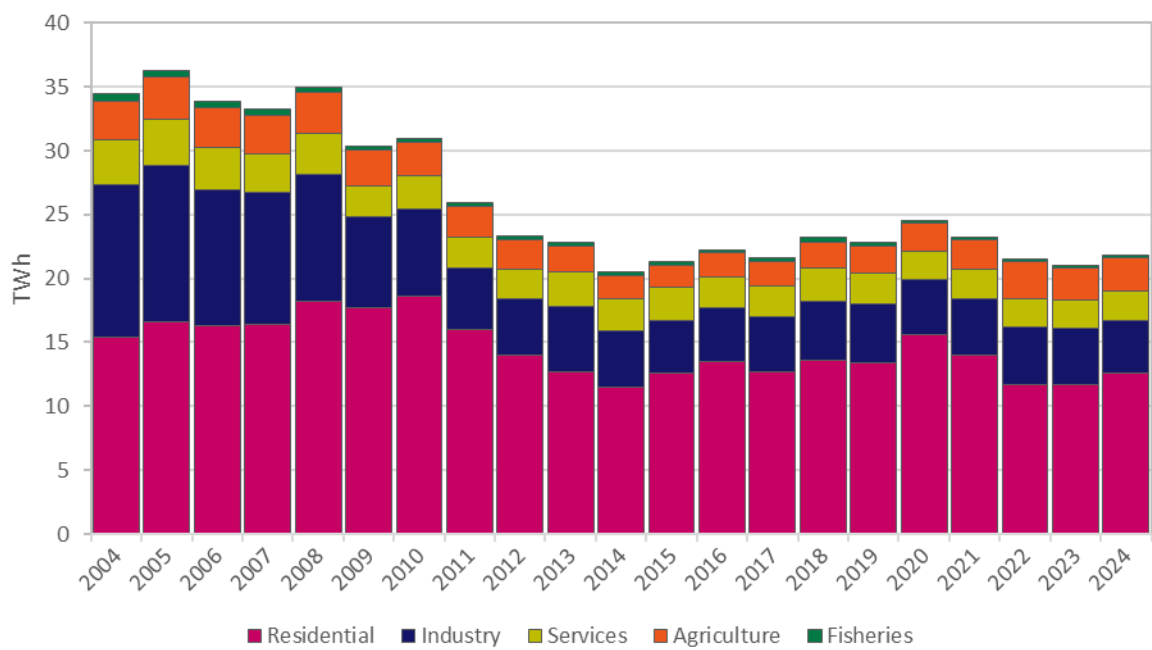
Table 6.3: Final energy in heat mode by energy type (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	20.47 (41.8%)	21.31 (41.6%)	22.22 (42.0%)	21.60 (41.4%)	23.19 (41.6%)	22.80 (41.7%)	24.55 (43.4%)	23.20 (42.2%)	21.55 (42.4%)	21.02 (43.5%)	21.84 (43.2%)
Natural gas	18.76 (38.3%)	19.88 (38.8%)	20.56 (38.8%)	20.89 (40.1%)	22.50 (40.4%)	22.69 (41.5%)	22.57 (39.9%)	22.46 (40.8%)	20.80 (41.0%)	19.07 (39.5%)	20.42 (40.4%)
Renewables	3.07 (6.3%)	3.17 (6.2%)	3.31 (6.3%)	3.45 (6.6%)	3.54 (6.4%)	3.44 (6.3%)	3.56 (6.3%)	3.55 (6.5%)	3.75 (7.4%)	4.04 (8.4%)	4.31 (8.5%)
Coal	3.88 (7.9%)	3.96 (7.7%)	4.08 (7.7%)	3.33 (6.4%)	3.59 (6.4%)	2.97 (5.4%)	3.06 (5.4%)	3.09 (5.6%)	2.15 (4.2%)	1.67 (3.5%)	1.66 (3.3%)
Peat	2.33 (4.8%)	2.34 (4.6%)	2.30 (4.3%)	2.20 (4.2%)	2.30 (4.1%)	2.13 (3.9%)	2.20 (3.9%)	2.10 (3.8%)	1.87 (3.7%)	1.63 (3.4%)	1.49 (2.9%)
Non-ren. waste	0.48 (1.0%)	0.51 (1.0%)	0.49 (0.9%)	0.66 (1.3%)	0.64 (1.1%)	0.66 (1.2%)	0.62 (1.1%)	0.63 (1.1%)	0.68 (1.3%)	0.87 (1.8%)	0.84 (1.7%)
Total	49.00 (100%)	51.17 (100%)	52.95 (100%)	52.14 (100%)	55.75 (100%)	54.69 (100%)	56.56 (100%)	55.02 (100%)	50.80 (100%)	48.30 (100%)	50.55 (100%)

Figure 6.6 profiles oil use for final energy in the heat mode broken down by sector. The largest and most consistent reduction in oil use for heat has come from the industry sector, with the service sector also seeing smaller reductions.

Oil consumption for heat in the residential sector fell by 40% from 2010 to 2014, during a period of (then) record high oil prices, before trending upwards in the second half of the decade. It fell in 2021 and again in 2022, coinciding with the end of COVID restrictions, milder weather and a return of higher prices. Oil consumption in households increased in 2024.

Figure 6.6: Final consumption of oil for heat by sector



6.3 Transport mode

The transport mode captures all energy consumption in transport, excluding electricity. The transport mode is equivalent to the transport sector (see section 5.2.2) minus electricity consumption in that sector.

6.3.1 Final energy in transport mode by subsector

Figure 6.7 and Table 6.4 show the trends and details of final energy use in the transport mode by subsector.

The road unspecified quantity in Figure 6.7 and Table 6.4 relates to measured consumption of transport fuels that cannot be definitively attributed to one of the road transport subsectors, and is composed entirely of petrol, road diesel and respective blended biofuels. See section 5.2.2 for more details.

Figure 6.7: Final energy in transport mode by subsector

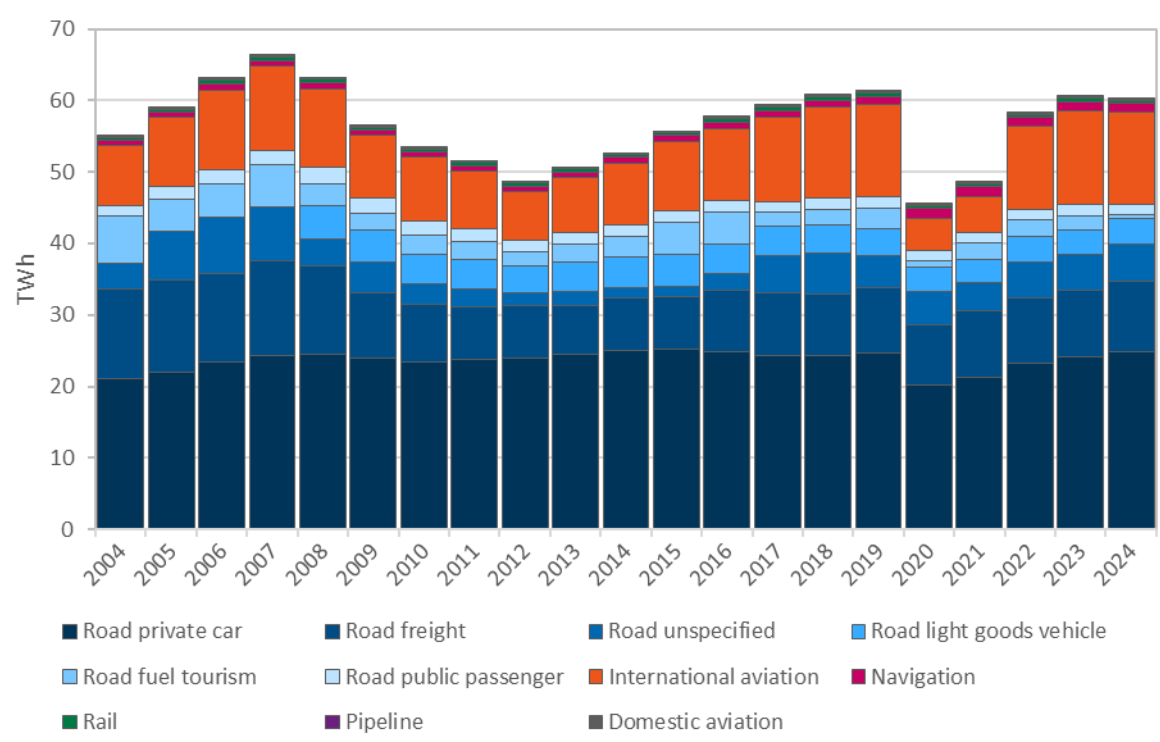


Table 6.4: Final energy in transport mode by subsector (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Road private car	25.08 (47.7%)	25.19 (45.3%)	24.81 (43.0%)	24.35 (41.0%)	24.34 (40.0%)	24.60 (40.1%)	20.11 (44.3%)	21.33 (43.9%)	23.20 (39.8%)	24.08 (39.8%)	24.77 (41.1%)
Road freight	7.23 (13.8%)	7.28 (13.1%)	8.55 (14.8%)	8.70 (14.7%)	8.53 (14.0%)	9.18 (15.0%)	8.43 (18.5%)	9.25 (19.0%)	9.19 (15.7%)	9.28 (15.3%)	10.01 (16.6%)
Road unspecified	1.52 (2.9%)	1.55 (2.8%)	2.41 (4.2%)	5.31 (9.0%)	5.68 (9.3%)	4.51 (7.4%)	4.68 (10.3%)	3.88 (8.0%)	5.07 (8.7%)	5.08 (8.4%)	5.07 (8.4%)
Road light goods vehicle	4.33 (8.2%)	4.39 (7.9%)	4.18 (7.2%)	4.10 (6.9%)	3.97 (6.5%)	3.82 (6.2%)	3.45 (7.6%)	3.20 (6.6%)	3.51 (6.0%)	3.45 (5.7%)	3.55 (5.9%)
Road fuel tourism	2.82 (5.4%)	4.51 (8.1%)	4.47 (7.7%)	1.92 (3.2%)	2.17 (3.6%)	2.89 (4.7%)	0.94 (2.1%)	2.41 (5.0%)	2.29 (3.9%)	2.03 (3.4%)	0.59 (1.0%)
Road public passenger	1.58 (3.0%)	1.55 (2.8%)	1.54 (2.7%)	1.51 (2.5%)	1.59 (2.6%)	1.59 (2.6%)	1.38 (3.0%)	1.40 (2.9%)	1.46 (2.5%)	1.48 (2.4%)	1.45 (2.4%)
International aviation	8.65 (16.5%)	9.79 (17.6%)	10.04 (17.4%)	11.82 (19.9%)	12.77 (21.0%)	12.91 (21.1%)	4.58 (10.1%)	5.12 (10.5%)	11.76 (20.2%)	13.18 (21.8%)	12.94 (21.4%)
Navigation	0.84 (1.6%)	0.83 (1.5%)	1.00 (1.7%)	0.88 (1.5%)	0.98 (1.6%)	1.04 (1.7%)	1.27 (2.8%)	1.36 (2.8%)	1.16 (2.0%)	1.22 (2.0%)	1.14 (1.9%)
Rail	0.41 (0.8%)	0.42 (0.7%)	0.42 (0.7%)	0.44 (0.7%)	0.44 (0.7%)	0.46 (0.8%)	0.37 (0.8%)	0.40 (0.8%)	0.45 (0.8%)	0.49 (0.8%)	0.52 (0.9%)
Pipeline	0.03 (0.1%)	0.05 (0.1%)	0.25 (0.4%)	0.24 (0.4%)	0.26 (0.4%)	0.20 (0.3%)	0.18 (0.4%)	0.18 (0.4%)	0.19 (0.3%)	0.17 (0.3%)	0.19 (0.3%)
Domestic aviation	0.06 (0.1%)	0.06 (0.1%)	0.07 (0.1%)	0.07 (0.1%)	0.06 (0.1%)	0.07 (0.1%)	0.05 (0.1%)	0.08 (0.2%)	0.08 (0.1%)	0.09 (0.1%)	0.10 (0.2%)
Total	52.54 (100%)	55.62 (100%)	57.74 (100%)	59.33 (100%)	60.80 (100%)	61.28 (100%)	45.43 (100%)	48.60 (100%)	58.34 (100%)	60.56 (100%)	60.33 (100%)

6.3.2 Final energy use in transport mode by energy type

Figure 6.8 and Table 6.5 show the final energy use in the transport mode by fuel type. To avoid double counting across the transport and electricity modes, energy provided by electricity (e.g. EVs) are counted in the electricity mode only, and so are excluded from this data.

In 2020, every single fuel type in the transport mode saw a reduction against its 2019 values, with an overall reduction in final energy use of 26% in transport. All fuels remained below pre-pandemic levels in 2021 and 2022 (with the exception of biodiesel). In 2023, biodiesel, bioethanol and jet kerosene surpassed pre-pandemic levels. Jet kerosene consumption reflects the growth in international aviation, while the increase in biodiesel and bioethanol reflect increased biofuel blending as well as the return of road transport energy consumption to near pre pandemic levels.

Petrol use fell each year from 2007 and continued into the COVID-impacted years. This was mostly driven by a sustained switch from petrol to diesel vehicles. Petrol use increased in 2021 and has continued to grow each year since, with diesel use decreasing since 2022.

Liquid biofuels accounted for 6.3% of final energy in the transport mode in 2024.

Figure 6.8: Final energy in transport mode by fuel

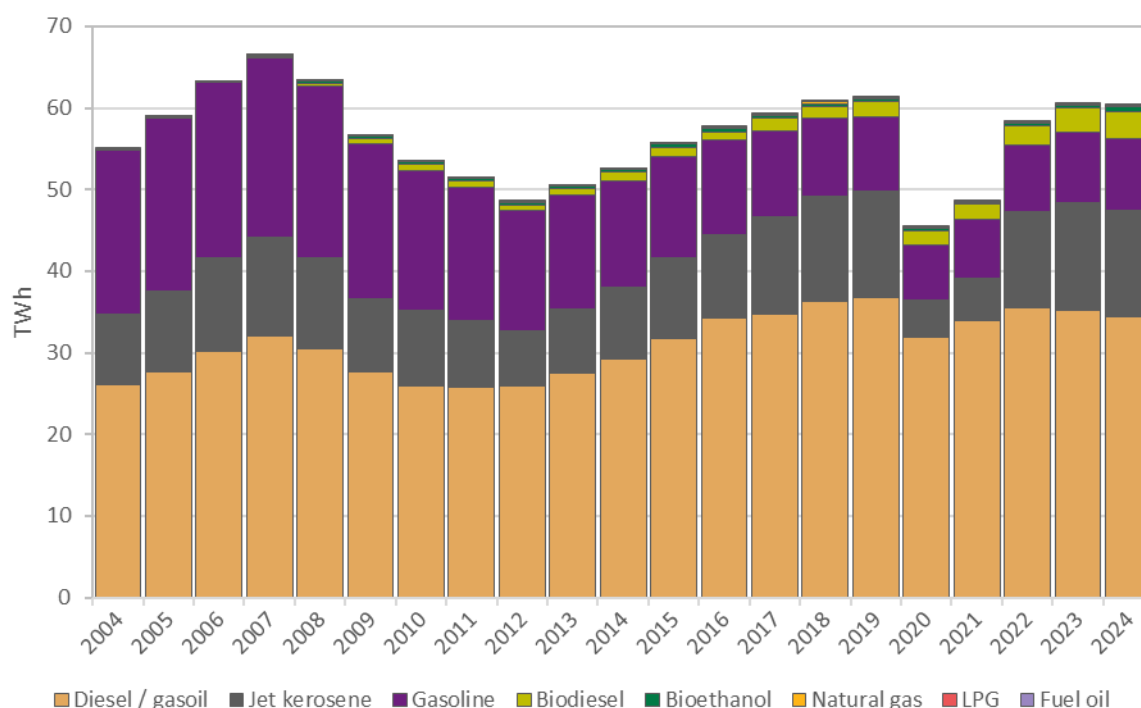


Table 6.5: Final energy in transport mode by fuel (share in brackets)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Diesel / gasoil	29.25 (55.7%)	31.72 (57.0%)	34.32 (59.4%)	34.80 (58.7%)	36.31 (59.7%)	36.80 (60.1%)	31.83 (70.1%)	33.98 (69.9%)	35.50 (60.9%)	35.16 (58.1%)	34.34 (56.9%)
Jet kerosene	8.70 (16.6%)	9.84 (17.7%)	10.10 (17.5%)	11.88 (20.0%)	12.83 (21.1%)	12.98 (21.2%)	4.63 (10.2%)	5.18 (10.7%)	11.84 (20.3%)	13.26 (21.9%)	13.03 (21.6%)
Gasoline	13.18 (25.1%)	12.50 (22.5%)	11.66 (20.2%)	10.52 (17.7%)	9.59 (15.8%)	9.08 (14.8%)	6.73 (14.8%)	7.13 (14.7%)	8.12 (13.9%)	8.65 (14.3%)	8.90 (14.8%)
Biodiesel	1.04 (2.0%)	1.14 (2.1%)	1.00 (1.7%)	1.52 (2.6%)	1.48 (2.4%)	1.90 (3.1%)	1.82 (4.0%)	1.87 (3.9%)	2.39 (4.1%)	2.89 (4.8%)	3.26 (5.4%)
Bioethanol	0.31 (0.6%)	0.35 (0.6%)	0.38 (0.7%)	0.34 (0.6%)	0.32 (0.5%)	0.30 (0.5%)	0.23 (0.5%)	0.24 (0.5%)	0.27 (0.5%)	0.38 (0.6%)	0.57 (0.9%)
Natural gas	0.03 (0.1%)	0.05 (0.1%)	0.25 (0.4%)	0.24 (0.4%)	0.26 (0.4%)	0.20 (0.3%)	0.18 (0.4%)	0.19 (0.4%)	0.20 (0.3%)	0.20 (0.3%)	0.21 (0.4%)
LPG	0.02 (0.0%)	0.03 (0.1%)	0.03 (0.1%)	0.03 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.02 (0.0%)	0.01 (0.0%)
Fuel oil	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	52.54 (100%)	55.62 (100%)	57.74 (100%)	59.33 (100%)	60.80 (100%)	61.28 (100%)	45.43 (100%)	48.60 (100%)	58.34 (100%)	60.56 (100%)	60.32 (100%)

6.4 Electricity mode

The electricity mode captures all final energy consumption of electricity, regardless of sector. The electricity mode does not include electricity consumed outside of final energy consumption: energy transformation processes (refining, pumped storage), own use in power stations or transmission/distribution losses.

6.4.1 Final energy use in electricity mode by sector

Figure 6.9 shows the trends and breakdown of final energy consumption in the electricity mode across the main sectors. 2024 was a record high for final energy demand for electricity. In order of largest to smallest, the sectoral consumers are services, residential, industry, agriculture and fisheries, and transport (barely visible in Figure 6.9).

Figure 6.9: Final consumption in electricity mode by sector

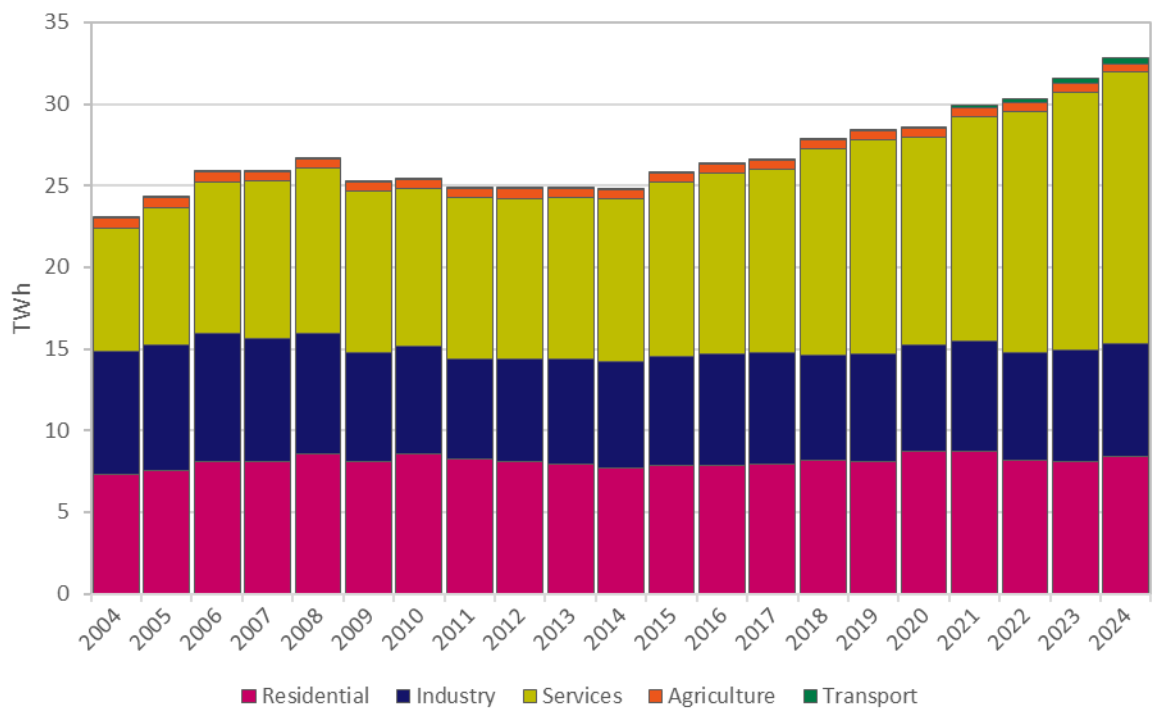


Table 6.6 provides the quantities and shares of final energy use of the electricity mode by sector over the last 11 years. Overall, the last 10 years have seen a significant increase in electricity consumption in services (+67%) and modest increases in residential (+9%) and industry (+6%). Consumption in agriculture remains low at 1.5% of the total. The percentage growth in using electricity in transport over the last 10- and 20-year period is significant, but the absolute quantity of electricity used in transport remains small. Electricity use in transport includes the DART rail system, the Luas light rail system and EVs on the road. Transport accounts for around 1.3% of consumption in the electricity mode.

Table 6.6: Final consumption in electricity mode by sector (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Services	9.97 (40.2%)	10.69 (41.4%)	11.05 (41.9%)	11.24 (42.2%)	12.62 (45.3%)	13.10 (46.1%)	12.73 (44.5%)	13.74 (45.9%)	14.77 (48.7%)	15.83 (50.2%)	16.63 (50.6%)
Residential	7.70 (31.0%)	7.88 (30.6%)	7.87 (29.8%)	7.95 (29.9%)	8.16 (29.3%)	8.11 (28.5%)	8.68 (30.3%)	8.74 (29.2%)	8.21 (27.1%)	8.10 (25.7%)	8.37 (25.5%)
Industry	6.54 (26.3%)	6.62 (25.7%)	6.85 (26.0%)	6.82 (25.6%)	6.47 (23.2%)	6.56 (23.1%)	6.54 (22.8%)	6.73 (22.5%)	6.58 (21.7%)	6.80 (21.6%)	6.93 (21.1%)
Agriculture	0.56 (2.2%)	0.56 (2.2%)	0.56 (2.1%)	0.56 (2.1%)	0.56 (2.0%)	0.55 (1.9%)	0.57 (2.0%)	0.56 (1.9%)	0.54 (1.8%)	0.49 (1.6%)	0.49 (1.5%)
Transport	0.04 (0.2%)	0.04 (0.2%)	0.05 (0.2%)	0.05 (0.2%)	0.07 (0.2%)	0.09 (0.3%)	0.10 (0.3%)	0.15 (0.5%)	0.22 (0.7%)	0.33 (1.0%)	0.43 (1.3%)
Fisheries	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Total	24.81 (100%)	25.79 (100%)	26.38 (100%)	26.63 (100%)	27.87 (100%)	28.40 (100%)	28.62 (100%)	29.92 (100%)	30.32 (100%)	31.57 (100%)	32.86 (100%)

6.5 Primary energy requirement by mode

Figure 6.10 shows the primary energy supply through the lens of the electricity, transport and heat modes. To avoid double counting, heat and transport energy provided by electricity is counted in the electricity mode only.

Figure 6.11 shows the historical trend in primary energy by mode. All three modes have a broadly similar share in primary energy. This differs from the modal split of final energy use (see section 6.1) where the electricity mode is approximately half that of heat and transport. This is because a significant amount of energy is lost in the thermal generation of electricity, and never reaches end users for final consumption. Therefore, the primary supply electricity mode is always substantially higher than the final use electricity mode. For more information on electricity generation inputs, outputs and efficiency, see section 4.2.

All three of the modes decreased during the economic downturn from 2008 to 2012. Transport grew to become the largest of the three modes from 2014 to 2019, before shrinking to become the smallest in 2020 due to the impact of the COVID-19 travel restrictions. Primary energy in transport overtook heat and electricity again since 2022

In the context of the electricity mode, increased demand for electricity was offset by increased electricity imports, continued growth in non-combustible renewables (solar PV and wind) and reduction in thermal generation, reducing the overall primary energy input to the electricity mode. Further details on electricity generation are provided in section 4.2.

Figure 6.10: Shares of primary energy by mode

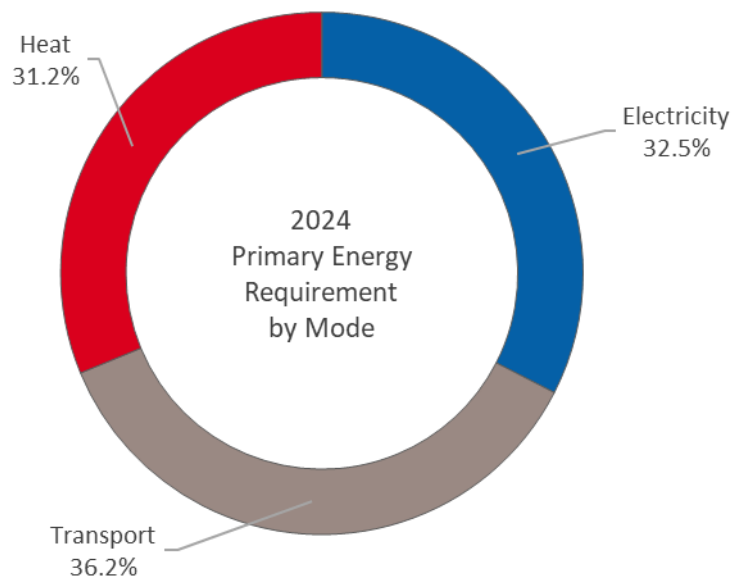
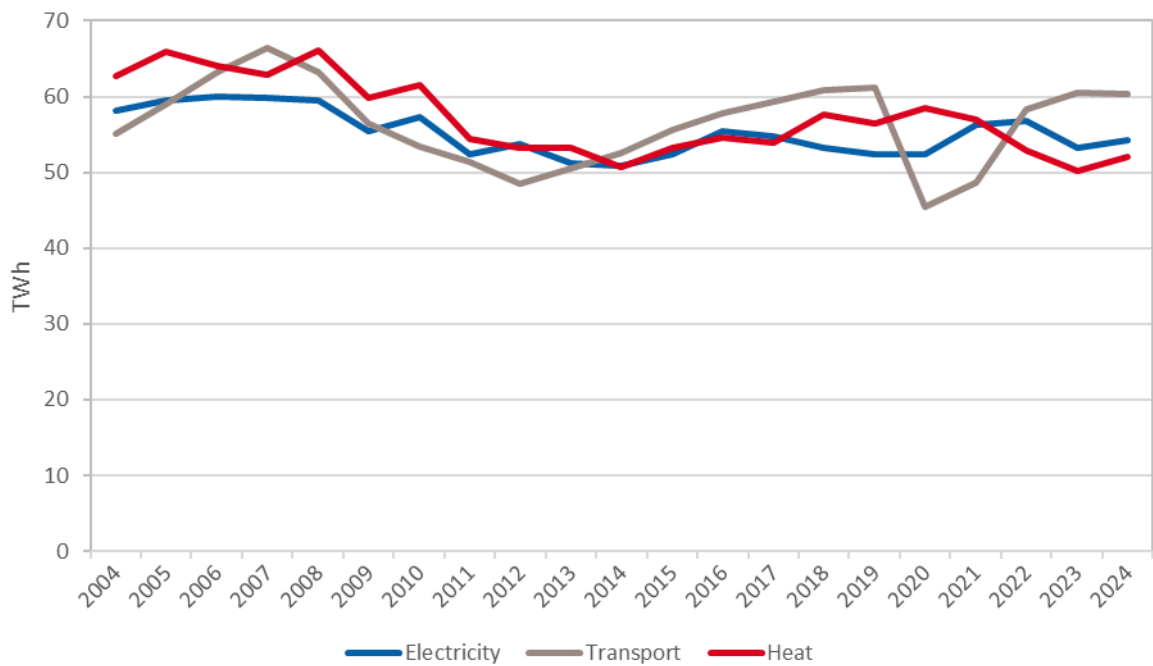


Figure 6.11: Primary energy by mode



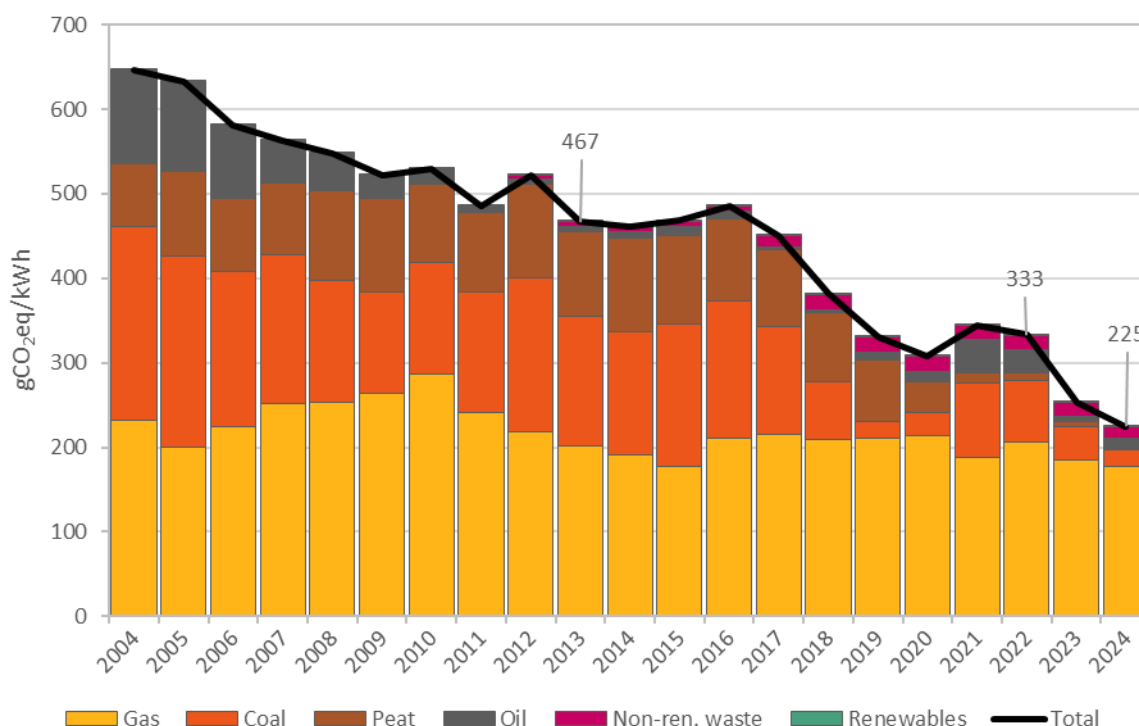
7 Energy-related greenhouse gas (GHG) emissions

In this section, GHG quantities from energy product combustion have been calculated using sector and energy product specific data from the EPA's National Inventory Submissions [3] (see section 2.3 for more details). GHG emissions associated with fugitive emissions (e.g. from natural gas production and distribution, oil refining and storage, disused coal mines (1990-present) and also from coal mining (1990-1995)) are provided separately by the EPA in Ireland's National Inventory Submissions (1990-2023) [3] and in Ireland's Provisional Greenhouse Gas Emissions (1990-2024). [16]

7.1 GHG intensity of electricity total consumption

Figure 7.1 shows the CO₂eq emission intensity of Ireland's electricity consumed, which is measured in gCO₂eq/kWh. The stacked bars show the share of GHG emissions by energy type for each kWh of electricity consumption in Ireland. It is important to note that the stacked bars in the graph represent the contributions of different energy types to the overall emission intensity of Ireland's electricity total consumption, not the emission intensity of the individual energy types themselves.

Figure 7.1: GHG emissions per kWh of electricity total consumption, with contribution by energy type



The CO₂eq intensity of electricity consumption fell to a historic low in 2020, before increasing slightly in 2021 due to an increase in emissions from coal and, to a lesser extent, oil. The CO₂eq intensity fell again in 2022, 2023 and 2024 due to a decrease the share of oil and coal in the generation mix. The dramatic overall improvements in annual CO₂eq emission intensity, as seen in Figure 7.1, are due to reductions in using coal for electricity generation, increased generation from renewable sources and increased net imports in 2023 and 2024. Coal input to electricity generation ended in June 2025.

Over the longer term, there has been a shift away from coal and oil, two of the fuels with the highest CO₂eq intensity. These fuels have been replaced by a combination of high efficiency gas combined cycle gas turbine (CCGT) generation and renewables. Imported electricity is also considered as zero carbon in the context of national emissions, as emissions are counted in the jurisdiction in which they are emitted.

7.2 Energy-related GHG emissions by sector

This subsection provides a breakdown of energy-related GHG emissions by sector of final energy consumption in the National Energy Balance. In addition to presenting the direct GHG emissions from combustion in each sector, GHG emissions associated with electricity consumption in that sector are presented (these include emissions associated with electricity transmission and distribution losses). Consequently, the resulting sectoral GHG emission totals provide an indication of the total energy related emissions that can be attributed to activity in each energy-use sector.

The approach used in this subsection differs to that used in the National GHG Inventory (along with the carbon budgets and sectoral emissions ceilings), where sectoral totals are based on emissions directly arising in each sector. Under the sectoral emissions ceiling, emissions arising from combustion of fuel in public electricity production are counted in under in the *electricity* sector regardless of where the electricity itself is consumed. Fugitive emissions captured under the electricity CAP sector are also not included in this section.

Figure 7.2, Figure 7.3 and Table 7.1 show the most recent and historical energy-related CO₂eq emissions split by sector. In this case, the emissions from electricity generation are shown separately from the emissions from direct fossil fuel use in the end-use sectors.

International aviation is also shown separately. It is included in the National Energy Balance (in line with international practice) and so is included in the figures for transport energy use in this report. However, it is not included in the National Greenhouse Gas Inventory (in line with international practice) and is also not included in Ireland's national greenhouse gas emissions reduction targets for 2030 and 2050.

Excluding international aviation, energy-related CO₂eq emissions reached a recent low point in 2020 before increasing slightly in 2021 and decreasing again in 2022, 2023 and 2024. Transport (excluding international aviation) provides the largest share of energy-related CO₂eq emissions, followed by electricity generation, the residential sector, industry, international aviation, services, agriculture, other and fisheries, in that order. The 'other' category includes emissions from oil refineries, natural gas refining and peat briquetting.

Figure 7.2: Current sectoral shares of energy-related CO₂eq

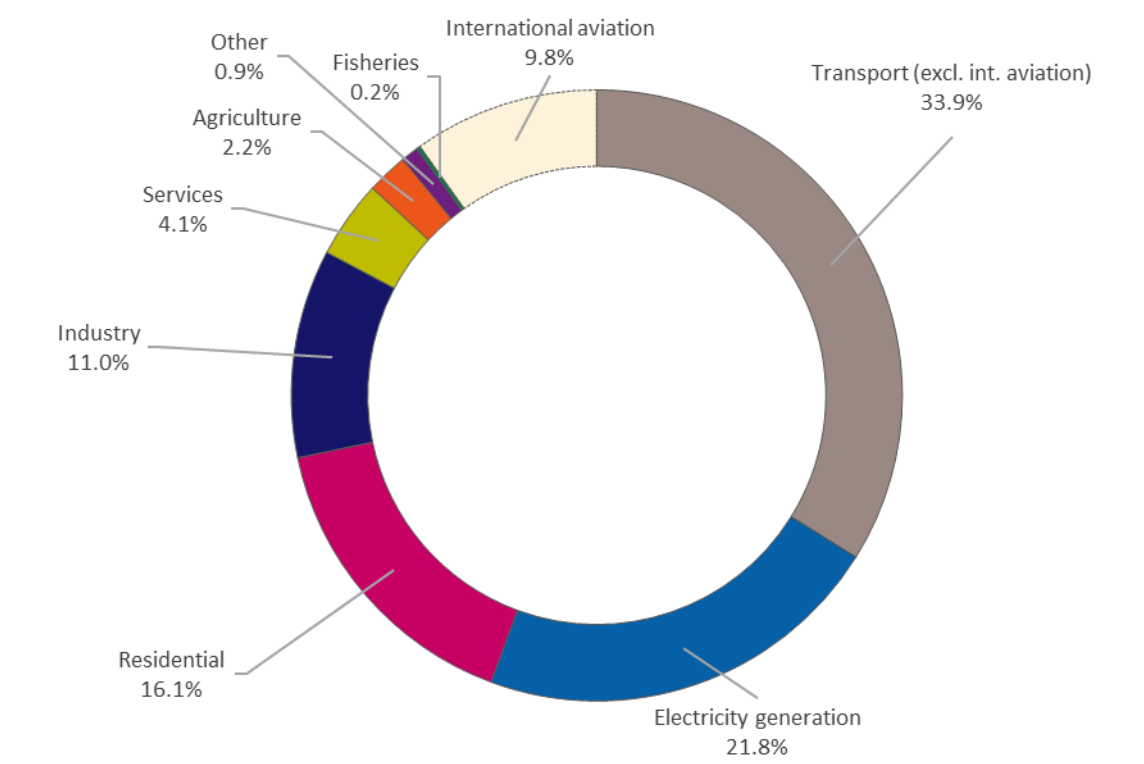
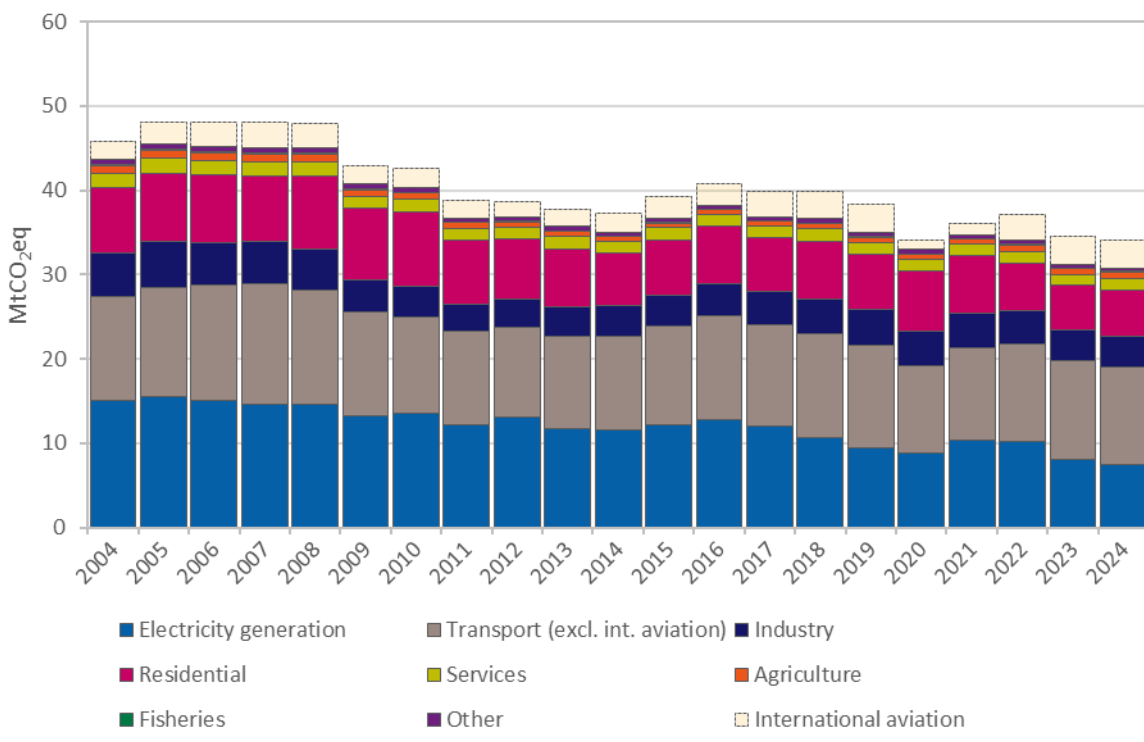


Figure 7.3: Energy-related annual CO₂eq emissions by sector¹⁹



¹⁹ Emissions for agriculture, as with all sectors, shown in the chart and tables are for energy-related emissions only.

Table 7.1: Energy-related CO₂eq by sector (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Transport (excl. int. aviation)	11.20 (30.1%)	11.69 (29.8%)	12.21 (29.9%)	12.05 (30.2%)	12.22 (30.6%)	12.22 (31.9%)	10.29 (30.1%)	10.97 (30.4%)	11.64 (31.4%)	11.69 (33.8%)	11.54 (33.9%)
Electricity generation	11.53 (30.9%)	12.13 (30.9%)	12.86 (31.5%)	12.04 (30.2%)	10.69 (26.8%)	9.45 (24.7%)	8.85 (25.9%)	10.35 (28.7%)	10.14 (27.3%)	8.02 (23.2%)	7.41 (21.8%)
Residential	6.25 (16.8%)	6.64 (16.9%)	6.89 (16.9%)	6.33 (15.9%)	6.82 (17.1%)	6.55 (17.1%)	7.19 (21.1%)	6.71 (18.6%)	5.62 (15.1%)	5.23 (15.1%)	5.48 (16.1%)
Industry	3.59 (9.6%)	3.65 (9.3%)	3.77 (9.2%)	3.97 (10.0%)	4.18 (10.5%)	4.14 (10.8%)	4.13 (12.1%)	4.17 (11.6%)	3.94 (10.6%)	3.75 (10.8%)	3.76 (11.0%)
Services	1.41 (3.8%)	1.50 (3.8%)	1.44 (3.5%)	1.37 (3.4%)	1.50 (3.8%)	1.48 (3.9%)	1.29 (3.8%)	1.39 (3.8%)	1.34 (3.6%)	1.30 (3.8%)	1.40 (4.1%)
Agriculture	0.53 (1.4%)	0.51 (1.3%)	0.54 (1.3%)	0.55 (1.4%)	0.59 (1.5%)	0.61 (1.6%)	0.65 (1.9%)	0.64 (1.8%)	0.85 (2.3%)	0.72 (2.1%)	0.74 (2.2%)
Other	0.44 (1.2%)	0.52 (1.3%)	0.42 (1.0%)	0.46 (1.2%)	0.52 (1.3%)	0.46 (1.2%)	0.48 (1.4%)	0.47 (1.3%)	0.47 (1.3%)	0.42 (1.2%)	0.32 (0.9%)
Fisheries	0.07 (0.2%)	0.07 (0.2%)	0.06 (0.1%)	0.07 (0.2%)	0.08 (0.2%)	0.07 (0.2%)	0.06 (0.2%)	0.06 (0.2%)	0.05 (0.1%)	0.06 (0.2%)	0.07 (0.2%)
Total (excl. int. aviation)	35.01 (94.0%)	36.71 (93.5%)	38.18 (93.6%)	36.86 (92.3%)	36.61 (91.7%)	34.97 (91.3%)	32.95 (96.5%)	34.75 (96.3%)	34.06 (91.8%)	31.20 (90.1%)	30.71 (90.2%)
International aviation	2.24 (6.0%)	2.54 (6.5%)	2.60 (6.4%)	3.06 (7.7%)	3.31 (8.3%)	3.34 (8.7%)	1.19 (3.5%)	1.32 (3.7%)	3.04 (8.2%)	3.41 (9.9%)	3.35 (9.8%)
Total (incl. int. aviation)	37.25 (100%)	39.24 (100%)	40.78 (100%)	39.92 (100%)	39.91 (100%)	38.31 (100%)	34.13 (100%)	36.07 (100%)	37.10 (100%)	34.61 (100%)	34.06 (100%)

7.2.1 Industry emissions

In recent years, approximately 61% of greenhouse gas emissions from industry have been energy related. The total sectoral energy-related CO₂eq emissions as shown here include estimations of upstream emissions for the electricity consumed by industry. Figure 7.4 shows the energy-related CO₂eq emissions from industry, detailing the on-site CO₂eq emissions associated with direct fuel combustion and the upstream emissions associated with electricity consumption. Table 7.2 shows the quantities and relative shares of energy-related CO₂eq emissions in industry by energy product.

Figure 7.4: Industry energy-related CO₂eq emissions by energy type

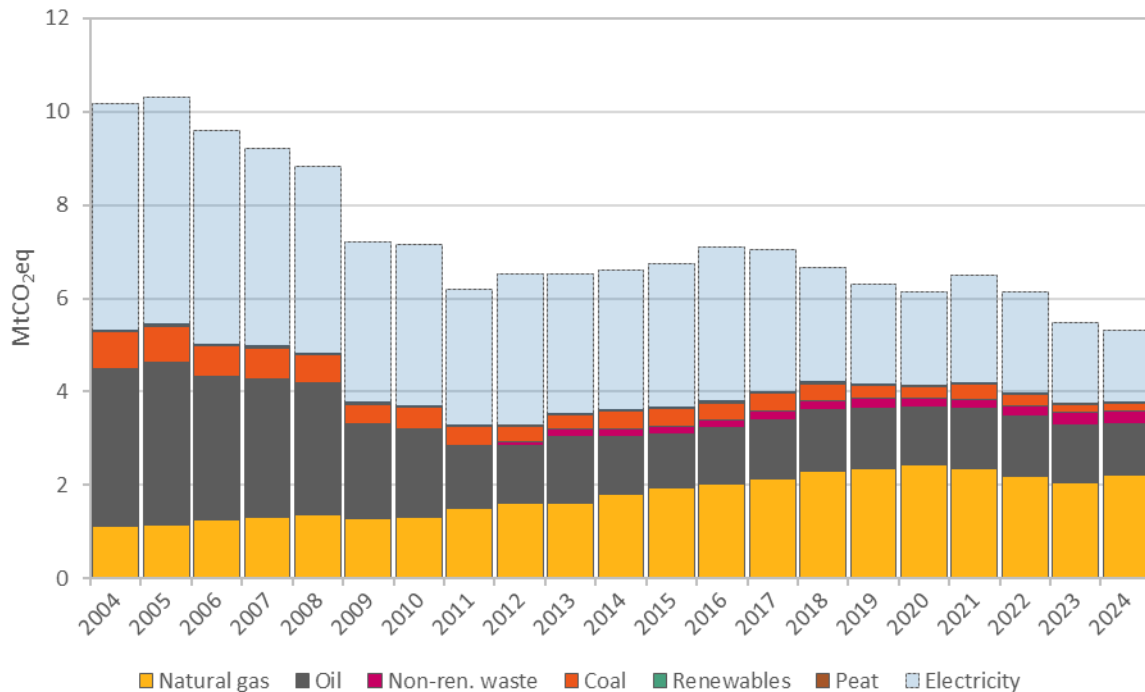


Table 7.2: Quantities and shares of energy-related CO₂eq emissions in industry (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Natural gas	1.80 (27.2%)	1.93 (28.6%)	2.03 (28.6%)	2.14 (30.4%)	2.29 (34.4%)	2.34 (37.1%)	2.44 (39.7%)	2.36 (36.4%)	2.18 (35.6%)	2.04 (37.4%)	2.22 (41.7%)
Electricity	3.02 (45.6%)	3.10 (45.9%)	3.33 (46.9%)	3.07 (43.6%)	2.47 (37.1%)	2.17 (34.4%)	2.01 (32.8%)	2.32 (35.7%)	2.19 (35.7%)	1.72 (31.5%)	1.56 (29.4%)
Oil	1.24 (18.7%)	1.16 (17.2%)	1.20 (16.9%)	1.26 (17.9%)	1.32 (19.9%)	1.31 (20.7%)	1.23 (20.0%)	1.28 (19.7%)	1.29 (21.0%)	1.24 (22.6%)	1.11 (20.9%)
Non-ren. waste	0.16 (2.4%)	0.18 (2.6%)	0.16 (2.3%)	0.20 (2.8%)	0.19 (2.9%)	0.20 (3.2%)	0.18 (3.0%)	0.20 (3.0%)	0.22 (3.6%)	0.28 (5.0%)	0.27 (5.0%)
Coal	0.39 (5.9%)	0.37 (5.5%)	0.37 (5.2%)	0.36 (5.1%)	0.36 (5.5%)	0.27 (4.4%)	0.27 (4.4%)	0.32 (4.9%)	0.24 (3.9%)	0.18 (3.3%)	0.15 (2.8%)
Renewables	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)	0.01 (0.2%)
Peat	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.1%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Total	6.61 (100%)	6.75 (100%)	7.10 (100%)	7.04 (100%)	6.65 (100%)	6.31 (100%)	6.14 (100%)	6.49 (100%)	6.13 (100%)	5.47 (100%)	5.32 (100%)

7.2.2 Transport emissions

Figure 7.5 and Table 7.3 show the energy-related CO₂eq emissions from transport, detailing emissions associated with direct fuel combustion and the upstream emissions associated with electricity consumption. Table 7.4 provides an emissions breakdown by subsector (includes direct and indirect emissions).

Figure 7.5: Transport energy-related emissions by energy type (including international aviation)

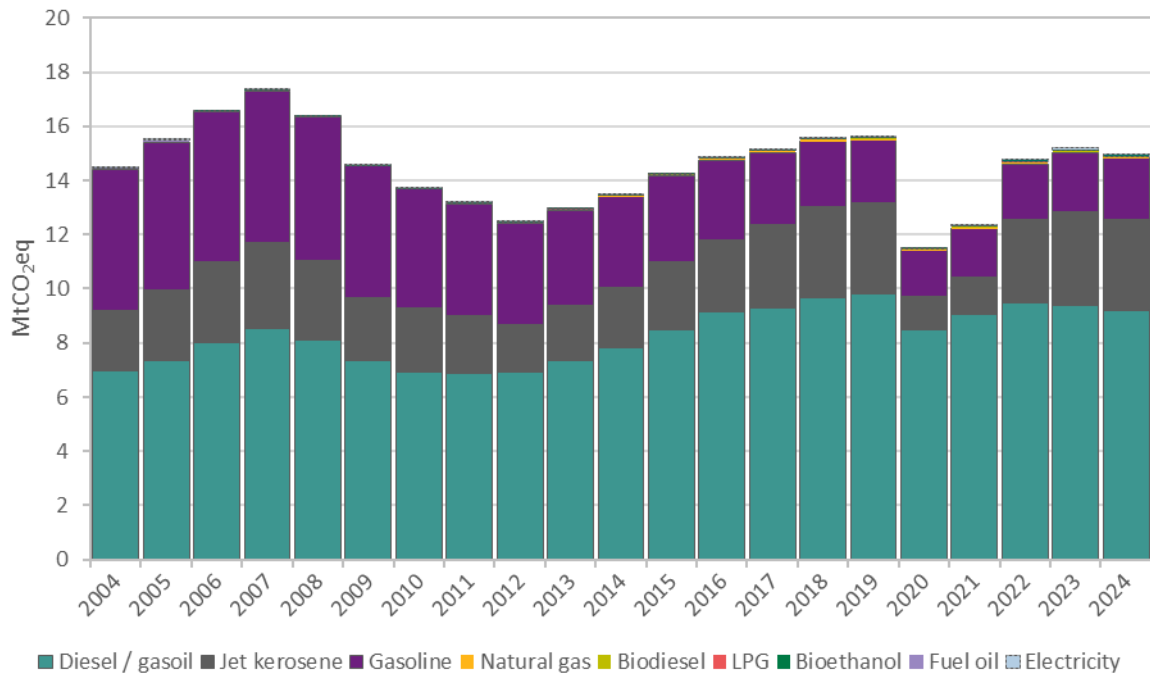


Table 7.3: Quantities and shares of energy-related CO₂eq emissions in transport (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Diesel / gasoil	7.80 (58.0%)	8.46 (59.4%)	9.16 (61.7%)	9.29 (61.4%)	9.69 (62.3%)	9.82 (63.0%)	8.50 (73.9%)	9.07 (73.5%)	9.48 (64.2%)	9.39 (61.9%)	9.18 (61.2%)
Jet kerosene	2.25 (16.7%)	2.55 (17.9%)	2.61 (17.6%)	3.07 (20.3%)	3.32 (21.4%)	3.36 (21.5%)	1.20 (10.4%)	1.34 (10.9%)	3.06 (20.8%)	3.43 (22.6%)	3.37 (22.5%)
Gasoline	3.35 (24.9%)	3.17 (22.3%)	2.96 (20.0%)	2.67 (17.6%)	2.43 (15.6%)	2.30 (14.8%)	1.70 (14.8%)	1.81 (14.6%)	2.06 (13.9%)	2.19 (14.4%)	2.26 (15.1%)
Electricity	0.02 (0.1%)	0.02 (0.1%)	0.02 (0.2%)	0.02 (0.2%)	0.03 (0.2%)	0.03 (0.2%)	0.03 (0.3%)	0.05 (0.4%)	0.07 (0.5%)	0.08 (0.6%)	0.10 (0.6%)
Natural gas	0.01 (0.1%)	0.01 (0.1%)	0.05 (0.3%)	0.05 (0.3%)	0.05 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)
Biodiesel	0.02 (0.1%)	0.02 (0.1%)	0.02 (0.1%)	0.03 (0.2%)	0.03 (0.2%)	0.03 (0.2%)	0.03 (0.3%)	0.03 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)
LPG	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Bioethanol	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Fuel oil	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Total	13.46 (100%)	14.24 (100%)	14.84 (100%)	15.14 (100%)	15.55 (100%)	15.59 (100%)	11.51 (100%)	12.35 (100%)	14.76 (100%)	15.19 (100%)	14.99 (100%)

Table 7.4: Quantities and shares of transport CO₂eq emissions by subsector, including GHG emissions from direct combustion and indirect GHG emissions from electricity use (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Road private car	6.35 (47.2%)	6.37 (44.7%)	6.30 (42.4%)	6.14 (40.6%)	6.16 (39.6%)	6.19 (39.7%)	5.05 (43.9%)	5.38 (43.6%)	5.82 (39.4%)	5.98 (39.4%)	6.07 (40.5%)
Road freight	1.86 (13.8%)	1.87 (13.2%)	2.22 (14.9%)	2.22 (14.7%)	2.19 (14.1%)	2.33 (15.0%)	2.13 (18.5%)	2.34 (19.0%)	2.30 (15.6%)	2.29 (15.1%)	2.44 (16.3%)
Road unspecified	0.38 (2.8%)	0.39 (2.7%)	0.60 (4.1%)	1.34 (8.9%)	1.44 (9.3%)	1.14 (7.3%)	1.17 (10.2%)	0.97 (7.9%)	1.27 (8.6%)	1.25 (8.2%)	1.23 (8.2%)
Road light goods vehicle	1.12 (8.3%)	1.13 (7.9%)	1.08 (7.3%)	1.05 (6.9%)	1.02 (6.5%)	0.97 (6.2%)	0.87 (7.6%)	0.81 (6.6%)	0.88 (6.0%)	0.85 (5.6%)	0.87 (5.8%)
Road fuel tourism	0.72 (5.4%)	1.16 (8.1%)	1.16 (7.8%)	0.49 (3.2%)	0.56 (3.6%)	0.73 (4.7%)	0.24 (2.1%)	0.61 (4.9%)	0.57 (3.9%)	0.50 (3.3%)	0.14 (1.0%)
Road public passenger services	0.40 (3.0%)	0.40 (2.8%)	0.40 (2.7%)	0.38 (2.5%)	0.41 (2.6%)	0.40 (2.6%)	0.35 (3.0%)	0.36 (2.9%)	0.37 (2.5%)	0.37 (2.4%)	0.36 (2.4%)
Navigation	0.22 (1.7%)	0.22 (1.6%)	0.27 (1.8%)	0.24 (1.6%)	0.26 (1.7%)	0.28 (1.8%)	0.34 (2.9%)	0.36 (2.9%)	0.31 (2.1%)	0.32 (2.1%)	0.30 (2.0%)
Rail	0.14 (1.0%)	0.14 (1.0%)	0.15 (1.0%)	0.15 (1.0%)	0.15 (1.0%)	0.15 (1.0%)	0.12 (1.1%)	0.13 (1.1%)	0.15 (1.0%)	0.15 (1.0%)	0.16 (1.1%)
Pipeline	0.01 (0.0%)	0.01 (0.1%)	0.05 (0.3%)	0.05 (0.3%)	0.05 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.3%)	0.04 (0.2%)	0.04 (0.3%)
Domestic aviation	0.01 (0.1%)	0.02 (0.1%)	0.02 (0.1%)	0.02 (0.1%)	0.02 (0.1%)	0.02 (0.1%)	0.01 (0.1%)	0.02 (0.2%)	0.02 (0.1%)	0.02 (0.2%)	0.03 (0.2%)
Total (excl. int. aviation)	11.22 (83.4%)	11.71 (82.2%)	12.23 (82.5%)	12.08 (79.8%)	12.25 (78.7%)	12.25 (78.6%)	10.32 (89.7%)	11.02 (89.3%)	11.72 (79.4%)	11.77 (77.5%)	11.64 (77.6%)
International aviation	2.24 (16.6%)	2.54 (17.8%)	2.60 (17.5%)	3.06 (20.2%)	3.31 (21.3%)	3.34 (21.4%)	1.19 (10.3%)	1.32 (10.7%)	3.04 (20.6%)	3.41 (22.5%)	3.35 (22.4%)
Total (incl. int. aviation)	13.46 (100%)	14.24 (100%)	14.84 (100%)	15.14 (100%)	15.55 (100%)	15.59 (100%)	11.51 (100%)	12.35 (100%)	14.76 (100%)	15.19 (100%)	14.99 (100%)

7.2.3 Residential emissions

Figure 7.6 shows the energy-related CO₂eq emissions from residential, detailing emissions associated with direct energy product combustion and the upstream emissions associated with electricity consumption. Recently, emissions reduced year-on-year between 2021-2023, reflecting the decrease in consumption in the sector due to milder weather and high energy prices. Residential sector energy-related emissions, returned to growth in 2024.

Figure 7.6: Residential energy-related CO₂eq by energy type

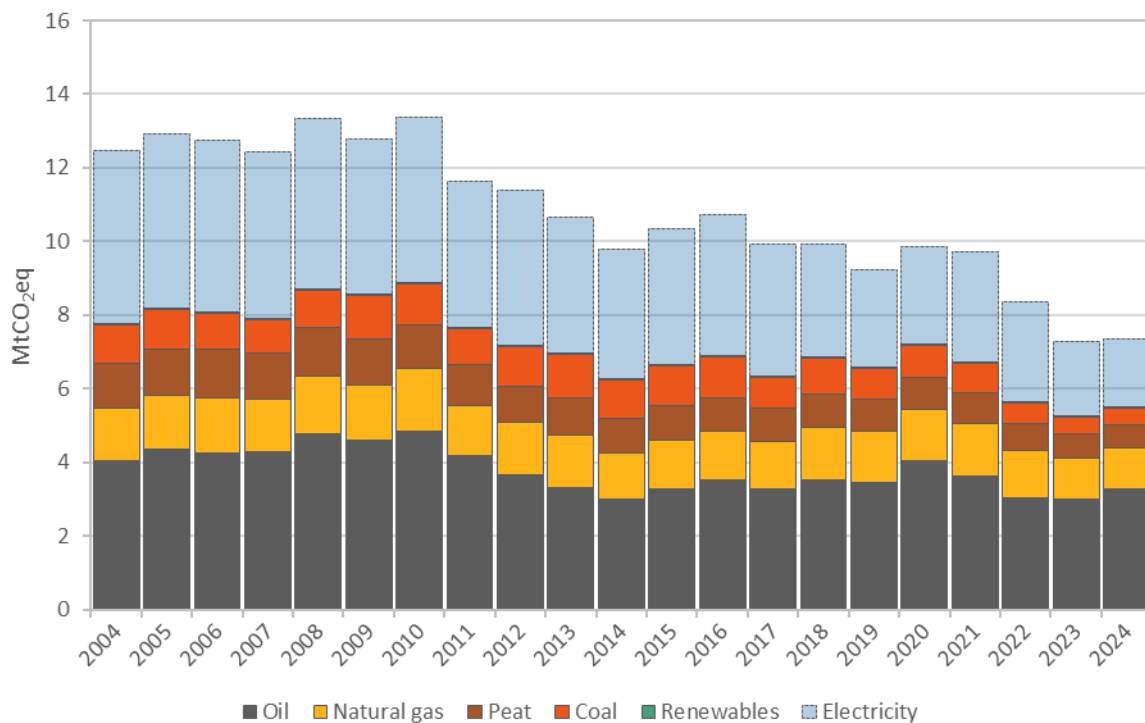


Table 7.5: Quantities and shares of energy-related CO₂eq emissions in the residential sector (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Oil	2.99 (30.5%)	3.28 (31.8%)	3.51 (32.8%)	3.28 (33.1%)	3.53 (35.5%)	3.47 (37.6%)	4.05 (41.0%)	3.63 (37.3%)	3.02 (36.2%)	3.01 (41.4%)	3.26 (44.3%)
Electricity	3.55 (36.3%)	3.69 (35.7%)	3.82 (35.7%)	3.58 (36.1%)	3.12 (31.4%)	2.68 (29.1%)	2.67 (27.1%)	3.01 (31.0%)	2.74 (32.7%)	2.05 (28.2%)	1.88 (25.6%)
Natural gas	1.28 (13.0%)	1.33 (12.8%)	1.32 (12.3%)	1.30 (13.1%)	1.42 (14.2%)	1.39 (15.1%)	1.39 (14.1%)	1.41 (14.5%)	1.28 (15.4%)	1.11 (15.2%)	1.14 (15.5%)
Peat	0.93 (9.5%)	0.93 (9.0%)	0.91 (8.5%)	0.88 (8.8%)	0.91 (9.2%)	0.85 (9.3%)	0.88 (8.9%)	0.84 (8.6%)	0.75 (9.0%)	0.66 (9.0%)	0.60 (8.2%)
Coal	1.04 (10.6%)	1.09 (10.5%)	1.13 (10.6%)	0.86 (8.7%)	0.96 (9.6%)	0.82 (8.9%)	0.86 (8.7%)	0.82 (8.4%)	0.55 (6.6%)	0.44 (6.1%)	0.47 (6.4%)
Renewables	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.2%)
Non-ren. waste	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Total	9.80 (100%)	10.33 (100%)	10.71 (100%)	9.91 (100%)	9.94 (100%)	9.23 (100%)	9.86 (100%)	9.72 (100%)	8.36 (100%)	7.28 (100%)	7.37 (100%)

7.2.4 Commercial and public services emissions

Figure 7.7 shows the primary energy-related CO₂eq emissions of the services sector, distinguishing between the on-site CO₂eq emissions associated with direct energy product combustion and the upstream emissions associated with electricity consumption.

Figure 7.7: Commercial and public services sector CO₂eq emissions by energy type

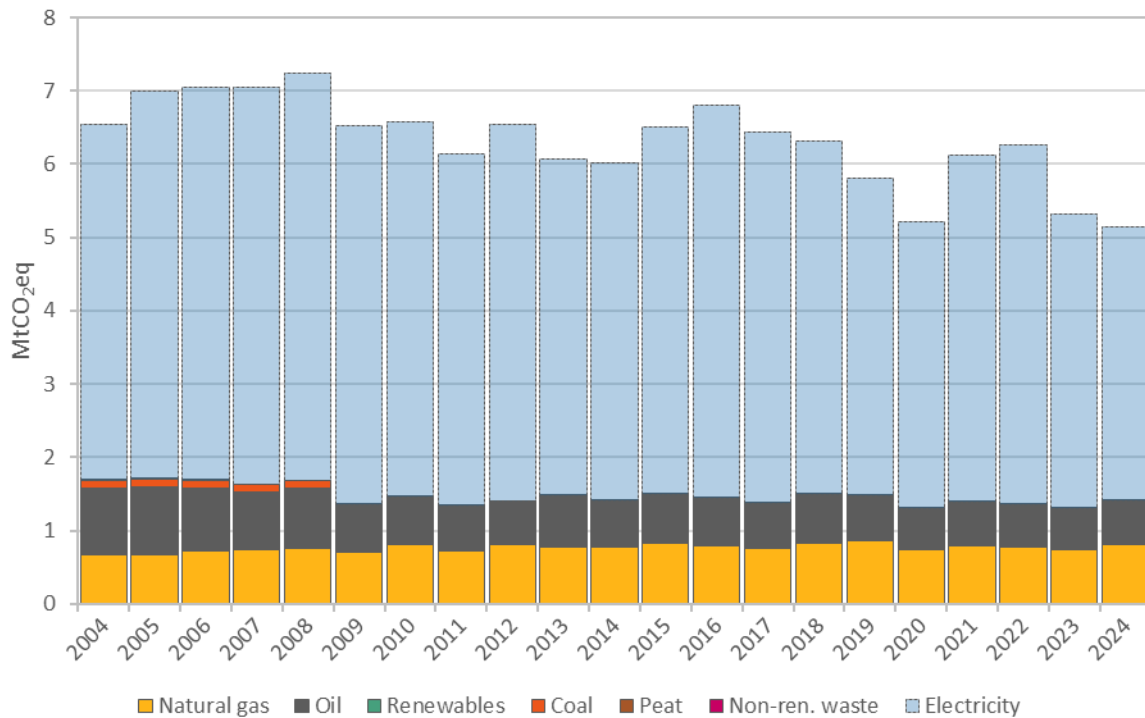


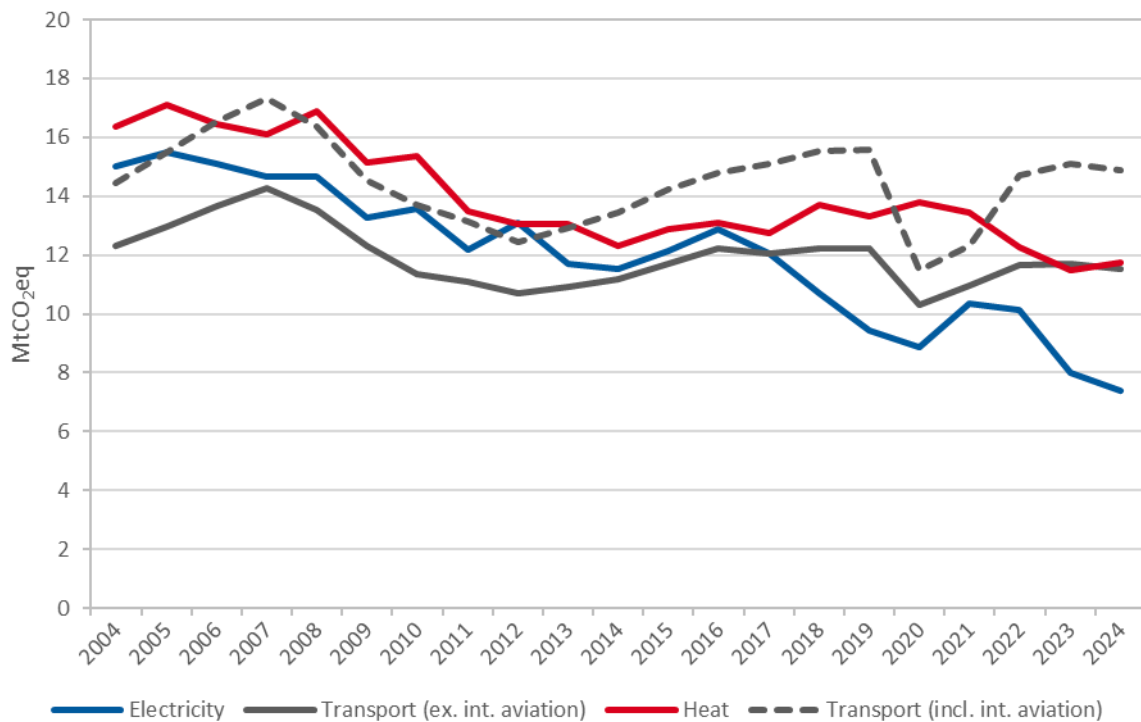
Table 7.6: Quantities and shares of CO₂eq emissions in commercial and public services (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Electricity	4.60 (76.5%)	5.00 (77.0%)	5.36 (78.8%)	5.06 (78.7%)	4.82 (76.3%)	4.34 (74.6%)	3.92 (75.2%)	4.74 (77.4%)	4.93 (78.6%)	4.01 (75.5%)	3.74 (72.7%)
Natural gas	0.77 (12.7%)	0.82 (12.7%)	0.79 (11.7%)	0.76 (11.8%)	0.83 (13.1%)	0.85 (14.7%)	0.74 (14.2%)	0.79 (13.0%)	0.78 (12.5%)	0.74 (14.0%)	0.81 (15.7%)
Oil	0.63 (10.5%)	0.67 (10.3%)	0.63 (9.3%)	0.60 (9.3%)	0.66 (10.5%)	0.61 (10.6%)	0.54 (10.4%)	0.58 (9.5%)	0.55 (8.8%)	0.55 (10.4%)	0.59 (11.4%)
Renewables	0.01 (0.2%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.1%)	0.01 (0.2%)
Coal	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Peat	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Non-ren. waste	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)
Total	6.01 (100%)	6.50 (100%)	6.80 (100%)	6.43 (100%)	6.31 (100%)	5.81 (100%)	5.21 (100%)	6.12 (100%)	6.27 (100%)	5.32 (100%)	5.15 (100%)

7.3 Energy-related GHG emissions by mode

Figure 7.8 shows energy-related CO₂eq emissions, divided into the three modes of electricity, transport and heat. This aggregates the direct energy-related emissions from industry, residential, commercial services, public services, agriculture and fisheries, along with emissions from oil refining, peat briquetting and natural gas production as shown in Figure 7.2, Figure 7.3 and Table 7.1, together as heat. The emissions for transport including international aviation, are shown for reference.

Figure 7.8: Energy-related CO₂eq emissions by electricity, transport and heat



Energy-related CO₂eq emissions in all three modes declined after 2007 during the recession, but transport returned to growth after 2012 with heat and electricity returning to growth after 2014.

From 2016, there was a dramatic reduction in CO₂eq emissions from electricity generation, due to the reduction in coal and peat (the most carbon-intensive fossil fuels) and an increase in electricity from renewable and zero-carbon sources. CO₂eq emissions from electricity generation reached a minimum in 2020, before increasing again in 2021. In 2022, 2023, and 2024 CO₂eq emissions from electricity generation decreased again, reaching another minimum in 2024.

In 2020, there was a dramatic reduction in CO₂eq emissions from transport due to travel restrictions imposed during the COVID-19 pandemic, with emissions from international aviation alone dropping 65%. Emissions from transport subsequently increased in 2021, almost returning to pre-pandemic levels in 2022. In contrast, CO₂eq emissions from heat increased slightly during 2020, due in part to more time spent at home during COVID-19 restrictions, before decreasing again in 2021, 2022 and 2023, due the end of restrictions, milder weather and high fuel prices. Heat energy-related emissions, returned to growth in 2024.

7.4 Avoided emissions from renewable energy

Using renewable energy displaces the use of fossil fuels, thereby avoiding greenhouse gas (GHG) emissions. This subsection presents an estimate of the GHG emissions avoided by use of renewable energy in Ireland.

The net emission of CO₂ from the combustion of biogenic carbon in sustainable biomass fuels (including solid biomass, biofuel, bioliquids and biogas) is counted as zero (in line with international practice), as it is assumed that the biomass itself absorbs an equal amount of CO₂ during growth from the fast domain carbon cycle as is emitted during combustion [12]. In contrast, fossil fuel extraction and combustion results in the transfer of carbon from the slow domain carbon cycle to the fast domain carbon cycle, resulting in greater quantities of carbon in our current atmosphere. While biomass fuels produce some GHGs during combustion, their net contribution to the overall quantity of GHGs in the current atmosphere is much less than that from the combustion of fossil fuels on a per unit energy basis when considering CO₂, CH₄ and N₂O together.

SEAI estimate the annual amount of GHG avoided by each renewable energy type based on the fossil fuels that are displaced. This analysis requires an estimate of the types of fossil fuels displaced, as well as the quantities, by each renewable energy type for each year from 1990 to 2024. This analysis is conducted for each of the modes (electricity, heat and transport) and the results are presented on this basis. Avoided emissions associated with electricity generated from renewables are included within the electricity mode.

The following simplifying assumptions are used in calculating the avoided emissions in the *electricity mode*:

- Renewables displace marginal fossil fuel generators that would otherwise have been required to produce the electricity – specifically natural gas, gas oil (distillate) and fuel oil.
- The efficiency and carbon intensity of the displaced marginal generation is equal to that of the public thermal power plants using gas oil, fuel oil or natural gas in the specific year. Consequently, the efficiency and carbon intensity of the displaced generation varies each year in line with the historical trend seen in public power generation from 1990 to present.
- Renewables do not displace electricity imports.
- Wind generation increases emissions from fossil fuel generation due to the effects of increased ramping and cycling. This is estimated using a factor, based on previous detailed electricity dispatch modelling [13, 14, 15].
- Renewables do not displace coal or peat generation. Historically, this was effectively the case as coal and peat plants operated as base load. However, in recent years peat generation has been phased out (ceasing entirely at the start of 2024) and coal generation has decreased significantly in the last ten years (ceasing at the end of June 2025). The assumption that renewables do not displace any peat or coal generation means that the estimated avoided emissions are conservatively low in recent years.

In the *heat mode*, avoided emissions are calculated for each renewable energy type in each sector, using the following assumptions:

- Renewables replace fossil fuel consumption based on the fossil fuel mix in that sector and year, except for the items listed below.
- Biomass in the *wood and wood products* subsector does not displace fossil fuel alternative(s) – this industry has traditionally used process waste and residue as a fuel.
- Renewable waste in the *other non-metallic mineral products* subsector displaces the fossil fuel mix of that subsector, which contains a higher proportion of solid fossil fuels (coal and petroleum coke) than the industry sector as a whole.
- Biodiesel displaces gas oil use in each sector.
- Avoided emissions associated with ambient heat use in heat pumps must take account of the total heat delivered by heat pumps, along with the carbon intensity of the associated electricity consumption.

In the *transport mode*, it is assumed that biofuel and biogas displace their fossil fuel equivalents. In this analysis, the renewable portion of electricity consumed in EVs or rail is not considered as avoided emissions in the transport mode – emissions avoided due to renewable generation are included in the electricity mode.

Figure 7.9 shows the trend in annual avoided GHG emissions from renewable energy in electricity, heat and transport from 2004 to present. The estimated emissions avoided through the use of renewable energy was 7.38 MtCO₂eq in 2024, more than any previous year, with:

- 5.79 MtCO₂eq avoided by use of renewables in electricity;
- 0.98 MtCO₂eq avoided by use of renewables in transport; and
- 0.61 MtCO₂eq avoided by use of renewables in heat.

Decarbonising the electricity system, combined with increased electrification of heat and transport through the use of electric vehicles (EVs) and heat pumps is an important part of the strategy for decarbonising the energy system as a whole. The use of renewable electricity ensures that switching to EVs and heat pumps results in less GHG emissions than the fossil fuel alternative. Electrification of heat and transport reduces direct fossil fuel use in the non-ETS sector, thereby contributing to meeting the non-ETS greenhouse gas emissions reduction target²⁰, see section 8.1.

Figure 7.9: Avoided emissions from renewable energy in electricity, heat and transport modes

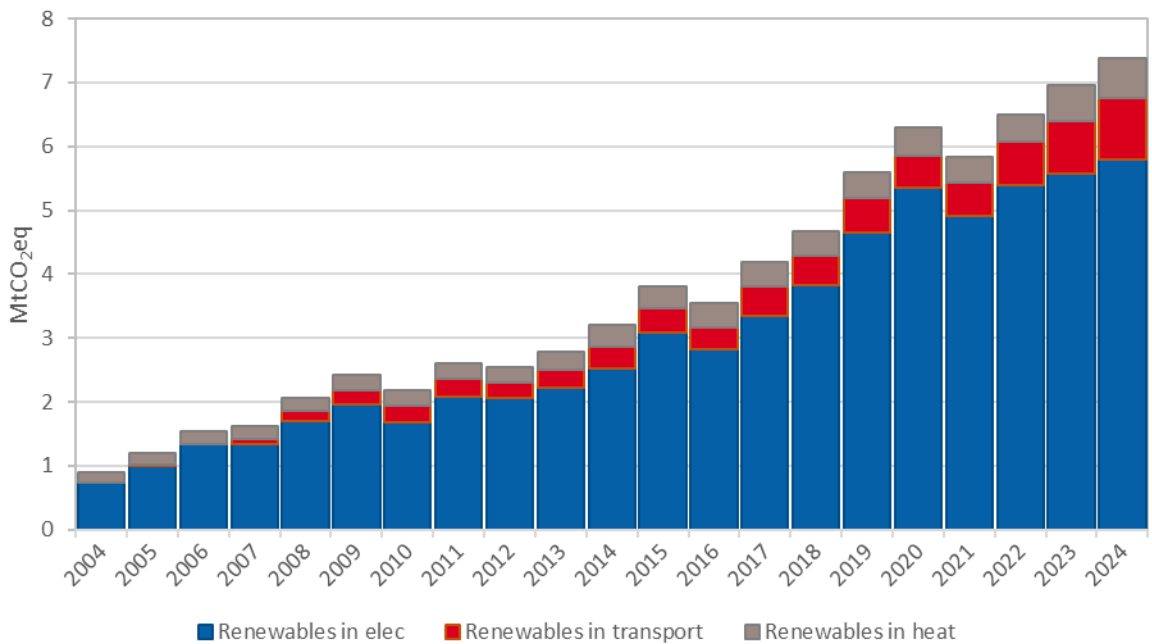
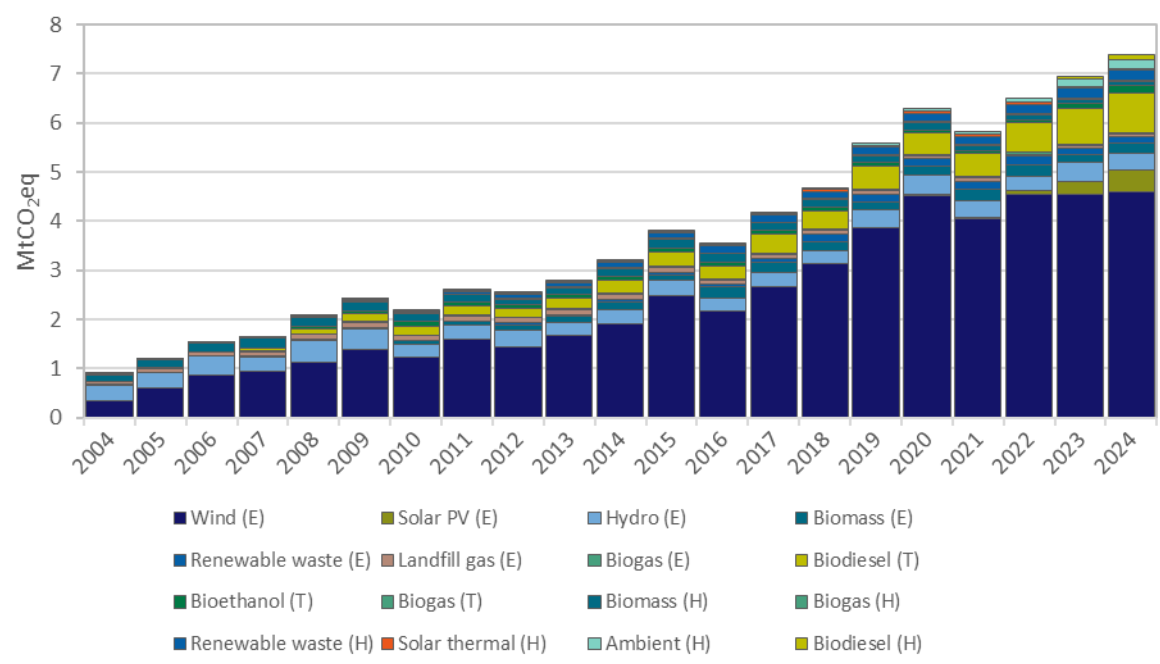


Figure 7.10 shows the breakdown in annual avoided GHG emissions by each renewable energy sub-type and mode of application. Wind accounted for 4.59 MtCO₂eq or 62.2% of the total avoided emissions in 2024. This was followed by biodiesel in transport (11.3%), solar PV in electricity (6.2%) and hydro in electricity (4.4%).

²⁰ Electricity generation is covered by the EU emissions trading system (ETS), therefore emissions savings achieved in electricity generation do not count directly towards Ireland’s EU targets to reduce greenhouse gas emissions outside of the ETS (non-ETS)

Figure 7.10: Avoid emissions by renewable energy sub-type²¹



²¹ Mode denoted in brackets: (E) for electricity, (T) for transport and (H) for heat

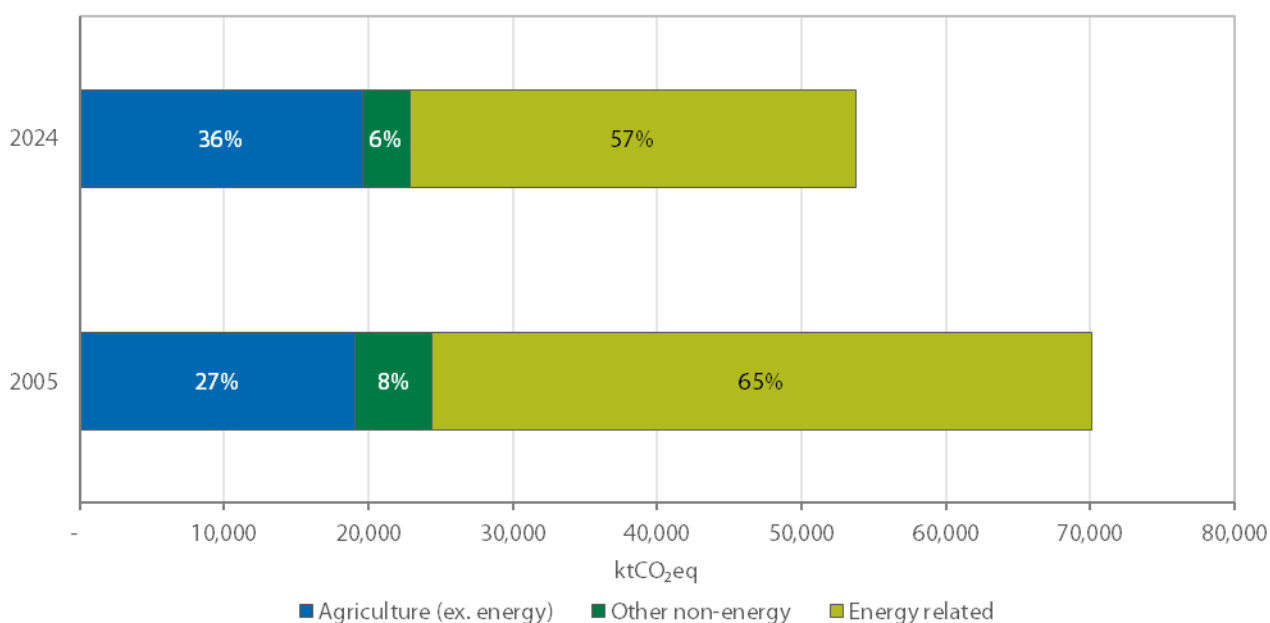
8 Energy targets and policy

This section examines areas that are a focus of national and international energy policy, including Ireland's progress towards its EU targets for renewable energy and greenhouse gas emissions, and also the issues of energy security and cost competitiveness.

8.1 Greenhouse gas emissions

Figure 8.1 shows greenhouse gas emissions by source for 2005 and provisional figures for 2024 (excluding land use and land use change), as reported by the EPA [16]. The share of greenhouse gas emissions from energy use has fallen in both percentage and absolute quantity since 2005. Ireland has a comparably large share of greenhouse gas emissions from agriculture when compared to the EU average. For the EU as a whole in 2023, the latest year for which data is available, 12% of greenhouse gas emissions were from agriculture, compared to 36% in Ireland [17]. Almost all (98%) of the energy-related greenhouse gas emissions are from CO₂, with the rest from methane (CH₄) and nitrous oxide (N₂O).

Figure 8.1: National total greenhouse gas emissions by source²²



Source: data from EPA

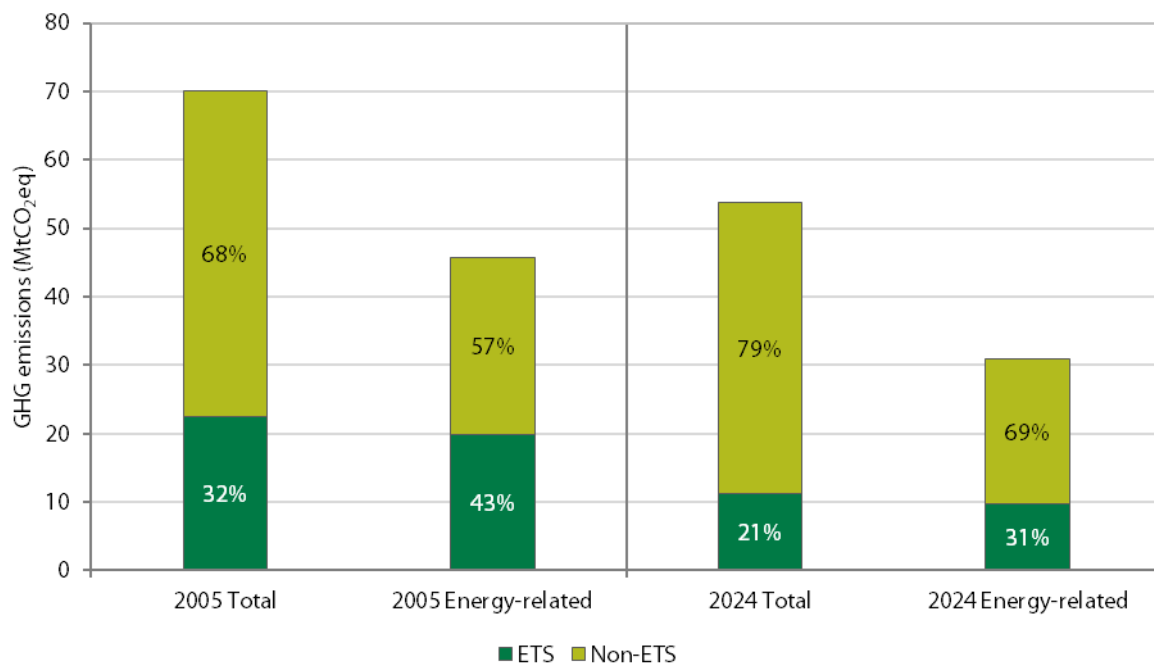
²² Emissions from international aviation are not included in national total greenhouse gas emissions.

8.1.1 Greenhouse gas emissions reductions targets

The EU 2030 climate and energy framework sets a target for the EU as a whole to achieve a 55% greenhouse gas emissions reduction by 2030 compared to 1990. The greenhouse gas emissions reductions targets are split across two categories. The first category covers large-scale carbon emitters, typically large industrial sites or electricity generation stations, but also including some bodies in the services sector and international aviation. These bodies are dealt with at EU level under the EU ETS. The second category covers all greenhouse gas emissions not covered by the ETS, known as the non-ETS sector. Achieving greenhouse gas emissions reductions in the non-ETS sector is the responsibility of national governments. The EU Effort Sharing Regulation (ESR) [18] and implementing decision [19] requires Ireland to reduce its non-ETS emissions by at least 42% by 2030, compared to a 2005 baseline. The ESR also limits the non-ETS emissions in the intervening years 2021-2029, further details are available in the EPA's Provisional Greenhouse Gas Emissions 1990-2024 report [16].

Figure 8.2 compares Ireland's total and energy emissions, split between ETS and non-ETS sectors, for the baseline year (2005) and 2024.

Figure 8.2: Energy-related and total GHG emissions in the ETS and non-ETS sectors for 2005 baseline and 2024

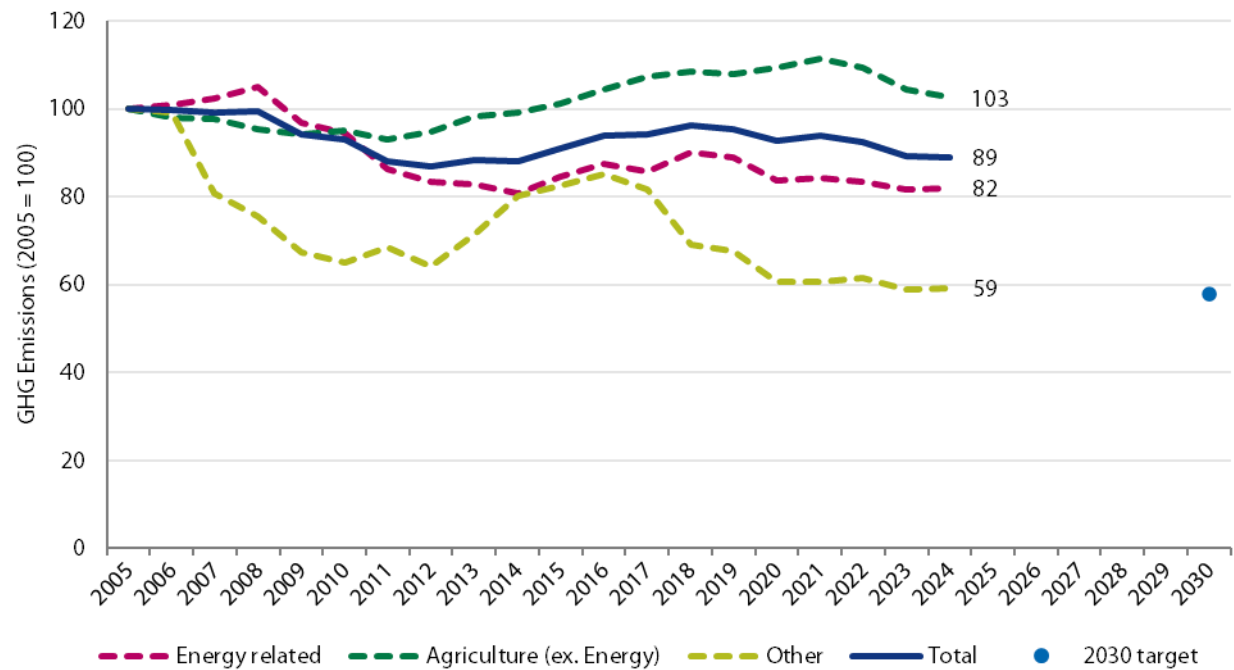


Source: data from EPA

The non-ETS sector includes most greenhouse gas emissions in the residential, transport and services sectors. It also includes most non-energy-related emissions, notably from agriculture.

Figure 8.3 shows the trend in non-ETS emissions relative to 2005 for all non-ETS emissions, and also separately for energy related non-ETS emissions and agriculture non-ETS emissions. The data are from the EPA and are provisional for 2024.

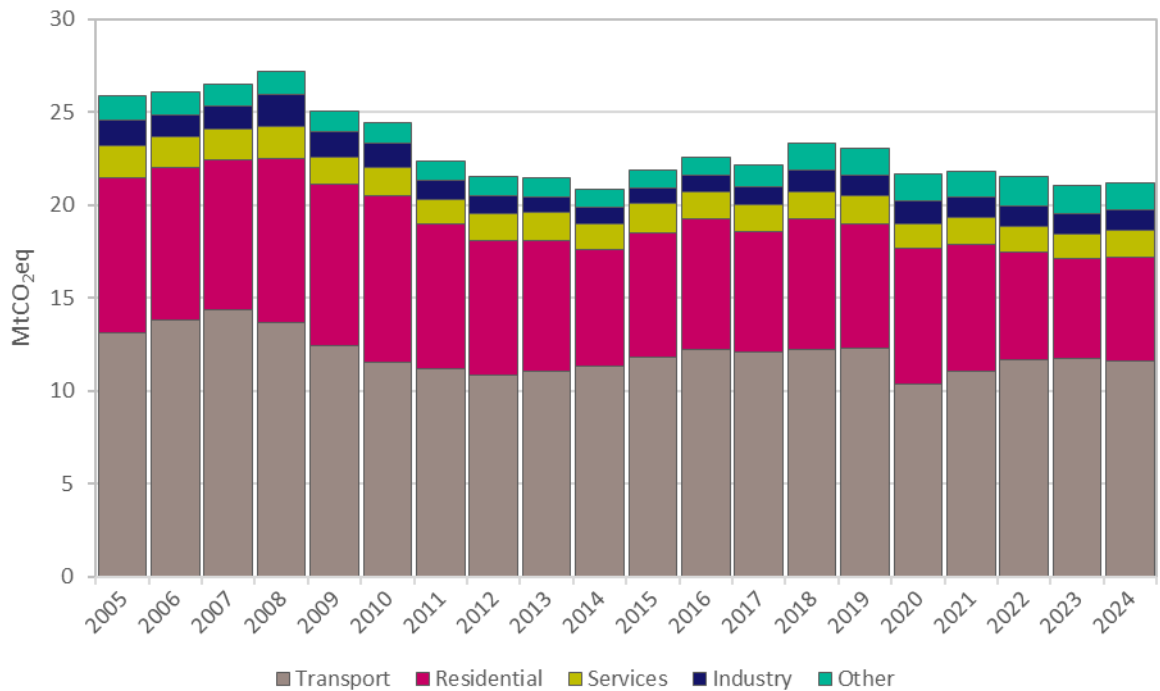
Figure 8.3: Index of non-ETS greenhouse gas emissions relative to 2005



Source: data from EPA

Figure 8.4 shows the trend in non-ETS energy-related greenhouse gas emissions split by sector.²³ Transport and households typically account for over 80% of energy-related non-ETS emissions.

Figure 8.4: Energy-related non-ETS greenhouse gas emissions

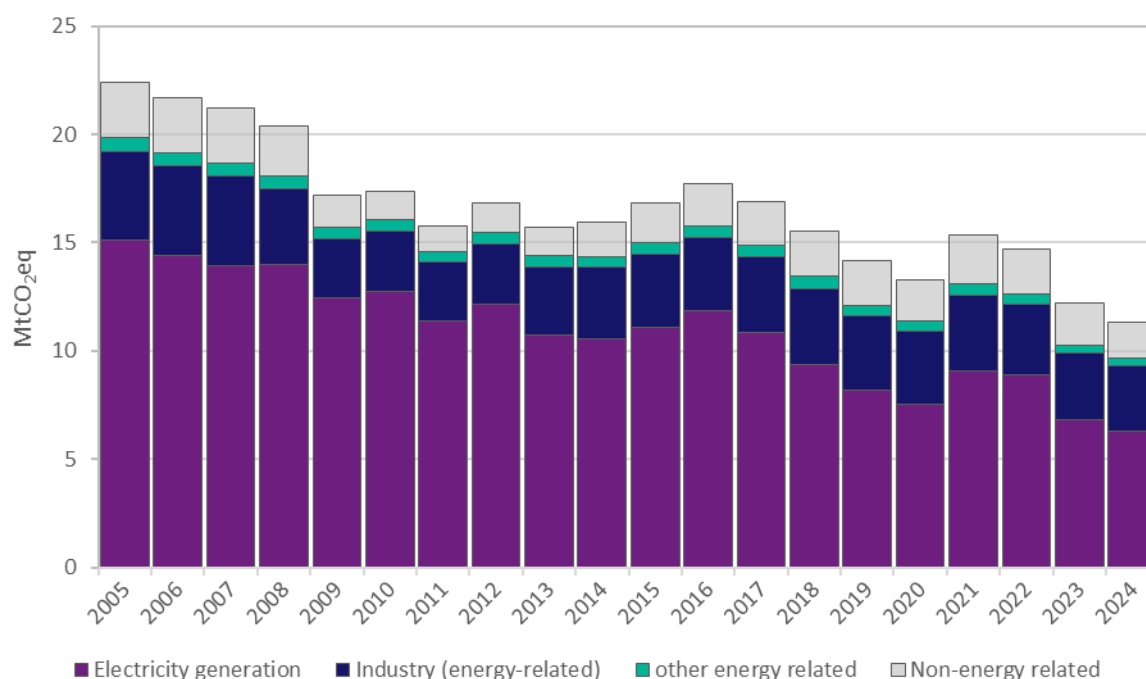


Source: data from EPA

Figure 8.5 shows the trend in emissions from fossil fuel combustion from those installations included in the EU ETS in Ireland after 2005. Most of the ETS emissions in Ireland are from electricity generation, and this is also where most of the reduction in greenhouse gas emissions has occurred.

²³ This excludes emissions associated with electricity use by these sectors – emissions from electricity generation are included in the EU ETS. It also excludes international aviation and the activity of organisation in the industry and services sectors that are within the ETS.

Figure 8.5: EU ETS greenhouse gas emissions in Ireland (excluding aviation)



Source: data from EPA

Table 8.1: Energy-related GHG emissions, ETS and non-ETS (share)

GHG [MtCO ₂ eq]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Energy-related ETS	14.33 (24.7%)	15.01 (24.9%)	15.78 (25.2%)	14.87 (24.0%)	13.44 (21.9%)	12.12 (20.3%)	11.40 (19.8%)	13.08 (21.8%)	12.64 (21.5%)	10.28 (18.7%)	9.67 (18.0%)
Non-energy related ETS	1.65 (2.8%)	1.83 (3.0%)	1.97 (3.1%)	2.04 (3.3%)	2.09 (3.4%)	2.06 (3.4%)	1.91 (3.3%)	2.26 (3.8%)	2.07 (3.5%)	1.93 (3.5%)	1.65 (3.1%)
Total ETS	15.98 (27.5%)	16.84 (27.9%)	17.75 (28.4%)	16.91 (27.3%)	15.54 (25.3%)	14.18 (23.8%)	13.31 (23.1%)	15.34 (25.5%)	14.71 (25.0%)	12.21 (22.3%)	11.33 (21.1%)
Energy-related non-ETS	20.87 (36.0%)	21.85 (36.2%)	22.59 (36.1%)	22.15 (35.8%)	23.31 (37.9%)	23.01 (38.6%)	21.65 (37.6%)	21.81 (36.3%)	21.56 (36.6%)	21.07 (38.4%)	21.21 (39.5%)
Non-energy related non-ETS	21.15 (36.5%)	21.64 (35.9%)	22.27 (35.6%)	22.77 (36.8%)	22.65 (36.8%)	22.47 (37.7%)	22.56 (39.2%)	22.96 (38.2%)	22.59 (38.4%)	21.56 (39.3%)	21.22 (39.5%)
Total non-ETS	42.02 (72.5%)	43.48 (72.1%)	44.86 (71.6%)	44.93 (72.7%)	45.96 (74.7%)	45.49 (76.2%)	44.21 (76.9%)	44.77 (74.5%)	44.16 (75.0%)	42.63 (77.7%)	42.42 (78.9%)
National total	58.00 (100%)	60.33 (100%)	62.61 (100%)	61.84 (100%)	61.49 (100%)	59.67 (100%)	57.52 (100%)	60.10 (100%)	58.86 (100%)	54.85 (100%)	53.75 (100%)

ETS is a 'cap and trade' system where an EU-wide limit or cap is set for participating installations. The cap is reduced over time so that total emissions fall. Within that limit, 'allowances' for emissions are auctioned or allocated for free (outside the power generation sector and depending on the nature of the installation). Individual operators must report the greenhouse gas emissions from their installations each year and surrender sufficient allowances to cover their emissions. If an operator exceeds its available allowances, it must purchase more. If an operator has reduced its emissions, it can sell its leftover allowances. The system is designed to bring about reductions in emissions at the lowest possible overall cost. The EPA is responsible for implementing the EU ETS in Ireland and administering the accounts on Ireland's domain in the Union Registry.

8.1.2 Carbon budgets and sectoral emission ceilings

The carbon budget represents the total greenhouse gas emissions, measured in tonnes of CO₂ equivalent (CO₂eq), that Ireland may emit in different periods. The Climate Action and Low Carbon Development Act [20] commits Ireland to a legally binding target of a 51% reduction in emissions by 2030, compared to 2018 levels. The carbon budget programme comprises three successive five-year carbon budgets [21]:

- **295 MtCO₂eq** between 2021-2025
- **200 MtCO₂eq** between 2026-2030
- **151 MtCO₂eq** between 2031-2035

In total, these budgets commit Ireland to an average overall reduction in emissions of -4.8% per annum from 2021 to 2025, and -8.3% per annum from 2026 to 2030. Sectoral ceilings [22] published in July 2022 set out the maximum amount of greenhouse gas emissions that are permitted in different sectors of the economy during a carbon budget period. Combined, the carbon budgets and sectoral ceilings tell us when and where we need to make the emission reductions to remain on track for our 2030 target.

In 2024, Ireland's national energy-related emissions were 30.9 ktCO₂eq, which accounted for 57% of Ireland's total greenhouse gas (GHG) emissions 54.8 MtCO₂eq²⁴. This means SEAI's National Energy Balance is a key input into the Environmental Protection Agency's (EPA) GHG inventory calculation for the formal calculation of emission results. Practically 100% of emissions from the electricity, transport, residential (buildings) and commercial (buildings) sectors are energy related; however, 66% of emissions from the industry sector and only about 4% of emissions from the agriculture sector are energy related. Most agriculture emissions are methane from livestock, and nitrous oxide from fertiliser and manure management.

Figure 8.6 summarises the sectoral ceilings within the first two carbon budgets, spanning 2021–2025 and 2026–2030. The five-year total CO₂eq emissions permitted by the different sectors in each budget period (squares). The 'indicative emission' value for a sector is a guide to the maximum annual emissions expected from that sector at the end of the budget period, but it is not a binding target. For example, the transport sector is permitted to emit a total of 54 MtCO₂eq in the five-year period between 2021 to 2025, with an indicative annual emission of 10 MtCO₂eq in the 2025 calendar year.

²⁴ As per the EPA's *Provisional Greenhouse Gas Emissions 1990-2024*, the national total was 53,752 ktCO₂eq in 2024, while the national total with LULUCF (land-use, land-use change and forestry) was 57,646 ktCO₂eq.

Figure 8.6: Sectoral ceilings first two carbon budgets, spanning 2021–2025 and 2026–2030

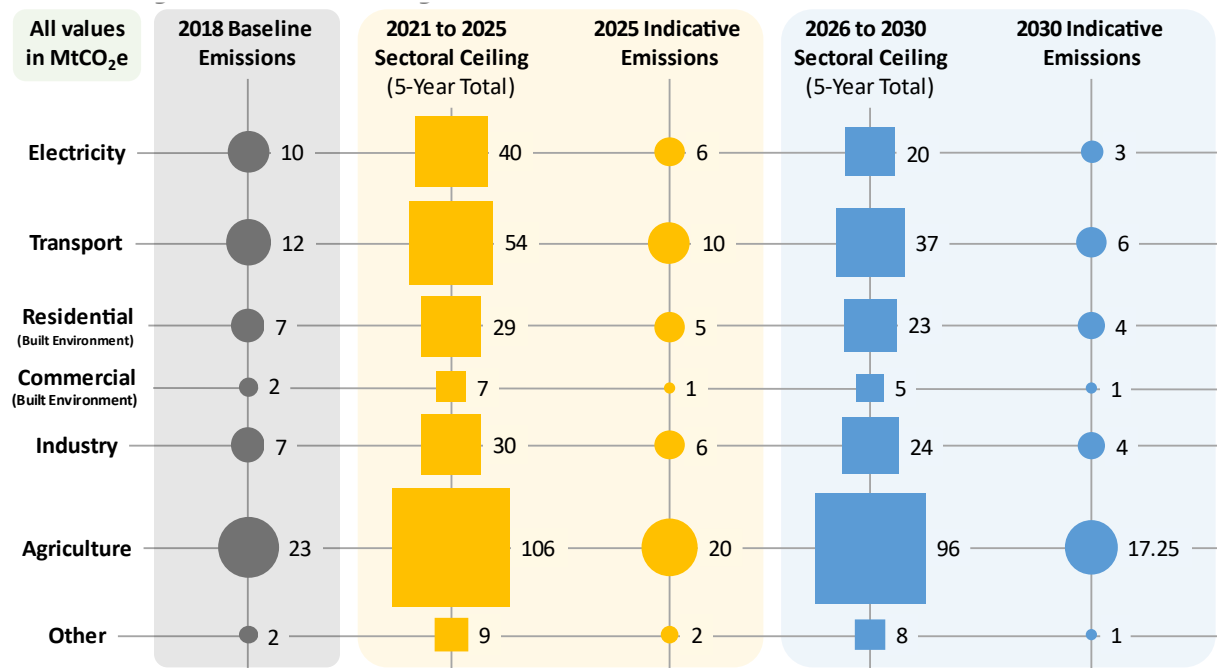
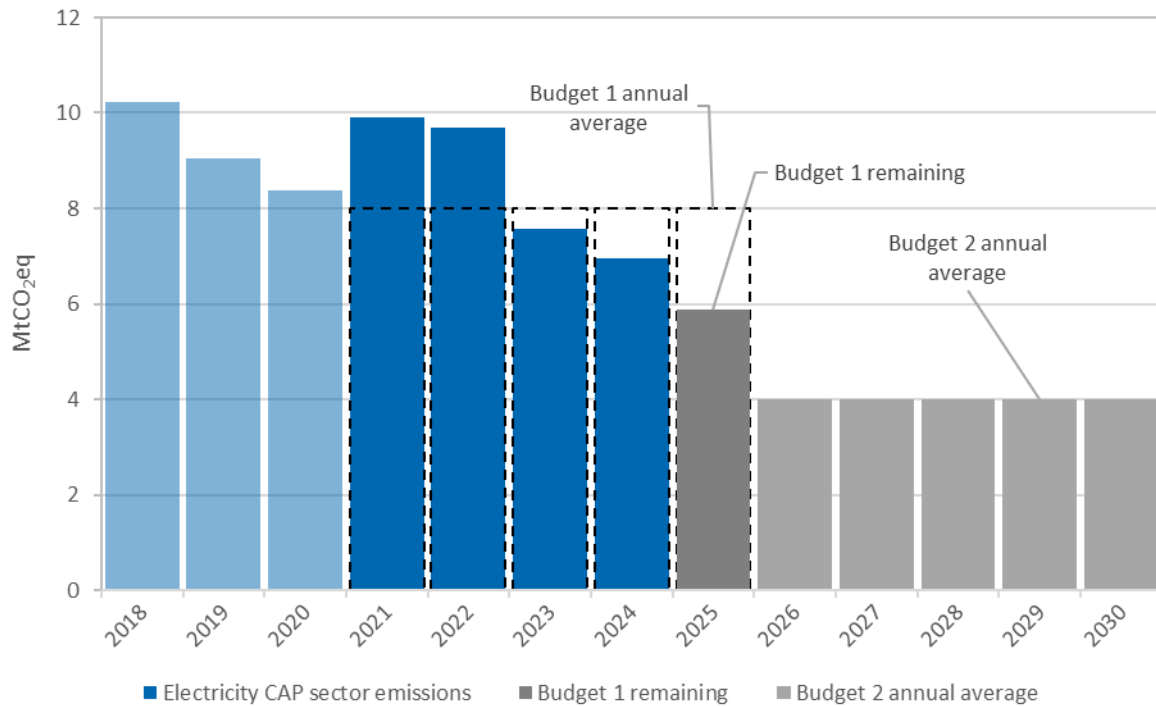


Figure 8.7 shows the emissions to the end of 2024 under the *electricity* sectoral emission ceiling, along with remaining budget for 2025 (*i.e.* the 2021–2025 period) and 2026–2030 period. The budget amounts for 2026–2030 are presented as annual averages. The translucent blue bars for 2018–2020 relate to GHG emissions emitted from the electricity sector, prior to carbon budget 1. The solid blue bars for 2021, 2022, 2023, 2024 relate to GHG emissions emitted from the electricity sector, to date, under carbon budget 1. It is worth noting that the remaining budget values (grey bars) are not targets. The dark grey bar for 2025 conveys the emissions compatible with satisfying the sectoral emission ceiling for carbon budget 1. The light grey bars convey the average annual emissions compatible with satisfying the sectoral emission ceiling in the 2026–2030 period related to carbon budget 2.

The 2021-2025 ceiling for emissions from the electricity sector is 40.0 MtCO₂eq in total, or 8.00 MtCO₂eq as an annual average (denoted by the dotted bars). Over the four years of the budget period, the total sectoral emissions were 34.1 MtCO₂eq, meaning that remaining budget is 5.89 MtCO₂eq for 2025.

Figure 8.7: Electricity sectoral emissions ceiling

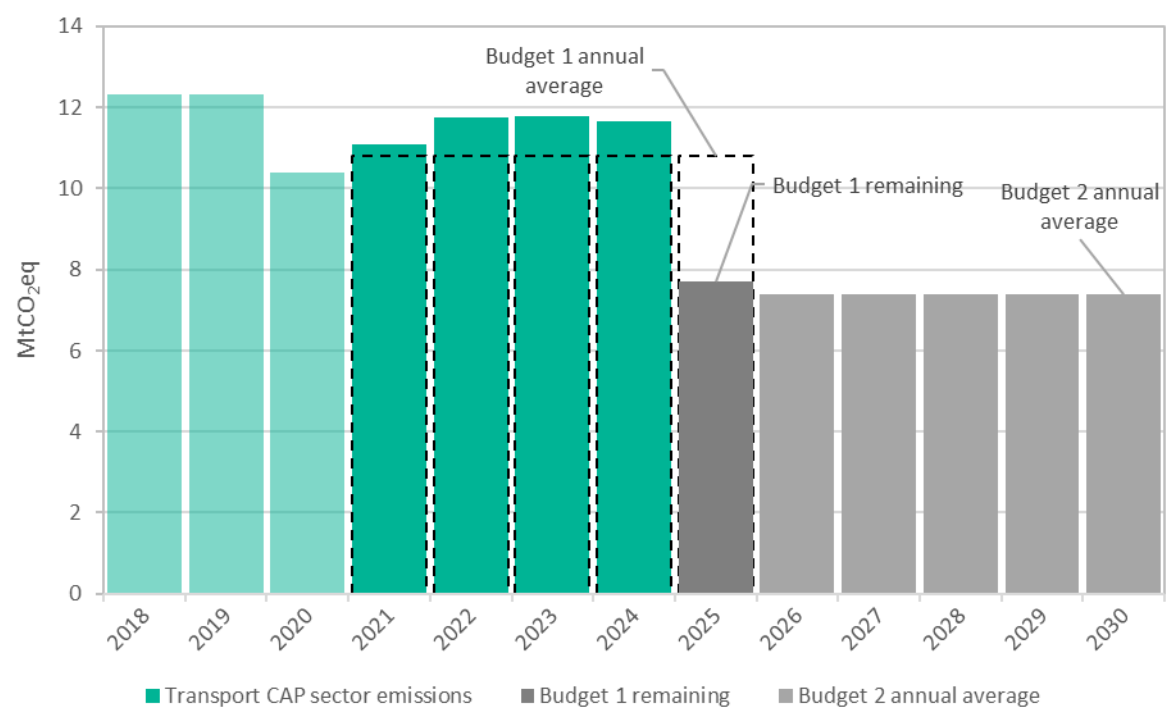


Source: emission data from EPA's GHG Provisional Emissions 1990-2024

Figure 8.8 shows the emissions to the end of 2024 under the *transport* sectoral emission ceiling, along with remaining budget for 2025 (*i.e.* the 2021-2025 period) and 2026-2030 period. The budget amounts for 2026-2030 are presented as annual averages. The translucent green bars for 2018-2020 relate to GHG emissions emitted from the electricity sector, prior to carbon budget 1. The solid green bars for 2021, 2022, 2023, 2024 relate to GHG emissions emitted from the transport sector, to date, under carbon budget 1. It is worth noting that the remaining budget values (grey bars) are not targets. The dark grey bar for 2025 conveys the emissions compatible with satisfying the sectoral emission ceiling for carbon budget 1. The light grey bars convey the average annual emissions compatible with satisfying the sectoral emission ceiling in the 2026-2030 period related to carbon budget 2.

The 2021-2025 ceiling for emissions from the transport sector is 54.0 MtCO₂eq in total, or 10.8 MtCO₂eq as an annual average (denoted by the dotted bars). Over the first four years of the budget period, the total sectoral emissions were 46.3 MtCO₂eq, meaning that remaining budget is 7.71 MtCO₂eq for 2025.

Figure 8.8: Transport sectoral emissions ceiling



Source: emission data from EPA's GHG Provisional Emissions 1990-2024

8.2 Renewable energy targets

8.2.1 Renewable Energy Directive and targets

The first Renewable Energy Directive (RED) [23] was the most important legislation influencing the growth of renewable energy in the EU and Ireland for the decade ending in 2020. From 2021, RED was replaced by the second (or recast) Renewable Energy Directive (RED II) [24], which continues to promote the growth of renewable energy out to 2030. RED set out two mandatory targets for renewable energy in Ireland to be met by 2020, while REDII sets new targets and criteria to be met by Ireland in 2030 and the interim.

Under the European Green Deal and REPowerEU plan, a significant revision [25] to RED II was finalised during 2023 and entered into force on 20 November 2023 (referred to as RED III). RED III sets more ambitious targets for 2030 and requires Member States to take additional measures to promote the use of renewable energy. RED III also redefines how progress towards the renewable energy targets is calculated, with significant changes made in how the renewable energy share in transport (RES-T) is determined.

The main target set out in the directives relates to overall renewable energy share (RES) and is commonly referred to as the **overall RES** target. The overall RES is the annual percentage of the gross final consumption (GFC)²⁵ of energy from renewable sources in the state and is calculated as the sum of:

- a. Gross final consumption of electricity from renewable sources - referred to as **RES-E**
- b. Gross final consumption of energy from renewable sources in heating and cooling – referred to as **RES-H**
- c. Final consumption of energy from renewable sources in the transport sector – referred to as **RES-T**

Overall RES

Under RED I, Ireland's target for the overall RES was 16% of gross final energy consumption in 2020. Ireland's overall RES in 2020 was 13.5%, meaning that Ireland was obligated to acquire statistical transfers of 3.3 TWh of renewable energy from other Member States to compensate for this shortfall.

RED II introduced a binding EU-wide target for overall RES of 32% in 2030 and required Member States to set their national contributions to the EU-wide target. As per the National Energy and Climate Plan (NECP) 2021-2030 [26], submitted in 2019, Ireland's overall RES target was set at 34.1% in 2030.

RED III increased the binding EU-wide target for overall RES to at least 42.5%. In July 2024, Ireland set its national contributions to this target in its updated NECP. Ireland's overall RES target is now 43.0% in 2030 [27].

RES-E

There is no RES-E target specified in the EU legislation. Ireland's national RES-E target was for 40% of gross electricity consumption to come from renewable sources in 2020. The actual RES-E achieved in 2020 was 39.1%, falling just short of the national target. Nevertheless, the development of renewable electricity was a major success in Ireland since 2005. Ireland's updated NECP 2021-2030 and Climate Action Plan 2025 (CAP 25) includes a target to increase the share of electricity generated from renewable sources to 80% in 2030 [27, 28].

²⁵ Gross final consumption of energy is an alternative to total final consumption and is the denominator used by the EU to track progress towards the targets in the RED. It is defined as "the energy commodities delivered for energy purposes to industry, transport, households, services including public services, agriculture, forestry and fisheries, the consumption of electricity and heat by the energy branch for electricity and heat production, and losses of electricity and heat in distribution and transmission."

RES-H

While the RES-H was calculated annually and formed a component of the overall RES, RED I & II did not specify binding targets for RES-H. RED II required Ireland to “endeavour to increase” the RES-H by an indicative 1.1 percentage points (pp) as an annual average calculated for the periods 2021–2025 and 2026–2030. However, RED III introduced a requirement for states to increase RES-H by at least 0.8 pp and 1.1 pp as annual averages calculated for 2021-2025 and 2026-2030, respectively.

RES-T

The second mandatory target set by the directive relates to the renewable energy share in the transport sector.²⁶ This is commonly referred to as the RES-T target. The 2020 RES-T target was for at least 10% of energy consumed in road and rail transport to come from renewable sources. The actual RES-T achieved in 2020 was 10.2%, meaning that Ireland did meet this target. RED II set a new RES-T target of 14% by 2030; however, RED III has increased the 2030 RES-T target to 29%, on an energy basis. Alternatively, a Member State may choose to follow a target to reduce the greenhouse gas intensity of energy in transport by 14.5%. RED III also slightly changed the scope of the RES-T to include additional forms of transport.

8.2.2 Biomass sustainability

Besides specifying new EU and national renewable energy targets, RED II also introduced new sustainability and verification criteria for biomass fuels (solid and gaseous) to be implemented from the transposition date (30 June 2021). Introducing these criteria led to circumstances where a portion of the biomass fuel consumed in Ireland in 2021 and 2022 could not be counted in the national renewable shares, specifically towards the overall RES, RES-E and RES-H.

From the beginning of July 2021, biomass fuel consumed in installations, above certain sizes, must fulfil various sustainability²⁷ and greenhouse gas saving criteria to be counted towards national renewable energy targets, or be eligible for financial supports. In addition, the biomass fuel must be subject to a verification procedure requiring economic operators to demonstrate that the sustainability and greenhouse gas saving criteria, as laid out in the directive, have been fulfilled. Verification must require economic operators to, among other things, maintain sustainability records of all consignments of biomass, applying a mass balance, and arrange for independent auditing of all information submitted to the competent authority.

National legislation [29] was introduced in July 2022 to transpose the biomass sustainability and verification requirements of RED II into Irish law. While a system of verification commenced in 2023 for operators of biomass installations under support schemes, no verification system was in place during 2021 or 2022. Consequently, any biomass consumed during the 2021 (from July onwards) or 2022 that was required to meet the sustainability criteria and undergo verification could not be included in the overall RES, RES-E and RES-H.

Only biomass fuel consumed in an installation with a rated thermal output above one of the following thresholds is required to meet the sustainability criteria:

- Equal to or exceeding 20 MW for solid biomass fuels (decreasing to 7.5 MW in 2025 under RED III); or
- Equal to or exceeding 2 MW for gaseous biomass fuels.

²⁶ In the context of RED and RED II, consumption in the transport sector relates only to energy in road and rail; however, renewable energy consumed in aviation and marine can contribute towards the RES-T.

²⁷ ‘Sustainability’, in the context of biomass under REDII, relates to: soil monitoring and management; protection of land with high biodiversity; protection of land with high carbon stock; protection of peatland; sustainable forest production; and land-use, land-use change and forestry (LULUCF).

In 2021, approximately 776 GWh of renewable final energy could not be counted towards the RES overall (equivalent to 0.6 pp). This increased to 1,517 GWh or 1.1 pp in 2022, but has since decreased to 152 GWh in 2023 and 123 GWh in 2024.

Most of this biomass fuel that could not contribute to the RES targets in 2021 or 2022 was consumed in EU ETS installations. For 2023 onwards, verification of the sustainability of biomass fuels was completed under the EU ETS system. Consequently, only a relatively small quantity of biomass fuel (and bioliquid) that was consumed in 2023 and 2024 could not contribute towards the RES targets:

- 0.1 pp in 2023
- 0.1 pp in 2024

Biomass fuels consumed in smaller installations are not required to meet the sustainability and verification criteria to be counted towards the renewable energy targets. Biomass, such as wood, consumed in the residential sector and in many smaller commercial/industrial boilers can be included the RES.

Where biomass fuel could not contribute to the RES results, due to the absence of verification, it still contributed to decarbonisation by displacing fossil fuel and reducing greenhouse gas emissions under EU ETS and Ireland's national emissions allocations.

8.2.3 RES Results

Table 8.2 shows Ireland's the overall RES figures, along with the RES-E/-H/-T, for 2020 to 2023. The values in the table are calculated under the EU legislation and Commission guidance that was active at the time: 2020 values were calculated under RED I, while 2021-2023 were calculated under RED II.

Table 8.2: Ireland's progress towards renewable energy share (RES) targets

Target [%]	2020	2021	2022	2023	2024	2030 Target	Note on Target
RES-E	39.05%	37.66%	37.39%	40.43%	41.34%	80%	National ²⁸
RES-T	10.19%	4.65%	5.82%	7.59%	8.66%	29%	RED III
RES-H	6.26%	5.42%	5.33%	7.78%	7.91%	15.8%	RED III ²⁹
Overall RES	13.53% (16.16% with stat transfers)	12.95%	13.02%	15.24%	16.06%	43.0%	RED III / NECP

²⁸ National target as set out in Climate Action Plans 2023, 2024 and 2025

²⁹ This is based on the 0.8 pp and 1.1 pp average annual increases for 2021-2025 and 2026-2030, as set out in RED III

Figure 8.9 shows the current split of renewable energy between the three modes. Figure 8.10 shows the contribution of renewable electricity, heat and transport to the overall RES target. Renewable electricity makes the largest contribution to the overall RES and has been responsible for most of the overall growth in renewable energy since 2005. Figure 8.10 also shows the share of heat and electricity generated by biomass fuels that was excluded from the overall RES (see section 8.2.2 for further details).

Figure 8.9: 2023 share of renewable energy by mode

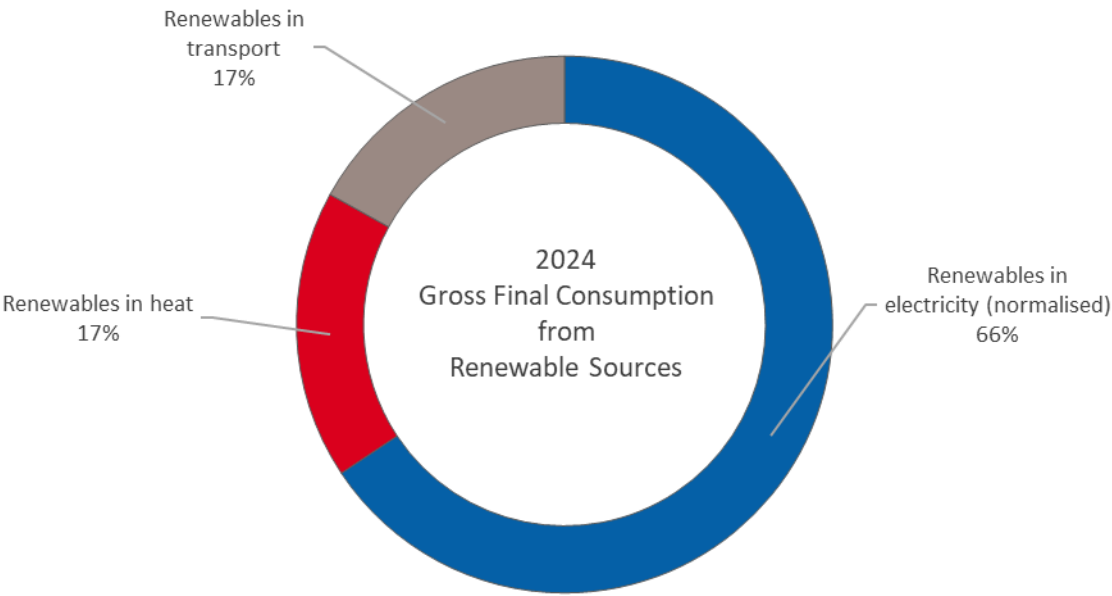


Figure 8.10: Renewable energy share (overall RES) with contribution by mode

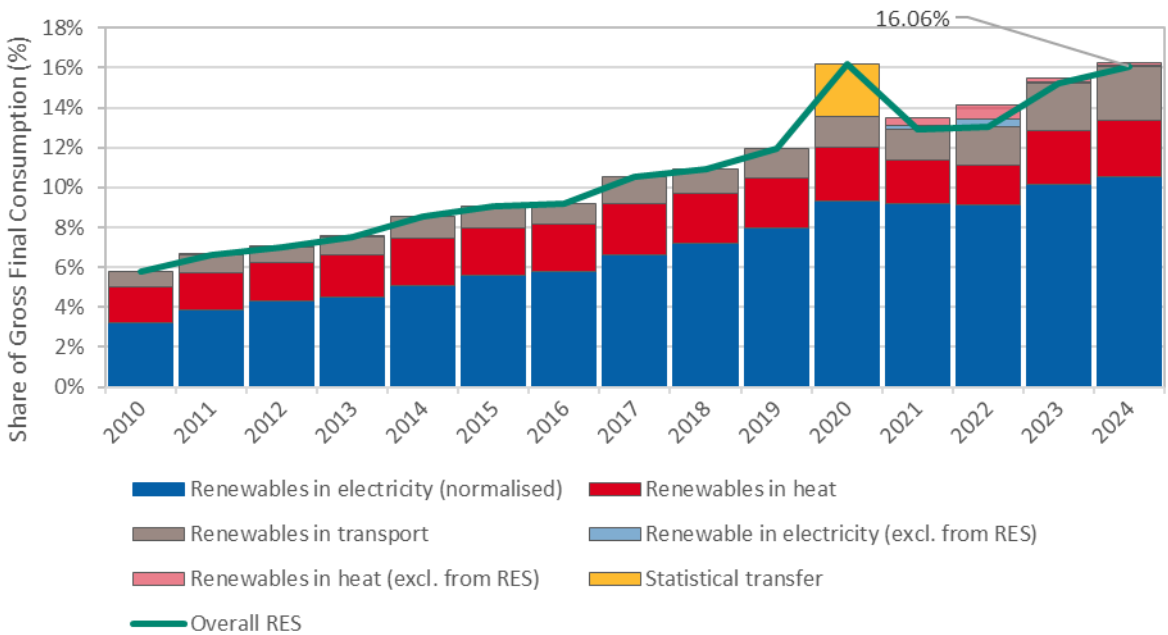


Figure 8.11 shows the overall RES from 2010 to 2022 along with Ireland’s current 2030 target of 43.0%. and the minimum trajectory set out in Ireland’s updated NECP, as per EU legislation [30].

Figure 8.11: Renewable energy share (overall RES) and trajectories to 2030 target

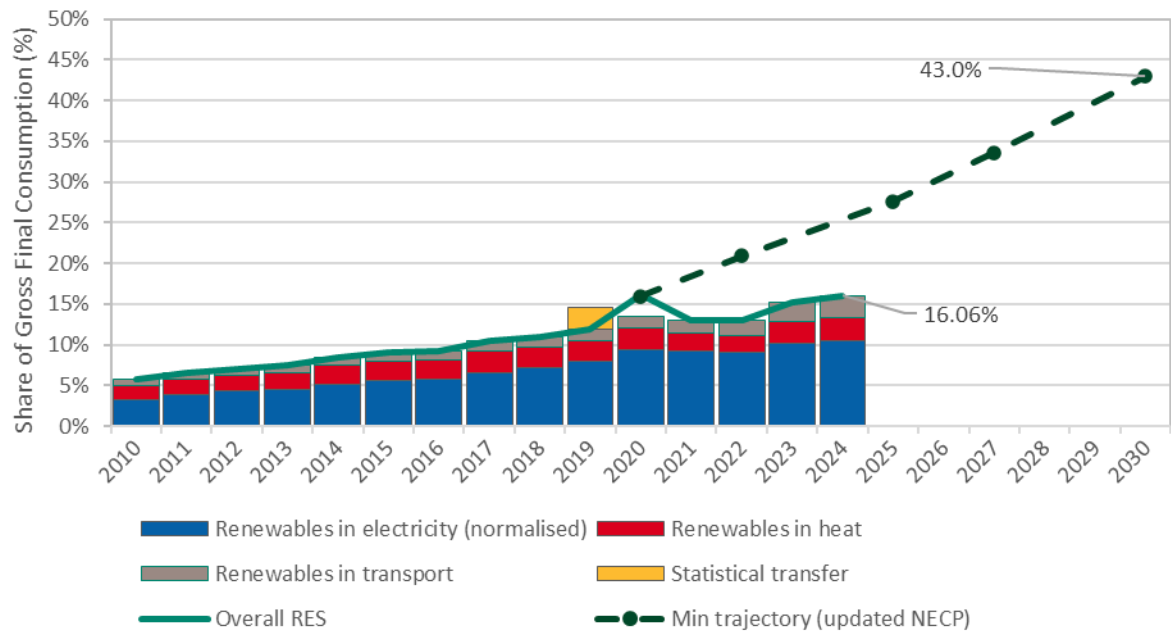


Figure 8.12 illustrates the 2024 gross final consumption (GFC) of energy in each of transport, heat and electricity according to the RED II calculation, and the portion of each mode that comes from renewable sources. This gives important context to the separate heat, transport and electricity targets. Although electricity has the highest share of renewable energy, it is the smallest of the modes in terms of GFC. The transport mode, which has the smallest renewable share, makes up the largest share of GFC, followed by heat.

Figure 8.12: Renewable and fossil gross final energy consumption by mode

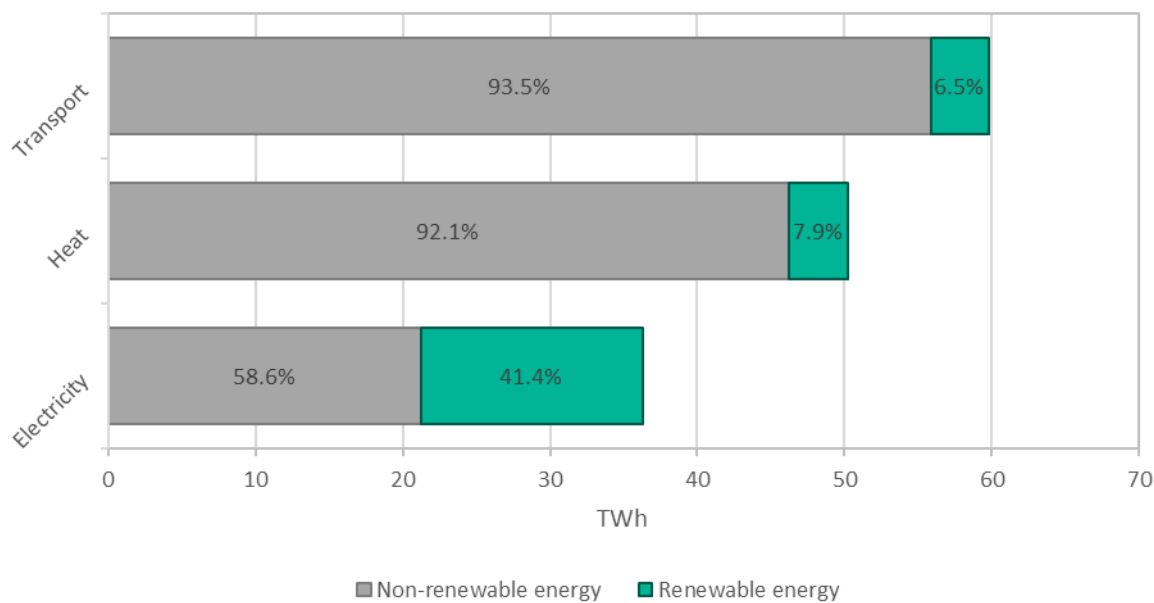


Figure 8.13 and Table 8.3 show the share and quantity of renewable energy used each year, split by source. Most of the growth in renewable energy has come from wind. Wind provides more than half of all renewable energy. Solid biomass and bioliquids were the next largest sources of growth. Bioenergy, including solid biomass, renewable wastes, landfill gas, biogas and bioliquids, collectively accounted for approximately one-third of Ireland’s renewable energy.

Figure 8.13 and Table 8.3 also show the quantities and shares of final energy from biomass and biogas that were excluded from the 2021-2023 RES figures as verification of sustainability was outstanding, see section 8.2.2 for further details.

Figure 8.13: Renewable energy share of gross final consumption by source

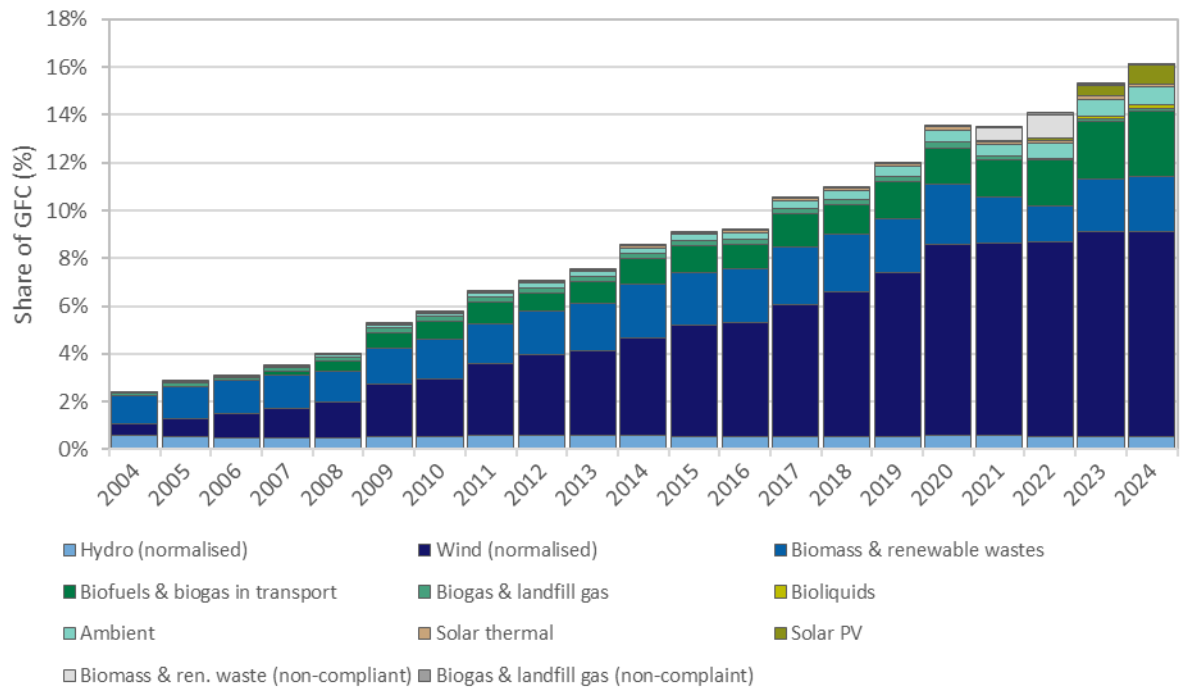


Table 8.3: Overall RES - Renewable energy in gross final consumption by energy type (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydro (normalised)	0.73 (0.6%)	0.72 (0.5%)	0.73 (0.5%)	0.72 (0.5%)	0.72 (0.5%)	0.74 (0.5%)	0.76 (0.6%)	0.76 (0.6%)	0.76 (0.5%)	0.76 (0.5%)	0.75 (0.5%)
Wind (normalised)	5.25 (4.1%)	6.20 (4.7%)	6.57 (4.8%)	7.63 (5.6%)	8.69 (6.1%)	9.80 (6.9%)	10.68 (8.0%)	10.97 (8.1%)	11.34 (8.1%)	11.88 (8.6%)	12.20 (8.6%)
Biomass & renewable wastes	2.93 (2.3%)	2.90 (2.2%)	3.16 (2.3%)	3.31 (2.4%)	3.48 (2.4%)	3.29 (2.3%)	3.35 (2.5%)	2.65 (1.9%)	2.05 (1.5%)	3.06 (2.2%)	3.28 (2.3%)
Biofuels & biogas in transport	1.35 (1.1%)	1.48 (1.1%)	1.38 (1.0%)	1.87 (1.4%)	1.79 (1.3%)	2.19 (1.5%)	2.03 (1.5%)	2.11 (1.5%)	2.68 (1.9%)	3.30 (2.4%)	3.87 (2.7%)
Biogas & LFG	0.30 (0.2%)	0.31 (0.2%)	0.32 (0.2%)	0.32 (0.2%)	0.30 (0.2%)	0.30 (0.2%)	0.33 (0.2%)	0.23 (0.2%)	0.14 (0.1%)	0.20 (0.1%)	0.23 (0.2%)
Bioliquids	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.10 (0.1%)	0.16 (0.1%)
Ambient	0.27 (0.2%)	0.31 (0.2%)	0.38 (0.3%)	0.45 (0.3%)	0.51 (0.4%)	0.61 (0.4%)	0.66 (0.5%)	0.66 (0.5%)	0.83 (0.6%)	1.02 (0.7%)	1.10 (0.8%)
Solar thermal	0.12 (0.1%)	0.13 (0.1%)	0.14 (0.1%)	0.15 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)	0.16 (0.1%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.0%)	0.04 (0.0%)	0.06 (0.0%)	0.08 (0.1%)	0.15 (0.1%)	0.65 (0.5%)	1.09 (0.8%)
Numerator Overall RES (Overall RES)	10.94 (8.5%)	12.06 (9.1%)	12.68 (9.2%)	14.45 (10.5%)	15.68 (10.9%)	17.13 (12.0%)	18.04 (13.5%)	17.63 (12.9%)	18.11 (13.0%)	21.13 (15.2%)	22.86 (16.1%)
Biomass & ren. waste (non- compliant)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.69 (0.5%)	1.33 (1.0%)	0.04 (0.0%)	0.03 (0.0%)
Biogas & LFG (non- compliant)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.09 (0.1%)	0.18 (0.1%)	0.12 (0.1%)	0.10 (0.1%)
Ren. energy not contributing to overall RES	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.78 (0.6%)	1.52 (1.1%)	0.15 (0.1%)	0.12 (0.1%)
Denominator Overall RES (GFC adjusted)	128.52 (100%)	132.73 (100%)	137.95 (100%)	137.40 (100%)	143.28 (100%)	143.00 (100%)	133.30 (100%)	136.21 (100%)	139.15 (100%)	138.59 (100%)	142.29 (100%)

8.2.4 Electricity from renewable energy sources (RES-E)

Ireland has no binding national target for RES-E for 2030, nor was there one for 2020, but RES-E forms the backbone of Ireland's strategy to achieve the overall renewable energy target for 2030. Ireland's Climate Action Plan 2025 (CAP 25) [28] includes a target to increase the share of electricity generated from renewable sources up to 80% by 2030.

Figure 8.14 and Table 8.4 show the renewable energy share in electricity (RES-E) by source. RES-E decreased from 39.1% in 2020 to 37.7% in 2021. This drop was primarily due to an increase in electricity consumption, while renewable generation remained stagnant, with around 0.8 pp of electricity generated from biomass or biogas that was ineligible (*i.e.* non-compliant) to be counted towards the RES-E. In 2022, the RES-E dropped to 37.4% with around 1.6 pp of electricity from biomass and biogas that was ineligible to contribute. The RES-E increased over the next two years and reached 41.3% in 2024, with solar PV contributing 3.0 pp and overtaking hydro.

Figure 8.14: RES-E - gross final consumption of electricity from renewable sources

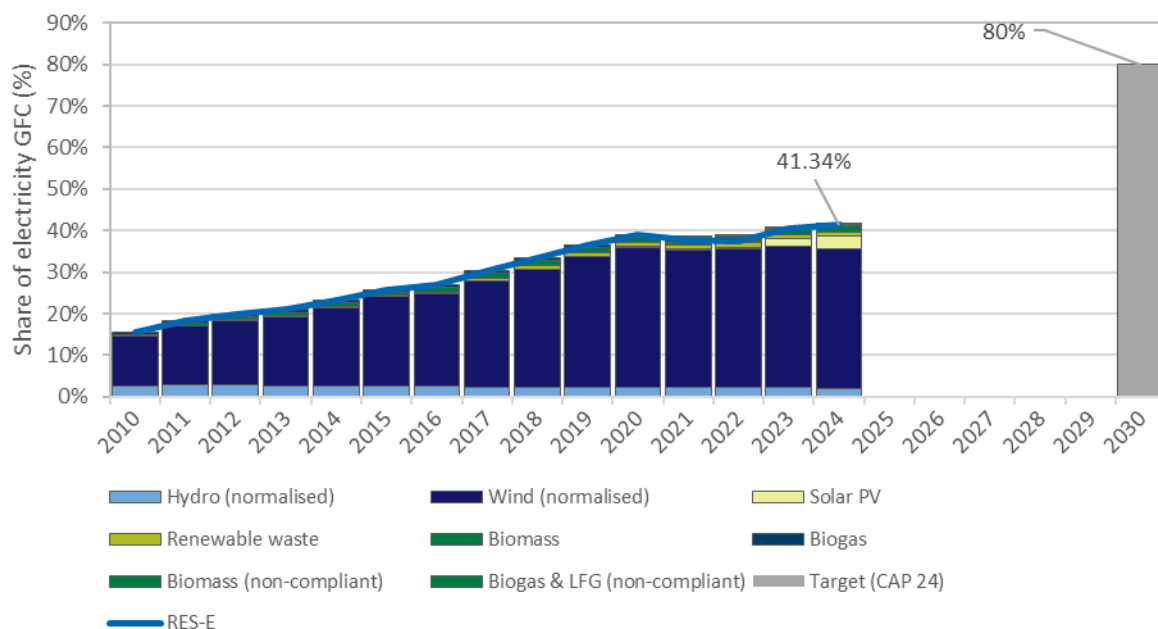


Table 8.4: RES-E – gross final consumption of electricity from renewable sources (share in brackets)

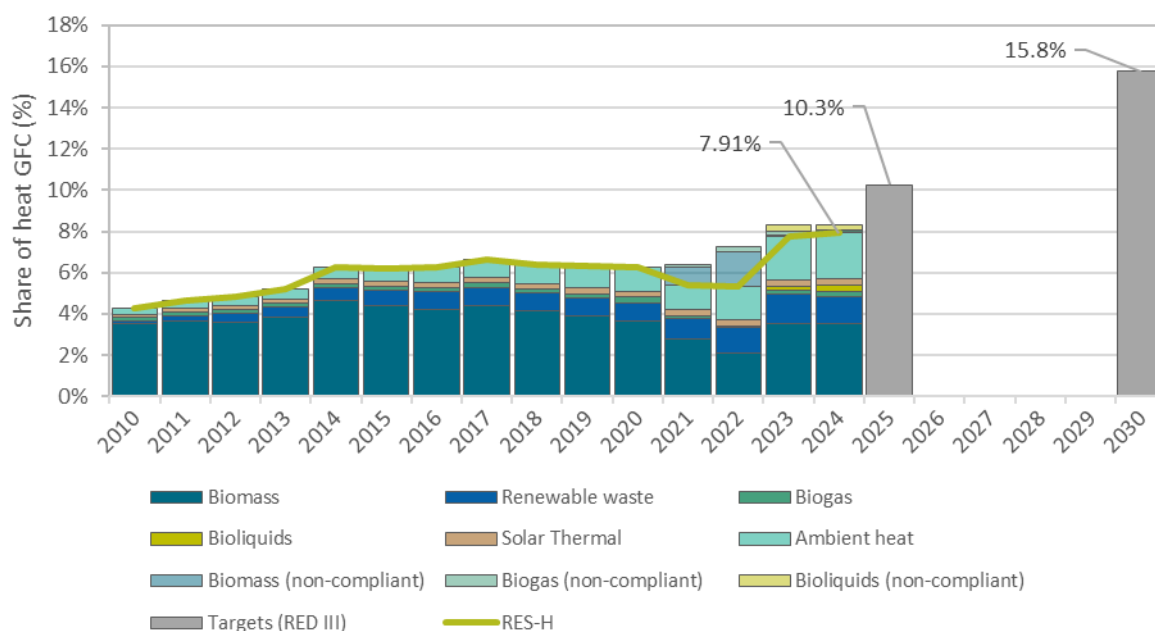
Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydro (normalised)	0.73 (2.6%)	0.72 (2.5%)	0.73 (2.5%)	0.72 (2.4%)	0.72 (2.3%)	0.74 (2.4%)	0.76 (2.4%)	0.76 (2.3%)	0.76 (2.2%)	0.76 (2.2%)	0.75 (2.1%)
Wind (normalised)	5.25 (18.8%)	6.20 (21.5%)	6.57 (22.3%)	7.63 (25.4%)	8.69 (28.2%)	9.80 (31.3%)	10.68 (33.5%)	10.97 (32.9%)	11.34 (33.3%)	11.88 (34.1%)	12.20 (33.6%)
Solar PV	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.0%)	0.01 (0.0%)	0.02 (0.1%)	0.04 (0.1%)	0.06 (0.2%)	0.08 (0.3%)	0.15 (0.4%)	0.65 (1.9%)	1.09 (3.0%)
Renewable waste	0.07 (0.3%)	0.08 (0.3%)	0.08 (0.3%)	0.15 (0.5%)	0.33 (1.1%)	0.32 (1.0%)	0.33 (1.0%)	0.35 (1.1%)	0.35 (1.0%)	0.33 (1.0%)	0.34 (0.9%)
Biomass (compliant)	0.26 (0.9%)	0.20 (0.7%)	0.40 (1.3%)	0.38 (1.3%)	0.33 (1.1%)	0.35 (1.1%)	0.43 (1.4%)	0.23 (0.7%)	0.03 (0.1%)	0.34 (1.0%)	0.51 (1.4%)
Biogas & LFG (compliant)	0.20 (0.7%)	0.20 (0.7%)	0.21 (0.7%)	0.20 (0.7%)	0.18 (0.6%)	0.19 (0.6%)	0.18 (0.6%)	0.15 (0.4%)	0.11 (0.3%)	0.12 (0.3%)	0.11 (0.3%)
Numerator RES-E	6.52 (23.3%)	7.40 (25.7%)	7.99 (27.1%)	9.09 (30.3%)	10.29 (33.3%)	11.43 (36.5%)	12.44 (39.1%)	12.55 (37.7%)	12.73 (37.4%)	14.08 (40.4%)	15.01 (41.3%)
Biomass (non- compliant)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.24 (0.7%)	0.48 (1.4%)	0.00 (0.0%)	0.00 (0.0%)
Biogas & LFG (non- compliant)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.02 (0.1%)	0.05 (0.2%)	0.05 (0.1%)	0.04 (0.1%)
Ren. energy excluded from numerator	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.26 (0.8%)	0.53 (1.6%)	0.05 (0.1%)	0.05 (0.1%)
Denominator RES-E (GFC of elec)	27.96 (100%)	28.78 (100%)	29.51 (100%)	29.99 (100%)	30.87 (100%)	31.36 (100%)	31.85 (100%)	33.33 (100%)	34.05 (100%)	34.83 (100%)	36.31 (100%)

8.2.5 Heat from renewable energy sources (RES-H)

Under RED III, which entered into force in November 2023 (2025 will be the first reporting year under RED III), Ireland must increase the share of renewable energy in heating and cooling by at least 0.8 pp and 1.1 pp as annual averages for the periods 2021-2025 and 2026-2030, respectively. Based on Ireland's 2020 RES-H figure of 6.3%, Ireland's RES-H would need to reach 10.3% in 2025 to achieve the target for 2021-2025.

Figure 8.15 and Table 8.5 show the share of heating provided by renewable energy as a share of overall heat (gross final consumption of heat). RES-H fell from 6.3% in 2020 to 5.42% in 2021 and then 5.33% in 2022; this decrease can be attributed to the transition from RED I to RED II and the introduction of new sustainability and verification criteria for biomass fuels, see section 8.2.2. In 2023 and 2024, the RES-H increased and reached 7.91% in 2024, a new high. This increase was due to the sustainability data relating to biomass and biogas becoming available from 2023 (*i.e.* less renewable energy excluded from the RES-H), an increase in the quantity of renewable energy and an overall decrease in consumption of energy in heat, compared to 2022 levels.

Figure 8.15: RES-H – gross final consumption of heat from renewable sources



Renewable heat energy is dominated by the use of solid biomass and renewable wastes in industry. Utilisation of ambient energy (*via* heat pumps) has grown four-fold between 2014 and 2024 and is now a significant source of renewable heat energy, accounting for approximately 27% of renewable heat energy in 2024. Table 8.5 shows the breakdown of renewable energy types used for heat.

Recent growth in renewable energy use for heat has been due to increased use of renewable wastes in industry and increased use of heat pumps delivering ambient energy in the residential and services sectors. The latter is mostly due to revisions to building regulations for new dwellings and the support of grant schemes.

Table 8.5: RES-H – gross final consumption of heat from renewable sources (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Biomass	2.29 (4.7%)	2.26 (4.4%)	2.24 (4.2%)	2.31 (4.4%)	2.35 (4.2%)	2.16 (3.9%)	2.09 (3.7%)	1.52 (2.8%)	1.05 (2.1%)	1.70 (3.5%)	1.76 (3.5%)
Renewable waste	0.30 (0.6%)	0.36 (0.7%)	0.44 (0.8%)	0.47 (0.9%)	0.47 (0.8%)	0.47 (0.8%)	0.50 (0.9%)	0.54 (1.0%)	0.63 (1.2%)	0.68 (1.4%)	0.67 (1.3%)
Biogas	0.09 (0.2%)	0.10 (0.2%)	0.11 (0.2%)	0.11 (0.2%)	0.11 (0.2%)	0.12 (0.2%)	0.15 (0.3%)	0.08 (0.1%)	0.03 (0.1%)	0.09 (0.2%)	0.11 (0.2%)
Bioliquids	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.10 (0.2%)	0.16 (0.3%)
Solar Thermal	0.12 (0.3%)	0.13 (0.3%)	0.14 (0.3%)	0.15 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.3%)	0.16 (0.3%)
Ambient heat	0.27 (0.5%)	0.31 (0.6%)	0.38 (0.7%)	0.45 (0.9%)	0.51 (0.9%)	0.61 (1.1%)	0.66 (1.2%)	0.66 (1.2%)	0.83 (1.6%)	1.02 (2.1%)	1.10 (2.2%)
RES-H numerator	3.07 (6.3%)	3.17 (6.2%)	3.31 (6.2%)	3.49 (6.6%)	3.60 (6.4%)	3.51 (6.3%)	3.57 (6.3%)	2.97 (5.4%)	2.70 (5.3%)	3.74 (7.8%)	3.97 (7.9%)
Biomass (non-compliant)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.45 (0.8%)	0.85 (1.7%)	0.03 (0.1%)	0.02 (0.0%)
Biogas (non-compliant)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.07 (0.1%)	0.13 (0.3%)	0.07 (0.1%)	0.05 (0.1%)
Bioliquids (non-compliant)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.16 (0.3%)	0.11 (0.2%)
Ren. energy excluded from numerator	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.51 (0.9%)	0.98 (1.9%)	0.26 (0.5%)	0.19 (0.4%)
RES-H denominator (GFC heat)	49.10 (100%)	51.32 (100%)	53.02 (100%)	52.73 (100%)	56.37 (100%)	55.31 (100%)	57.06 (100%)	54.77 (100%)	50.63 (100%)	48.12 (100%)	50.23 (100%)

8.2.6 Transport energy from renewable sources (RES-T)

RED I established a mandatory minimum target for the share of renewable energy sources in transport (RES-T) by 2020: 10% of all petrol, diesel, biofuels and electricity consumed in road and rail transport. Ireland's RES-T reached 10.2% in 2020.

RED II requires Ireland, along with all Member States, to set an obligation on fuel suppliers to ensure that the share of renewable energy within the final consumption of energy in transport is at least 14% by 2030. This target has increased to 29% by 2030 under RED III.

It should be noted that renewable energy consumed in any mode of transport contributes towards the RES-T, but that only certain energy types consumed in *road & rail* are counted in the denominator:

$$\text{RES-T (RED II)} = \frac{\text{Final consumption of renewable energy in all transport}}{\text{Final consumption of energy in road \& rail}}$$

The only fossil fuels counted in the denominator are petrol, diesel and natural gas; the relatively small quantity of fossil LPG consumed in road transport is not included in RES-T denominator, but is included in the overall RES.

In accordance with Eurostat guidance and reporting requirements for RES data, the above RED II definition will be used for calculating RES-T up to and including 2024. For 2025 onwards the scope of the RES-T calculation will change to that defined by RED III, which will include all transport modes including marine transport:

$$\text{RES - T (RED III)} = \frac{\text{Final consumption of renewable energy in all transport}}{\text{Final consumption in all transport}}$$

To maintain consistency with Ireland's official 2024 RES data reported to the European Commission (Eurostat) in November 2025, the 2021-2024 RES-T presented in this subsection are calculated under the RED II definitions and methodology.

RED II, as with RED I before it, specifies several weightings or multipliers that are applied to certain fuels or energy for the calculation of RES-T. These weightings help to incentivise the use of advanced biofuels and biofuels from wastes over crop-based fuels, generally promoting those with lower life-cycle greenhouse gas intensities. Table 8.6 sets out the current multipliers under RED II and how they have changed from those used for RED I. Approximately 88% of the biofuel and biogas used in transport is produced from feedstocks listed in Annex IX of RED II, including used cooking oil (UCO) and tallow (category 1 & 2), and consequently count twice towards the RES-T target.

These multipliers do not apply to the overall RES. Prior to the transition to RED II, there was a significant difference between the RES-T value and the share of renewable energy in transport, as a component of the overall RES.

Table 8.6: Multipliers used in the calculation of RES-T

	RED I	RED II
Biofuels & biogas from Annex IX feedstocks	2× (numerator only)	2× (numerator & denominator)
Renewable electricity in road	5× (numerator only)	4 × (numerator & denominator)
Renewable electricity in rail	2.5 × (numerator & denominator)	1.5 × (numerator & denominator)
Aviation & maritime fuels (excl. fuels from food & feed crops)	-	1.2 × (numerator)

In addition to the change in multipliers used in the RES-T calculation, RED II also includes three limits on biomass fuels produced from certain feedstocks, see Table 8.7. These limits are defined as percentages of energy consumed in transport. Although two of these limits can technically apply to bioliquids or biomass fuels consumed outside of the transport sector, they are currently only relevant to fuels used in the transport sector as no bioliquids from food or feed crops have been recorded outside of the transport sector.

The first limit relates to the share of used cooking oil (UCO) and animal fat (tallow) category 1 & 2 and is the most consequential change to the calculation of Ireland's RES-T arising from the transition to RED II. UCO and tallow (category 1 & 2) accounted for 5.3% of final energy consumption in road and rail transport (i.e. the RES-T denominator) in 2024; under RED II, this biofuel is limited to 1.7% of road and rail transport energy for the purposes of RES-T. In 2024, approximately 47% of the biofuel and biogas in transport consumed in Ireland (before application of multipliers) could not be counted towards RES-T due to this limit.

The second limit relates to fuels produced from food and feed crops (e.g. bioethanol from maize, biodiesel from sunflower). Biofuels from food and feed crops only constituted 0.9% of final consumption of energy in road and rail transport in 2024.

The third limit applies to biofuels (as well as to bioliquids and biomass fuels) produced from crops with a 'high indirect land-use change risk' (high ILUC risk), specifically fuels produced from palm oil. The limit is based on the share of biofuels or biogas from high ILUC risk feedstocks in Ireland in 2019, i.e. 0.028%. No biofuels or biogas produced from such feedstocks were consumed in 2024.

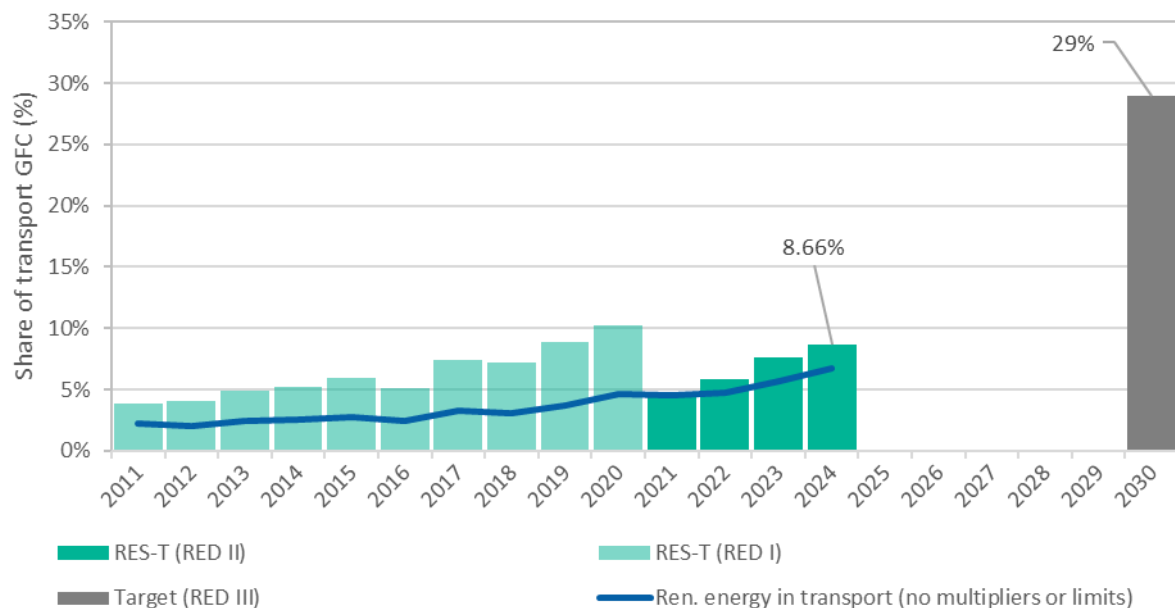
Table 8.7: Limits on biofuels from certain feedstocks

Feedstocks	Fuel type	Scope	Limit	2024 share
Used cooking oil (UCO) and animal fat (tallow cat. 1 & 2)	- Biofuels - Biogas	RES-T only	1.7% of transport energy	5.3%
Food and feed crops	- Biofuels - Bioliqids - Biomass fuels in transport	Overall RES & RES-T	2% of transport energy	0.9%
High ILUC-risk food & feed crops	- Biofuels - Bioliqids - Biomass fuels	Overall RES & RES-T	0.028% of transport energy	0%

Figure 8.16 shows the annual RES-T from 2011 to present and the share of renewable energy in transport, as it contributes to the overall RES. The RES-T increased from 7.59% in 2023 to 8.66% in 2024. The decrease in RES-T from 10.2% in 2020 to 4.65% in 2021 was due to the change in calculation methodology between RED I and RED II.

The share of renewable energy in transport, without multipliers or limits, increased from 5.70% in 2023 to 6.73% in 2024.

Figure 8.16: RES-T – final consumption in transport from renewable sources



The Renewable Transport Fuel Obligation (RTFO), formerly the Biofuel Obligation Scheme (BOS), requires fuel suppliers to include 25% (by energy) renewable transport fuel in their annual sales of transport fuels [31].

The RTFO is a certificate-based mandate that grants one certificate for each MJ of renewable transport fuel placed on the market in Ireland; two certificates are granted to fuels produced from feedstocks listed in Annex IX of RED II.

Oil companies are required to apply to NORA for certificates and to demonstrate that the quantities of renewable transport fuel for which they are claiming certificates are accurate. Since the Sustainability Regulations (SI 33 of 2012) [32] were introduced, companies are also required to demonstrate that the renewable transport fuel being placed on the market is sustainable, fulfilling the requirements of RED II.

Since the obligation was established in 2010, it has increased as follows:

- 4% by volume in 2010;
- 6% by volume in 2013;
- 8% by volume in 2017;
- 10% by volume in 2019;
- 11% by volume in 2020;
- 13% by volume in 2022;
- 16.985% by energy in 2023;
- 21% by energy in 2024; and
- 25% by energy in 2025.

Further details on the RTFO are available in NORA's RTFO Annual Reports [33].

Figure 8.17 and Table 8.8 show the final energy from renewable sources used in transport in absolute terms, without multipliers or limits applied and including biomethane supplied through the gas network. Liquid biofuels provided almost all of the renewable energy in transport, 95.2% in 2024, with renewable electricity providing 3.98% and biomethane (also referred to as bio compressed natural gas) contributing 0.861%. The biomethane quantities shown here include pure biomethane supplied directly to vehicles, without injection to the gas network, as well as biomethane that was supplied via the gas network, *i.e.* based on mass balancing and transfers of sustainability certificates.

Figure 8.17: Renewable energy in transport by energy type (without multipliers)

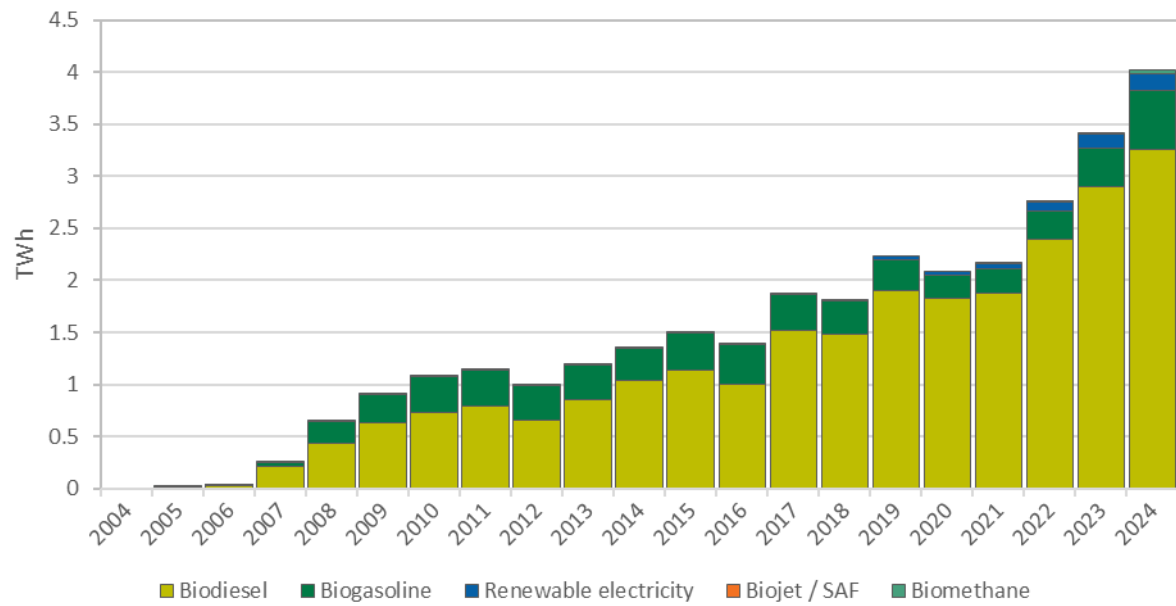


Table 8.8: Renewable energy for transport by source - without multipliers or limits (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Biodiesel	1.04 (76.6%)	1.14 (76.2%)	1.00 (71.8%)	1.52 (81.0%)	1.48 (81.5%)	1.90 (85.2%)	1.82 (87.6%)	1.87 (86.4%)	2.39 (86.4%)	2.89 (84.5%)	3.26 (81.1%)
Biogasoline	0.31 (22.8%)	0.35 (23.2%)	0.38 (27.4%)	0.34 (18.3%)	0.32 (17.5%)	0.30 (13.7%)	0.23 (10.9%)	0.24 (10.9%)	0.27 (9.8%)	0.38 (11.1%)	0.57 (14.1%)
Renewable electricity	0.01 (0.6%)	0.01 (0.6%)	0.01 (0.8%)	0.01 (0.7%)	0.02 (1.0%)	0.03 (1.2%)	0.03 (1.5%)	0.05 (2.5%)	0.09 (3.1%)	0.12 (3.6%)	0.16 (4.0%)
Biojet / SAF	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.01 (0.3%)	0.00 (0.0%)	0.00 (0.0%)
Biomethane	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.2%)	0.01 (0.4%)	0.02 (0.7%)	0.03 (0.9%)
Total	1.36 (100%)	1.50 (100%)	1.39 (100%)	1.88 (100%)	1.82 (100%)	2.23 (100%)	2.08 (100%)	2.17 (100%)	2.77 (100%)	3.42 (100%)	4.02 (100%)

8.3 Energy prices to industry

Energy use is an important part of economic activity and therefore the price paid for energy is a determining factor in the economy's competitiveness. Ireland has a high import dependence on oil and gas and is essentially a price-taker on these commodities. The EU has introduced competition into the electricity and gas markets through the liberalisation process to reduce energy costs to final consumers.

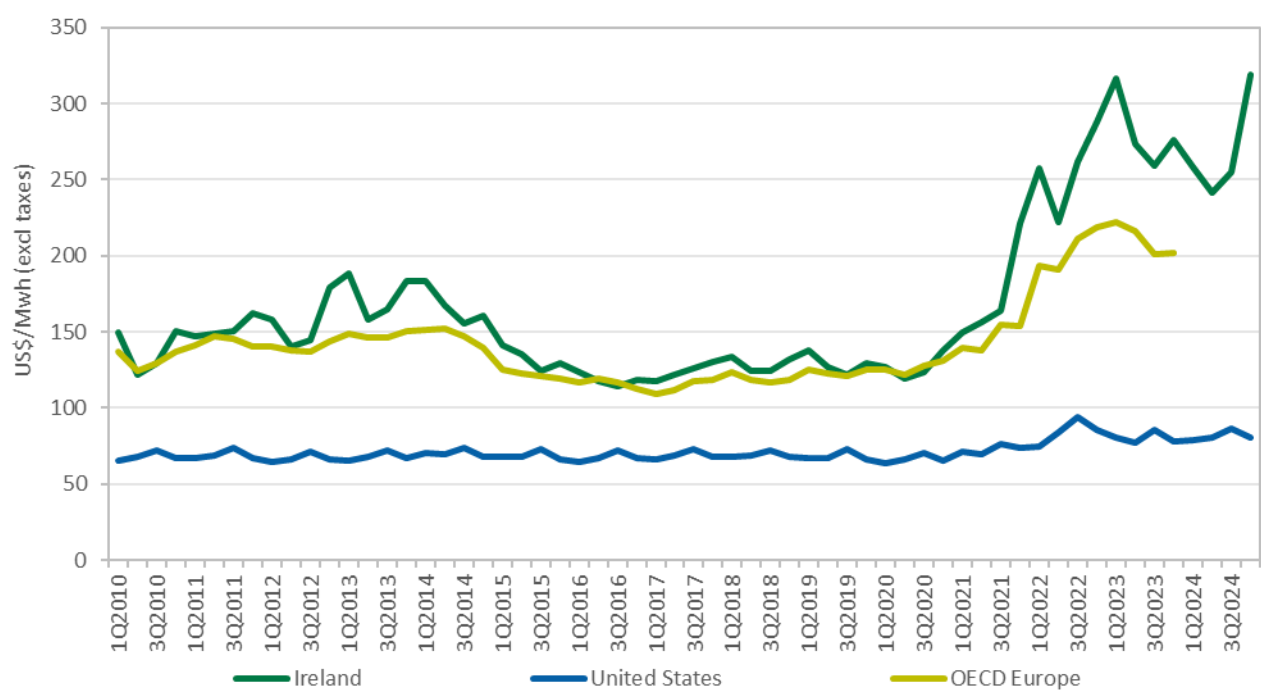
SEAI publishes annual reports based on data collected under EU legislation [34] on the transparency of gas and electricity prices. These reports focus specifically on gas and electricity prices using data published by Eurostat and are a useful reference on cost competitiveness and cover both business and households.

This section focuses on business energy prices. It presents comparisons of the cost of energy in various forms in Ireland and compares prices in OECD Europe and the US. The source of the data presented here is the International Energy Agency's (IEA) Energy Prices and Taxes [35]. This data source was chosen because it is produced quarterly and contains data to the end of 2024. Prices shown are in US dollars and are in current (nominal) money. Relative price increases since 2020, however, are tabulated for EU countries and the US in index format in both nominal and real terms.

Nominal value refers to the current value expressed in money terms in a given year, whereas real value adjusts nominal value to remove the effects of price changes and inflation, to give the constant value over time indexed to a reference year.

Figure 8.18 and Table 8.9 show that electricity prices to Irish industry from the first quarter of 2010 to the last quarter of 2024. The fuel mix for electricity generation is one factor that has a key bearing on the variation in the price of electricity. Compared to the EU, Ireland has a high overall dependency for electricity generation on fossil fuels including gas generation.

Figure 8.18: Electricity prices to industry²⁶



Source: Energy Prices and Taxes ©OECD/IEA

Table 8.9: Electricity price to industry change since 2020

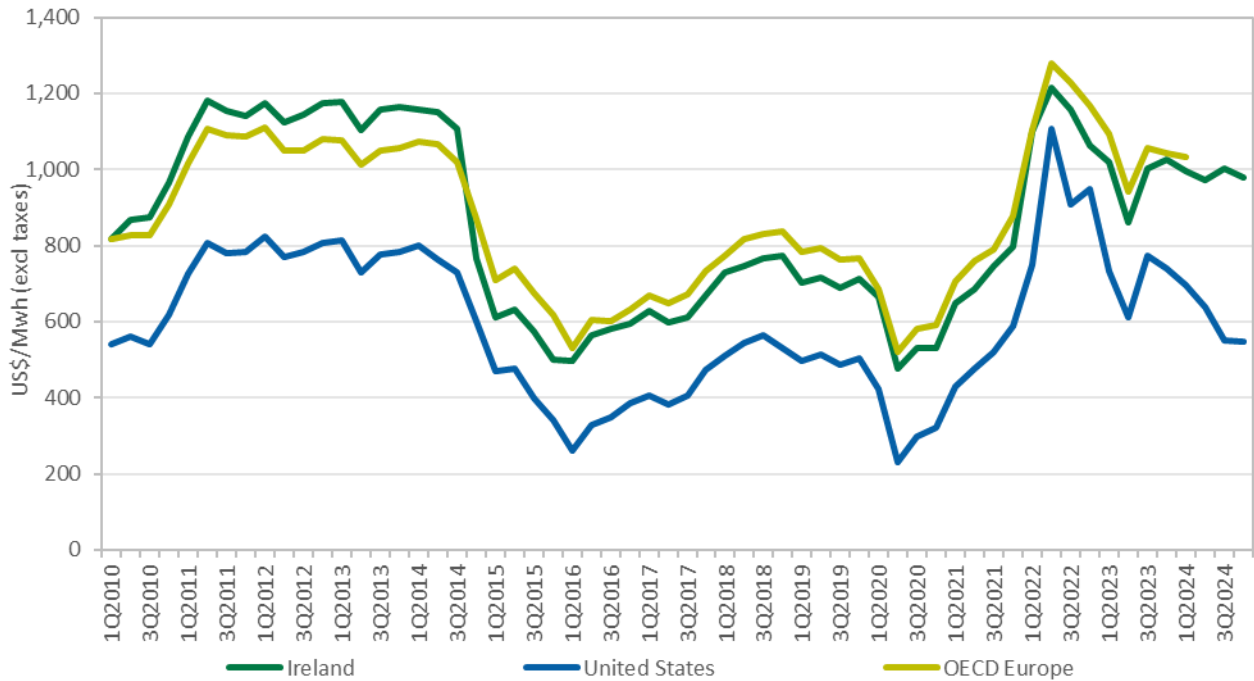
Index 2020 = 100	OECD Europe ³⁰	Austria	Belgium	Denmark	Finland	France	Germany	Greece	Ireland	Italy	Luxembourg	Netherlands	Portugal	Spain	Sweden	United Kingdom	United States
4th quarter 2024 (nominal)	n/a	209	175	177	128	197	133	179	231	157	252	234	104	145	156	213	113
4th quarter 2024 (real)	n/a	148	105	114	78	118	106	135	250	115	162	153	92	96	108	159	93

Source: Energy Prices and Taxes ©OECD/IEA

³⁰ No data available for Q1 2024 onwards for OECD Europe, data is not currently available from IEA database (27th November 2025).

Table 8.10 shows that oil prices to industry in Ireland were 33% higher in real terms in Q4 2024 than in 2020. The average oil price in Europe and the US also increased. Crude oil prices averaged around \$81/barrel in 2024, compared with \$82/barrel on average in 2023.

Figure 8.19: Oil prices to industry²⁶



Source: *Energy Prices and Taxes* ©OECD/IEA

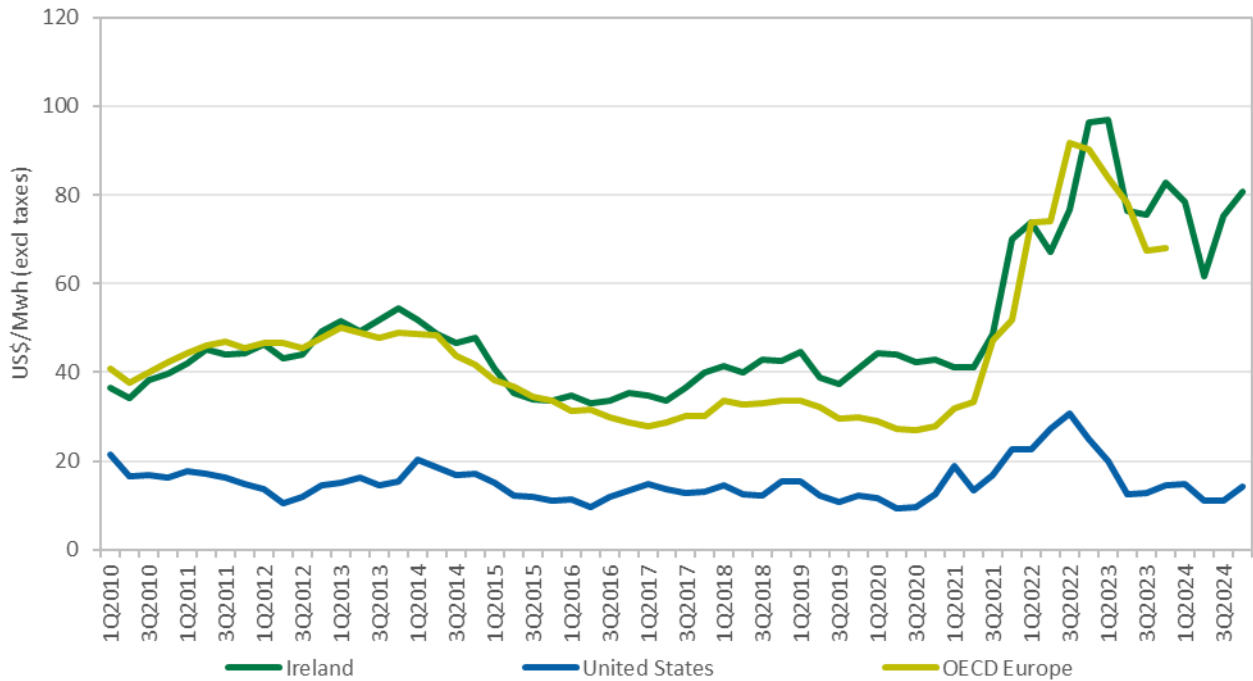
Table 8.10: Oil prices to industry change since 2020

Index 2020 = 100	OECD Europe ²⁶	Austria	Belgium	Denmark	Finland	France	Germany	Greece	Ireland	Italy	Luxembourg	Netherlands	Portugal	Spain	Sweden	United Kingdom	United States
4th quarter 2024 (nominal)	n/a	115	128	142	149	137	149	132	143	125	150	113	130	136	148	128	139
4th quarter 2024 (real)	n/a	99	91	88	114	100	112	94	133	91	106	87	101	95	111	97	113

Source: *Energy Prices and Taxes* ©OECD/IEA

Figure 8.20 and Table 8.11 show that natural gas prices to Irish industry since the first quarter of 2010. In the fourth quarter of 2024, the price of gas to industry in Ireland was 85% above 2020 levels in real terms. The figure also shows the gap between gas prices in Europe and the USA.

Figure 8.20: Natural gas prices to industry²⁶



Source: *Energy Prices and Taxes* ©OECD/IEA

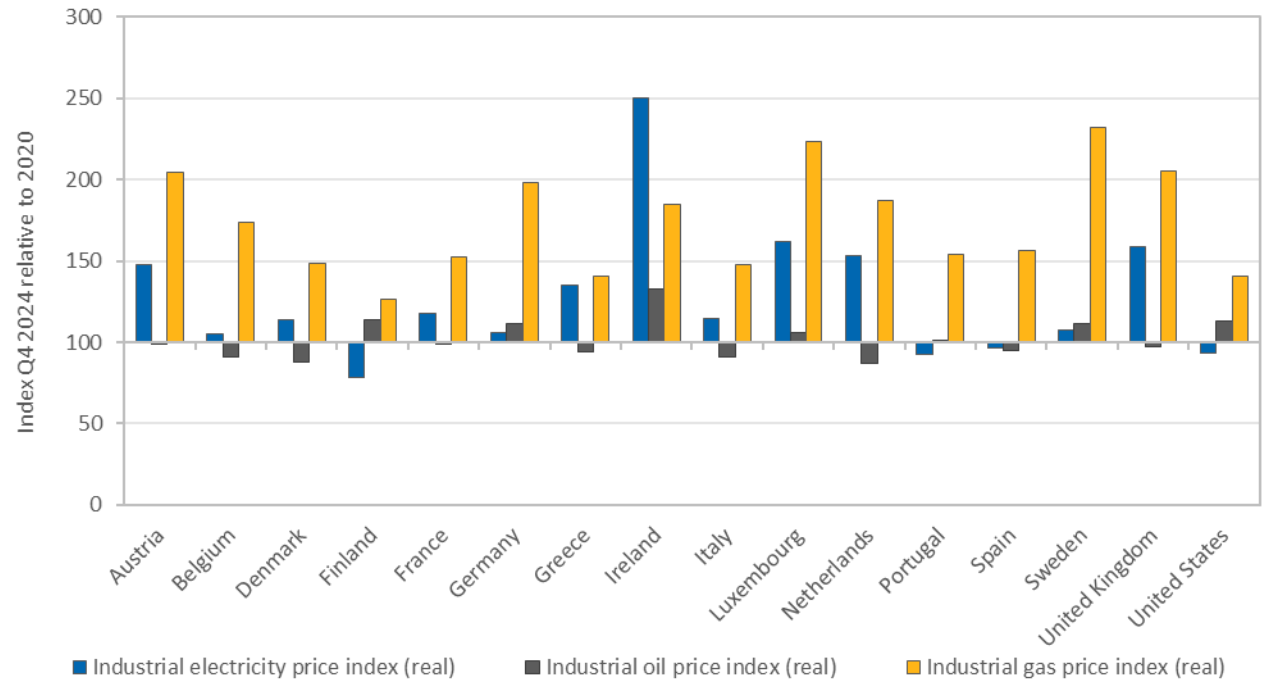
Table 8.11: Natural gas prices to industry change since 2020

Index 2020 = 100	OECD Europe ²⁶	Austria	Belgium	Denmark	Finland	France	Germany	Greece	Ireland	Italy	Luxembourg	Netherlands	Portugal	Spain	Sweden	United Kingdom	United States
4th quarter 2024 (nominal)	n/a	238	243	241	165	209	264	197	198	202	315	244	198	224	309	270	174
4th quarter 2024 (real)	n/a	205	174	149	127	153	198	140	185	148	224	187	154	156	232	206	141

Source: *Energy Prices and Taxes* ©OECD/IEA

Figure 8.21 summarises the data presented in Table 8.9, Table 8.10 and Table 8.11. The IEA publishes an overall energy price index (real) for industry, which shows that the overall energy price to Irish industry between 2020 and the fourth quarter of 2024 increased for all fuels.

Figure 8.21: Real energy price changes to industry since 2020 (index)



Source: *Energy Prices and Taxes* ©OECD/IEA

9 Drivers of energy demand

This section takes a high-level view of the trends in the economy, weather, energy use and energy-related greenhouse gas emissions since 2001.

9.1 Energy, economy and emissions

Energy supply responds to the level of demand for energy services (heating, transportation and electricity) and how end users want that energy demand satisfied. Energy service demand is driven primarily by economic activity and by the energy end-use technologies employed in undertaking such activity.

The relationship between economic activity and energy demand is less straightforward in Ireland than it is for most other countries. Gross Domestic Product (GDP) is the most widely accepted measure of economic activity internationally, but Ireland's GDP is strongly influenced by the revenue and profits reported by multinational companies. Some economic activity of these companies results in large amounts of value added (see Appendix 4), but with little energy consumption. This was illustrated clearly in 2015, when Irish GDP increased suddenly by +25% from 2014, due to the transfer of intellectual property from multinational companies. Care must be taken when comparing macro-economic indicators, such as energy per unit GDP, across countries.

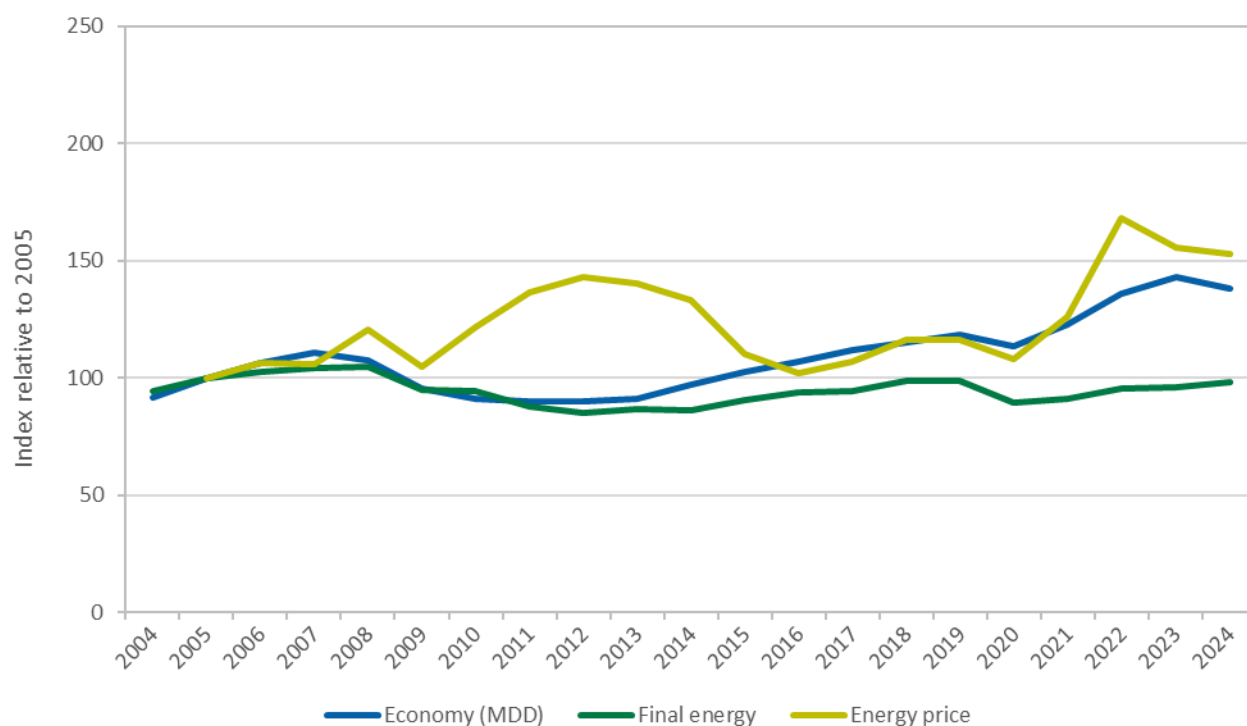
The CSO has developed alternative indicators to GDP that more accurately reflect the level of economic activity in the domestic economy, and to remove the distorting effects of globalisation. Modified domestic demand³¹ (MDD) was first published in the Quarterly National Accounts [36] results for Q1-2017 and excluded trade by aircraft leasing companies, exports and imports of R&D services, and exports and imports of R&D-related IP products. For comparison, Ireland's MDD grew by +5.3% from 2014 to 2015, vs. +25% for GDP.

Figure 9.1 shows the historical trends for MDD, energy prices and final energy, each expressed as an index relative to 2005. This figure illustrates changes in economic growth and shows the effect of the economic downturn between 2008 and 2012 (and the subsequent return to growth after 2013) and the dip in 2020 due to the impact of COVID-19 on economic activity.

Table 9.1 gives the GDP and MDD values for the economy, primary energy, final energy and energy-related CO₂eq emissions for the period.

³¹ Modified **Final** Domestic Demand, comprises of: (a) personal consumption of goods and services; (b) net expenditure by central and local government; (c) modified gross domestic fixed capital formation – excluding investment in aircraft related to leasing, and R&D service imports and trade in IP. Modified **Total** Domestic Demand also includes the value of physical changes in stocks and net acquisitions of valuables. MDD in this report refers to Modified **Total** Domestic Demand. Previous editions of this report presented another economic indicator, modified gross national income (GNI*), as an alternative to GDP. For more information on the differences between GDP, GNI* and MDD refer to the CSO.

Figure 9.1: Index of modified domestic demand, final energy demand and energy price



Source: SEAI, CSO and IEA

Table 9.1: GDP, MDD, final energy, primary energy and energy-related CO₂eq

Energy type	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
GDP [Billion €2023]	256.01	319.05	322.95	355.60	382.94	401.98	430.74	500.77	538.38	524.73	538.37
MDD [Billion €2023]	187.34	198.46	206.90	216.34	223.69	229.61	220.77	237.90	262.21	278.18	278.90
Final energy [TWh]	126.36	132.58	137.07	138.09	144.42	144.38	130.61	133.54	139.46	140.42	143.74
Primary energy [TWh]	154.07	161.83	168.84	167.55	171.24	169.93	155.61	160.69	166.80	163.86	168.04
Energy related GHG [ktCO ₂ eq]	35,194	36,860	38,371	37,026	36,750	35,134	33,055	34,888	34,202	31,352	30,880

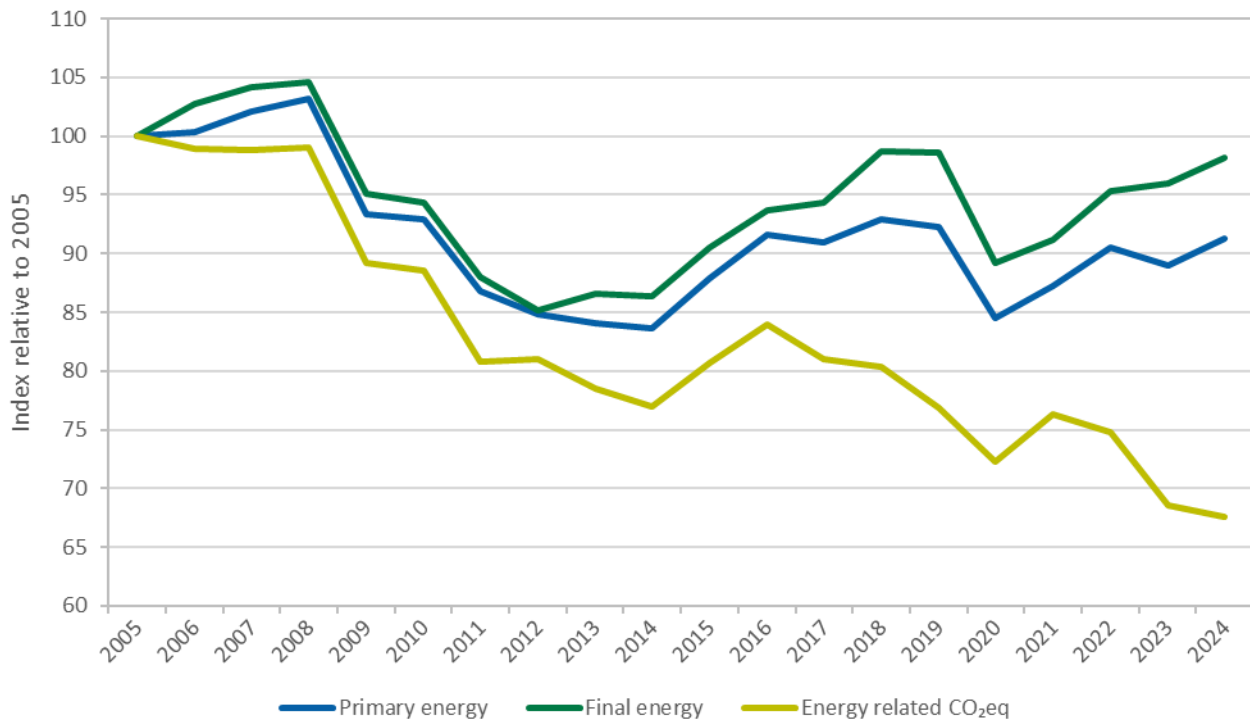
Source: CSO, EPA and SEAI

In 2020, final energy use, energy-related CO₂eq emissions and MDD fell, due largely to the impact of COVID-19 on economic activity (while GDP increased). Indicators returned to growth in 2021 and continued into 2022 and 2023 except for GDP which fell by 2.5% in 2023, driven by a contraction in multinational-dominated sectors.

Figure 9.2 shows the relationship between final energy demand, primary energy use and energy-related CO₂eq emissions, expressed as an index relative to 2005. The difference between the trends in final energy use and primary energy supply arises from improvements in the efficiency of energy transformations, particularly electricity generation.

The overall efficiency of electricity generation has increased over the period. These improvements are driven by introducing higher efficiency CCGT gas generators, reductions in inefficient coal generation, and the increased supply of wind-generated electricity (considered 100% efficient).

Figure 9.2: Index of final energy, primary energy and energy-related CO₂eq



Source: EPA and SEAI

9.2 Energy and the weather

Weather variations from year to year can have a significant effect on the energy demand of a country, particularly on the portion of the energy demand associated with space heating. A method to measure the weather, or climatic variation, is the use of 'degree days'.

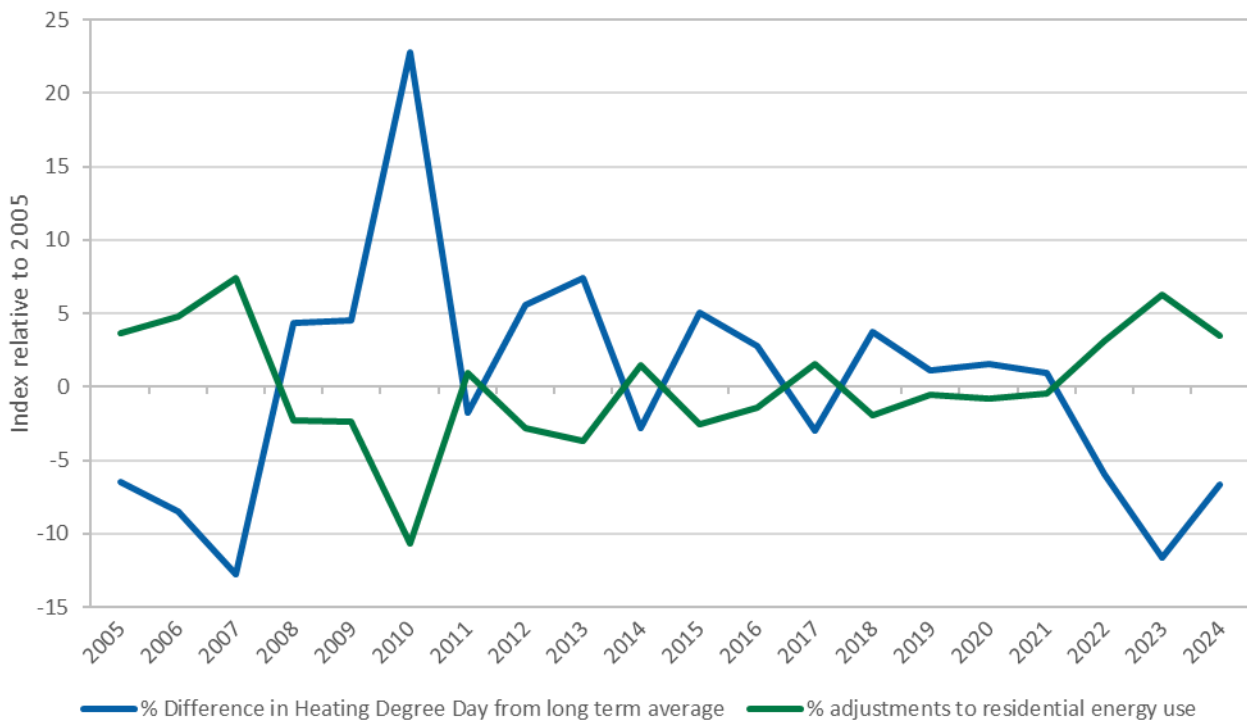
Degree days are the measure or index used to consider the severity of the weather when looking at energy use in terms of heating (or cooling) load on a building. A degree day is a measure of how cold (or warm) it is outside, relative to a day on which little or no heating (or cooling) would be required. Thus, it is a measure of the cumulative temperature deficit (or surplus) of the outdoor temperature, relative to a neutral target temperature (base temperature) at which no heating or cooling would be required. The larger the number of heating degree days, the colder the weather. For example, if the outdoor temperature for a particular day is 10 degrees lower, on average, than the base temperature (15.5 degrees), this would contribute 10 degree days to the annual or monthly total. The typical heating season in Ireland is October to May.

Met Éireann calculates degree day data for each of its synoptic weather stations. SEAI calculates a population weighted average of these data to arrive at a meaningful degree day average for Ireland that is related to the heating energy demand of the country.

Figure 9.3 shows the percentage deviation in the number of heating degree days from the long-term average between 2005 and 2024. Over that period, 2010 was the coldest year recorded and 2007 was the warmest. The portion of each fuel assumed to be used for heating is adjusted by multiplying it by the ratio of the long-term average number of degree days to the number of degree days in the given

year. This adjustment yields a lower normalised energy consumption in cold years, and a higher normalised consumption in mild years. Typically, the weather adjustment is within $\pm 6\%$ of the actual energy consumption. The largest correction over the period was for 2010, an exceptionally cold year, where the weather-corrected energy consumption was over 10% less than the actual energy consumption.

Figure 9.3: Deviation from average heating degree days (2005 to 2024) and resulting weather adjustment



Source: Met Éireann and SEAI

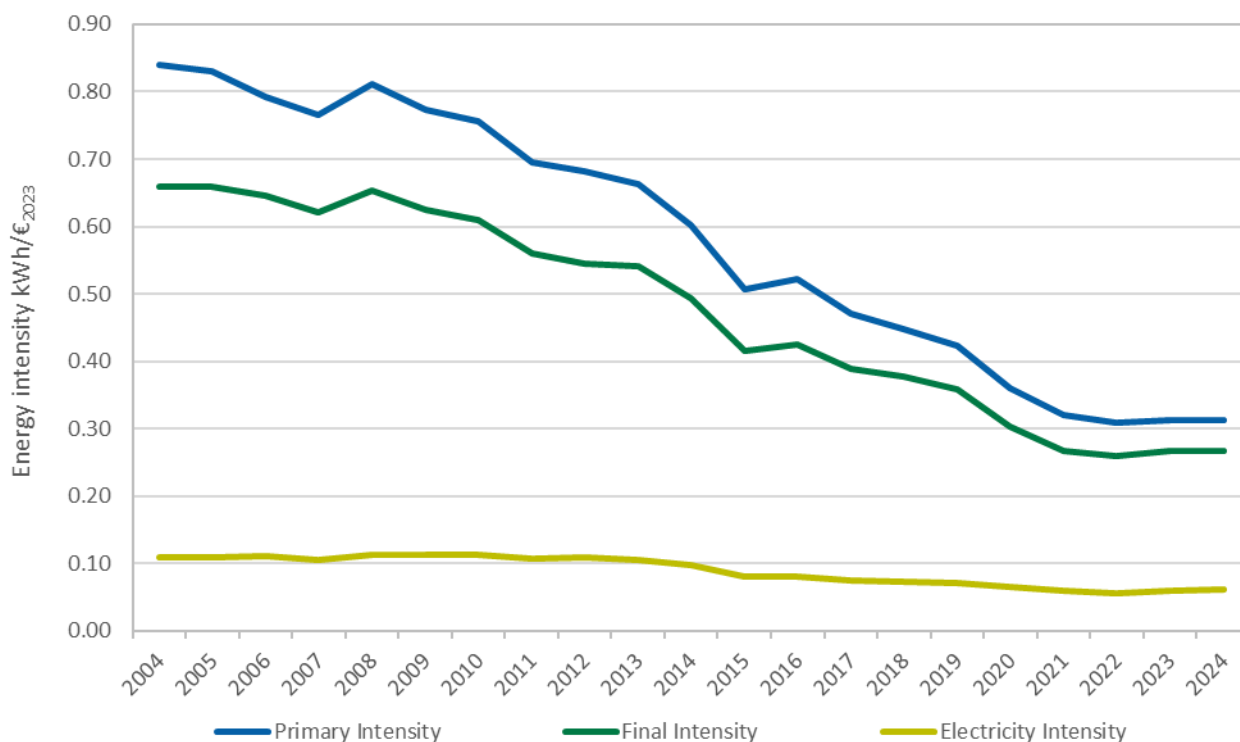
9.3 Economic energy intensities

Energy intensity is defined as the amount of energy required to produce a functional output. For the economy, the measure of output is generally taken to be the GDP. As mentioned in section 9.1, MDD is a more meaningful indicator of economic activity in Ireland, but GDP is still the standard international metric; therefore, here we present energy intensity in terms of GDP. We use GDP measured in constant prices to remove the influence of inflation.

Figure 9.4 shows the trend in primary energy intensity (primary energy divided by GDP) and final energy intensity (final energy consumption divided by GDP) (at constant 2023 prices). The difference between these two trends reflects the amount of energy lost in the transformation of primary energy into final energy, mostly for electricity generation. The primary and final energy intensity of the economy has generally fallen since 2002, except in 2008 and 2016, which can be attributed to a combination of increased energy efficiency and increases in GDP. The graph also shows electricity intensity (final electricity consumption divided by GDP). In 2024 GDP grew by 2.6%.

The sharp fall in the energy intensity of the economy in 2015 of 16% must be understood in the context of the 25% increase in GDP, which resulted from transferring assets into Ireland and had little or no effect on energy consumption. This change should be viewed as an adjustment rather than a reduction in intensity. This is a good example of why energy intensity is not a good measure of energy efficiency progress, especially in Ireland.

Figure 9.4: Primary, final and electricity intensities



Source: SEAI

There are many factors that contribute to how trends in energy intensity of the economy evolve. These factors include technological efficiency and the fuel mix, particularly in relation to electricity generation; economies of scale in manufacturing; and, not least, the structure of the economy. The structure of the economy in Ireland has changed considerably over the past 20 to 30 years. It has shifted toward the high value-added sectors, such as pharmaceuticals, electronics and services. Relative to traditional 'heavier' industries, such as car manufacturing and steel production, these growing sectors are not highly energy intensive. Examples of changes to the structure of the industry sector include the cessation of steel production in 2001, of fertiliser production in late 2002, and of sugar production in 2007.

The energy intensity of the economy will continue to decrease if, as expected, the economy becomes increasingly dominated by high value added, low energy-consuming sectors. This results in a more productive economy from an energy perspective but does not necessarily mean that the actual processes used are more energy efficient, or that less energy is being used overall in the economy.

10 Further sectoral analysis

This section explores in more detail the changes in energy use in each of the main sectors: industry, transport, residential and services.

10.1 Primary energy requirement by sector

Figure 10.1 shows how Ireland's primary energy supply ultimately services the energy needs of different sectors of the economy. Where primary energy is used directly by end users in a particular sector, then allocation is straightforward. An example is the use of natural gas in the residential sector. Where fuels undergo a transformation process before final use by an end-user, then the full primary energy required to satisfy that final use is allocated to the sector. For example, for the electricity used in the residential sector, the fuels used to generate that electricity (gas, wind, coal, peat and oil, *etc.*) are allocated to the residential sector.

Figure 10.1: Total primary energy requirement by sector

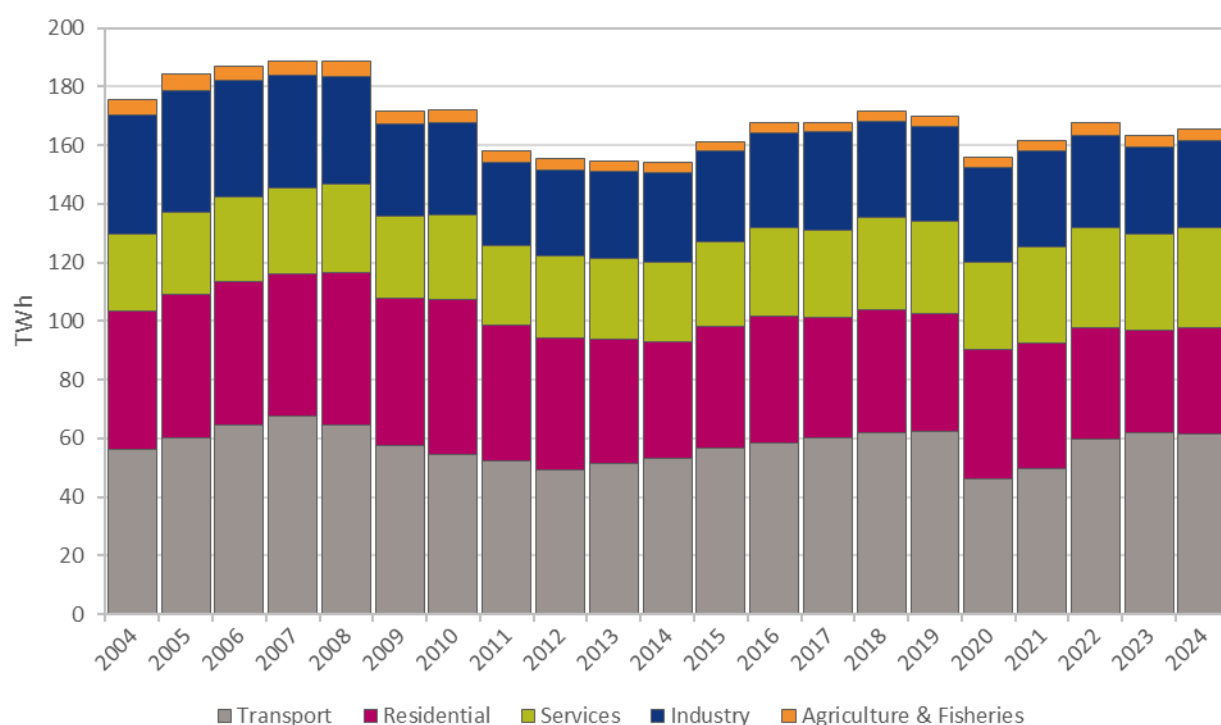


Table 10.1 details the quantities, shares and trends of primary energy supply across economic sectors. The total primary supply is split across the transport, residential, services, industry, and agriculture and fisheries sectors. Sectoral energy-related CO₂eq emissions are presented in section 7.2.

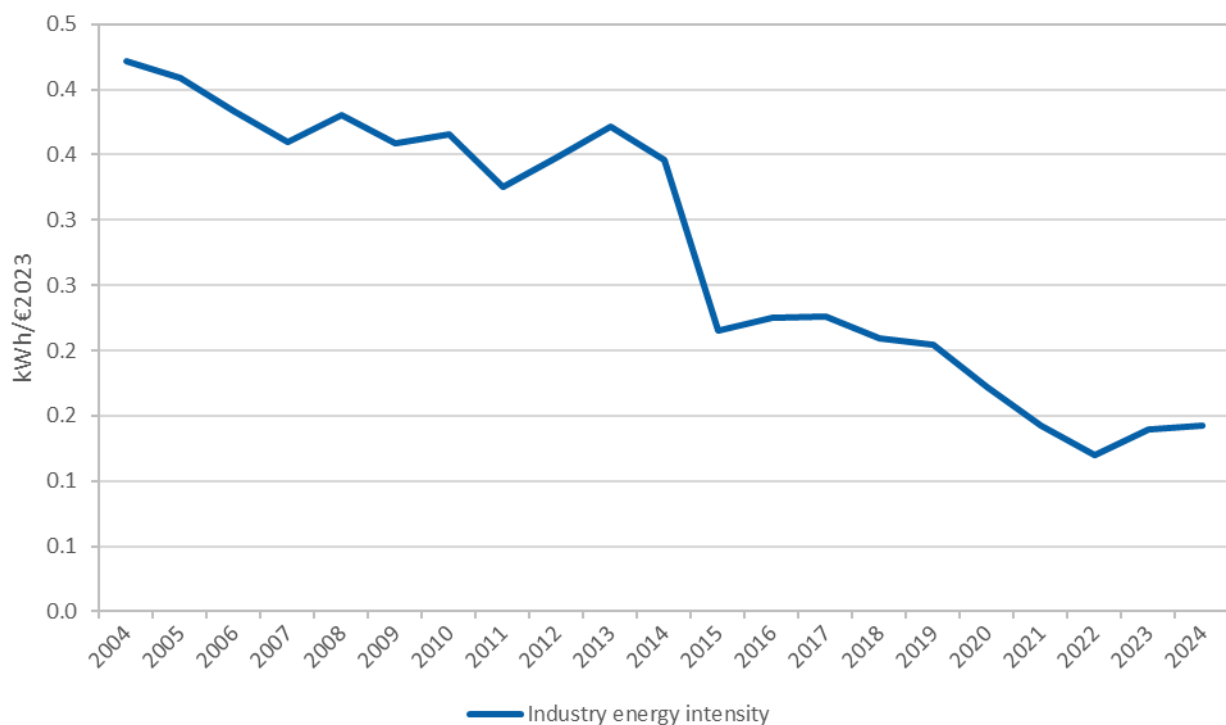
Table 10.1: Primary energy by sector compared with previous years (share)

Energy [TWh]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Transport	53.15 (34.5%)	56.55 (35.1%)	58.52 (34.9%)	60.23 (35.9%)	61.75 (36.0%)	62.12 (36.6%)	46.31 (29.7%)	49.66 (30.7%)	59.76 (35.6%)	62.03 (38.0%)	61.62 (37.3%)
Residential	39.60 (25.7%)	41.52 (25.8%)	42.98 (25.7%)	41.00 (24.4%)	42.21 (24.6%)	40.63 (23.9%)	44.09 (28.3%)	42.91 (26.6%)	37.98 (22.6%)	34.86 (21.4%)	36.12 (21.8%)
Services	27.35 (17.8%)	28.94 (17.9%)	30.22 (18.0%)	29.88 (17.8%)	31.43 (18.3%)	31.37 (18.5%)	29.69 (19.0%)	32.68 (20.2%)	34.29 (20.4%)	32.93 (20.2%)	33.98 (20.5%)
Industry	30.62 (19.9%)	31.05 (19.3%)	32.54 (19.4%)	33.39 (19.9%)	32.74 (19.1%)	32.39 (19.1%)	32.37 (20.7%)	32.78 (20.3%)	31.44 (18.8%)	29.74 (18.2%)	30.04 (18.2%)
Agriculture & Fisheries	3.27 (2.1%)	3.17 (2.0%)	3.27 (2.0%)	3.36 (2.0%)	3.46 (2.0%)	3.42 (2.0%)	3.56 (2.3%)	3.53 (2.2%)	4.22 (2.5%)	3.60 (2.2%)	3.63 (2.2%)
Total	153.99 (100%)	161.22 (100%)	167.52 (100%)	167.86 (100%)	171.60 (100%)	169.92 (100%)	156.02 (100%)	161.58 (100%)	167.68 (100%)	163.17 (100%)	165.38 (100%)

10.2 Industry energy intensity

Industrial energy intensity is the amount of energy required to produce a unit of value added, measured in constant money values. Figure 10.2 shows the industrial energy intensity over the period expressed in kilograms of oil equivalent per euro of industrial value added at 2023 money value (kWh/€2023). Over the period, industrial energy consumption fell, while value added increased, resulting in a reduction in intensity.

Figure 10.2: Industry energy intensity



Value-added output from industry was 74% higher in 2015 compared with 2014. The large increase in gross value added in 2015 is explained by several one-off factors, such as transferring assets into Ireland, and what are known as reverse takeovers. This increase in gross value added incurred no additional energy consumption.

The step-change in industry energy intensity in 2015 illustrates the fact that energy intensity is not a good indicator of energy efficiency, as variations may result from many factors, such as structural changes, or changes to the fuel mix or activity.

10.3 Transport

10.3.1 Private car activity

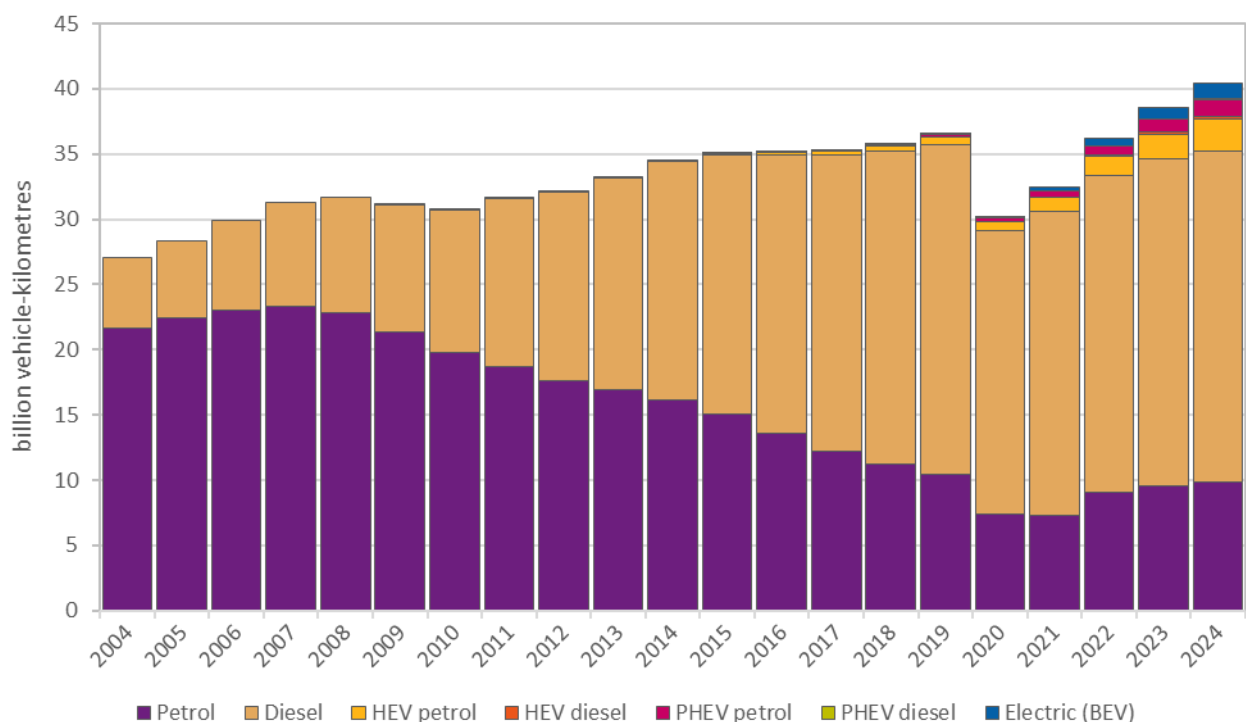
Figure 10.3 shows the total kilometres driven by private cars in Ireland each year. This is based on an analysis of NCT data for all years, except 2020 and 2021. The NCT methodology assumes that the kilometres driven between the last two dates in a car which has had an NCT are split evenly across that time periods, which can be 4, 2 or 1 years. In normal times, this is a reasonable assumption, but if there is a sudden change the activity pattern and the functioning of the NCT is disrupted, both of which occurred during COVID-19, then this method no longer provides good results. Therefore, SEAI developed an alternative approach for 2020 and 2021; this estimated the reduction in activity of the average petrol car in these years from the observed drop in petrol use (most of which occurs in private cars). It was then assumed that the average diesel car's activities were reduced by the same proportion as the average petrol car. From 2022, we returned to the NCT methodology for estimating vehicle kilometres.

NCT data was unavailable to SEAI for 2024 and so the average kilometres travelled per vehicle type from 2023 was used with updated vehicle stock number for 2024.

The total number of vehicle-kilometres travelled declined following the economic crash (during 2009 and 2010) but returned to growth soon after in 2011. Total vehicle-kilometres continued to grow until the dramatic fall in 2020 due to travel restrictions during COVID-19.

Annual vehicle-kilometres for electric vehicles (EVs) have been estimated since 2009 and have grown every year since. Hybrid electric vehicles (HEV) and Plug-in hybrid electric vehicles (PHEV) have been estimated since 2015.

Figure 10.3: Private car total annual vehicle-kilometres



Source: Based on data from NCT and Dept. Transport, Vehicle Registration Unit

10.3.2 CO₂ intensity of new private cars

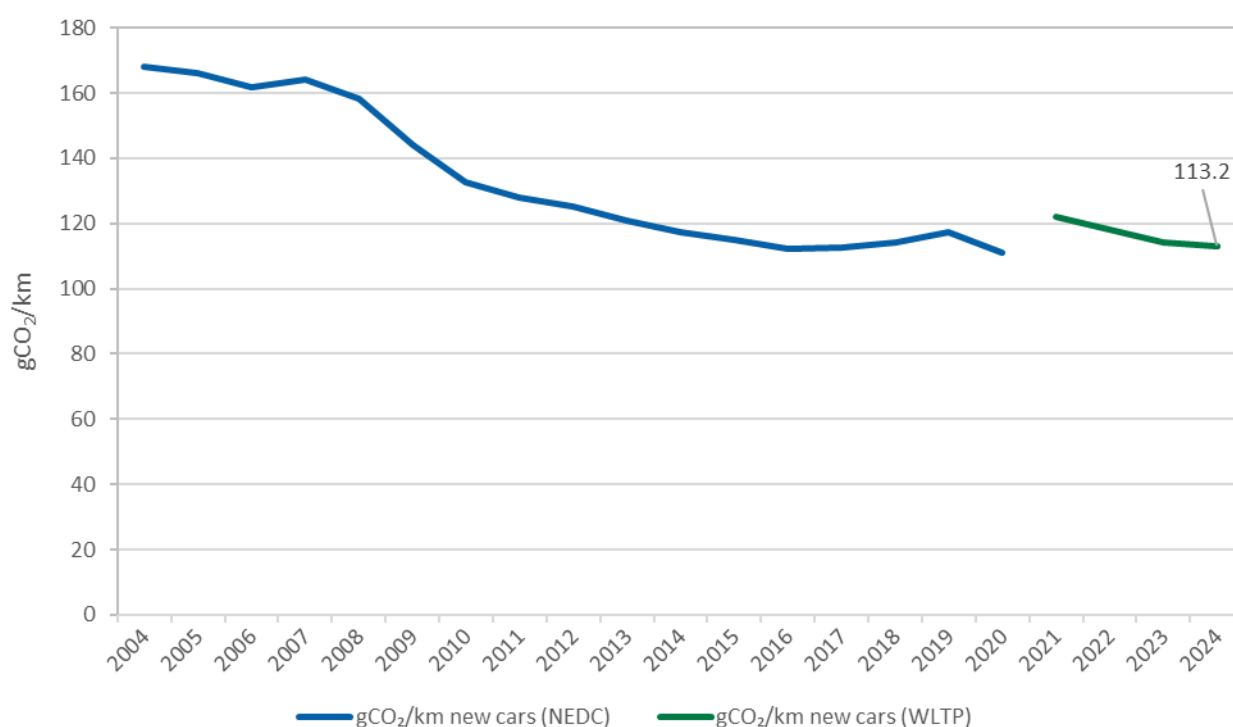
Figure 10.4 shows the change in the weighted average specific CO₂ emissions of new cars licensed for the first time (excluding battery EVs) over the period, according to standardised testing procedures. This does not include imported, second-hand cars.

Since 2008, the combined effect of the EU legislation obligating manufacturers to reduce average fleet emissions and the changes to the Irish taxation system for private cars has been to shift new car purchases from higher to lower CO₂ emissions bands, and to reduce the average specific CO₂ emissions of new cars.

The standardised testing procedures are known to underestimate the fuel use and CO₂ emissions of cars, compared to typical real-world driving conditions. The difference between the test emissions and the emissions produced in real-world driving conditions is referred to as the on-road factor, or the performance gap. Several reports by the International Council on Clean Transportation highlighted that the performance gap between test results and real-world driving increased dramatically after 2008, and that the real-world fuel consumption and carbon emissions of new vehicles are increasingly higher than the reported values under standardised testing procedures.

The data up to 2020 is based on the results of a standardised laboratory test procedure called the New European Driving Cycle (NEDC). From January 2021, a new test methodology called the Worldwide Harmonised Light Vehicle Test Procedure (WLTP) came into force for all new cars to better reflect real-world driving profiles.

Figure 10.4: Specific CO₂ emissions of new cars, excluding battery electric vehicles

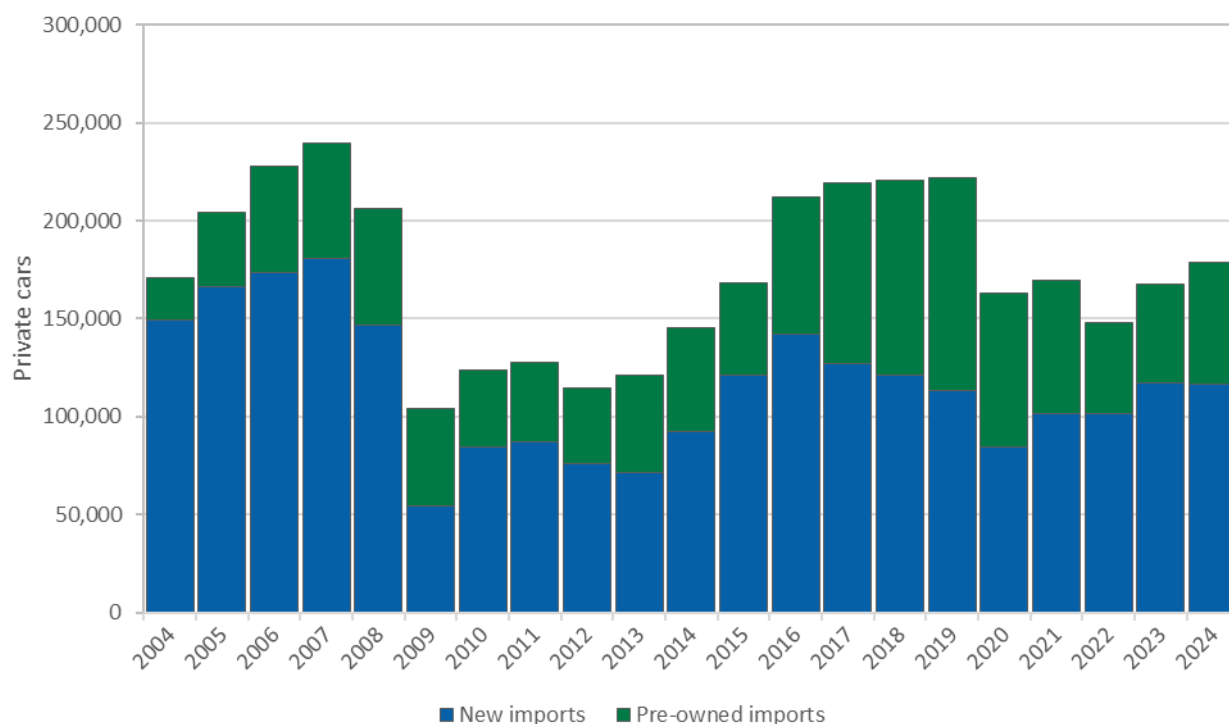


Source: Based on data from Dept. Transport, Vehicle Registration Unit

10.3.3 Penetration of zero emissions vehicles

There are two sources of cars for the Irish market: brand new imports and pre-owned imports, shown in Figure 10.5. The importance of the pre-owned imports market varies over time, but in the lead up to the UK leaving the EU, its share had increased until 2019 and has fallen since then. This is important as the profile of pre-owned imports tends to differ to that of new car imports.

Figure 10.5: Share of private cars licensed for the first time that are new or pre-owned imports

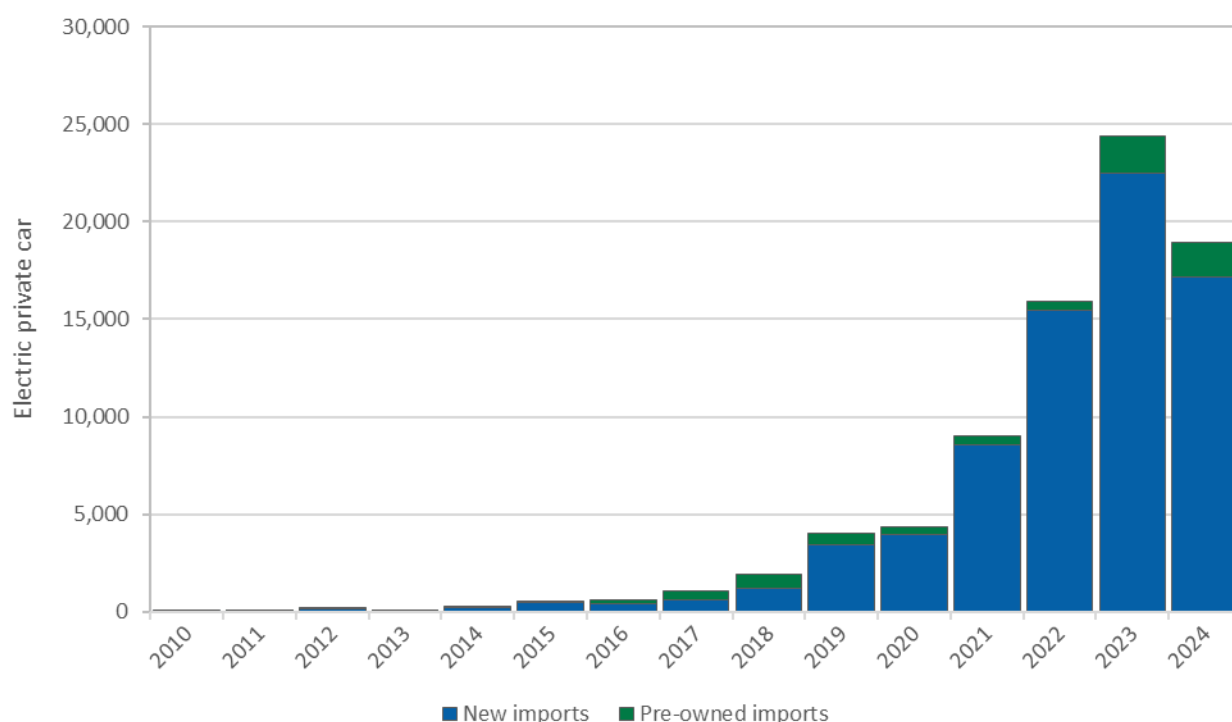


Source: CSO and Dept. Transport, Vehicle Registration Unit

Figure 10.6 and Table 10.2 show the share of private cars added to the Irish car stock each year that are EVs, split into new imports and pre-owned imports.

This is showing strong growth, but still from a low base. As 85% of all vehicles licensed for the first time in 2024 have an internal combustion engine and given that the typical lifespan of a car is around 15 years, it will be well into the next decade before there is a significant phasing out of cars with internal combustion engines.

Figure 10.6: Private cars licensed for the first time that are battery electric vehicles



Source: CSO and Dept. Transport, Vehicle Registration Unit.

Table 10.2: Electric private cars licensed for the first time (share)

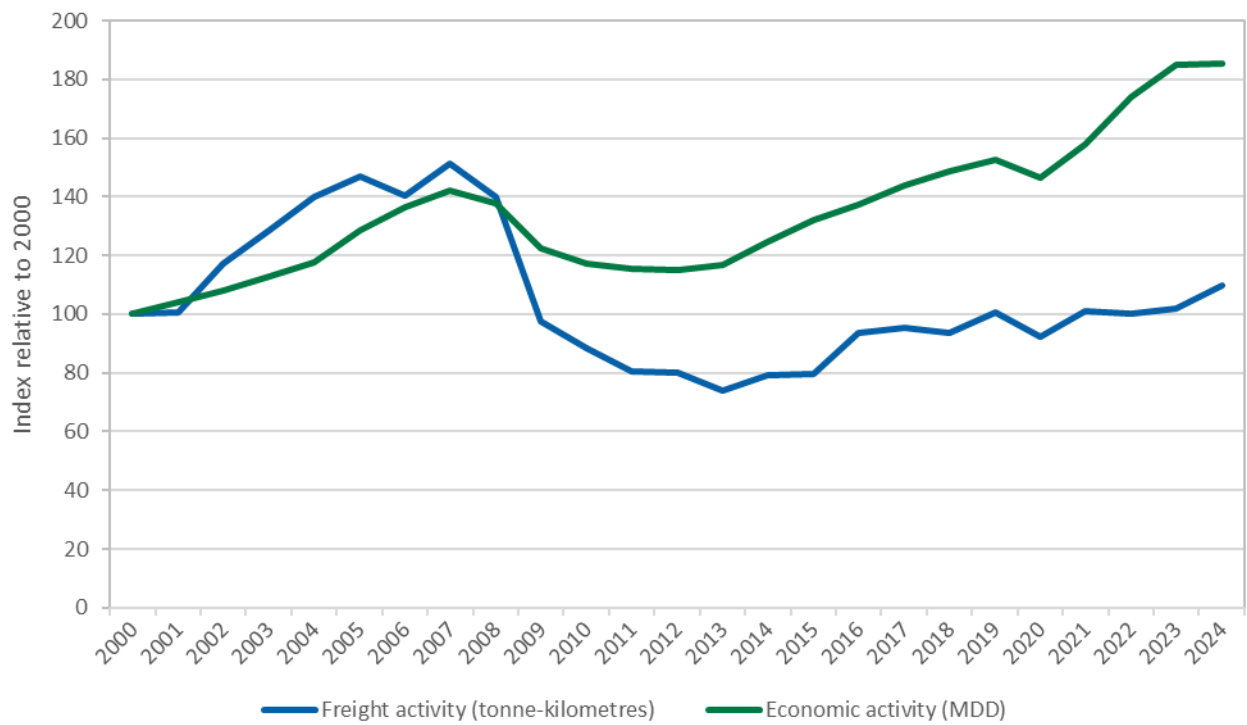
Private Cars	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
New imports	222 (81.3%)	476 (83.4%)	392 (64.5%)	623 (57.1%)	1,222 (63.6%)	3,443 (84.9%)	3,940 (90.2%)	8,554 (94.9%)	15,462 (97.3%)	22,493 (92.3%)	17,191 (90.9%)
Pre-owned imports	51 (18.7%)	95 (16.6%)	216 (35.5%)	469 (42.9%)	700 (36.4%)	611 (15.1%)	428 (9.8%)	455 (5.1%)	435 (2.7%)	1,865 (7.7%)	1,714 (9.1%)
Total electric cars licensed for the first time	273 (100%)	571 (100%)	608 (100%)	1,092 (100%)	1,922 (100%)	4,054 (100%)	4,368 (100%)	9,009 (100%)	15,897 (100%)	24,358 (100%)	18,905 (100%)

10.3.4 Heavy goods vehicle activity

The main metric used to measure activity in the road freight sector is tonne-kilometres. This is the total weight of material transported, multiplied by the distance over which it is transported. Figure 10.7 and Table 10.3 present data on road freight tonne-kilometres, along with data on economic growth, as measured by MDD. In Figure 10.7, the data are presented as an index with respect to 2000. The data are taken from the CSO's Road Freight Transport Survey, [37] which considers, for example, vehicles taxed as goods vehicles, those weighing over two tonnes unladen and those which are actually used as goods vehicles, rather than for service-type work. We estimate the energy use of HGVs based on the activity, as measured by tonne-kilometres, and the energy consumption per tonne-kilometre, based on the EU average.

Although HGV activity was less affected by COVID-19 travel restrictions than private cars or aviation, the number of tonne-kilometres still fell in 2020. This was nearly twice the reduction seen in total economic activity, as measured by MDD. HGV activity returned to growth in 2021.

Figure 10.7: Road freight activity



Source: CSO

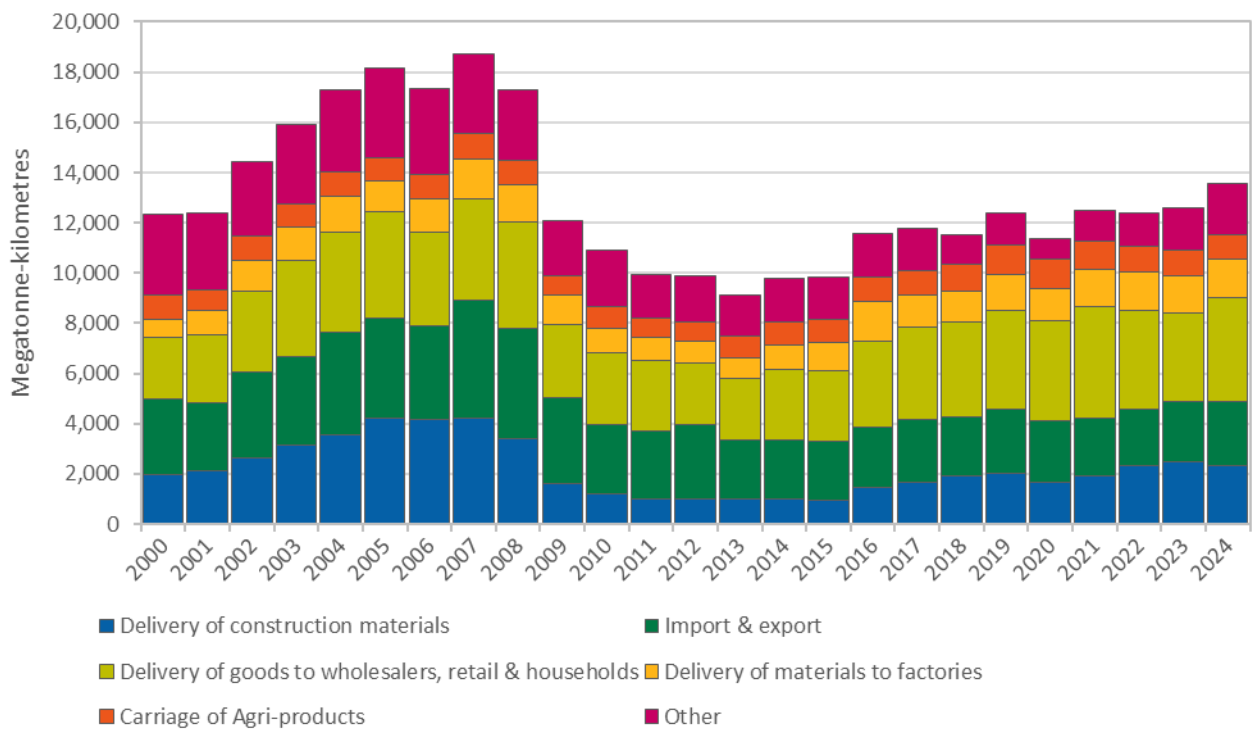
Table 10.3: Road freight activity

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Mega tonne-kilometres	9,772	9,844	11,564	11,759	11,537	12,403	11,383	12,485	12,384	12,596	13,575
MDD [billion €2023]	187.3	198.5	206.9	216.3	223.7	229.6	220.8	237.9	262.2	278.2	278.9

Source: CSO

The CSO also provides data on HGV activity, classed by main type of work done. Figure 10.8 shows the trends for tonne-kilometres in each category between 2000 and 2024.

Figure 10.8: Road freight activity by main type of work done



Source: CSO

Between 2007 and 2013, the category 'Delivery of construction materials' experienced both the largest absolute decrease and the largest percentage decrease (77%). It was responsible for the largest share of the total reduction in activity from 2007 to 2013, accounting for 34%. This corresponded to the collapse of activity in the construction sector during this period.

Despite the recovery of the economy between 2012 and 2019, the HGV activity in most categories did not recover to 2007 levels. HGV activity fell by 8.2% in 2020 due to the COVID-19 pandemic, returning to 2019 levels in 2021. In 2024, HGV activity increased by 7.8%, driven by increases in the delivery of goods and other activities.

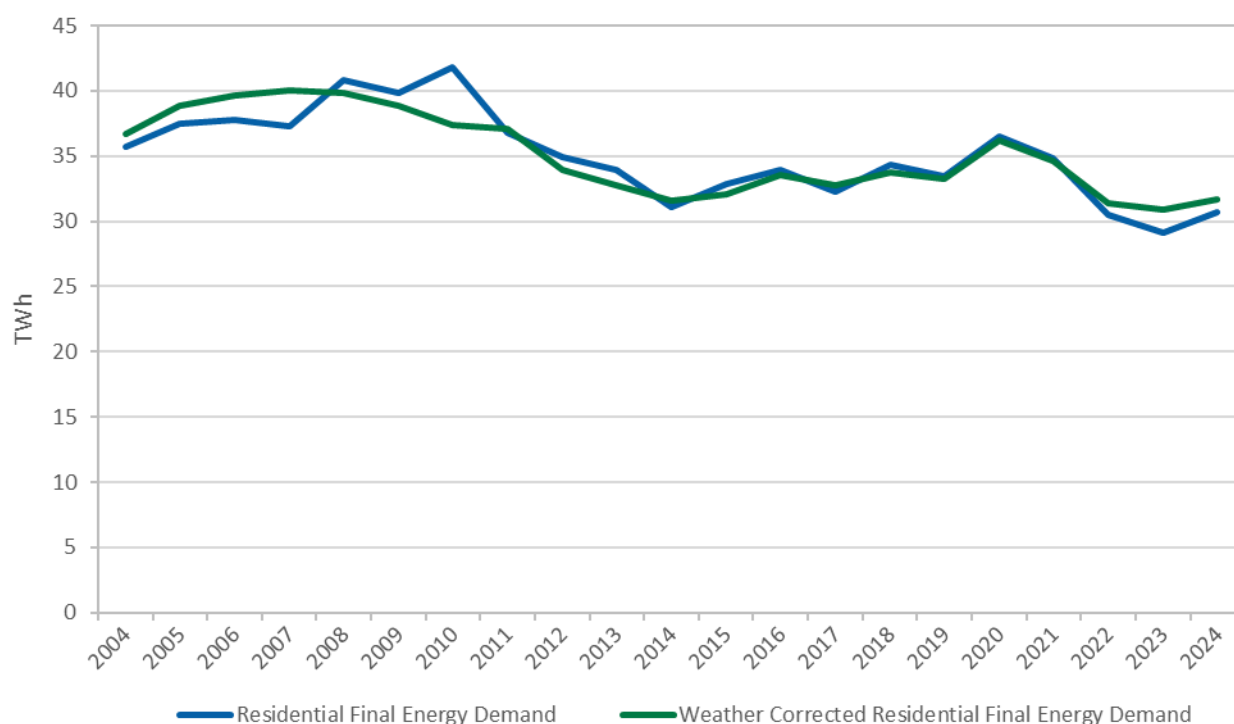
10.4 Residential

10.4.1 Weather correction

Figure 10.9 shows the trend for the residential sector's final energy consumption between 2002 and 2023, with and without weather correction. Weather correction yields a lower normalised energy consumption in cold years (e.g. 2010), and a higher normalised consumption in mild years (e.g. 2023 and 2024).

Annual variations in weather affect the space heating requirements of occupied buildings. Weather correction involves adjusting the energy used for space heating by benchmarking the weather in a particular year with that of a long-term average, measured in terms of numbers of degree days.

Figure 10.9: Residential final energy

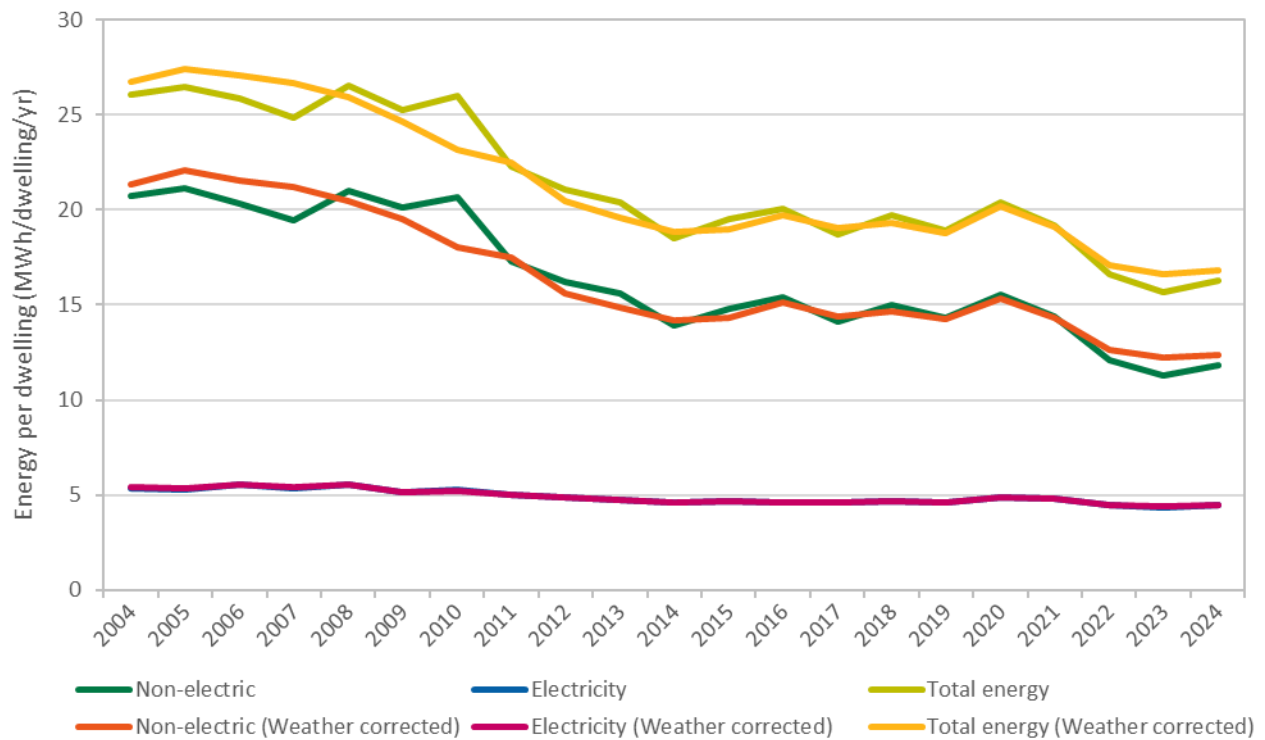


Source: Based on SEAI and Met Éireann data

10.4.2 Energy consumption per dwelling

Figure 10.10 shows the trend in final energy consumption per dwelling with and without weather correction.

Figure 10.10: Energy per dwelling (permanently occupied)



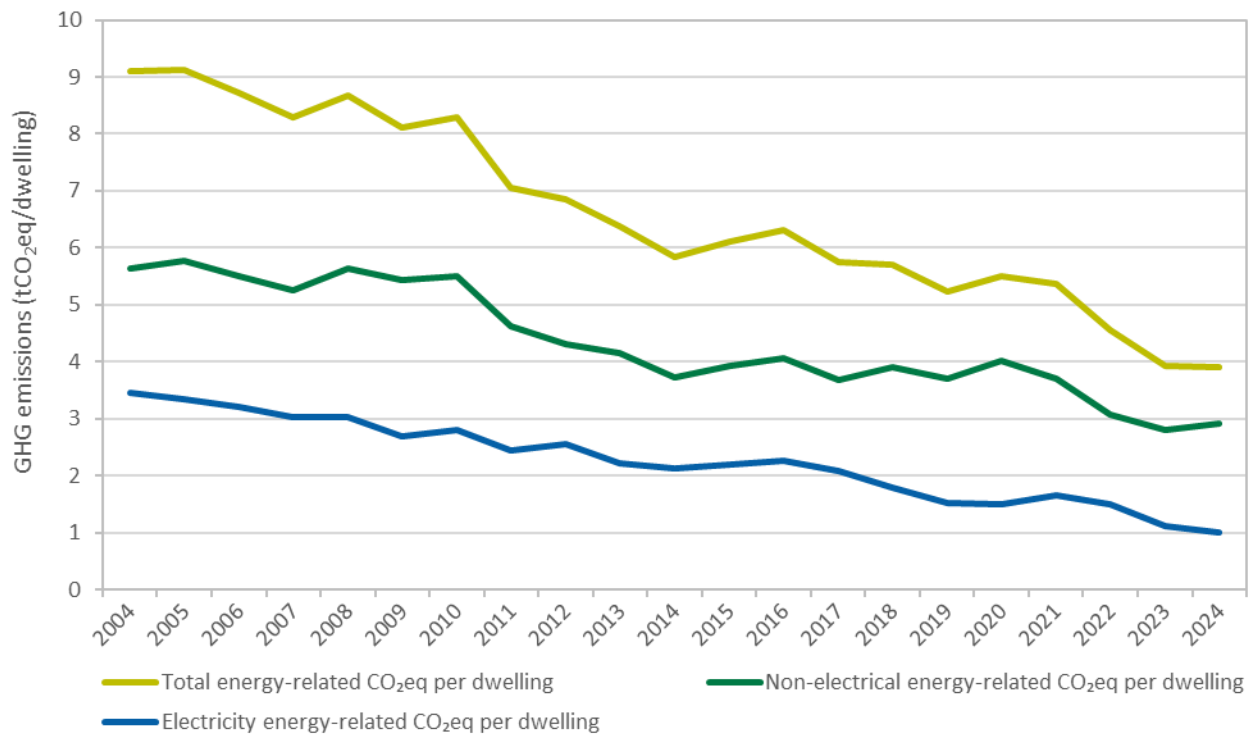
Source: Based on SEAI, CSO and Met Éireann data

Table 10.4: Quantities of energy consumption and CO₂eq emissions per dwelling (share)

Energy [kWh/dw]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Non-electric	13,943 (75.2%)	14,821 (76.0%)	15,384 (76.8%)	14,115 (75.3%)	15,028 (76.2%)	14,326 (75.7%)	15,510 (76.2%)	14,371 (74.9%)	12,121 (73.1%)	11,299 (72.2%)	11,835 (72.7%)
Electricity	4,590 (24.8%)	4,669 (24.0%)	4,637 (23.2%)	4,621 (24.7%)	4,681 (23.8%)	4,587 (24.3%)	4,850 (23.8%)	4,820 (25.1%)	4,468 (26.9%)	4,357 (27.8%)	4,445 (27.3%)
Total energy	18,533 (100%)	19,490 (100%)	20,021 (100%)	18,737 (100%)	19,709 (100%)	18,913 (100%)	20,359 (100%)	19,191 (100%)	16,589 (100%)	15,656 (100%)	16,280 (100%)
Non-electric (weather corrected)	14,204 (75.5%)	14,353 (75.5%)	15,110 (76.6%)	14,397 (75.6%)	14,668 (75.9%)	14,222 (75.6%)	15,355 (76.0%)	14,287 (74.8%)	12,610 (73.7%)	12,228 (73.5%)	12,372 (73.4%)
Electricity (weather corrected)	4,603 (24.5%)	4,645 (24.5%)	4,625 (23.4%)	4,635 (24.4%)	4,663 (24.1%)	4,582 (24.4%)	4,842 (24.0%)	4,816 (25.2%)	4,495 (26.3%)	4,409 (26.5%)	4,475 (26.6%)
Total energy (weather corrected)	18,807 (100%)	18,998 (100%)	19,735 (100%)	19,032 (100%)	19,332 (100%)	18,804 (100%)	20,197 (100%)	19,103 (100%)	17,105 (100%)	16,636 (100%)	16,847 (100%)
GHG [tCO ₂ eq/dw]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Non-electrical energy-related GHG	3.7 (63.7%)	3.9 (64.3%)	4.1 (64.3%)	3.7 (63.9%)	3.9 (68.6%)	3.7 (70.9%)	4.0 (72.9%)	3.7 (69.0%)	3.1 (67.3%)	2.8 (71.8%)	2.9 (74.4%)
Electricity energy-related GHG	2.1 (36.3%)	2.2 (35.7%)	2.3 (35.7%)	2.1 (36.1%)	1.8 (31.4%)	1.5 (29.1%)	1.5 (27.1%)	1.7 (31.0%)	1.5 (32.7%)	1.1 (28.2%)	1.0 (25.6%)
Total energy-related GHG	5.8 (100%)	6.1 (100%)	6.3 (100%)	5.8 (100%)	5.7 (100%)	5.2 (100%)	5.5 (100%)	5.4 (100%)	4.6 (100%)	3.9 (100%)	3.9 (100%)

Source: Based on SEAI, CSO and Met Éireann data

Figure 10.11: Unit energy-related greenhouse gas emissions per dwelling



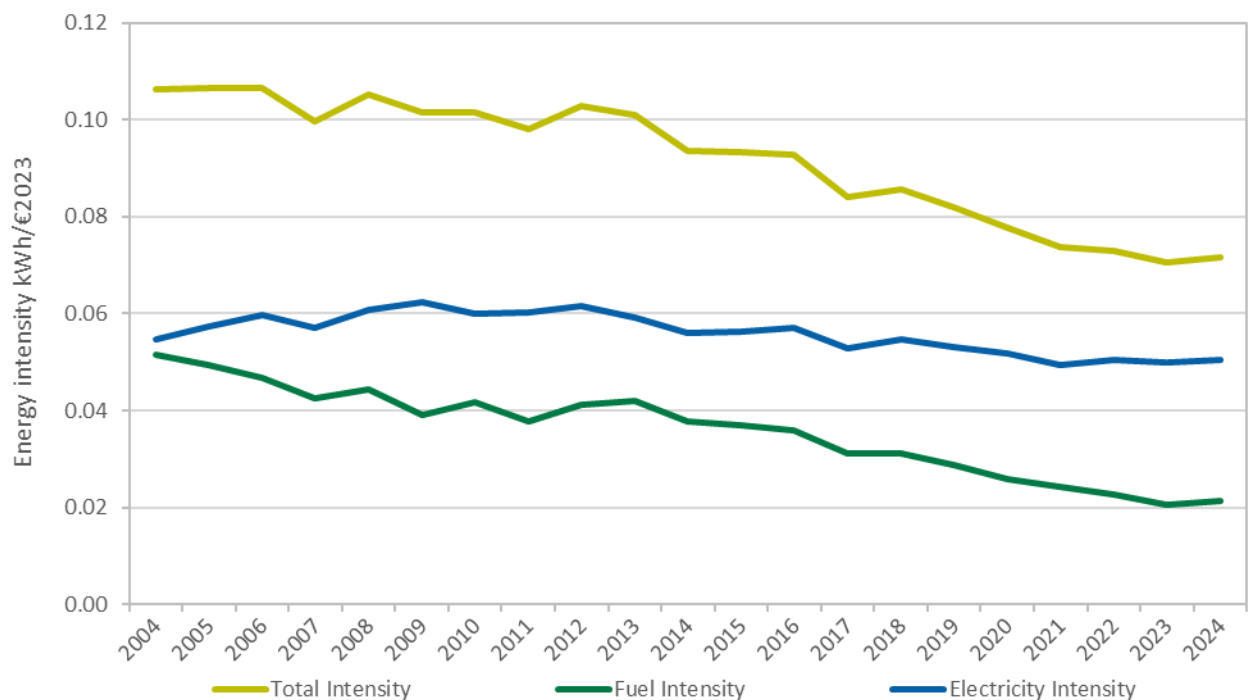
Source: Based on SEAI and CSO data

10.5 Commercial and public services

10.5.1 Energy intensity of the commercial and public services sector

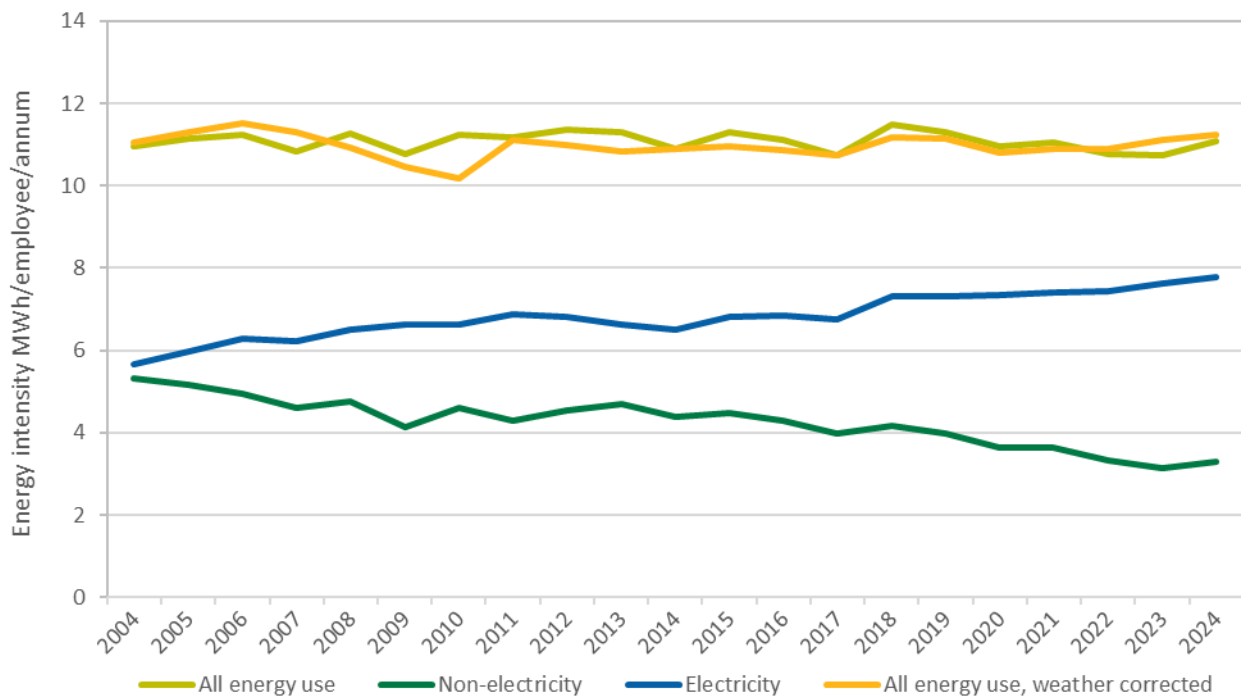
The energy intensity of the services sector is generally measured in relation to the value added generated by services activities. As shown in Figure 10.12, this intensity is flatter than that of industry.

Figure 10.12: Energy intensity of the commercial and public services sector



Two other indicators in this sector are energy use per unit of floor area and per employee. The consumption of oil and gas is mainly for space heating purposes and is likely to be more related to the floor area heated, rather than to the number of people occupying a building at a given time. Due to an absence of data on floor area in the services sector, it is not currently possible to calculate the consumption per unit of floor area.

Figure 10.13: Energy per employee in the commercial and public services sector



Electricity use per employee is used as an indicator of energy use in the services sector because, usually, there is a correlation between electricity use and the number of employees. Figure 10.13 shows electricity per employee has been increasing since 2020.

Table 10.5: Quantities of energy per employee in commercial and public services (share)

Energy [kWh/employee]	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Electricity	6,513 (59.7%)	6,820 (60.4%)	6,830 (61.5%)	6,745 (62.9%)	7,312 (63.7%)	7,325 (64.8%)	7,327 (66.8%)	7,395 (67.0%)	7,435 (69.1%)	7,620 (70.9%)	7,789 (70.3%)
Non-electricity energy	4,389 (40.3%)	4,475 (39.6%)	4,283 (38.5%)	3,987 (37.1%)	4,168 (36.3%)	3,980 (35.2%)	3,642 (33.2%)	3,645 (33.0%)	3,331 (30.9%)	3,130 (29.1%)	3,285 (29.7%)
Total energy	10,902 (100%)	11,295 (100%)	11,113 (100%)	10,732 (100%)	11,480 (100%)	11,305 (100%)	10,969 (100%)	11,040 (100%)	10,766 (100%)	10,750 (100%)	11,075 (100%)

10.5.2 Public sector developments

The public sector comprises approximately 4,000 separate public bodies, about 3,650 of which are individual schools. The other 350 comprise, among other things, Government departments, non-commercial State bodies, State-owned companies and local authorities. Each 'public body' is a stand-alone organisation and can range in size from very small (for example, a small rural school or a five-person agency) to very large (for example, the Health Service Executive or An Garda Síochána). The vast majority of energy is consumed by the 100 largest organisations.

Public services energy consumption comprises two main classes of energy consumer:

- **Public sector buildings** (offices, hospitals, clinics, nursing homes, schools, prisons, barracks, Garda stations, *etc.*), which primarily consume electricity, natural gas and oil-based fuels in addition to smaller amounts of renewable and solid fuels.
- **Public sector utilities**, which primarily consume electricity, such as wastewater treatment plants, water treatment facilities, pumping stations and street lighting (~400,000 units).

In addition, the energy consumed by public bodies also includes some consumption counted in the transport sector in the National Energy Balance. For example, this includes public transport fleets (rail, bus, *etc.*), as well as other transport fleets operated by public bodies such as ambulances, local authority vehicles, Garda fleet, Defence Forces' vehicles, *etc.*

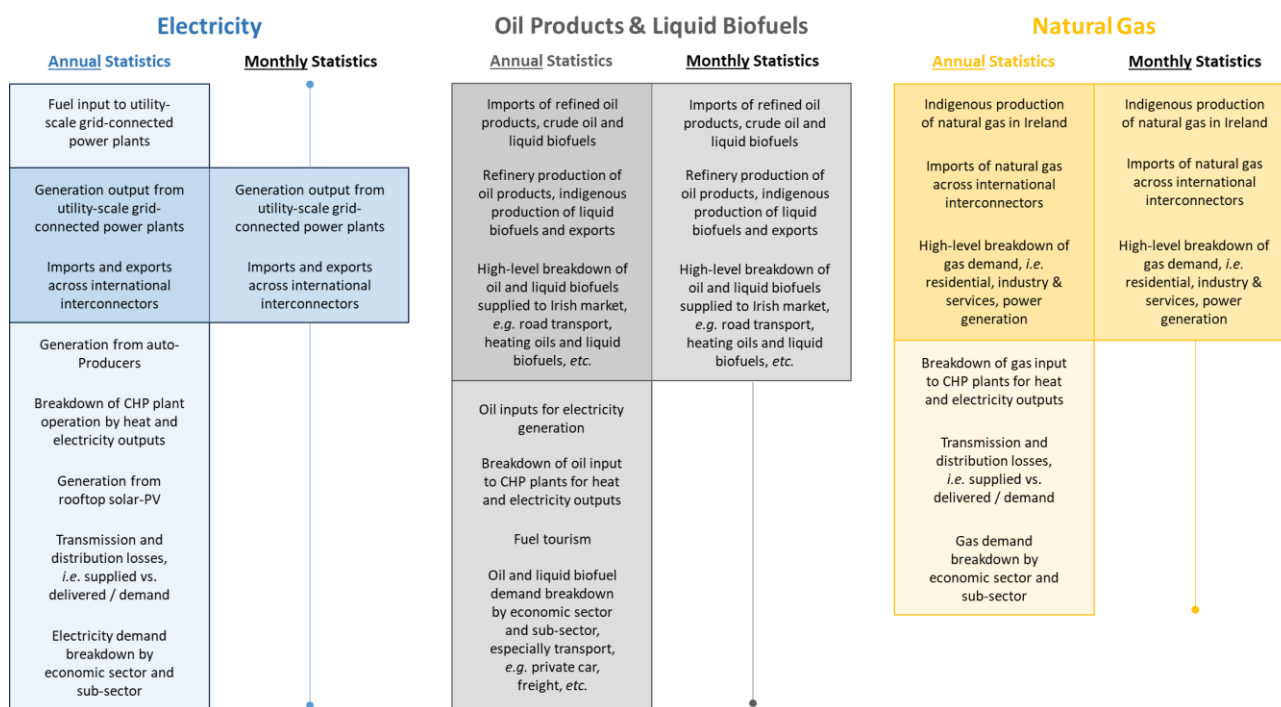
Under the Climate Action Plan, every public sector organisation must reduce energy-related emissions by 51% by 2030 and achieve a 50% improvement in energy efficiency in the public sector by 2030.

Since 1 January 2011, public sector bodies have been required to report to the Government annually on their energy use and the actions they have taken to reduce consumption. The reporting system includes a national public sector energy database, which includes all public sector electricity and natural gas meter numbers. Over time, the monitoring and reporting system will build a comprehensive bottom-up picture of energy consumption in the sector through the population of the national public sector energy database.

SEAI have published the Annual Report 2024 on Public Sector Energy Efficiency Performance [38]. It noted that 344 public sector bodies and 2,946 schools completed reports on energy, and these represented 99% of total public sector energy consumption. The total primary energy consumption in 2023 of these bodies was 9,543 GWh (primary energy), which comprised 4,164 GWh of electricity, 3,260 GWh of thermal energy and 2,119 GWh of transport energy. The combined total primary energy consumption in 2023 of these bodies was 9,543 GWh, and their total energy-related CO₂ was 1,582 ktCO₂. The total energy spend was estimated at €1,087 million in 2023. The report also noted that the improvement in energy efficiency amounts to 34.9% on business as usual, equivalent to 5,571 GWh of avoided primary energy consumption. Non-electricity CO₂ emissions had decreased by 6.3% in 2023 since the baseline, while total CO₂ emissions had decreased by 24.1%.

11 Provisional energy data from monthly surveys

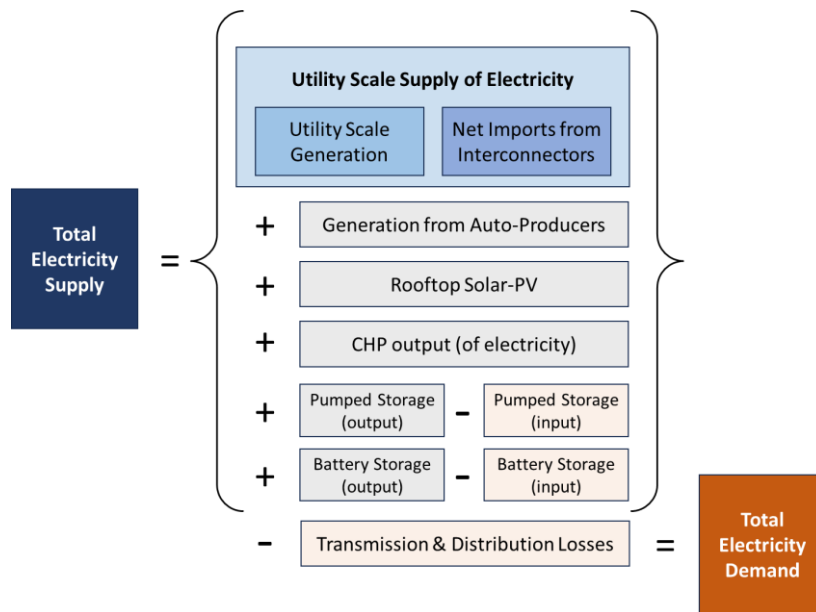
The trends observed in provisional monthly data are not as definitive or as inclusive as those based on the annual energy data in SEAI's National Energy Balance, or the annual emissions data in EPA's National Greenhouse Gas Emissions Inventory. But the trends in the monthly data can often be identified just a few weeks after the fact, providing much timelier insights. The different scopes of monthly and annual energy statistics are summarised below:



Insights from provisional monthly data are best regarded as useful early indications of shifts and trends in energy that will later be more fully quantified and verified by definitive annual data. The complementary use of both fast-provisional monthly data, and slower-definitive annual data, helps balance the competing needs of timeliness and authority in Ireland's energy data.

11.1 Monthly utility scale electricity data to 2025

The monthly data currently available to SEAI on electricity supply covers only the *utility scale supply* of electricity, i.e. the electricity supplied by large-scale grid-connected sites, and through international interconnectors. As a result, the monthly data does not capture instances of mini- or micro-generation of electricity, for example the electricity generated in rooftop solar panels. The monthly data also does not capture the small number of auto-producers active in Ireland that generate electricity for their own consumption.

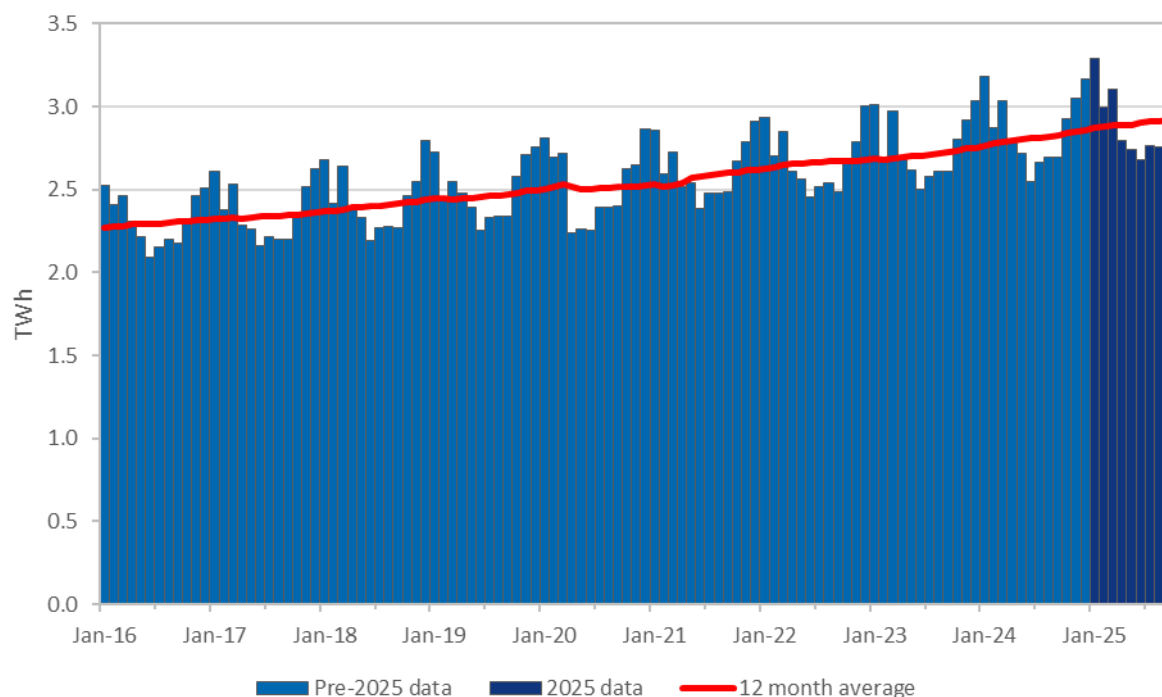


Nonetheless, the fast-provisional total utility scale supply of electricity calculated from monthly data serves as an excellent proxy for identifying developing trends in Ireland's total electricity supply, and the total electricity demand driving that supply. The utility scale supply of electricity in recent years has accounted for approximately 95% of total electricity supply. Small-scale electricity generation, as well as transmission, distribution, and storage losses, are fully accounted for in SEAI's National Energy Balance, which benefit from more definitive annual surveys and cross-agency reconciliations of data. The monthly data is provided by EirGrid in megawatt hours (MWh) in terms of a net electricity production basis and converted to terawatt hours (TWh) by SEAI.

11.1.1 Seasonality in utility scale monthly electricity supply

The bar chart in Figure 11.1 shows the utility scale electricity supply per month from January 2016 to September 2025. For clarity and ease of comparison, 2025 is coloured dark blue, whereas, previous years are coloured in a lighter shade of blue. The red line shows a 12-month moving average of those monthly values and helps to identify underlying long-term trends. While the seasonality of electricity supply is somewhat apparent in this 'raw' monthly data, it can be made much clearer through cross-year averaging and normalisation for number of days in a given month, as shown in subsequent figures.

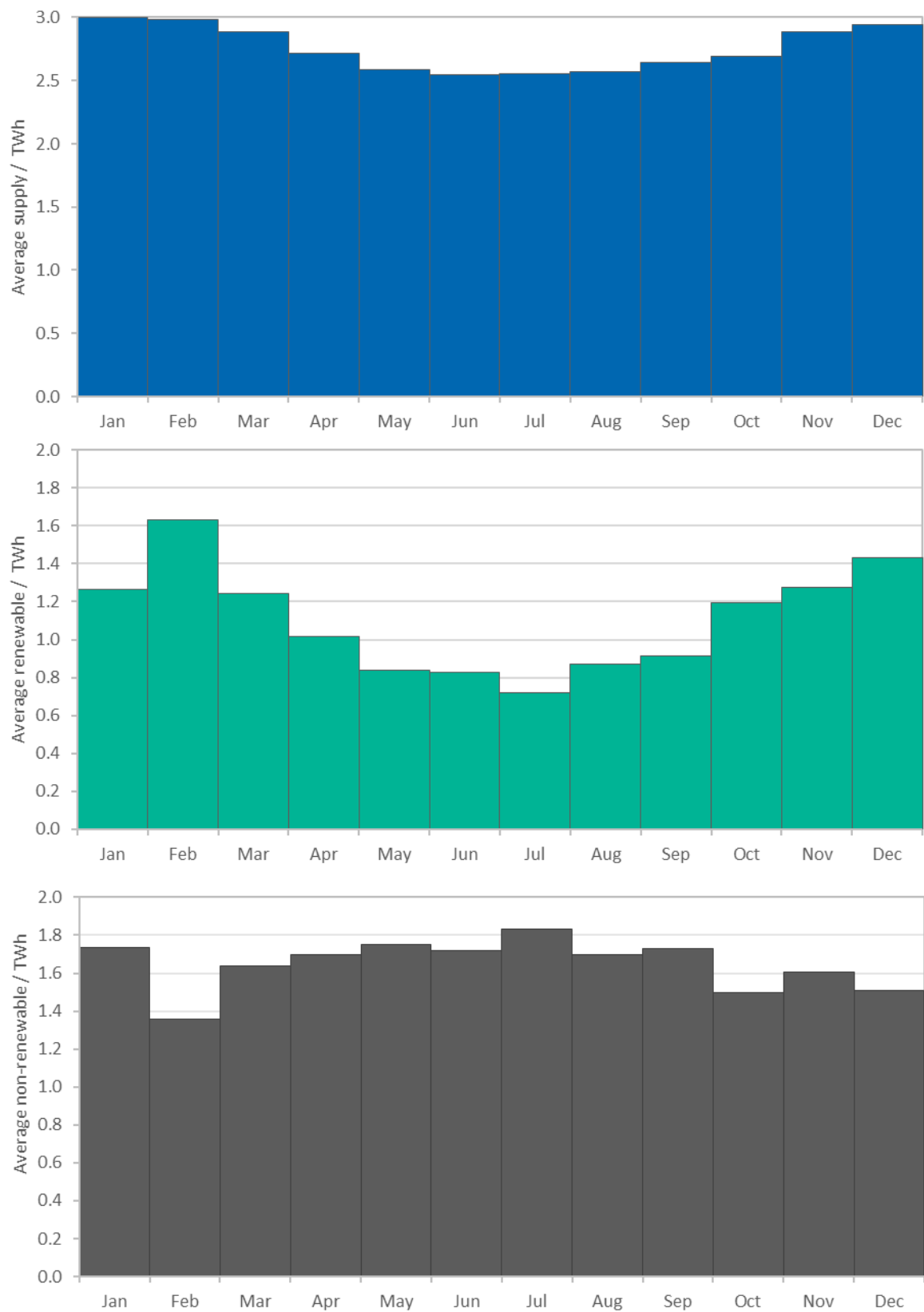
Figure 11.1: Monthly utility scale electricity supply – January 2016 to September 2025



Source: data from EirGrid

Figure 11.2 shows the average monthly utility scale electricity supply by month, where each month's value is the average of that month's electricity supply (total [top], renewable [middle] or non-renewable [bottom]), normalised for number of days in a month, over the previous 5 years. For example, the January value is the average of January in 2021, 2022, 2023, 2024 and 2025 and the October value is the average of October in 2020, 2021, 2022, 2023 and 2024. It shows the seasonality of our electricity supply whereby the electricity supply is higher in the winter months and lower in the summer months. The renewable supply shows a pronounced seasonal variation, oscillating around an average value of approximately 1.1 TWh. It has a maximum of approximately 1.4 TWh from December to February, and a minimum of approximately 0.8 TWh from June to August. The seasonal variation of renewable electricity generation is therefore approximately $\pm 29\%$ of the annual average. This variation is largely driven by the strong seasonality of wind generation, which makes up a considerable fraction of our renewable electricity generation (see below for more details). Fortunately, the seasonality of our renewable electricity generation matches the seasonality of our demand – the wind generally blows hardest in the winter months, and this is when energy demand for light and heat is highest. The non-renewable supply is relatively flat at approximately 1.6 TWh per month, but shows higher non-renewable electricity supply in the summer months, and lower non-renewable electricity supply in the winter months. Non-renewable electricity supply is anti-correlated to renewable supply – when renewable supply is high, non-renewable electricity is low, and vice versa. This is most clearly illustrated by comparing the average values of February and July, where the particularly high and low renewable supply values match particularly low and high non-renewable supply values.

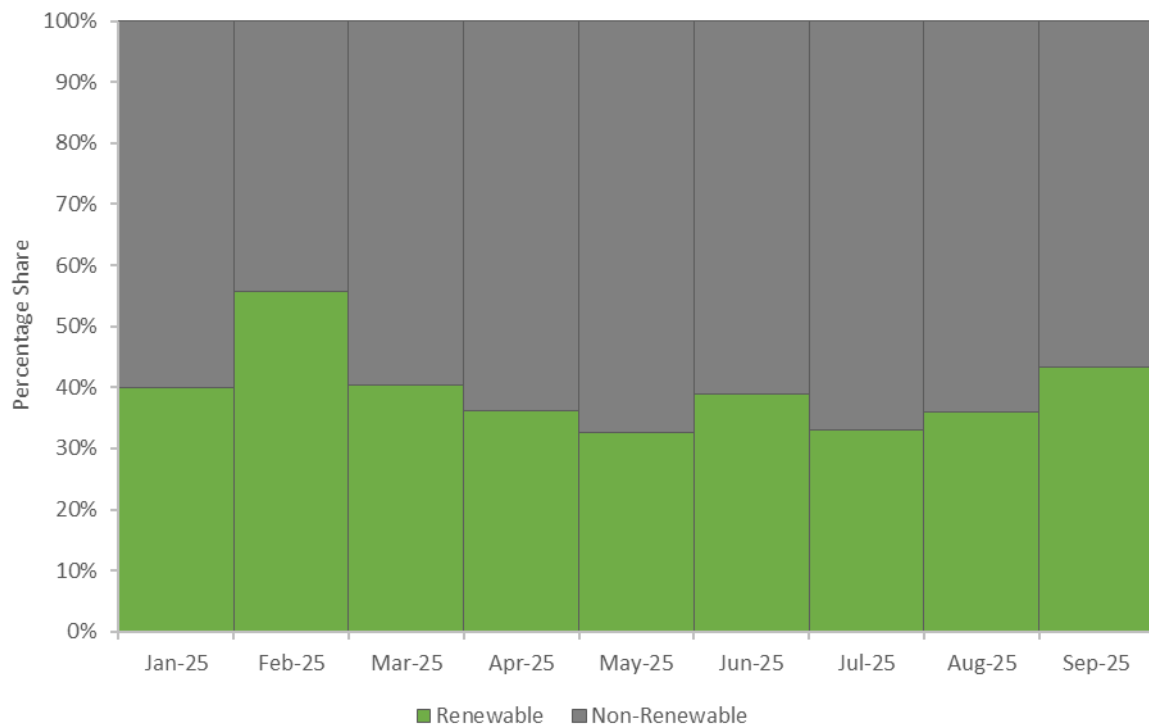
Figure 11.2: Average monthly utility scale electricity supply over last 5 years (normalised for number of days per month)



Source: data from EirGrid

Figure 11.3 shows utility scale monthly electricity supply (*i.e.* total generation plus net imports) from January to September in 2025, split by electricity supply from renewable generation and non-renewable sources. Although this value can be regarded as a leading indicator of Ireland's renewable share of energy in electricity (RES-E), it is not directly equivalent to the RES-E value because the methodology for the RES-E calculation additionally includes (1) mini- and micro- generation of renewable electricity, (2) the electricity generated by auto-producers (including CHP plants), (3) multi-year normalisation of wind and hydro generations, and (4) a formal accounting for the sustainability status of various biomass fuels.

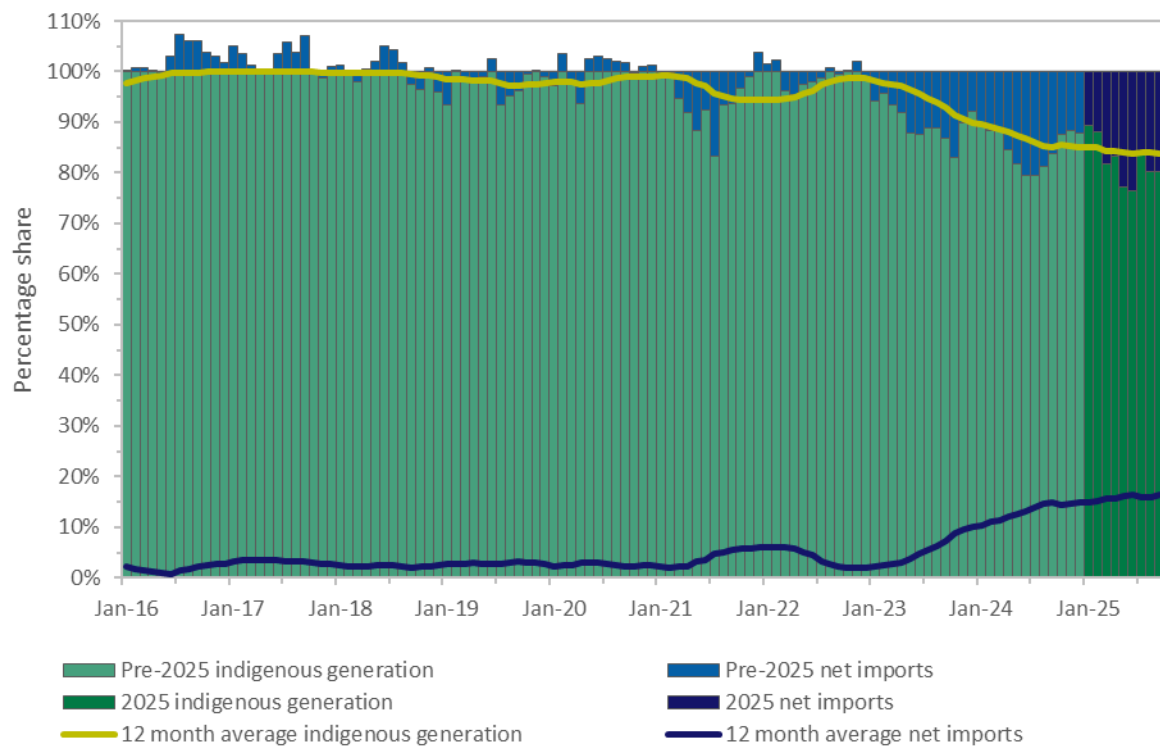
Figure 11.3: Share of renewable and non-renewable utility scale electricity supply – January to September 2025



Source: data from EirGrid

The bar chart in Figure 11.4 shows the share of utility scale monthly electricity supply, split by electricity supply from indigenous generation (green) and net imports (blue) between January 2016 to September 2025. For clarity and ease of comparison, 2025 is coloured in darker shades, whereas previous years are coloured in lighter shades. In months with percentage shares above 100%, this represents periods when there was a net export of electricity outside of Ireland's national boundaries, across international interconnectors. The lines show two 12-month moving averages of the monthly data and help to identify underlying long-term trends in monthly indigenous electricity production (green) and electricity net imports (blue) data.

Figure 11.4: Share of utility scale electricity supply split by indigenous generation and net imports

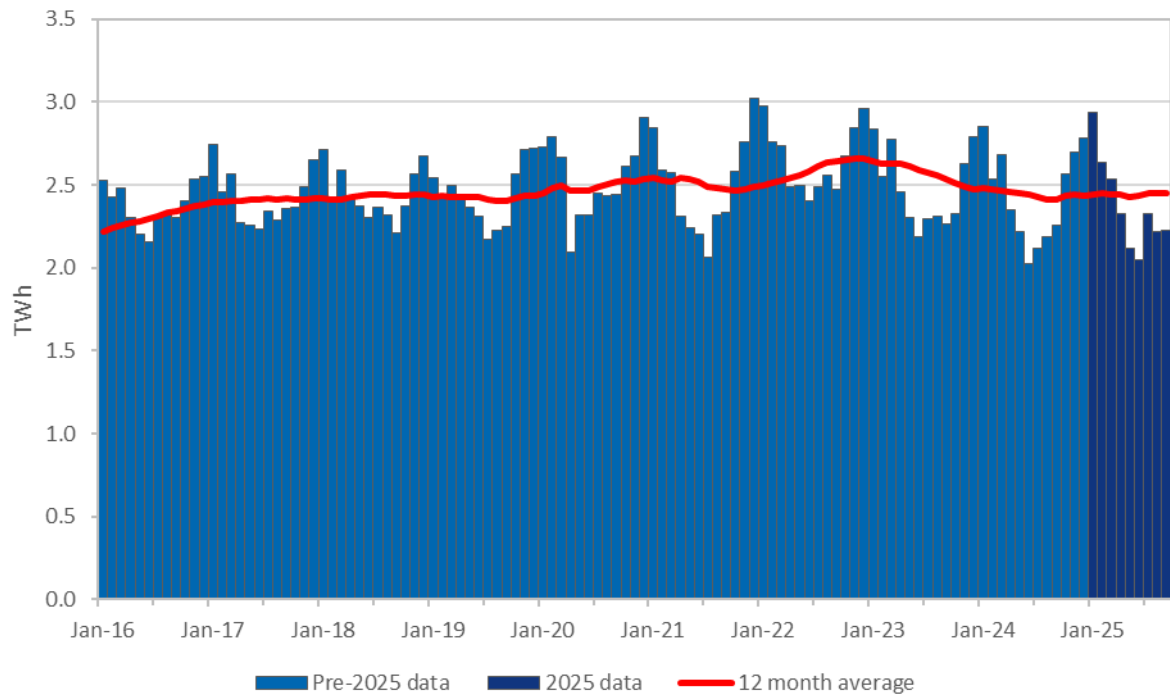


Source: data from EirGrid

11.1.2 Monthly utility scale electricity generated

The bar chart in Figure 11.5 shows utility scale electricity generated and exported to the grid per month from January 2016 to September 2025. For clarity and ease of comparison, 2025 is coloured dark blue, whereas previous years are coloured in a lighter shade of blue. The red line shows a 12-month moving average of monthly values and helps to identify underlying long-term trends.

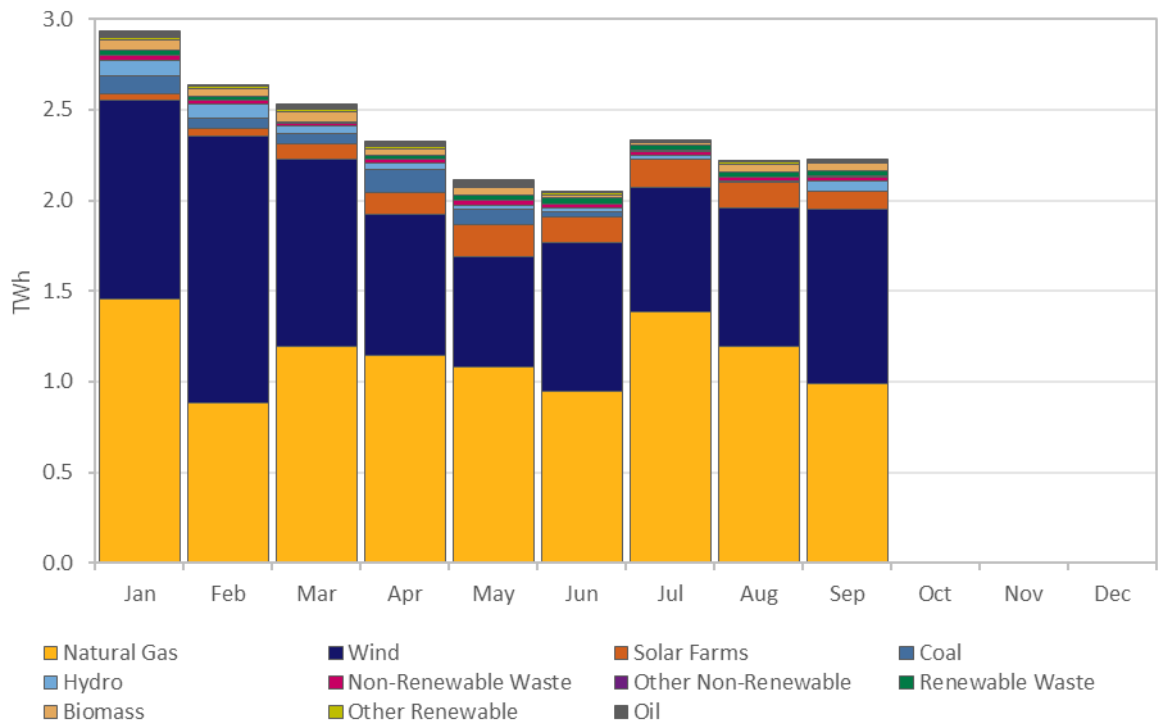
Figure 11.5: Monthly utility scale electricity generation – January 2016 to September 2025



Source: data from EirGrid

Figure 11.6 shows the monthly electricity generation to the grid to date in 2025, broken down by input fuel or energy type. Note that this figure shows the electricity generated due to each fuel/energy input, rather than the quantity of fuel/energy needed to generate that electricity. The seasonal variations in electricity generated from wind and gas are visible, with wind accounting for 56% of utility scale electricity generation or 49% of utility scale electricity supply in February 2025.

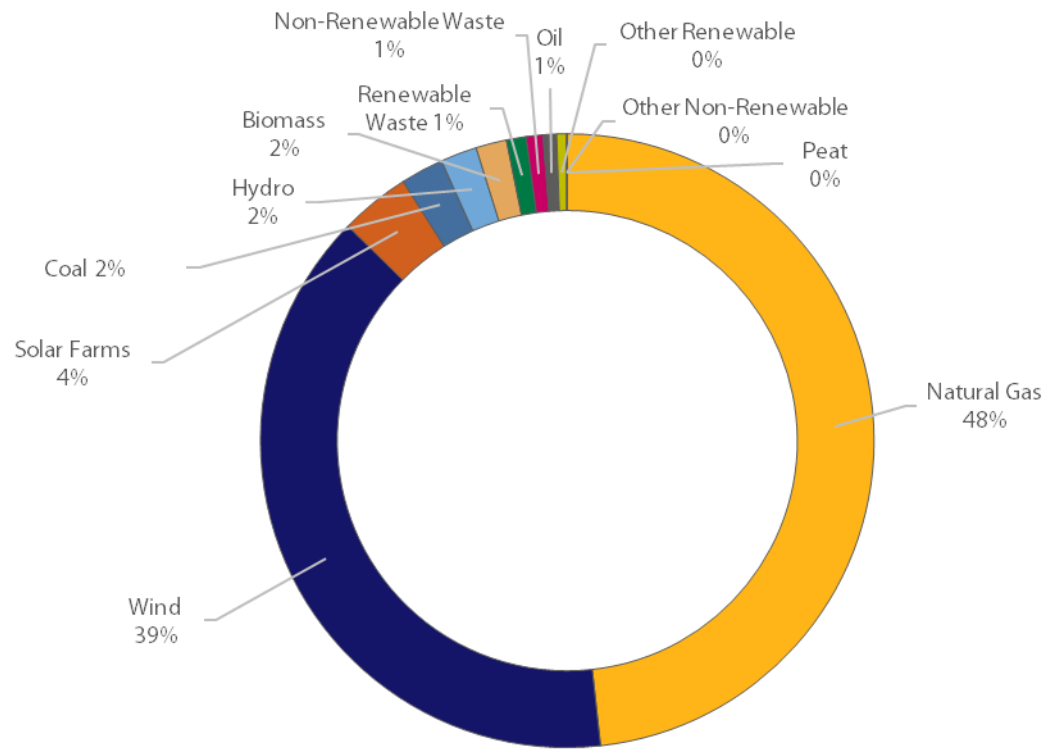
Figure 11.6: Monthly electricity generation by source in 2025



Source: data from EirGrid

Figure 11.7 shows the breakdown of the generated electricity by input fuel or energy type, averaged over the last 12 calendar months, e.g. from October 2024 to September 2025. This 12-month approach is more appropriate than simply averaging over the year-to-date, because electricity generation from October to December has higher seasonal renewable contributions.

Figure 11.7: Sources of electricity generation – 12-month rolling average to September 2025



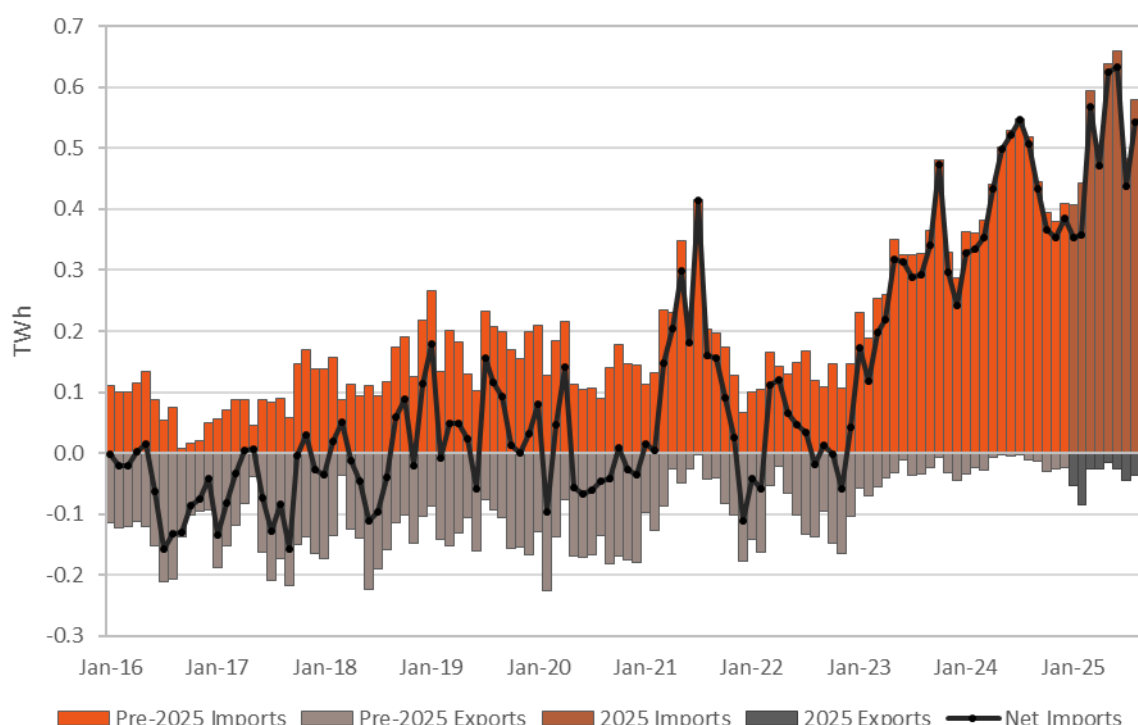
Source: data from EirGrid

11.1.3 Monthly electricity imported and exported in 2025

The bar chart in Figure 11.8 shows the import (orange) and export (grey) of electricity to-and-from Ireland through international interconnections each month. The resulting 'net import' of electricity through the interconnections is shown by the black data points joined with a line to guide the eye. For clarity and ease of comparison, 2025 is coloured in darker shades, whereas previous years are coloured in lighter shades. A positive net import value for a given month means that Ireland imported more electricity than it exported that month, while a negative net import value for a given month means that Ireland exported more electricity than it imported that month.

The Greenlink interconnector, connecting Wexford with Wales, entered its 'commercial operations' phase at the beginning of 2025, with capacity available through the implicit SEM-GB 'Intra Day 1 Auction' from 29th January 2025.

Figure 11.8: Monthly imported and exported electricity (with net imports) – January 2016 to September 2025



Source: data from EirGrid

11.2 Gas

Ireland's national gas supply comes from the sum *indigenous production* at the Corrib Gas Field and *international imports* from the UK through a pair of international interconnectors to Moffat in Scotland.

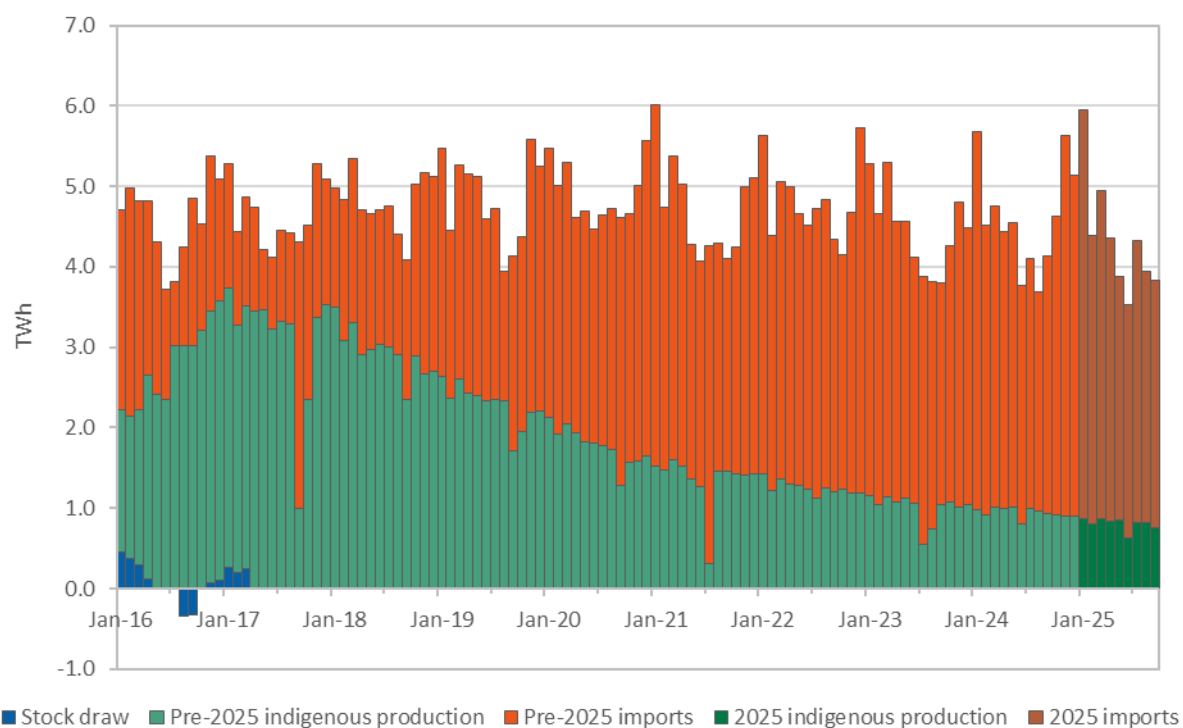
SEAI uses public data from the Gas Network Ireland dashboard, and monthly survey returns from the Corrib Gas Field, to determine Ireland's national gas supply, and its split across indigenous production and imports flows.

The gas monthly data is expressed in terms of gross calorific value (GCV), while SEAI's national Energy Balance expresses all energy in terms of net calorific value (NCV). The net calorific value can be calculated from the gross calorific value by multiplying by a factor of 0.902.

11.2.1 Seasonality in monthly gas supply

Figure 11.9 shows the monthly supply of grid gas over the last ten years, split into imported gas (orange), indigenous gas (green) and stock draw (blue). For clarity and ease of comparison, 2025 is coloured in darker shades, whereas previous years are coloured in lighter shades. There was a large step-change increase in indigenous production in January 2016 due to the connection of the Corrib gas field to the national gas grid. Indigenous production peaked in 2017 and has slowly been decreasing since then, which is commensurate with the natural life cycle of a gas field. The sharp drops in indigenous production in September 2017, July 2021 and July-August 2023 correspond to periods of maintenance at the Corrib gas field or its connection to the grid. In these periods of low indigenous production, the gas interconnectors between Ireland and Scotland stepped-up delivery of imports to satisfy demand. The monthly supply of gas to the grid can be 'spikey' because a substantial fraction of total gas supply is used for electricity generation which backs intermittent wind generation. In periods of low wind, gas supply to gas-fired electricity plants needs to increase, and vice versa.

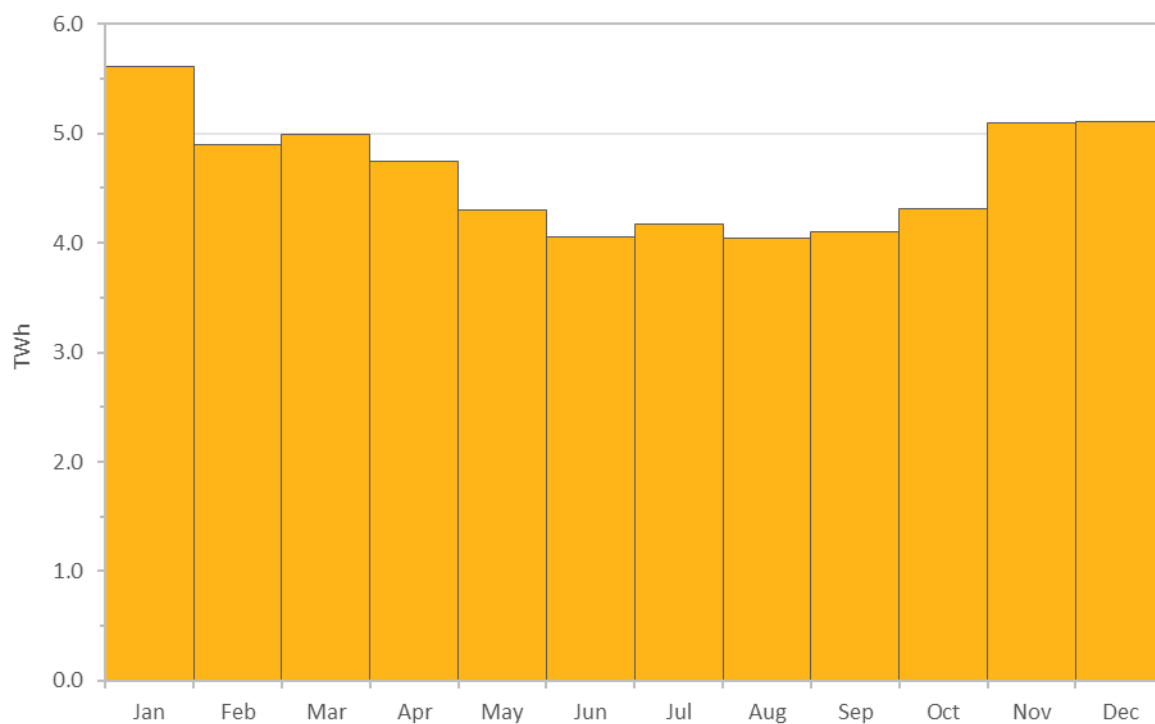
Figure 11.9: Monthly gas supply trends – January 2016 to September 2025



Source: data from GNI & SEAI

Figure 11.10 shows the total gas supply to the grid by month, where each month's value is the average value of that month's gas supply, normalised for number of days in a month, over the previous 5 years. For example, the January value is the average of January in 2021, 2022, 2023, 2024 and 2025 and the October value is the average of October in 2020, 2021, 2022, 2023 and 2024. While certain sectors have pronounced seasonal variations, particularly the residential sector's gas demand for space heating, the overall seasonal variation in total gas supply is relatively small. Market-shifts act to flatten the profile of monthly gas supply across the year. In winter months, gas supply to the residential sector is higher for heating, but gas supply to electricity generation is lower (because wind generation has a seasonal peak). Conversely, in summer months, gas supply to the residential sector is lower, but gas supply to electricity generation is higher (because wind generation has a seasonal low).

Figure 11.10: Average monthly gas supply over the last 5 years (normalised for number of days per month)

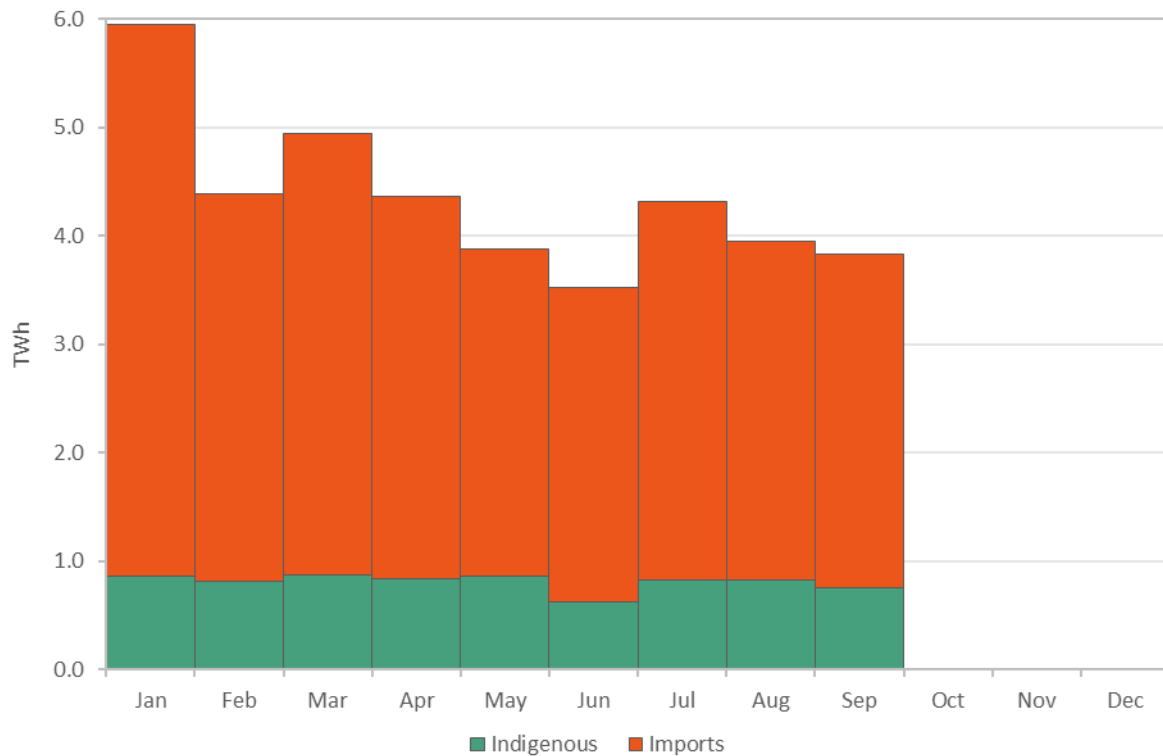


Source: data from GNI & SEAI

11.2.2 Monthly gas supply in 2025

As currently collected, the monthly gas supply data shown in Figure 11.11 is broken down into indigenous gas and imported gas. In general, indigenous supply is relatively constant on a month-to-month basis (outside of periodic maintenance in June 2025 [39]) and imported gas is adjusted to match demand patterns, especially those arising from gas-fired electricity generation plants stepping-up and -down to back renewable generation of electricity.

Figure 11.11: 2025 monthly gas supply

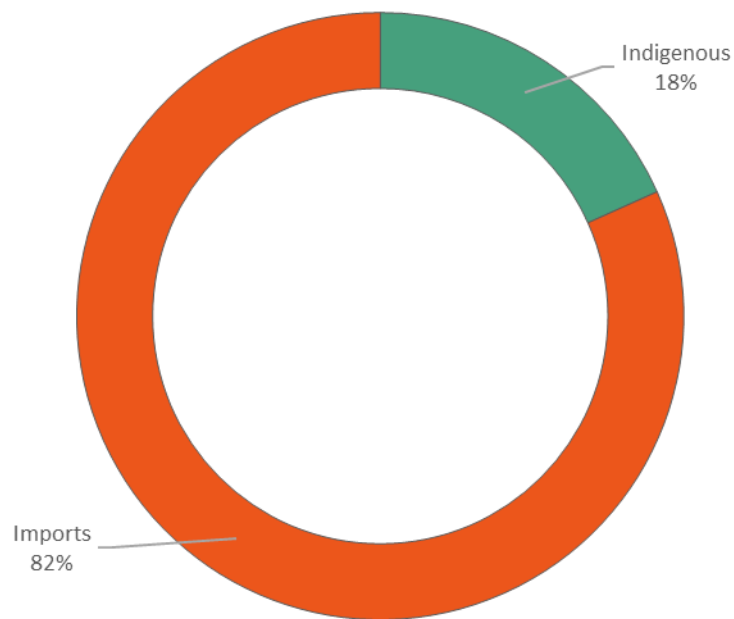


Source: data from GNI & SEAI

Figure 11.12 shows the breakdown of monthly gas supply data in 2025 by indigenous gas and imported gas, averaged over the last 12 calendar months, *i.e.* from October 2024 to September 2025. This approach is more appropriate than simply averaging over the year-to-date to September 2025, because electricity generation from October to December has higher seasonal renewable contributions. In the last 12-months, the proportion of electricity generation to the grid was:

- Imported gas 82%
- Indigenous gas 18%

Figure 11.12: Source of gas supply – 12-month rolling average to September 2025



Source: data from GNI & SEAI

11.3 Oil and liquid biofuels

Monthly data on the delivery of both oil products and biofuel products to the Irish market is made available to SEAI through its access to the national Oil Levy Application (OLA) database, which is maintained and operated by the Department of Climate, Energy and the Environment (DCEE). SEAI uses data from DCEE's OLA database to satisfy several of Ireland's energy reporting obligations to the European Commission under the Energy Statistics Regulation 1099/2008 [2], and to populate the national Energy Balance for Ireland.

Relevant obligated parties, for example large oil and liquid biofuel suppliers operating in the Irish market, are required to make monthly submission to the OLA database. Despite occasional corrections and revisions by these suppliers, the running total of monthly *gross inland deliveries* provides an excellent fast-provisional proxy for identifying trends in Ireland's demand for oil products and biofuel blending.

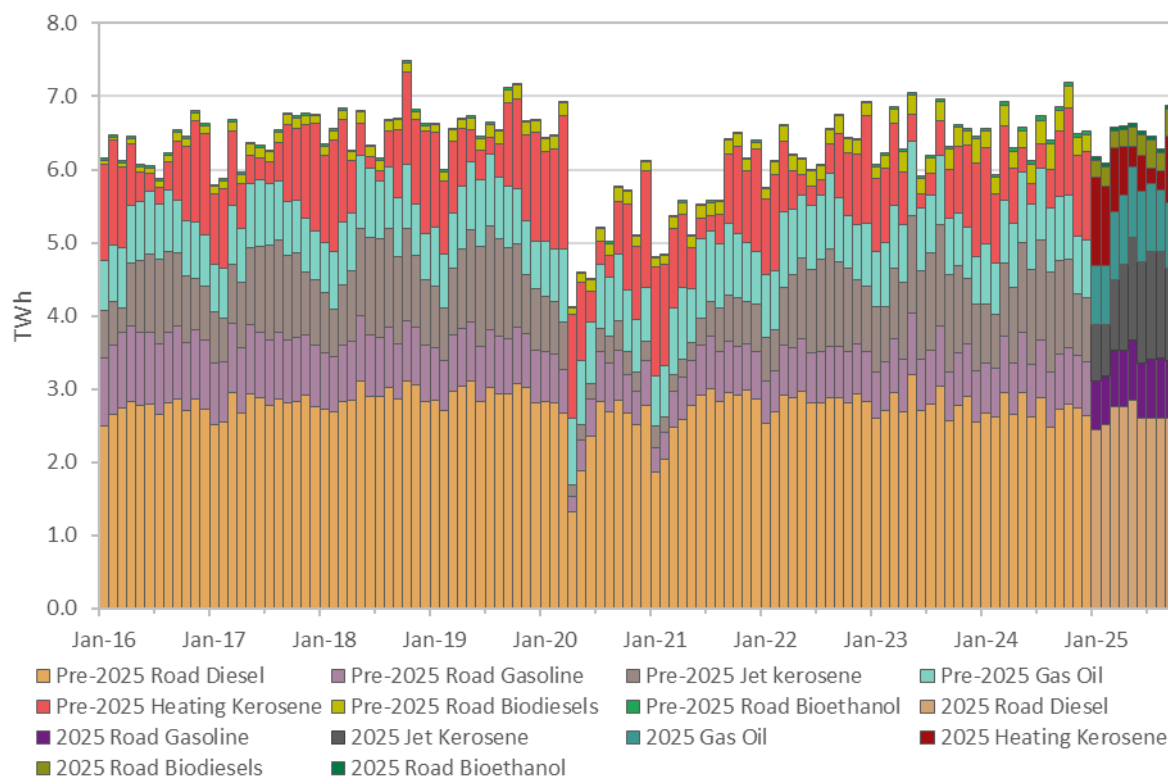
Monthly oil and biofuel supply is captured in terms of gross inland deliveries (GIDs), or "The observed delivery of finished petroleum products from primary sources (*e.g.* refineries, blending plants *etc.*) to the inland market" and published in terms of net calorific value (NCV). GID is the quantity of oil and biofuel products flowing through the market from production to delivery, now ready for sale or consumption:

Gross Inland Delivery = Primary product receipts + Refinery gross output + Recycled products - Refinery fuel + Imports - Exports - International marine bunkers + Inter-product transfers - Products transferred - Stocks changes

11.3.1 Seasonality and COVID impacts in monthly oil and road liquid biofuel deliveries

Figure 11.13 shows the sum of monthly oil and road liquid biofuel deliveries from road diesel (blue), road gasoline (purple), jet kerosene (grey), gas oil (teal), heating kerosene (red), road biodiesels (lime green) and road bioethanol (dark green) from January 2016 to September 2025. For clarity and ease of comparison, 2025 is coloured in darker shades, whereas previous years are coloured in lighter shades.

Figure 11.13: Oil and road liquid biofuel supply trends – January 2016 to September 2025

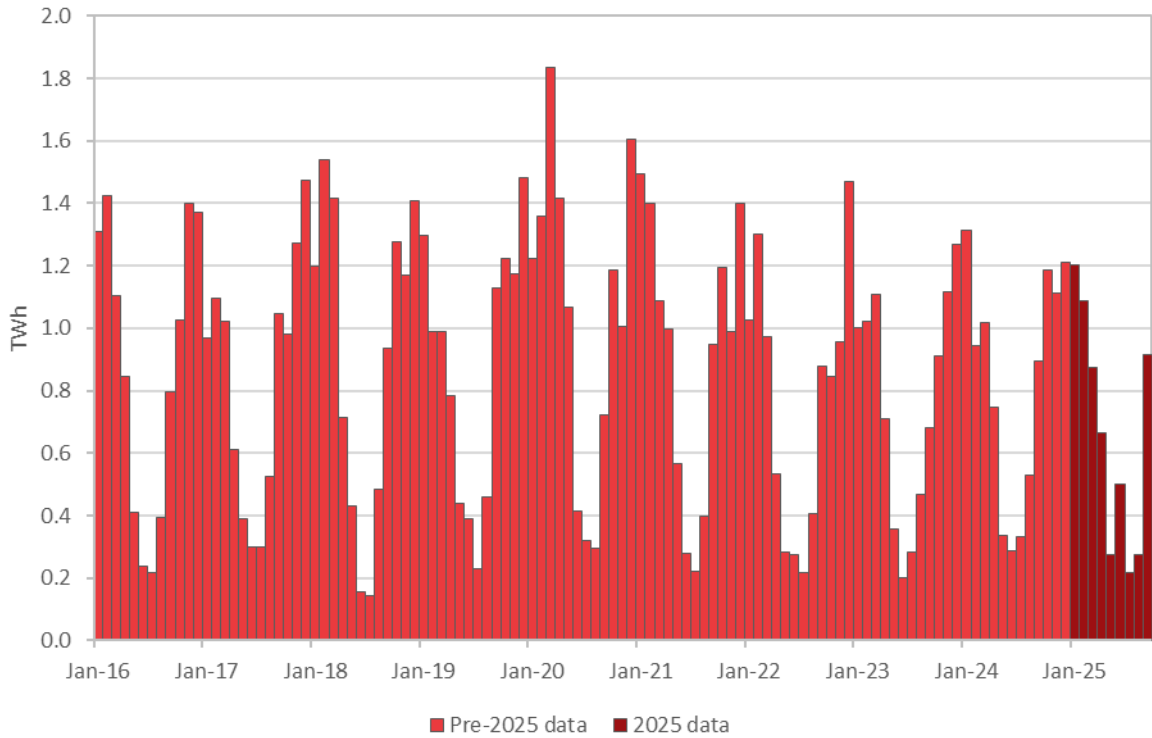


Source: data from Dept. of Climate, Energy and the Environment and SEAI

Figure 11.14 to Figure 11.17 breakout the monthly delivery of different oil products to better show their seasonal variation, and any impacts from COVID. For clarity and ease of comparison, 2025 is coloured in darker shades, whereas previous years are coloured in lighter shades.

Figure 11.14 shows the monthly deliveries of heating kerosene. It has a strong seasonal pattern with demand peaking in the height of winter and falling to about 16% of peak values in the summer months.

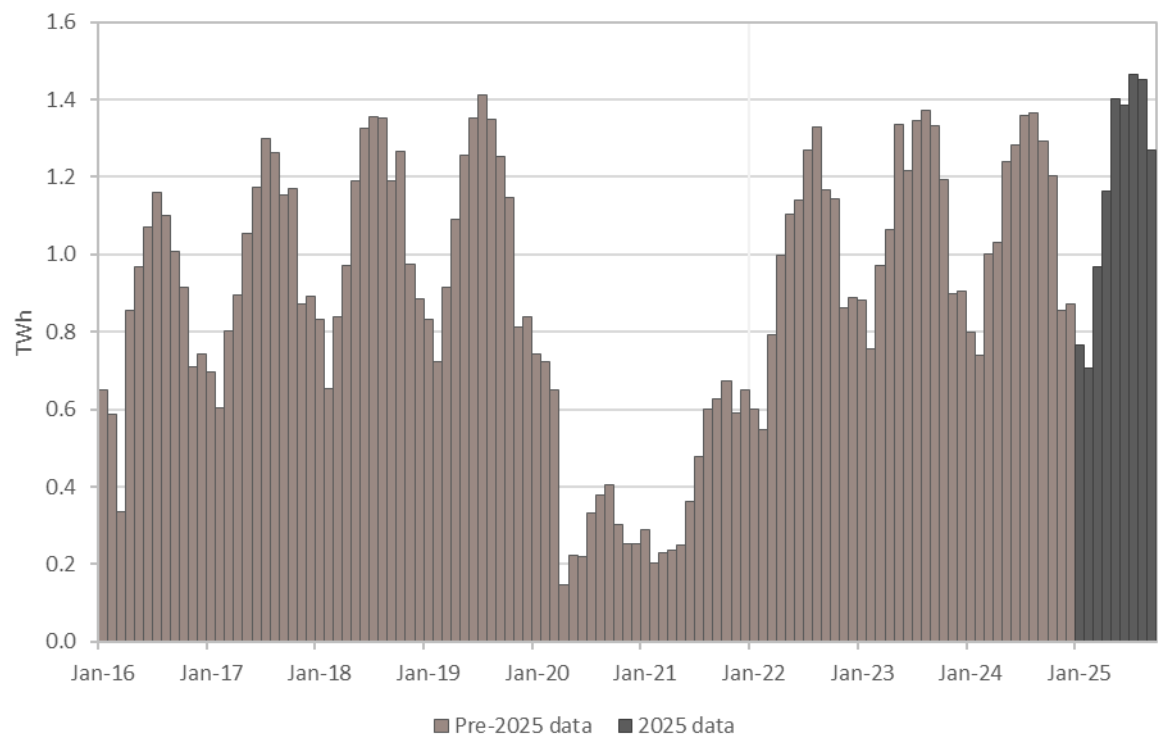
Figure 11.14: Trend in heating kerosene – January 2016 to September 2025



Source: data from Dept. of Climate, Energy and the Environment and SEAI

Figure 11.15 shows the monthly deliveries of jet kerosene. Deliveries of jet kerosene show peaks in summer months and in the Christmas / New Years period, due to the high volume of flights to and from Ireland for holidays and tourism. In April 2020, delivery of jet kerosene fell to just 10% of its peak value in 2019, as COVID travel restrictions were rolled out.

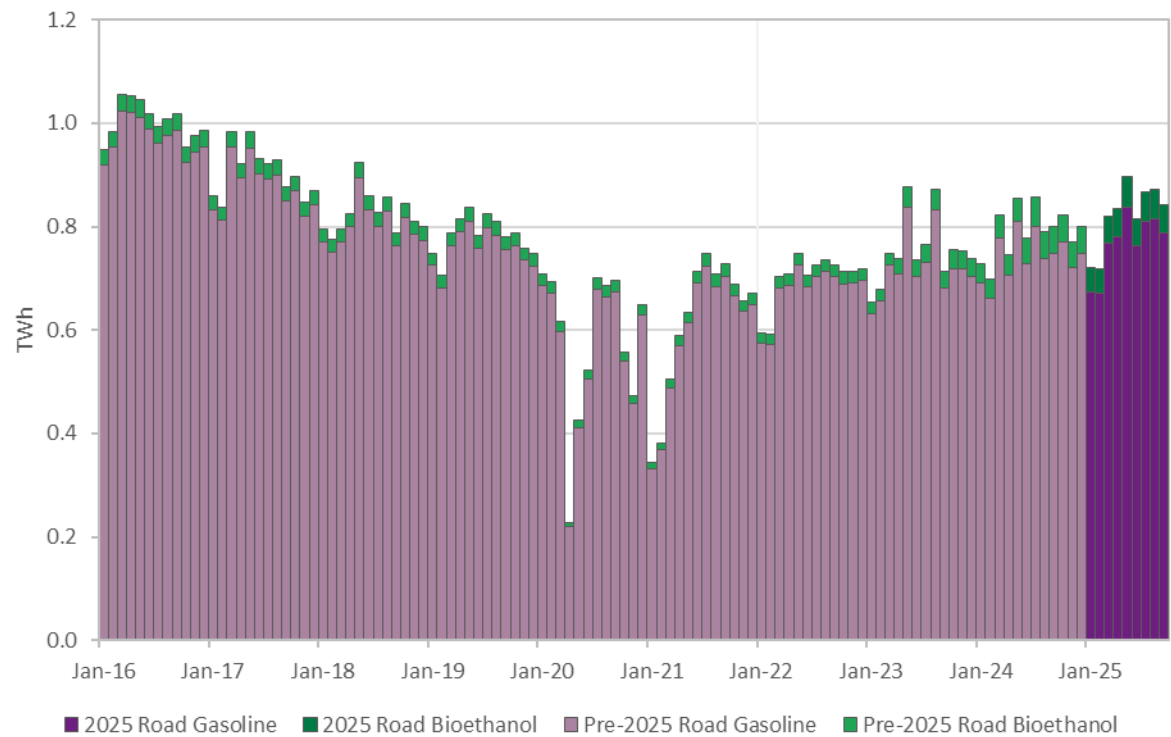
Figure 11.15: Trend in jet kerosene – January 2016 to September 2025



Source: data from Dept. of Climate, Energy and the Environment and SEAI

Figure 11.16 shows the monthly deliveries of road gasoline and bioethanol. It shows a gradual, but noticeable decline from 2016 to 2020, before being disrupted by COVID impacts. This led to significant dips in April 2020 and January 2021 following the announcement of national lockdown restrictions on 27 March and 31 December 2020, respectively. Recently, there were step change increases to the monthly bioethanol blend rate into road gasoline between April 2023 and August 2024. Since then, the bioethanol blend rate has remained relatively constant.

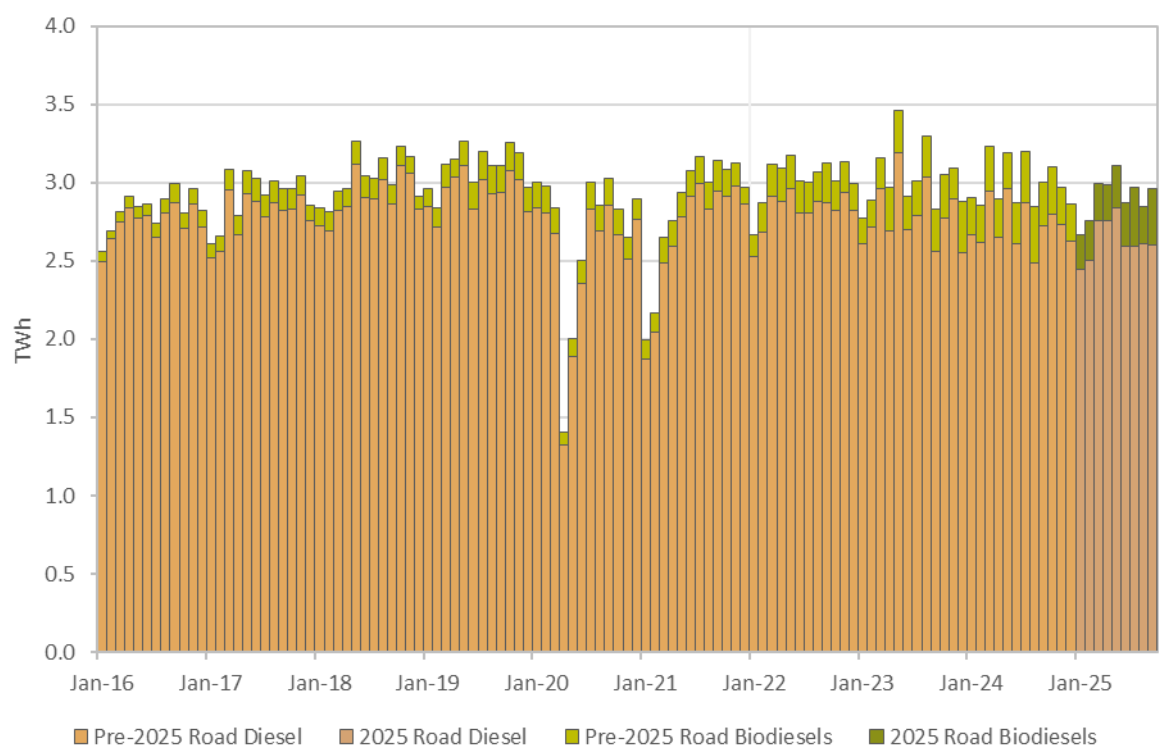
Figure 11.16: Trend in road gasoline and road bioethanol – January 2016 to September 2025



Source: data from Dept. of Climate, Energy and the Environment and SEAI

Figure 11.17 shows the monthly deliveries of road diesel. Road diesel deliveries show the same sharp COVID features as motor petrol. However, in contrast to motor petrol, road diesel shows a gradual but noticeable increase from January 2016 to January 2020. Since 2019, on average, there has been an increase in the monthly biodiesels blend rate into road diesel.

Figure 11.17: Trend in road diesel and road biodiesels – January 2016 to September 2025

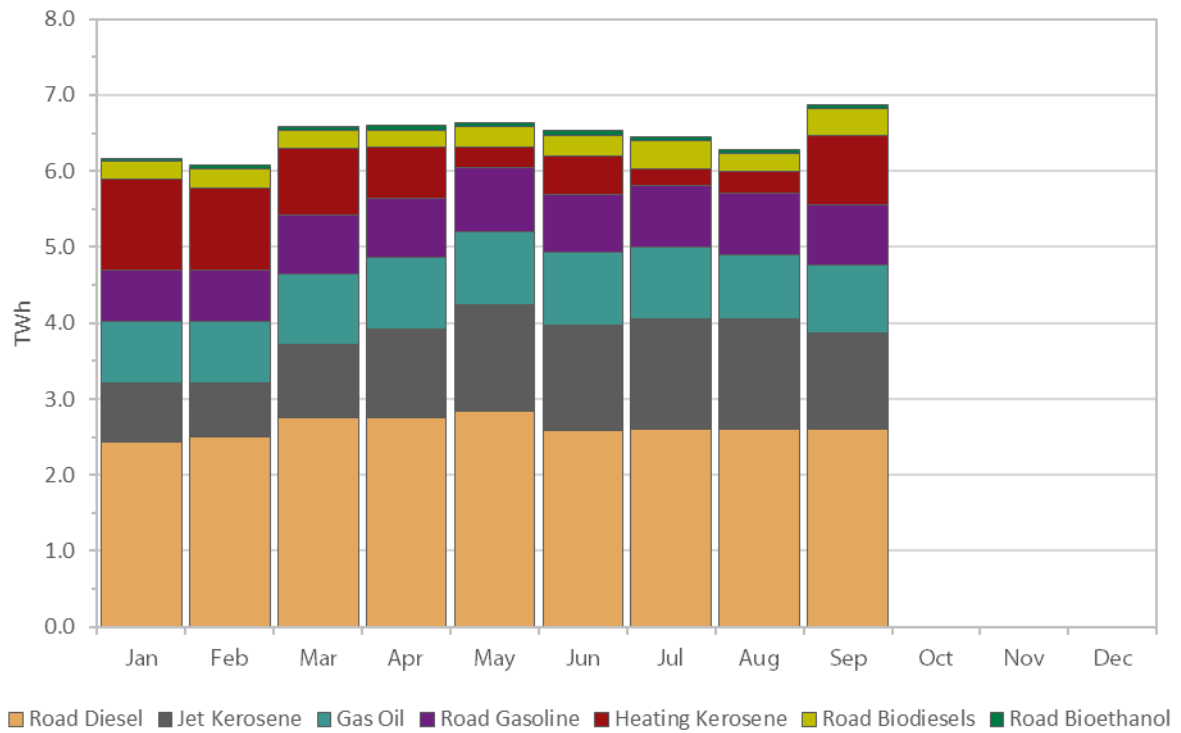


Source: data from Dept. of Climate, Energy and the Environment and SEAI

11.3.2 Monthly oil and road biofuel supply in 2025

Figure 11.18 shows the breakdown of monthly major oil product and road liquid biofuel deliveries, by type in 2025 to date. The seasonal variations in heating kerosene and jet kerosene are visible and tend to compensate for each other in the total oil deliveries, *i.e.* deliveries of jet kerosene are highest in summer, when deliveries of heating kerosene are lowest.

Figure 11.18: Monthly major oil product and road biofuel supply – 2025 to date

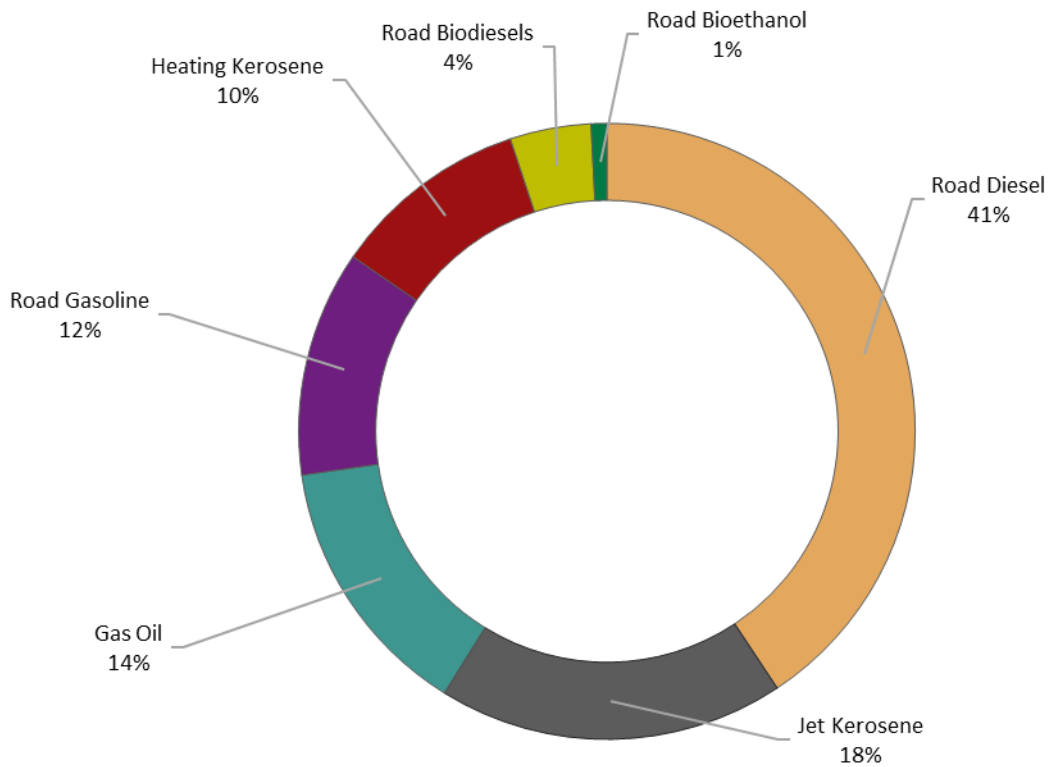


Source: data from Dept. of Climate, Energy and the Environment and SEAI

Figure 11.19 shows the percentage breakdown of major monthly oil and road biofuel deliveries by type, to date in 2025:

- Road diesel 41%
- Jet kerosene 18%
- Gas oil 14%
- Road gasoline 12%
- Heating kerosene 10%
- Road biodiesels 4%
- Road bioethanol 1%

Figure 11.19: Breakdown of major oil product and road liquid biofuel supply – 2025 to date



Source: data from Dept. of Climate, Energy and the Environment and SEAI

Appendix 1: List of figures

Figure 1.1: Key high-level insights on Ireland's greenhouse gas (GHG) emissions, the national energy mix, and Ireland's energy import dependency	10
Figure 1.2: Time-series of Ireland's national energy-related emissions since 1990.....	11
Figure 1.3: Waterfall plot showing 2018 to 2024 changes in national energy-related emissions.....	12
Figure 1.4: Breakout of 2024 national energy-related emissions across high-level sectors.....	12
Figure 1.5: Waterfall plot showing 2023 to 2024 changes in national energy-related emissions.....	12
Figure 1.6: Waterfall plot showing 2023 to 2024 changes in Ireland's gross electricity supply	13
Figure 1.7: Reported and estimated electricity sector emissions in the first carbon budget	14
Figure 1.8: Reported and estimated transport sector emissions in the first carbon budget	14
Figure 1.9: Illustration of the relationship between Ireland's total primary energy requirement (i.e. energy supply) and Ireland's total final energy consumption (i.e. energy demand)	15
Figure 1.10: Breakout of 2024 energy supply across energy products. The term 'NRW' stands for non-renewable wastes	16
Figure 1.11: Waterfall plot showing 2023 to 2024 changes in Ireland's energy supply. The term 'NRW' stands for non-renewable wastes	16
Figure 1.12: Breakout of 2024 renewable energy supply across energy sub-products	17
Figure 1.13: Waterfall plot showing 2023 to 2024 changes in Ireland's renewable energy supply	17
Figure 1.14: Breakout of 2024 energy demand across economic sectors	18
Figure 1.15: Waterfall plot showing 2023 to 2024 changes in Ireland's energy demand by sector.....	18
Figure 1.16: Breakout of 2024 energy demand across energy products. The term 'NRW' stands for non-renewable wastes	19
Figure 1.17: Waterfall plot showing 2023 to 2024 changes in Ireland's energy demand by product. The term 'NRW' stands for non-renewable wastes.....	19
Figure 1.18: Renewable energy content of road diesel, road petrol, and grid gas in 2024	20
Figure 1.19: Breakout of 2024 gross electricity supply by energy sub-products and type. The 'other' energy type includes non-fossil, non-renewable energy, like non-renewable wastes (NRW) and net imports of electricity across international interconnectors	21
Figure 1.20: Waterfall plot showing 2023 to 2024 changes in gross electricity supply by sub-product. The term 'NRW' stands for non-renewable wastes.....	22
Figure 1.21: Monthly imports, exports, and net imports of electricity across international interconnectors with Northern Ireland and Great Britain.....	22
Figure 1.22: Monthly imports, exports, and net imports of electricity in 2025 across the North-South, East-West, and Greenlink interconnectors	23
Figure 1.23: Monthly utility scale electricity supply in Ireland broken out by indigenous generation and net imports across international interconnectors.....	24
Figure 1.24: Installed wind capacity with climate action plan (CAP) targets for 2025 and 2030, and 'with additional measures' (WAM) capacity projections.....	24
Figure 1.25: Year-to-year increases in installed wind capacity to 2024, and year-to-year increases in WAM capacity projections of capacity from 2025 to 2030	25
Figure 1.26: Annual wind generation and dispatch down values figures.....	25
Figure 1.27: Time-series of Ireland's solar PV generation and a breakout of 2024 solar PV generation ..	26
Figure 1.28: Monthly solar PV generation from utility-scale grid-connected solar farms. The blue line is a 12-month moving point average (MPA) to illustrate the average trend	27
Figure 1.29: Breakout of 2024 electricity demand across economic sectors.....	28
Figure 1.30: Waterfall plot showing 2023 to 2024 changes in electricity demand by sector	28
Figure 1.31: Breakout of Ireland's electricity demand to illustrate the role of data centre (DC) electricity demand	29
Figure 1.32: Breakout of 2024 transport demand by sub-sector and type.....	30
Figure 1.33: Time-series of Ireland's energy demand for private car use and road freight	30
Figure 1.34: Monthly jet kerosene deliveries for aviation energy demand. The blue line is a 12-month moving point average (MPA) to illustrate the average trend	31

Figure 1.35: Breakout of 2024 transport demand by energy sub-product and type.....	31
Figure 1.36: Ireland's annual biofuel energy blending rates into road diesel and road petrol. The 2025 value, highlighted in yellow text, is a best estimate from SEAI, based on monthly data from January to September 2025.....	32
Figure 1.37: Breakout of 2024 overall heat demand across economic sectors.....	33
Figure 1.38: Waterfall plot showing 2023 to 2024 changes in heat demand by sector.....	33
Figure 1.39: Breakout of 2024 overall heat demand across energy products.....	34
Figure 1.40: Time-series breakout of Ireland's overall heat demand by energy type. The 'other' energy type includes non-fossil, non-renewable energy, like non-renewable wastes.....	34
Figure 1.41: Time-series breakout of Ireland's renewable heat demand by energy sub-type.....	34
Figure 1.42: Breakout of 2024 residential heat demand across energy products, and a breakout of residential electricity demand by consumption activity.....	35
Figure 1.43: Time-series of Ireland's residential heat demand.....	36
Figure 1.44: Comparison of year-to-year changes in residential heat demand and year-to-year changes in the number of heating degree days (HDDs).....	36
Figure 1.45: Ireland's overall renewable energy share (RES-Overall) results and targets.....	38
Figure 1.46: Ireland's renewable energy share of Electricity (RES-E) results and target.....	39
Figure 1.47: Ireland's total final energy consumption under the EU's energy efficiency directive (EED) and Ireland's 2030 Article 4 target.....	40
Figure 2.1: Main energy flows in Ireland.....	42
Figure 2.2: Energy Flow in Ireland in 2024 (TWh).....	42
Figure 3.1: Share of primary energy production by energy type.....	46
Figure 3.2: Indigenous primary energy production by energy type with production / supply ratio.....	46
Figure 3.3: Imports (positive) and exports (negative) by energy type (with total net imports).....	48
Figure 3.4: Share of primary energy requirement by energy type.....	50
Figure 3.5: Primary energy requirement by energy type.....	51
Figure 3.6: Import dependency of Ireland and the EU-27.....	53
Figure 4.1: Energy inputs to transformation processes.....	54
Figure 4.2: Flow of energy in electricity generation and consumption (TWh).....	55
Figure 4.3: Energy input to electricity generation by energy type.....	56
Figure 4.4: Change in input energy type to electricity generation in 2024 compared with 2023.....	58
Figure 4.5: Gross electricity production from all producers, excluding storage, by energy sub-type.....	59
Figure 4.6: Share of gross electricity production by producer category.....	61
Figure 4.7: Main activity producer - gross electricity production by energy type, excluding storage.....	62
Figure 4.8: Autoproducers - gross electricity production by energy type.....	64
Figure 4.9: Electricity imports and exports.....	65
Figure 4.10: Gross electricity supply by energy type.....	66
Figure 4.11: Change in gross electricity supply by energy type in 2024 compared with 2023.....	68
Figure 4.12: Pumped hydroelectric storage (Turlough Hill).....	69
Figure 4.13: Battery generation capacity [MW] at year end.....	70
Figure 4.14: Battery storage – energy storage capacity (MWh) at year end.....	71
Figure 4.15: CHP fuel input and thermal/electricity output.....	72
Figure 4.16: CHP electricity as percentage of total electricity generation.....	73
Figure 4.17: Installed solar PV capacity (AC) at year end.....	74
Figure 4.18: Map of grid-scale solar PV (solar farms) connected in Ireland.....	75
Figure 4.19: Installed wind generation capacity.....	76
Figure 4.20: Map of grid-connected wind farms in Ireland.....	77
Figure 4.21: Map of grid-connected hydroelectric generators in Ireland (pumped hydro not shown).....	78
Figure 4.22: Outputs from oil refining.....	80
Figure 4.23: Biomethane injected into the natural gas network.....	81
Figure 4.24: Annual production of peat briquettes.....	82
Figure 5.1: Final energy consumption by energy type.....	84

Figure 5.2: Total final consumption by energy type	85
Figure 5.3: Share final energy consumption by sector	87
Figure 5.4: Final energy consumption by sector	88
Figure 5.5: Shares of energy types in industry final energy	89
Figure 5.6: Final energy in industry by energy type.....	90
Figure 5.7: Shares of energy sub-types in transport final energy	93
Figure 5.8: Final energy in transport sector by energy sub-type	93
Figure 5.9: Shares of subsectors in transport final energy	95
Figure 5.10: Final energy in transport by subsector.....	96
Figure 5.11: Road transport – final energy by energy sub-type.....	97
Figure 5.12: International and domestic aviation – final energy by energy sub-type.....	98
Figure 5.13: Navigation, rail and pipeline – final energy by energy sub-type	99
Figure 5.14: Shares of energy types in residential final energy	100
Figure 5.15: Final energy in residential by energy type	101
Figure 5.16: Shares of energy types in commercial and public services final energy	103
Figure 5.17: Final energy in commercial and public services by energy type	103
Figure 5.18: Shares of final energy in commercial services by subsector	106
Figure 5.19: Final energy in commercial services by subsector (breakdown not available prior to 2009)	106
Figure 5.20: Shares of final energy in public services by subsector	107
Figure 5.21: Final energy in public services by subsector (breakdown not available prior to 2009)	107
Figure 5.22: Final energy consumption in agriculture by energy type.....	108
Figure 5.23: Final energy in fisheries by energy type	109
Figure 6.1: Current split of final energy by mode of application	110
Figure 6.2: Final energy in electricity, transport and heat modes	111
Figure 6.3: Final energy use for heat, actual and weather corrected	112
Figure 6.4: Final energy in heat mode by sector	113
Figure 6.5: Final energy in heat mode by fuel	114
Figure 6.6: Final consumption of oil for heat by sector	115
Figure 6.7: Final energy in transport mode by subsector.....	116
Figure 6.8: Final energy in transport mode by fuel	118
Figure 6.9: Final consumption in electricity mode by sector.....	119
Figure 6.10: Shares of primary energy by mode.....	121
Figure 6.11: Primary energy by mode	121
Figure 7.1: GHG emissions per kWh of electricity total consumption, with contribution by energy type	122
Figure 7.2: Current sectoral shares of energy-related CO ₂ eq	124
Figure 7.3: Energy-related annual CO ₂ eq emissions by sector	124
Figure 7.4: Industry energy-related CO ₂ eq emissions by energy type.....	126
Figure 7.5: Transport energy-related emissions by energy type (including international aviation)	127
Figure 7.6: Residential energy-related CO ₂ eq by energy type	129
Figure 7.7: Commercial and public services sector CO ₂ eq emissions by energy type	130
Figure 7.8: Energy-related CO ₂ eq emissions by electricity, transport and heat	131
Figure 7.9: Avoided emissions from renewable energy in electricity, heat and transport modes	133
Figure 7.10: Avoid emissions by renewable energy sub-type	134
Figure 8.1: National total greenhouse gas emissions by source	135
Figure 8.2: Energy-related and total GHG emissions in the ETS and non-ETS sectors for 2005 baseline and 2024.....	136
Figure 8.3: Index of non-ETS greenhouse gas emissions relative to 2005	137
Figure 8.4: Energy-related non-ETS greenhouse gas emissions.....	138
Figure 8.5: EU ETS greenhouse gas emissions in Ireland (excluding aviation)	139
Figure 8.6: Sectoral ceilings first two carbon budgets, spanning 2021–2025 and 2026–2030	141

Figure 8.7: Electricity sectoral emissions ceiling	142
Figure 8.8: Transport sectoral emissions ceiling.....	143
Figure 8.9: 2023 share of renewable energy by mode.....	147
Figure 8.10: Renewable energy share (overall RES) with contribution by mode.....	147
Figure 8.11: Renewable energy share (overall RES) and trajectories to 2030 target	148
Figure 8.12: Renewable and fossil gross final energy consumption by mode	148
Figure 8.13: Renewable energy share of gross final consumption by source	149
Figure 8.14: RES-E - gross final consumption of electricity from renewable sources	151
Figure 8.15: RES-H – gross final consumption of heat from renewable sources	153
Figure 8.16: RES-T – final consumption in transport from renewable sources	157
Figure 8.17: Renewable energy in transport by energy type (without multipliers).....	159
Figure 8.18: Electricity prices to industry ²⁵	161
Figure 8.19: Oil prices to industry ²⁵	162
Figure 8.20: Natural gas prices to industry ²⁵	163
Figure 8.21: Real energy price changes to industry since 2020 (index)	164
Figure 9.1: Index of modified domestic demand, final energy demand and energy price	166
Figure 9.2: Index of final energy, primary energy and energy-related CO ₂ eq	167
Figure 9.3: Deviation from average heating degree days (2005 to 2024) and resulting weather adjustment.....	168
Figure 9.4: Primary, final and electricity intensities.....	169
Figure 10.1: Total primary energy requirement by sector	170
Figure 10.2: Industry energy intensity.....	171
Figure 10.3: Private car total annual vehicle-kilometres.....	172
Figure 10.4: Specific CO ₂ emissions of new cars, excluding battery electric vehicles.....	173
Figure 10.5: Share of private cars licensed for the first time that are new or pre-owned imports	174
Figure 10.6: Private cars licensed for the first time that are battery electric vehicles.....	175
Figure 10.7: Road freight activity.....	176
Figure 10.8: Road freight activity by main type of work done	177
Figure 10.9: Residential final energy.....	178
Figure 10.10: Energy per dwelling (permanently occupied).....	179
Figure 10.11: Unit energy-related greenhouse gas emissions per dwelling.....	181
Figure 10.12: Energy intensity of the commercial and public services sector	181
Figure 10.13: Energy per employee in the commercial and public services sector	182
Figure 11.1: Monthly utility scale electricity supply – January 2016 to September 2025	186
Figure 11.2: Average monthly utility scale electricity supply over last 5 years (normalised for number of days per month).....	187
Figure 11.3: Share of renewable and non-renewable utility scale electricity supply – January to September 2025.....	188
Figure 11.4: Share of utility scale electricity supply split by indigenous generation and net imports	189
Figure 11.5: Monthly utility scale electricity generation – January 2016 to September 2025	190
Figure 11.6: Monthly electricity generation by source in 2025	191
Figure 11.7: Sources of electricity generation – 12-month rolling average to September 2025	192
Figure 11.8: Monthly imported and exported electricity (with net imports) – January 2016 to September 2025	193
Figure 11.9: Monthly gas supply trends – January 2016 to September 2025	194
Figure 11.10: Average monthly gas supply over the last 5 years (normalised for number of days per month).....	195
Figure 11.11: 2025 monthly gas supply.....	196
Figure 11.12: Source of gas supply – 12-month rolling average to September 2025	197
Figure 11.13: Oil and road liquid biofuel supply trends – January 2016 to September 2025	199
Figure 11.14: Trend in heating kerosene – January 2016 to September 2025.....	200
Figure 11.15: Trend in jet kerosene – January 2016 to September 2025	201

Figure 11.16: Trend in road gasoline and road bioethanol – January 2016 to September 2025202

Figure 11.17: Trend in road diesel and road biodiesels – January 2016 to September 2025203

Figure 11.18: Monthly major oil product and road biofuel supply – 2025 to date204

Figure 11.19: Breakdown of major oil product and road liquid biofuel supply – 2025 to date.....205

Appendix 2: List of tables

Table 3.1: Primary energy production by energy type (share in brackets)	47
Table 3.2: Imports by energy type (share in brackets)	48
Table 3.3: Exports by energy type (share in brackets)	49
Table 3.4: Net imports by energy type (share in brackets)	49
Table 3.5: Primary energy requirement by energy type (share in brackets)	52
Table 4.1: Energy input to electricity generation by energy type (share in brackets)	57
Table 4.2: Energy input to electricity generation from renewables and non-renewables (share in brackets)	57
Table 4.3: Gross electricity production from all producers, excluding storage, by energy sub-type (share in brackets)	60
Table 4.4: Main activity producer - gross electricity production by energy type (share in brackets)	63
Table 4.5: Autoproducers - gross electricity production by energy type (share in brackets)	64
Table 4.6: Electricity imports and exports	65
Table 4.7: Gross electricity supply by energy type (share in brackets)	67
Table 4.8: Gross electricity supply from renewables, non-renewables and net imports (share)	67
Table 4.9: Battery generation capacity at year end	70
Table 4.10: Battery storage capacity at year end	71
Table 4.11: Battery storage consumption and generation	71
Table 4.12: Operational CHP electrical capacity by fuel (share in brackets)	72
Table 4.13: Electricity exported to the grid from CHP	73
Table 4.14: Installed solar PV capacity (AC) at year end	74
Table 4.15: Wind capacity	76
Table 4.16: Annual capacity factor for wind and hydro	79
Table 4.17: Outputs from oil refining (share)	80
Table 4.18: Biomethane injected into the natural gas network	81
Table 5.1: Final energy consumption by energy type (share in brackets)	85
Table 5.2: Final energy consumption disaggregated by energy sub-type (share in brackets)	86
Table 5.3: Final energy consumption by sector (share in brackets)	88
Table 5.4: Final energy in industry by energy type (share in brackets)	90
Table 5.5: Final energy consumption in industry disaggregated by energy sub-type (share in brackets)	91
Table 5.6: Final energy in transport sector by energy sub-types (share in brackets)	94
Table 5.7: Transport final energy by subsector (share in brackets)	96
Table 5.8: Road transport – final energy by energy sub-type	97
Table 5.9: International and domestic aviation – final energy by energy sub-type (share in brackets)	98
Table 5.10: Navigation, rail and pipeline – final energy by energy sub-type (share in brackets)	99
Table 5.11: Final energy in residential sector by energy type (share in brackets)	101
Table 5.12: Final energy in residential sector disaggregated by energy sub-type (share in brackets)	102
Table 5.13: Final energy in commercial and public services (share in brackets)	104
Table 5.14: Final energy in commercial and public services disaggregated by energy sub-type (share in brackets)	105
Table 5.15: Final energy in agriculture by energy sub-type (share in brackets)	108
Table 5.16: Final energy in fisheries by energy sub-type (share in brackets)	109
Table 6.1: Final energy in electricity, transport and heat compared (share)	111
Table 6.2: Final energy in heat mode by sector (share)	113
Table 6.3: Final energy in heat mode by energy type (share)	114
Table 6.4: Final energy in transport mode by subsector (share)	117
Table 6.5: Final energy in transport mode by fuel (share in brackets)	118
Table 6.6: Final consumption in electricity mode by sector (share)	120
Table 7.1: Energy-related CO ₂ eq by sector (share)	125
Table 7.2: Quantities and shares of energy-related CO ₂ eq emissions in industry (share)	126
Table 7.3: Quantities and shares of energy-related CO ₂ eq emissions in transport (share)	127

Table 7.4: Quantities and shares of transport CO ₂ eq emissions by subsector, including GHG emissions from direct combustion and indirect GHG emissions from electricity use (share).....	128
Table 7.5: Quantities and shares of energy-related CO ₂ eq emissions in the residential sector (share).....	129
Table 7.6: Quantities and shares of CO ₂ eq emissions in commercial and public services (share).....	130
Table 8.1: Energy-related GHG emissions, ETS and non-ETS (share).....	139
Table 8.2: Ireland's progress towards renewable energy share (RES) targets	146
Table 8.3: Overall RES - Renewable energy in gross final consumption by energy type (share).....	150
Table 8.4: RES-E – gross final consumption of electricity from renewable sources (share in brackets) ..	152
Table 8.5: RES-H – gross final consumption of heat from renewable sources (share).....	154
Table 8.6: Multipliers used in the calculation of RES-T.....	156
Table 8.7: Limits on biofuels from certain feedstocks	157
Table 8.8: Renewable energy for transport by source - without multipliers or limits (share).....	159
Table 8.9: Electricity price to industry change since 2020	161
Table 8.10: Oil prices to industry change since 2020	162
Table 8.11: Natural gas prices to industry change since 2020	163
Table 9.1: GDP, MDD, final energy, primary energy and energy-related CO ₂ eq	166
Table 10.1: Primary energy by sector compared with previous years (share)	171
Table 10.2: Electric private cars licensed for the first time (share).....	175
Table 10.3: Road freight activity	176
Table 10.4: Quantities of energy consumption and CO ₂ eq emissions per dwelling (share).....	180
Table 10.5: Quantities of energy per employee in commercial and public services (share)	182

Appendix 3: Glossary of abbreviations

Abbreviation	Explanation
BEUS	Business Energy Use Survey
CAP	Climate action plan
CCGT	Combined cycle gas turbine
CHP	Combined heat and power
CO ₂	Carbon dioxide
CO ₂ eq	Carbon dioxide equivalent
CSO	Central Statistics Office
DART	Dublin Area Rapid Transit
DCEE	Department of Climate, Energy and the Environment
EPA	Environmental Protection Agency
ETS	EU Emission Trading Scheme
EV	Electric vehicle
FEC	Final energy consumption
GCV	Gross calorific value
GDP	Gross domestic product
GFC	Gross final consumption (of energy)
GHG	Greenhouse gas
GNI*	Modified gross national income
GNP	Gross national product
goe	Gramme of oil equivalent
GWh	Gigawatt hour
HGV	Heavy goods vehicle
HVO	Hydrotreated vegetable oil
ICT	Information and communications technology
IEA	International Energy Agency
IP	Intellectual property
IPCC	Intergovernmental Panel on Climate Change
ktoe	Kilotonne of oil equivalent
LFG	Landfill gas
LNG	Liquefied natural gas
MDD	Modified domestic demand

Abbreviation	Explanation
NCT	National Car Testing
NCV	Net calorific value
NECP	National Energy and Climate Plan
NEDC	New European Driving Cycle
NEEAP	National energy efficiency action plan
NORA	National Oil Reserves Agency
NREAP	National renewable energy action plan
NRW	Non-renewable waste
OECD	Organization for Economic Co-operation and Development
PV	Photovoltaic
R&D	Research and development
RED	Renewable Energy Directive
RED I	First Renewable Energy Directive - Directive 2009/28/EC
RED II	Recast Renewable Energy Directive – Directive (EU) 2018/2001
RED III	Updated recast Renewable Energy Directive – Directive (EU) 2018/2001 as amended by Directive (EU) 2023/2413
RES	Renewable energy share
RES-E	Renewable energy share in electricity
RES-H	Renewable energy share in heat
RES-T	Renewable energy share in transport
SEAI	Sustainable Energy Authority of Ireland
TFC	Total final energy consumption
TJ	Terajoule
toe	Tonne of oil equivalent
TPER	Total primary energy requirement
TWh	Terawatt hour
UCO	Used cooking oil
UNFCCC	United Nations Framework Convention on Climate Change
VOC	Volatile organic compounds
WLTP	Worldwide Harmonised Light Vehicle Test Procedure

Appendix 4: Glossary of terms

Autoproducer: Regulation (EC) No 1099/2008 defines autoproducers as private or publicly owned producers that generate electricity and/or heat wholly or partly for their own use which supports their primary activity. Examples of autoproducers are rooftop solar PV on homes or CHP units at manufacturing sites. Autoproducers may or may not export electricity to the grid.

Biomass: Directive (EU) 2018/2001 defines biomass as the biodegradable fraction of products, waste and residues from biological origin from agriculture, including vegetal and animal substances, from forestry and related industries, including fisheries and aquaculture, as well as the biodegradable fraction of waste, including industrial and municipal waste of biological origin.

Carbon dioxide (CO₂): A compound of carbon and oxygen formed when carbon is burned. Carbon dioxide is one of the main greenhouse gases. Units used in this report are t CO₂ – tonnes of CO₂, kt CO₂ – kilotonnes of CO₂ (10³ tonnes) and MtCO₂ – megatonnes of CO₂ (10⁶ tonnes).

Carbon dioxide equivalent (CO₂eq): Used as a measure of the global warming impact of greenhouse gases on the basis of their global warming potential. Quantities of gases other than CO₂ are converted into equivalent quantities of CO₂ based on their global warming potential.

Carbon intensity (gCO₂/kWh): This is the amount of CO₂ that will be released per kWh of energy of a given fuel. For most fossil fuels the value of this is almost constant, but in the case of electricity it will depend on the fuel mix used to generate the electricity and also on the efficiency of the technology employed.

Combined heat and power (CHP) plants: CHP refers to plants which are designed to produce both heat and electricity, for own use only or third-party owned and selling electricity and heat on site as well as exporting electricity to the grid.

Energy intensity: The amount of energy used per unit of activity. Examples of activity used in this report are gross domestic product (GDP), value added, number of households, employees, etc. Where possible, the monetary values used are in constant prices.

Gross and net calorific value (GCV and NCV): The GCV gives the maximum theoretical heat release during combustion, including the latent heat of condensation of the water vapour produced during combustion. This water is produced by the combustion of the hydrogen in the fuel, or in some cases from the evaporation of water already present in the fuel. The NCV excludes this heat of condensation because it cannot be recovered in conventional boilers. For natural gas, the difference between GCV and NCV is about 10%, for oil it is approximately 5%.

Gross domestic product (GDP): The GDP represents the total output of the economy over a period.

Gross electricity production: Gross electricity production is measured at the output terminals of the main generator or inverter; it therefore includes the energy taken by station auxiliaries and losses in transformers that are considered integral parts of the station. The difference between gross and net production is the amount of own use of electricity in the generation plants.

Gross final consumption (GFC): Directive (EU) 2018/2001 defines gross final consumption of energy as *“the energy commodities delivered for energy purposes to industry, transport, households, services including public services, agriculture, forestry and fisheries, the consumption of electricity and heat by the energy branch for electricity and heat production, and losses of electricity and heat in distribution and transmission.”*

Gross inland energy consumption: Sometimes abbreviated as gross inland consumption, is the total energy demand of a country or region. It represents the quantity of energy necessary to satisfy inland consumption of the geographical entity under consideration.

Heating degree days: 'Degree days' is the measure or index used to take account of the severity of the weather when looking at energy use in terms of heating (or cooling) 'load' on a building. A degree day is an expression of how cold (or warm) it is outside, relative to a day on which little or no heating (or cooling) would be required. It is thus a measure of the cumulative temperature deficit (or surplus) of the outdoor temperature relative to a neutral target temperature (base temperature) at which no heating or cooling would be required.

Main activity producer: Regulation (EC) No 1099/2008 defines main activity producers as a privately or publicly owned producers that generate electricity and/or heat for sale to third parties as their principal activity.

Modified gross national income (GNI*): Modified gross national income (or GNI*) was introduced by the CSO in 2017 to assess the level of activity in the Irish economy excluding the effects of globalisation that disproportionately affect the Irish economic results. GNI* is defined as GNI less the effects of the profits of re-domiciled companies and the depreciation of intellectual property products and aircraft leasing companies.

Nominal and real values: Nominal value refers to the current value expressed in money terms in a given year, whereas real value adjusts nominal value to remove effects of price changes and inflation to give the constant value over time indexed to a reference year.

Total final consumption (TFC): This is the energy used by the final consuming sectors of industry, transport, residential, services, agriculture and fisheries. It excludes the energy sector: electricity generation, oil refining, *etc.* Also referred to as total final energy consumption.

Total primary energy requirement (TPER): This is the total requirement for all uses of energy, including energy used to transform one energy form to another (such as burning fossil fuel to generate electricity) and energy used by the final consumer.

Value added: Value added is an economic measure of output. The value added of industry, for instance, is the additional value created by the production process through the application of labour and capital. It is defined as the value of industry's output of goods and services less the value of the intermediate consumptions of goods (raw materials, fuel, *etc.*) and services.

Wastes (non-renewable): The non-renewable portion of wastes used as an energy source.

Weather correction: Annual variations in weather affect the space heating requirements of occupied buildings. Weather correction involves adjusting the energy used for space heating by benchmarking the climate in a particular year with that of a long-term average measured in terms of number of degree days.

Appendix 5: Energy units and conversion factors

Energy conversion factors

From:	To:	toe	MWh	GJ
	Multiply by			
toe		×1	×11.63	×41.868
MWh		÷11.63	×1	×3.6
GJ		÷41.868	÷3.6	×1

Energy units

Joule (J): Joule is the international (SI) unit of energy.

Kilowatt hour (kWh): The conventional unit of energy that electricity is measured by and charged for commercially.

Tonne of oil equivalent (toe): This is a conventional standardised unit of energy. One tonne of oil equivalent is defined as having a net calorific value (NCV) of 41.868 GJ. A related unit is the kilogram of oil equivalent (kgoe), where 1 kgoe = 10⁻³ toe.

Decimal prefixes

deca (da)	10 ¹	deci (d)	10 ⁻¹
hecto (h)	10 ²	centi (c)	10 ⁻²
kilo (k)	10 ³	milli (m)	10 ⁻³
mega (M)	10 ⁶	micro (μ)	10 ⁻⁶
giga (G)	10 ⁹	nano (n)	10 ⁻⁹
tera (T)	10 ¹²	pico (p)	10 ⁻¹²
peta (P)	10 ¹⁵	femto (f)	10 ⁻¹⁵
exa (E)	10 ¹⁸	atto (a)	10 ⁻¹⁸

Calorific values

Fuel	toe/t (NCV)	MJ/m ³ (NCV)	MJ/kg (NCV)
Crude oil	1.023	-	42.81
Gasoline / petrol (100% petroleum)	1.065	-	44.59
Kerosene	1.056	-	44.20
Jet kerosene	1.053	-	44.10
Gas oil / diesel (100% petroleum)	1.034	-	43.31
Residual fuel oil (heavy oil)	0.985	-	41.24
Milled peat (2024)	0.130	-	5.43
Sod peat	0.313	-	13.10
Peat briquettes	0.443	-	18.55
Coal	0.665	-	27.84
LPG	1.126	-	47.16
Petroleum coke (2024)	0.754	-	31.57
Natural gas (2024)	-	35.20	-

Emission factors

CO₂ emission factors from fuel combustion are listed below. These emission factors include direct CO₂ emissions from fuel combustion and do not include indirect or upstream emissions, such as those from extraction, cultivation, transportation or processing of fuels. To calculate GHG quantities from fuel combustion, SEAI use sector and fuel specific data from the EPA's National Inventory Submissions [3] (see section 2.3 for more details).

Liquid Fuels	t CO ₂ /TJ (NCV)	g CO ₂ /kWh (NCV)
Motor Spirit (Gasoline)	70.0	251.9
Jet Kerosene	71.4	257.0
Other Kerosene	71.4	257.0
Gas/Diesel Oil	73.3	263.9
Residual Oil	76.0	273.6
LPG	63.7	229.3
Solid Fuels and Derivatives		
Petroleum Coke (2024)	93.37	336.1
Coal	94.6	340.6
Milled Peat (2024)	136.58	491.7
Sod Peat	104.0	374.4
Peat Briquettes	98.9	355.9
Gas		
Natural Gas (2024)	56.52	203.5
Electricity		

See SEAI website: <https://www.seai.ie/data-and-insights/seai-statistics/conversion-factors>

Appendix 6: Data sources

SEAI gratefully acknowledges the co-operation of all the organisations, agencies, energy suppliers and distributors that provide data and respond to its questionnaires and surveys throughout the year:

- Applus+ (National Car Test)
- Bus Eireann
- Central Statistics Office
- Department of Agriculture, Food and the Marine
- Department of Climate, Energy and the Environment
- Department of Housing, Local Government, and Heritage
- Department of Transport (Vehicle Registration Unit)
- EirGrid
- Environmental Protection Agency
- ESB Networks
- European Commission DG ENER
- European Heat Pump Association
- EU-funded ODYSSEE Project
- Eurostat
- Gas Networks Ireland
- International Energy Agency
- Irish Rail
- Met Éireann
- National Oil Reserves Agency (NORA)
- Revenue Commissioners
- Road Safety Authority
- Vehicle Certification Agency UK
- Survey respondents:
 - Electricity generators
 - CHP operators
 - Biogas & biomass operators
 - Landfill gas operators
 - Wind autoproducers
 - Wood and biomass suppliers
 - Solid fuel (coal) suppliers
 - Oil product suppliers
 - Electricity suppliers
 - Natural gas suppliers

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Appendix 8: Energy statistics revisions

Some changes, revisions and corrections to the historic energy balance data were implemented during 2025. The most significant of these are listed below.

Total Final Energy Consumption

- Sectoral split revised to align with the latest data from the 2023 CSO Business Energy Use Survey (BEUS), including historic revisions. Please see full methodological changes from the CSO at <https://www.cso.ie/en/releasesandpublications/ep/p-beu/businessenergyuse2023/backgroundnotes/>

Bituminous Coal

- Calorific values for bituminous coal used in industry was updated to align to data available from the EU Emissions Trading Scheme.

Wind

- For 2014 onwards, gross electricity production from wind has been calculated from metered net electricity production data (from EirGrid) and estimates of average losses and own consumption at wind farms.

Biodiesel

- Biodiesel consumption added to industry and services from 2023 onwards.

Ambient Heat

- The ambient heat calculation has been revised from 2018, to reflect an update to the stock model of domestic heat pumps, based on the latest data from the BER database.

Gas oil and kerosene consumption in agriculture, residential and marine

SEAI Energy Statistics receives monthly data on the sales of each petroleum product for consumption within Ireland; it uses a several sources of data and statistical methods to apportion this consumption between the final use sectors of the economy in the National Energy Balance.

Department of Climate, Energy and the Environment's Oil Levy Application (OLA) database is the primary source of data on the import, export, stock change and sales of oil products within the State. Suppliers report monthly volumes for each of their oil products into the OLA database. SEAI use aggregated data from all suppliers to determine the monthly and annual supply of each oil product.

While the OLA database provides data on the total consumption of each oil products, it does not provide a breakdown of sales/consumption at a sectoral or activity level, as required for the national energy statistics. Certain oil products, e.g. motor gasoline or road diesel, are used in only one sector, e.g. transport. However, gas oil and kerosene are used across multiple sectors. Gas oil is used as fuel in engines, generators and boilers across every sector, including international marine bunkers and electricity generation.

Annual consumption of gas oil and kerosene is then apportioned between the sectors of the economy based on, amongst other data, the CSO's Business Energy Use Survey, tax rebate data on marine fuels, EU-ETS data.

SEAI is currently undertaking a programme to develop and implement new processes to collect annual quantitative data on the consumption of gas oil and kerosene within the residential and agriculture sectors, along with additional disaggregated data on the consumption of gas oil in international marine bunkers and domestic navigation.

Once the new data-collection processes are implemented, SEAI is likely to revise the timeseries for gas oil and kerosene consumption in the agriculture and residential sectors. Updates to the gas oil timeseries for international marine bunkers and domestic navigation may also be necessary. SEAI expects the programme to deliver new sectoral data in 2026.

Appendix 9: Electricity and gas price tables

SEAI collects and analyses electricity and gas prices every 6-months, in line with EU Regulation 2016/1952. SEAI calculates the effective unit price of energy (the revenue collected for energy delivered, divided by the total quantity of energy delivered) across different consumption bands, and then calculate the weighted average price, based on the market-share of each consumption band. Using data available from Eurostat, SEAI ranks the effective unit price paid by Irish consumers, compared to their European counterparts.

The tables below update the effective electricity and gas energy prices paid by residential and business consumers in Ireland for January – June 2025. The tables breakdown the effective unit price of energy in each band, the changes in that price over the last 12- and 24-months, Ireland's EU ranking of the price in that band, with a ranking of 1 meaning highest price, and the market share of each band.

Ireland introduced several measures to alleviate the impact of elevated energy prices experienced since 2021. Domestic electricity customers, including 'pay as you go' customers, have to-date received total energy account credits of €1,500 off their bills, broken down as:

- four credits of €200 in April 2022, November 2022, January 2023 and March 2023
- three credits of €150 in December 2023, January 2024 and March 2024
- two credits of €125 in November 2024 and January 2025.

These rebates are administered by electricity suppliers and are accounted for in the residential electricity prices for the relevant semester. They act to reduce the effective unit price of residential electricity. A further measure to tackle rising energy costs is a cut on VAT on electricity bills, which was reduced from 13.5% to 9% from 1st of May 2022.

Electricity and Gas Price Comparisons

For residential electricity and gas prices, SEAI provide Ireland's energy prices and EU ranking in terms of both Euro (EUR) and purchasing power standard (PPS) units.

PPS is a 'virtual' unit of currency, calculated by Eurostat, in which national 'account aggregates' are normalised to recognise price-level differences between different countries. Notionally, one unit of PPS currency allows someone to purchase the same amount of goods and services in any country (which is not true for one unit of EUR, because of the varying costs of living in Eurozone countries).

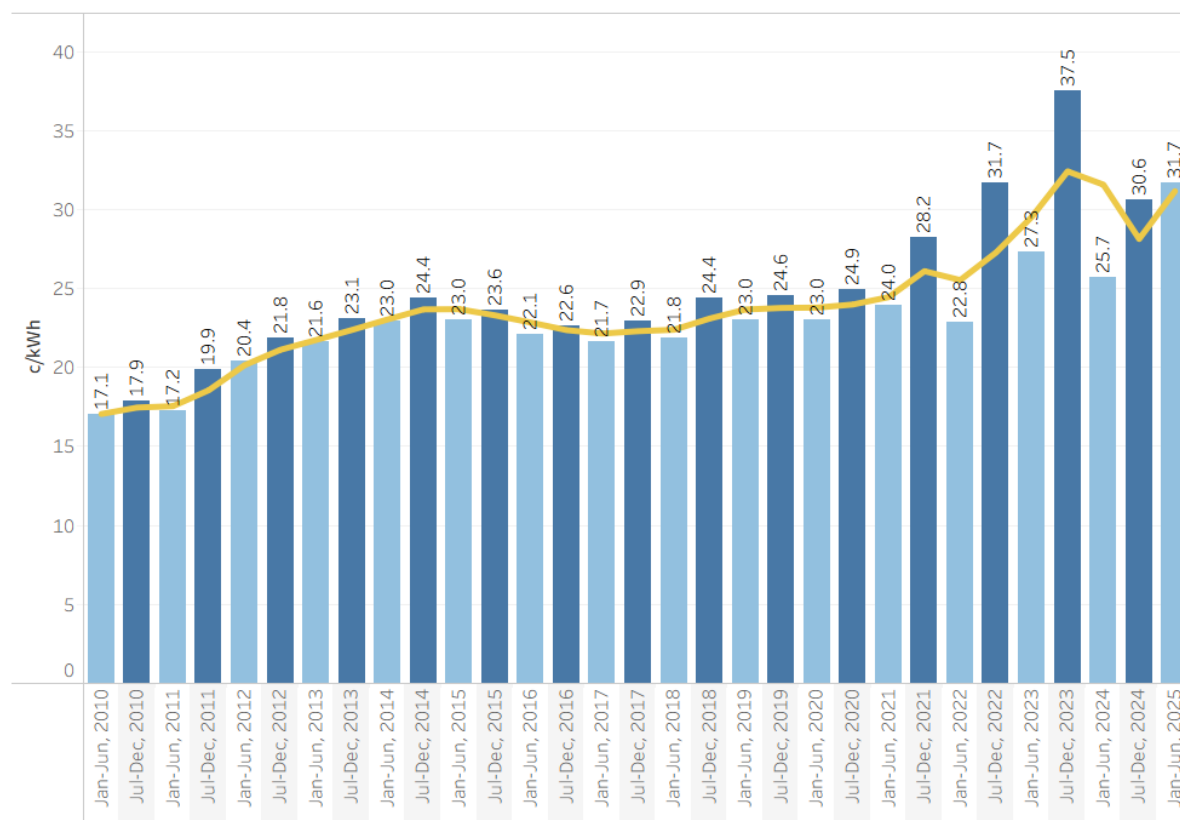
When comparing the cost of energy to households, PPS-ranking can offer more nuanced insights than a EUR-ranking, because PPS considers the relative cost of living and the inflation rates of countries. While the normalisation applied in calculating the PPS prices includes consideration of the gross domestic product (GDP) of a country, it is not limited to a GDP correction – the PPS calculation involves product level, product group level, and aggregation level normalisations, to help ensure its robustness.

Household Prices - Electricity

Table A9.1: Household Electricity Prices – January to June 2025

Household electricity prices (all taxes included) weighted average across all suppliers	c/kWh S1 2025	Change in 12 months	Change in 24 months	Band Share of Market
Band DA Consumption < 1,000 kWh	33.5	+93%	+152%	8.6%
Band DB 1,000 kWh < Consumption < 2,500 kWh	36.5	+79%	+90%	10.8%
Band DC 2,500 kWh < Consumption < 5,000 kWh	32.6	+26%	+19%	33.6%
Band DD 5,000 kWh < Consumption < 15,000 kWh	30.0	+8%	+0%	39.4%
Band DE Consumption > 15,000 kWh	27.7	-4%	-6%	7.5%
Weighted Average	31.7	+24%	+16%	

Figure A9.1: Effective Unit Price (EUP) of Household Electricity in Ireland



*weighted average across all suppliers, and all consumption bands. Yellow line shows 12-month moving average to smooth-out seasonal effects

Table A9.2: Household Electricity Prices – EU Ranking – January to June 2025

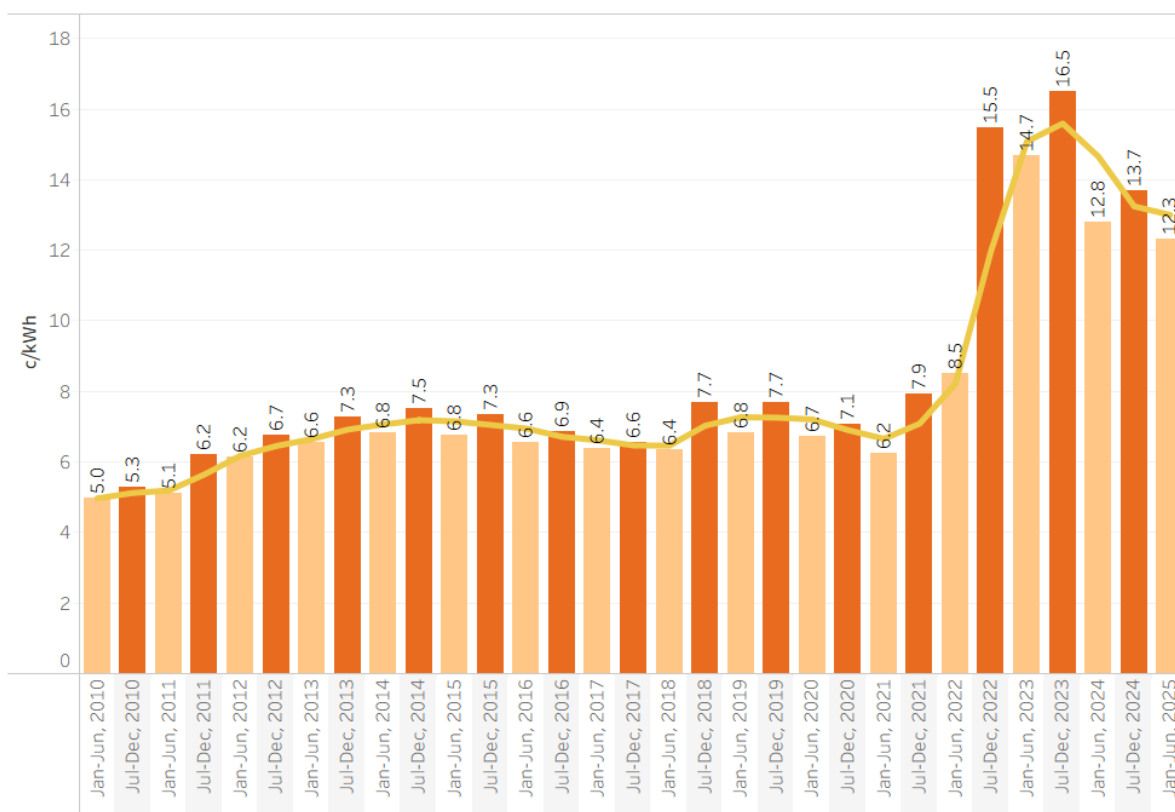
Household electricity prices (all taxes included) weighted average across all suppliers	c/kWh S1 2025	Ranking EU (EUR)	PPS/kWh S1 2025	Ranking EU (PPS)
Band DA Consumption < 1,000 kWh	33.5	15	28.3	22
Band DB 1,000 kWh < Consumption < 2,500 kWh	36.5	6	30.9	9
Band DC 2,500 kWh < Consumption < 5,000 kWh	32.6	5	27.5	12
Band DD 5,000 kWh < Consumption < 15,000 kWh	30.0	4	25.4	12
Band DE Consumption > 15,000 kWh	27.7	6	23.4	16
Weighted Average	31.7			

Household Prices – Gas

Table A9.3: Household Gas Prices – January to June 2025

Household gas prices (all taxes included) weighted average across all suppliers	c/kWh S1 2025	Change in 12 months	Change in 24 months	Band Share of Market
Band D1 Consumption < 20 GJ	14.7	+3%	-7%	7.8%
Band D2 20 GJ < Consumption < 200 GJ	12.2	-4%	-17%	89.7%
Band D3 Consumption > 200 GJ	10.2	+5%	-11%	2.5%
Weighted Average	12.3	-4%	-16%	

Figure A9.2: Effective Unit Price (EUP) of Household Gas in Ireland



*weighted average across all suppliers, and all consumption bands. Yellow line shows 12-month moving average to smooth-out seasonal effects

Table A9.4: Household Gas Prices – EU Ranking – January to June 2025

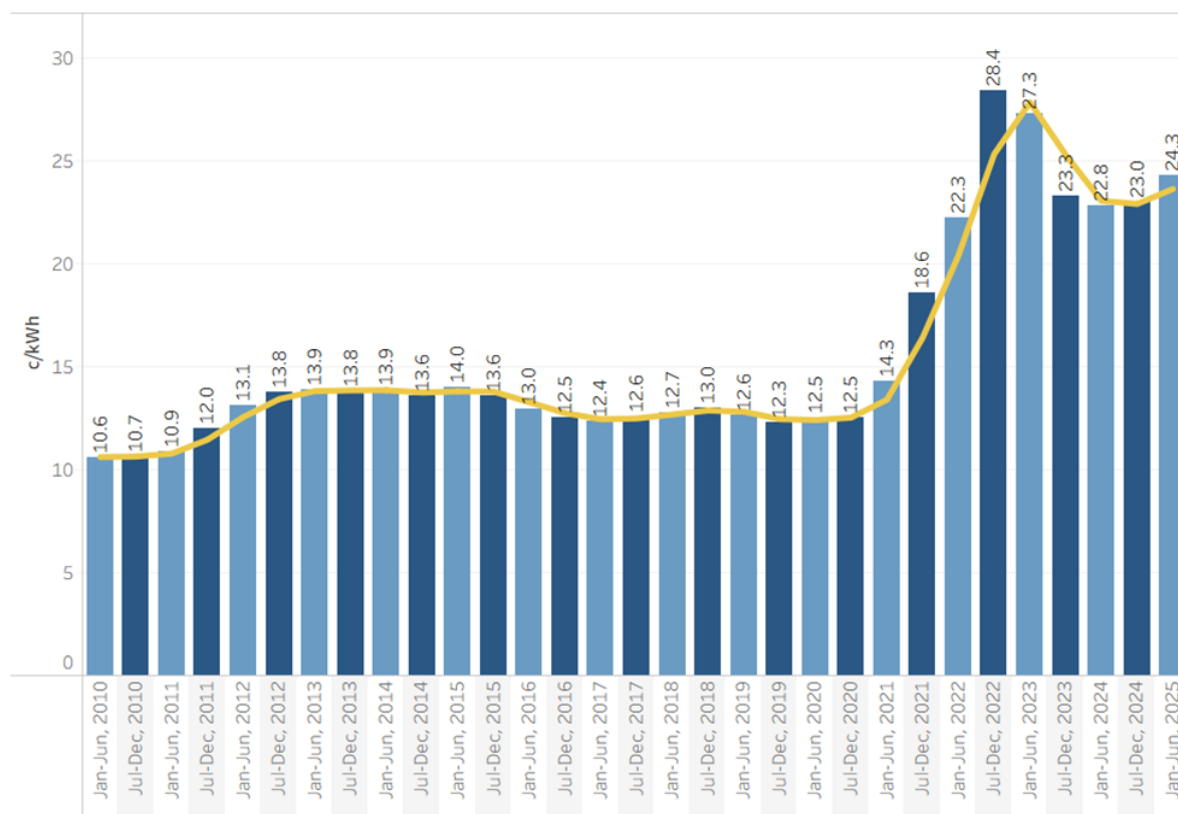
Household gas prices (all taxes included) weighted average across all suppliers	c/kWh S1 2025	Ranking EU (EUR)	PPS/kWh S1 2025	Ranking EU (PPS)
Band D1 Consumption < 20 GJ	14.7	12	12.4	14
Band D2 20 GJ < Consumption < 200 GJ	12.2	8	10.3	12
Band D3 Consumption > 200 GJ	10.2	9	8.6	17
Weighted Average	12.3			

Business Prices – Electricity

Table A9.5: Business Electricity Prices (EUR) – January to June 2025

Business electricity prices (ex VAT) weighted average across all suppliers	c/kWh S1 2025	Change in 12 months	Change in 24 months	Ranking (EU)	Band Share of Market
Band IA Consumption < 20 MWh	31.3	-3%	-16%	6	6.4%
Band IB 20 MWh < Consumption < 500 MWh	30.7	+1%	-14%	1	19.4%
Band IC 500 MWh < Consumption < 2,000 MWh	27.3	+6%	-4%	1	10.3%
Band ID 2,000 MWh < Consumption < 20,000 MWh	22.8	+5%	-5%	1	24.5%
Band IE 20,000 MWh < Consumption < 70,000 MWh	21.7	+5%	-5%	2	9.5%
Band IF 70,000 MWh < Consumption < 150,000 MWh	19.7	+0%	-18%	1	6.4%
Band IG > 150,000 MWh	19.5	+19%	-8%	2	23.6%
Weighted Average	24.3	+6%	-11%		

Figure A9.3: Effective Unit Price (EUP) of Business Electricity in Ireland



*weighted average across all suppliers, and all consumption bands. Yellow line shows 12-month moving average to smooth-out seasonal effects

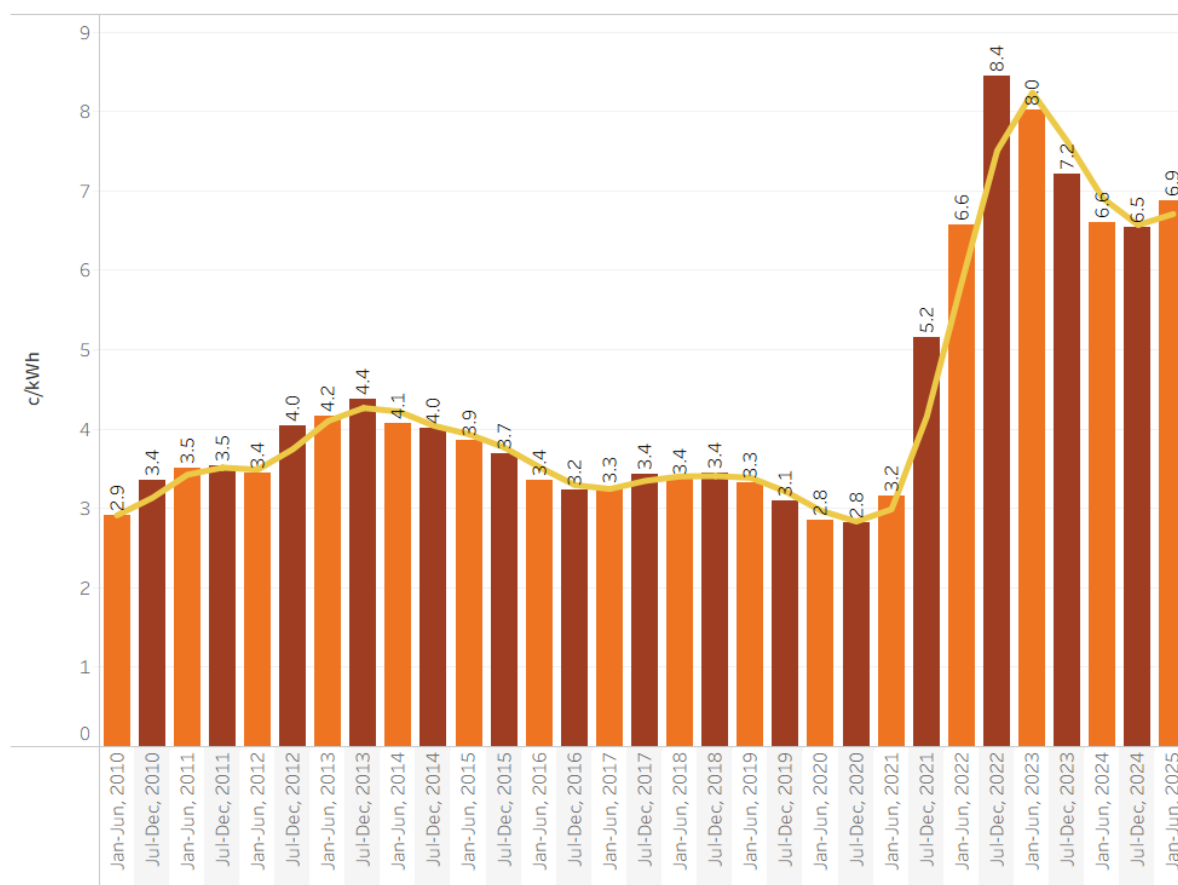
Business Prices – Gas

Table A9.6: Business Gas Prices (EUR) – January to June 2025

Business gas prices (ex VAT) weighted average across all suppliers	c/kWh S1 2025	Change in 12 months	Change in 24 months	Ranking (EU)	Band Share of Market
Band I1 Consumption < 1,000 GJ	9.8	-11%	-24%	7	c
Band I2 1,000 GJ < Consumption < 10,000 GJ	8.2	+5%	-15%	9	21.1%
Band I3 10,000 GJ < Consumption < 100,000 GJ	7.1	+10%	-8%	7	23.6%
Band I4 100,000 GJ < Consumption < 1,000,000 GJ	5.3	+5%	-16%	12	39.2%
Band I5 1,000,000 GJ < Consumption < 4,000,000 GJ	c	c	c	c	c
Weighted Average	6.9	+4%	-14%		

c = confidential (statistically suppressed)

Figure A9.4: Effective Unit Price (EUP) of Business Gas in Ireland



*weighted average across all suppliers, and all consumption bands. Yellow line shows 12-month moving average to smooth-out seasonal effects

Appendix 10: Energy in Ireland data and charts

The data and charts used in the preparation of this report, including timeseries data back to 1990, are available in spreadsheet format on the SEAI website:

<https://www.seai.ie/data-and-insights/seai-statistics/key-publications/energy-in-ireland>

Complete versions of the National Energy Balance for 1990 to 2024 are published on the SEAI website:

<https://www.seai.ie/data-and-insights/seai-statistics/key-publications/national-energy-balance>

Condensed versions of the 2024 National Energy Balance are provided in Table A10.2, in TWh, and Table A10.3, in ktoe.

Table A10.1: Energy products and sub-types in the National Energy Balance

Energy Type	Energy Sub-Type / Product	Note	Eurostat Product Code
Coal	Anthracite & Manufactured Ovoids	Anthracite and manufactured ovoids quantities are aggregated to avoid statistical disclosure	C0110
Coal	Bituminous Coal		C0129
Coal	Coke	Not used in Ireland since 1995	C0121 & C0311
Coal	Lignite / Brown Coal Briquettes		C0330
Electricity	Electricity		E7000
Natural Gas	Natural Gas		G3000
Non-Renewable Waste	Non-Renewable Waste		W6100_6220
Oil	Bitumen	Not used for energy purposes	O4695
Oil	Crude		O4100
Oil	Diesel / Gas Oil		O4671XR5220B
Oil	Fuel Oil		O4680
Oil	Gasoline		O4652XR5210B
Oil	Jet Kerosene		O4661XR5230B
Oil	Kerosene		O4669
Oil	LPG	Liquefied petroleum gases, includes propane and butane	O4630
Oil	Lubricants	Not used for energy purposes	O4692
Oil	Naphtha		O4640
Oil	Petroleum Coke		O4694
Oil	Refinery Feedstocks		O4300
Oil	Refinery Gas		O4610
Oil	White Spirit	Not used for energy purposes	O4691
Peat	Briquettes		P1210
Peat	Milled Peat		P1120
Peat	Sod Peat		P1110
Renewables	Ambient Heat	Ambient heat captured by heat pumps	RA600
Renewables	Biodiesel	This includes conventional biodiesel (fatty acid methyl ester), as well as hydrotreated vegetable oil (HVO) and co-processed hydrotreated vegetable oil (CHVO).	R5220P & R5220B
Renewables	Bioethanol		R5210P & R5210B
Renewables	Biogas		R5300
Renewables	Biomass	Also referred to as <i>primary solid biofuels</i> in international reporting	R5110-5150_W60000RI
Renewables	Hydro		RA100
Renewables	Landfill Gas		R5311
Renewables	Renewable Waste	The renewable (biogenic) portion of waste	W6210
Renewables	Solar Photovoltaic		RA420
Renewables	Solar Thermal		RA410
Renewables	Wind		RA300

Table A10.2: Short energy balance 2024 (TWh)

Energy [TWh]	Coal	Peat	Oil	Natural Gas	Renewables	Non-Renew. Waste	Electricity	Total
Indigenous Production	-	1.49	-	10.19	20.97	1.81	-	34.46
Imports	1.90	-	99.47	39.51	3.67	-	5.27	149.82
Exports	0.16	-	12.79	-	0.14	-	0.21	13.30
Mar. Bunkers	-	-	1.42	-	-	-	-	1.42
Stock Change	1.54	0.00	-0.86	-	-0.02	-	-	0.66
Primary Energy Supply (incl. non-energy)	3.28	1.49	84.40	49.70	24.49	1.81	5.06	170.23
Primary Energy Requirement (excl. non-energy)	3.28	1.49	82.21	49.70	24.49	1.81	5.06	168.04
Transformation Input	1.87	-	26.46	28.47	2.90	0.97	0.60	61.26
Public Thermal Power Plants	1.87	-	1.73	25.40	2.72	0.97	-	32.68
Combined Heat and Power Plants	-	-	0.01	2.76	0.18	-	-	2.95
Pumped Storage Consumption	-	-	-	-	-	-	0.54	0.54
Briquetting Plants	-	-	-	-	-	-	-	-
Oil Refineries & other energy sector	-	-	24.72	0.31	-	-	0.06	25.09
Transformation Output	-	-	24.67	-	-	-	18.07	42.74
Public Thermal Power Plants	-	-	-	-	-	-	15.82	15.82
Combined Heat and Power Plants	-	-	-	-	-	-	1.89	1.89
Pumped Storage Generation	-	-	-	-	-	-	0.36	0.36
Briquetting Plants	-	-	-	-	-	-	-	-
Oil Refineries	-	-	24.67	-	-	-	-	24.67
Exchanges and transfers	0.20	-	-0.90	0.04	-13.55	-	13.51	-0.70
Electricity	-	-	-	-	-13.51	-	13.51	-
Heat	-	-	-	-	-	-	-	-
Other	0.20	-	-0.90	0.04	-0.04	-	-	-0.70
Own Use and Distribution Losses	-	-	0.81	0.65	-	-	3.43	4.89
Available Final Energy Consumption	1.61	1.49	80.90	20.63	8.04	0.84	32.61	146.13
Non-Energy Consumption	-	-	2.19	-	-	-	-	2.19
Final Non-Energy Consumption	-	-	2.19	-	-	-	-	2.19
Total Final Energy Consumption	1.66	1.49	78.12	20.63	8.15	0.84	32.86	143.74
Industry	0.44	-	4.09	10.89	2.19	0.84	6.93	25.38
Non-Energy Mining	-	-	0.06	0.00	0.01	-	0.20	0.27
Food, Beverages and Tobacco	-	-	0.97	3.64	0.16	-	1.19	5.96
Textiles and Textile Products	-	-	0.05	0.02	0.00	-	0.03	0.09
Wood and Wood Products	-	-	0.03	0.07	1.17	-	0.27	1.53
Pulp, Paper, Publishing and Printing	-	-	0.01	0.05	0.00	-	0.08	0.15
Chemicals and Man-Made Fibres	-	-	0.11	1.48	0.00	-	0.99	2.58
Rubber and Plastic Products	-	-	0.04	0.08	0.00	-	0.26	0.39
Other Non-Metallic Mineral Products	0.44	-	1.40	0.27	0.72	0.84	0.72	4.41
Basic Metals and Fabricated Metal Products	-	-	0.09	4.40	0.01	-	0.63	5.12
Machinery and Equipment n.e.c	-	-	0.04	0.11	0.00	-	0.17	0.33
Electrical and Optical Equipment	-	-	0.04	0.28	0.00	-	1.43	1.74
Transport Equipment Manufacture	-	-	0.01	0.02	0.00	-	0.03	0.06
Other Manufacturing	-	-	0.10	0.43	0.00	-	0.62	1.15
Construction	-	-	1.14	0.04	0.12	-	0.30	1.60
Transport	-	-	56.28	0.21	3.84	-	0.43	60.76
Road Freight	-	-	9.07	0.03	0.91	-	-	10.01
Light Goods Vehicle	-	-	3.24	-	0.32	-	0.01	3.56
Road Private Car	-	-	22.79	-	1.98	-	0.35	25.13
Public Passenger Services	-	-	1.32	-	0.13	-	0.01	1.46
Rail	-	-	0.50	-	0.02	-	0.05	0.57
Domestic Aviation	-	-	0.10	-	-	-	-	0.10
International Aviation	-	-	12.94	-	-	-	-	12.94
Fuel Tourism	-	-	0.54	-	0.05	-	-	0.59
Navigation	-	-	1.13	-	0.01	-	-	1.14
Unspecified (other road and pipeline)	-	-	4.64	0.19	0.43	-	-	5.26
Residential	1.22	1.49	12.60	5.58	1.41	-	8.37	30.66
Commercial/Public Services	0.00	-	2.35	3.95	0.71	-	16.63	23.65
Commercial Services	0.00	-	1.11	2.09	0.47	-	13.54	17.21
Public Services	-	-	1.24	1.86	0.25	-	3.09	6.44
Agricultural	-	-	2.56	-	-	-	0.49	3.05
Fisheries	-	-	0.24	-	-	-	-	0.24
Statistical Difference	-0.05	0.00	0.60	0.00	-0.10	-	-0.25	0.20

Table A10.3: Short energy balance 2024 (ktoe)

Energy [ktoe]	Coal	Peat	Oil	Natural Gas	Renewables	Non-Renew. Waste	Electricity	Total
Indigenous Production	-	128	-	877	1803	156	-	2963
Imports	163	-	8553	3397	316	-	453	12883
Exports	14	-	1100	-	12	-	18	1144
Mar. Bunkers	-	-	122	-	-	-	-	122
Stock Change	132	0	-74	-	-2	-	-	56
Primary Energy Supply (incl. non-energy)	282	128	7257	4274	2106	156	435	14637
Primary Energy Requirement (excl. non-energy)	282	128	7069	4274	2106	156	435	14449
Transformation Input	161	-	2275	2448	249	83	52	5267
Public Thermal Power Plants	161	-	149	2184	234	83	-	2810
Combined Heat and Power Plants	-	-	1	237	16	-	-	254
Pumped Storage Consumption	-	-	-	-	-	-	47	47
Briquetting Plants	-	-	-	-	-	-	-	-
Oil Refineries & other energy sector	-	-	2126	26	-	-	5	2157
Transformation Output	-	-	2121	-	-	-	1554	3675
Public Thermal Power Plants	-	-	-	-	-	-	1360	1360
Combined Heat and Power Plants	-	-	-	-	-	-	162	162
Pumped Storage Generation	-	-	-	-	-	-	31	31
Briquetting Plants	-	-	-	-	-	-	-	-
Oil Refineries	-	-	2121	-	-	-	-	2121
Exchanges and transfers	17	-	-77	3	-1165	-	1162	-60
Electricity	-	-	-	-	-1162	-	1162	-
Heat	-	-	-	-	-	-	-	-
Other	17	-	-77	3	-3	-	-	-60
Own Use and Distribution Losses	-	-	70	56	-	-	295	420
Available Final Energy Consumption	138	128	6956	1774	691	72	2804	12565
Non-Energy Consumption	-	-	188	-	-	-	-	188
Final Non-Energy Consumption	-	-	188	-	-	-	-	188
Total Final Energy Consumption	143	128	6717	1774	700	72	2825	12360
Industry	38	-	352	936	188	72	596	2182
Non-Energy Mining	-	-	5	0	1	-	18	23
Food, Beverages and Tobacco	-	-	84	313	14	-	102	513
Textiles and Textile Products	-	-	4	1	0	-	3	8
Wood and Wood Products	-	-	2	6	100	-	23	131
Pulp, Paper, Publishing and Printing	-	-	1	5	0	-	7	13
Chemicals and Man-Made Fibres	-	-	10	127	0	-	85	222
Rubber and Plastic Products	-	-	3	7	0	-	23	33
Other Non-Metallic Mineral Products	38	-	121	24	62	72	62	379
Basic Metals and Fabricated Metal Products	-	-	8	378	1	-	54	440
Machinery and Equipment n.e.c	-	-	4	10	0	-	14	28
Electrical and Optical Equipment	-	-	3	24	0	-	123	150
Transport Equipment Manufacture	-	-	1	2	0	-	3	5
Other Manufacturing	-	-	9	37	0	-	53	99
Construction	-	-	98	3	10	-	26	138
Transport	-	-	4839	18	330	-	37	5224
Road Freight	-	-	780	2	78	-	-	861
Light Goods Vehicle	-	-	278	-	27	-	1	306
Road Private Car	-	-	1960	-	170	-	31	2161
Public Passenger Services	-	-	114	-	11	-	1	126
Rail	-	-	43	-	2	-	5	49
Domestic Aviation	-	-	9	-	-	-	-	9
International Aviation	-	-	1113	-	-	-	-	1113
Fuel Tourism	-	-	46	-	5	-	-	51
Navigation	-	-	97	-	1	-	-	98
Unspecified (other road and pipeline)	-	-	399	16	37	-	-	452
Residential	105	128	1083	480	121	-	720	2636
Commercial/Public Services	0	-	202	340	61	-	1430	2034
Commercial Services	0	-	95	180	40	-	1164	1480
Public Services	-	-	107	160	21	-	266	554
Agricultural	-	-	220	-	-	-	42	262
Fisheries	-	-	21	-	-	-	-	21
Statistical Difference	-4	0	52	0	-9	-	-21	17

Appendix 11: Version control

Version	Date	Notes
0.1	11/12/2025	ES draft
0.9	15/12/2025	Pre-release embargoed version
1.0	16/12/2025	Publication
1.1	18/12/2025	Spelling error in foreword and formatting of subscripts in MtCO ₂ eq corrected.



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